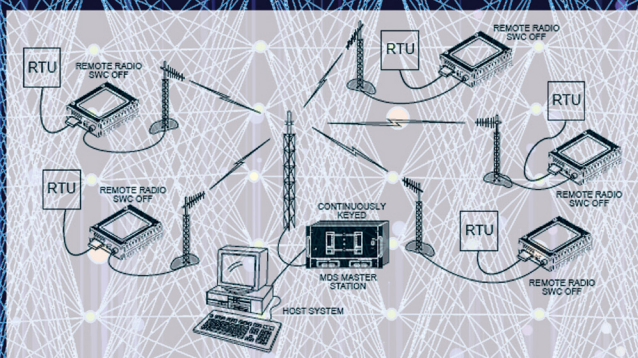


Artificial Intelligence in Wireless Sensors and Instruments Networks and Applications



Halit Eren



CRC Press
Taylor & Francis Group

Artificial Intelligence in Wireless Sensors and Instruments

This book heralds a new era in instrumentation and measurements. It combines artificial intelligence (AI) and wireless communications technologies with instrumentation and measurement systems to function as a single unit. AI has advanced considerably due to deep learning utilizing artificial neural networks, availability of large and curated datasets, implementation of a new generation of fast processors having millions of transistors in chips, advanced algorithms, competitive commercial interests, and interests of governments to gain advantages. At the same time, new and highly advanced wireless technologies open new frontiers in communication systems, both technologically and in terms of application aspects. Advanced technologies such as 5G and 6G networks enable easy use of communication systems by billions of people as well as by billions of machine-to-machine systems.

In this book, the communication principles are explained and the implementation of AI on wireless networks is discussed. Many examples are provided. The author discusses instruments and instrumentation networks, modern sensors, and transducers in detail.

AI is the technology humans have created where the machines do not only assist us but also think for us creatively in some cases, excelling humans thinking and reasoning. This book includes a chapter explaining how this is done, backed up with more than 50 figures. The security issues, fairness, efficiency, and social impact and acceptance of AI are highlighted. As explained in this book, AI and wireless communications are changing our lives in many ways, including entertainment, games, social interactions, medicine and healthcare, R&D, automated living, intelligent transport systems, finance and economy, and the Internet of Things.



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Preface

There is hardly a day that passes by, and we do not hear about intelligence. The recent progress in digital hardware and software advanced artificial neural network (ANN) to form the basis of advanced artificial intelligence (AI) through deep learning (DL) algorithms. Most of the mathematical foundations are known for decades, but those methods combined with the skillfully developed algorithms led to AI. AI is changing our lives in many ways in the economy, entertainment, games, social interactions, healthcare, R&D, transport, automated living, and so on. With its powerful capabilities, its functionalities have been extended to be almost ubiquitous.

AI has advanced considerably due to the availability of large mostly curated datasets. The preparation and curation of datasets requires complex data handling. Availability of fast computers, cloud, and fog computing allow us to use share computing for AI implementations. Also, advanced processors such as GPUs (Graphics Processing Units) ease the proliferated use on the AI.

AI has a wide spectrum of applications ranging from technology and medicine to economic and political decision making. Applications of AI are truly transforming our world and impacting all facets of society. As explained in this book, AI is also impacting measurements, instruments, instrumentation, and communication systems.

Wireless communication has advanced over the past two decades to find ubiquitous applications all around the globe. Initially, the technology concentrated for person-to-person communications, but today, there are millions of devices connected via internet and the wireless communication forms the backbone of it.

Instruments are essential in industrial operations, environmental applications, research and development, transportation, military, border security, health and medicine, space explorations, and in our daily lives. Instruments can be classified to be analog, digital, the combination of the two, and virtual instruments. Digital instruments are based on digital processors, which provide signal processing and data handling which can be achieved by software and hardware implementations. An advantage of digital systems is that data generated by instruments can be managed by computers and other digital platforms. This provides a wide range of possibilities for complex data processing, ease in communications and data transfer, multiplexing, data storage, visual displaying, efficient calibrations, enabling broadcasting of information, application of internet of things (IoTs), data mining, and AI.

This book explains the chain of technology that involves sensors, instruments, wireless communications, and AI in a holistic manner. This book has three sections: each being dedicated to the major group of topics. [Chapter 1](#) discusses AI and the supporting technologies which involve data handling, classification, clustering, regression, data efficiency, and AI in society. [Chapter 2](#) is dedicated wireless technologies starting from the basic principles to complex networks such as the 6G and space-based communication networks. [Chapter 3](#) concentrates on instruments, measurements, and sensors bringing all the three topics together as a single comprehensive entity. Some applications are given briefly in [Chapter 4](#) leading to needs of a series of books on the applications of AI, Wireless networks, and measurement systems.



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I would like to thank to my colleagues at Hacettepe University, Türkiye, for making the library and internet facilities available. Combining of state-of-art-of-knowledge in that university has laid the foundation background of this book. I would like to express my special gratitude to Professor Atilla Yilmaz for his encouragement to write this book. I met him at Wisconsin University during his sabbatical leave to find out that he narrowly missed to be my student in early years of my academic career and his first years of his studies.

I would like to thank my countless number of colleagues who listened to me patiently as the writing of this book progressed. My undergraduate and postgraduate fellows graduated from Sheffield University formed a WhatsApp group, and I am proud to be a member of that group. I have posted sections of this book and they supplied valuable comments for improvement in contents, text, and writing styles. In particular, I would like to thank my colleague Cahit Buckun who had deep interest in AI, and traced many up-to-date scholarly papers on the related topics. Many thanks to Professor Lance Fung who currently is the IEEE Asian Pacific Region (R10) Director, 2023–2024, for providing valuable discussions and support in between his extremely busy time schedules. Dr. Deniz Savas and Dr. Halit Sanli constantly checked on my progress and encouraged me to push ahead relentlessly. I would like to thank Denise Ann Grisdale for making suggestions on the style of writing. Finally, thanks to my kids Emre Eren and Pelin Eren, both trained engineers, for patiently listening to me and advising me on related experiences in their workplaces.

I am deeply saddened for the absence of late Professor John G. Webster with whom we collaborated for many years and undoubtedly would have made valuable contributions in writing this book. We had numerous successful books published together on measurements, instrumentation, sensors, and related topics.

Disclaimer: I would like to express that no form of AI (such as the ChatGPT or others) has been consulted or used in producing text in any part of this book.



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Dr. Halit Eren received BEng 1973, MEng 1975 in electrical engineering, and a PhD in 1978 in control engineering all at the University of Sheffield, UK. He obtained an MBA from Curtin University in 1999, majoring in International Management. Dr. Eren worked in Etibank as project engineer on automation and instrumentation in copper plants. He held the lecturer positions in Hacettepe University-Beytepe Campus, 1980–1981, and assistant professor at the Middle East Technical University-Gaziantep Campus 1982. He worked at the Curtin University between 1983 and 2015 first at the School of Mines Kalgoorlie, and then School of Electrical Engineering, Bentley Campus, Perth, Western Australia. His duties involved research and teaching in Control Systems, Digital Transmission and Interface Design, Process Control and Instrumentation, and Engineering Management. Dr. Eren has attracted numerous research grants from the governments and industry. He also acted as consultant for many private and governmental organizations. Dr. Eren supervised and graduated PhD and Masters-Degree students. He served as head of Department of Electronic and Electrical Engineering at Curtin University. He has held the position of adjunct senior research fellow since 2015.



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List of Abbreviations

AAI	All Round Artificial Intelligence
AC	Actor Critic
AC	Approximate Computation
AC	Audio Classification
ACK	Acknowledge
ACO	Ant Colony Optimization
AE	Auto-Encoder
AE-DNN	Auto-Encoder DNN
A/D	Analog-to-digital converters
AH	Agglomerative Hierarchical
AGI	Artificial General Intelligence
AGC	Automatic Gain Control
ALOHA	Areal Locations of Hazardous Atmospheres
ALU	Arithmetic logic unit
AI	Artificial Intelligence
ANI	Artificial Narrow Intelligence
ANN	Artificial Neural Network
AM	Amplitude Modulation
ANSI	American National Standards Institute
APSK	Amplitude Phase Shift Keying
AR	Auto Regressive
ARMA	Auto Regressive and Moving Average
ASI	Artificial Super Intelligence
ASIC	Application Specific Integrated Chips
ASK	Amplitude Shift Keying
BGAN	Broadband Global Area Networks
BiCMOS	Bipolar Complementary Metal Oxide Semiconductor
BIR	Balanced Iterative Reduction
BJT	Bipolar Junction Transistor
BLE	Bluetooth Low Energy
BLR	Bayesian Linear Regression
BM	Boltzmann Machine
BN	Bayesian Network
BP	Back Propagation
BP-CNN	Belief Propagation-CNN
bps	Bits per second
BPSK	Binary Phase Shift Keying
BT-WPAN	Bluetooth Wireless Personal Area Network
BW	Bandwidth
CBC	Centroid-Based Clustering
CBRP	Cluster Based Routing Protocol
CBS	Constraint-Based Classification

CCD	Charge-Coupled Device
CCK-OFDM	Complimentary Code Keying Orthogonal Frequency Division Multiplexing
CDM	Code Division Multiplexing
CDMA	Code Division Multiple Access
CI	Computational Intelligence
CI	Cyber Infrastructure
CIM	Computing-in-Memory
CISC	Complex Instruction Set Computer
CM	Channel Measurement
CMOS	Complementary Metal Oxide Semiconductors
CNN	Convolutional Neural Network
CPU	Central Processor Units
CR	Cognitive Radio
CRC	Cyclic Redundancy Check
CRN	Cognitive Radio Network
CS	Compressed Sensing
CSI	Channel State Information
CSMA	Carrier Sense Multiple Access
CSMA/CD	Carrier Sense Multiple Access with Collision Detection
CU	Centralized Unit
CSV	Comma Separated Value
CVD	Chemical Vapor Deposition
CVNN	Continuous Valued NN
CW	Continuous-Wave
DAI	Distributed Artificial Intelligence
DA	Deep Autoencoder
D/A	Digital-to-Analog Converter
DBM	Deep Boltzmann Machine
DBN	Deep Belief Networks
DBC	Density-Based Clustering
DBSC	Density-Based Special Clustering
DBC	Distribution-Based Clustering
DC	Document Classification
DDBM	Denoising Diffusion Probabilistic Model
DDL	Device Description Language
DE	Differential Equation
DEFT	Double Ended Tuning Fork
DFF	Deep Feed Forward
DFS	Depth First Search
DFT	Digital Fourier Transforms
DL	Deep Learning
DLL	Dynamic Link Libraries
DM	Diffusion Model
DM	Data Memory
DM	Discriminative Methods

DMA	Direct Memory Access
DNN	Deep Neural Network
DPSK	Differential Phase-Shift Keying
DRO	Dielectric Resonator Oscillator
DRL	Deep Reinforcement Learning
DT	Decision Tree
DQPSK	Differential QPSK
DSB	Double Sideband
DSCDMA	Direct Sequence CDMA
DSMA	Direct Sequence Multiple Access
DSMA	Data Sense Multiple Access
DSN	Deep Space Network
DSN	Distributed Sensor Networks
DSP	Digital Signal Processor
DSS	Dynamic Spectrum Sharing
DSSS	Direct Sequence Spread Spectrum
DSSS	Deep Space Surveillance Systems
DT	Decision Tree
DTR	Decision Tree Regression
DU	Distributed Unit
EA	Encoder and Autoencoder
EC	Edge Computing
ECG	Electrocardiogram
EDAS	Ethernet Data Acquisition System
EDGE	Enhanced Data GSM Environment
EGSM	Extended GSM
EED	End to end Encoder/Decoder
EEoI	Early Exit of Inference
EHF	Extremely High Frequency
ELN	Extreme Learning Machine
eMBB	Enhanced Mobile Broadband
EM	Expectation Maximization
EMC	Electromagnetic Compatibility
EMI	Electromagnetic Interference
EN	Elastic Net
FB	Feature Based
FC	Fuzzy Clustering
FCC	Fuzzy C-Mean Clustering
FCC	Federal Communications Commission
FDD	Frequency Division Duplexing
FDM	Frequency Division Multiplexing
FDMA	Frequency Division Multiple Access
FFD	Full Functional Device
FFBPNN	Feedforward Back-Propagation Neural Network
FeFET	Ferroelectric Field Effect Transistor
FET	Field-Effect Transistors

FFT	Fast Fourier Transforms
FHMA	Frequency Hopped Multiple Access
FHR	Frequency Hopping Recognition
FHSS	Frequency Hopped Spread Spectrum
FIR	Finite Impulse Response
FM	Frequency Modulation
FNN	Feedforward Neural Network
FSDMA	Hybrid FDMA/CDMA
FSK	Frequency Shift Keying
GA	Genetic Algorithm
GAN	Generative Adversarial Networks
GAN	Global Area Network
Gbps	Giga bits per second
GCN	Graph Convolutional Network
GEO	Geostationary Orbit
GFSK	Gaussian Frequency-Shift Keying
GM	Generative Methods
GNB	Gaussian Naïve Bayes
GMM	Gaussian Mixture Model
GO	Genetic Optimization
GPRS	General Packet Radio Services
GPS	Global Positioning System
GPT	Generative Pretrained Transformer
GPU	Graphic Processing Unit
GRU	Gated Recurrent Unit
GSO	Geostationary Satellite Orbits
GSM	Global System for Mobile Communications
HAS	Harmony Search Algorithm
HBC	Hierarchical Based Clustering
HC	Hierarchical Clustering
HDL	Hardware Description Language
HF	High Frequency
HCI	Host Controller Interface
HIPERLAN	High Performance Radio Local Area Networks
HITL	Human-in-the Loop
HN	Hopfield Network
HNN	Hopfield Neural Network
HPO	Hyper Parameter Optimization
IC	Image Classification
IC	Integrated Circuit
ICM	Intelligent Channel Measurement
ID	Identifications
IEEE	Institute of Electrical and Electronic Engineers
IF	Intermediate Frequency
I²C	Inter Integrated Circuit
IIR	Infinite Impulse Response

Internet	INTERconnected NETwork
I/O	Input/Output
IP	Internet Protocol
IoT	Internet of Things
ITU	International Telecommunications Union
KAA	Kernel Approximation Algorithm
KD	Knowledge Distillation
KF	Kalman Filter
KM	K-Means
KNN	K-Nearest Neighbor
IR	Infrared
IrDA	Infrared Data Association
ISFET	Ion-Sensitivity Field-Effect Transistors
ISOMAP	Isometric Mapping
ISP	Internet Server Provider
IS²	Integrated Smart Sensor
LAN	Local Area Network
LDA	Linear Discrimination Analysis
LDMOS	Lateral-Diffused Metal-Oxide Semiconductor
LED	Light-Emitting Diode
LEO	Low Earth Orbit
LF	Low Frequency
LIDAR	Laser Based Light Detection
Li-Fi	Light Fidelity
LMI	Limited Memory Intelligence
LMMSE	Linear Minimum Mean Square Error
LMP	Link Management Protocol
LNA	Low-Noise Amplifiers
LNN	Logic Learning Machine
LoRa	Long Range
LO	Local Oscillator
LOS	Line-of-Sight
LR	Lasso Regression
LR	Logistic Regression
LSB	Least significant bit
LSTM	Long Short-Term Memory
LWIM	Low-Power Wireless Integrated Microsensors
MAB	Multi Armed Bandit
MAC	Medium Access Control
MAN	Metropolitan Area Networks
MANET	Mobile Area Network
Mb	Megabit
Mbps	Mega bits per second
MCU	Microcontroller Units
MDP	Markov Decision Process
MEC	Mobile Edge Computing

MEMS	Micro-Electro-Mechanical Systems
MEM	Maximum Entropy Model
MEOMS	Micro-Electro-Optical Systems
MESFET	Metal Semiconductor FET
MF	Medium Frequency
MIL-STD	Military Standard
MIM	Metal-Insulator Metal
MIMO	Multiple Input Multiple Output
m-MIMO	massive Multiple Input Multiple Output
MIPS	Million Instructions Per Second
MKC	Multiple Kernel Clustering
MKL	Multiple Kernel Learning
ML	Machine Learning
MLP	Multi-Layer Perceptron
MMs	Markov Models
MMDS	Multipoint and/or Multichannel Distribution System
MMTC	Mass Machine Type Communication
MNB	Multinomial Naïve Bayes
Modem	MOdulator-DEModulator
MOFs	Metal Organic Frameworks
MOS	Metal Oxide Semiconductor
MOSFET	Metal Oxide Semiconductor Field Effect Transistors
MR	Modulation Recognition
MRI	Magnetic Resonance Imaging
MSB	Most Significant Bit
MSE	Mean Square Error
MSK	Minimum Shift Keying
MSP	Multi-step Prediction
MST	Micro Systems Technology
MTC	Machine Type Communication
M2M	Machine to Machine
NACK	Not Acknowledged
NAS	Neural Architecture Search
NB	Naïve Bayes
NBR	Native Binomial Regression
NCAP	Network Capable Application Processor
NEON	National Ecological Observatory Network
NF	Noise Figure
NFC	Near Field Communication
NN	Neural Network
NOS	Network Operating Systems
NTP	Normal Temperature and Pressure
NVM	Non-Volatile Memory
OBEX	Object Exchange
OFDM	Orthogonal Frequency Division Multiplexing
OFDMA	Frequency Division Multiple Access

O-OFDM	Optical Orthogonal Frequency Division Multiplexing
OOI	Ocean Observation initiatives
OOK	On-Off Keying
Op amp	Operational Amplifier
OPSK	Octonary Phase-Shift Keying
OQPSK	Offset QPSK
OR	Ordinary Regression
ORAN	Open Radio Access Networks
OS	Operating System
OSAT	Semiconductor Assembly and Test
PA	Power Amplifiers
PAI	Predictive AI
PAM	Pulse Amplitude Modulation
PAN	Personal Area Networks
PBL	Probability Bayes Learning
PCA	Principal Component Analysis
PCB	Printed Circuits Boards
PCM	Pulse Code Modulation
PCR	Principal Component Regression
PCR	Polymerize Chain Reaction
PDA	Personal Digital Assistance
PDCH	Packet Data Channel
PDM	Pulse Duration Modulation
PDF	Portable Document Format
PHEMT	Pseudomorph High Electron Mobility Transistor
PIC	Power IC
PIM	Process-in-Memory
PIR	Passive Infrared
PLL	Phase-Locked Loop
PM	Power Management
PM	Program Memory
PM	Phase Modulation
PMP	Post-Metal Programming
PnP	Plug-and-Play
PN	Pseudo-Noise
PPM	Pulse Phase Modulation
PPP	Point to Point Protocol
PR	Packet Radio
PR	Poisson Regression
PR	Polynomial Regression
PROM	Programmable Read-Only-Memory
PSD	Position Sensitive Detector
PSK	Phase Shift Keying
PSO	Particle Swam Optimization
PSoC	Programmable System-On-Chip
PWM	Pulse Width Modulation

PZT	Lead Zirconate Titanate.
QDA	Quadratic Discrimination Analysis
QoD	Quality of Data
QoR	Quality of Results
QoS	Quality of Service
QPSK	Quaternary Phase-Shift Keying
QR	Quantile Regression
RAM	Read Access Memory
RAN	Radio Access Networks
RBF-NN	Radial Basis Function Neural Network
RBM	Restricted Boltzmann Machines
RBN	Radical Basis Networks
R&D	Research and Development
RF	Random Forest
RF	Radio Frequency
RFCMOS	Radiofrequency CMOS
RFD	Reduced Functional Device
RFI	Radio Frequency Interference
RFID	Radiofrequency Identification
RFR	Random Forest Regression
RISC	Reduced Instruction Set Computer
RL	Reinforcement Learning
RLHF	Reinforcement Learning from Human Feedback
RM	Reinforcement Methods
RMI	Reactive Machine Intelligence
RNN	Recurrent Neural Network
ROM	Read Only Memory
RR	Ridge Regression
RTD	Resistive Temperature Detector
RTL	Register Transfer Level
SAI	Self-Aware Intelligence
SAW	Surface Acoustic Wave
SC	Subtractive Clustering
SCADA	Supervisory Control and Data Acquisition
SCI	Serial Communications Interface
SDE	Stochastic Differential Equation
SDMA	Space Division Multiple Access
SDN	Software Defined Network
SDP	Service Discovery Protocol
SDS	Smart Distributed System
SGD	Stochastic Gradient Descent
SGF	Super High Frequency
S/H	Sample-and-Hold
SI	Système International d'Unités
SIG	Special Interest Group
SL	Supervised Learning

S/N	Signal-to-Noise
SNR	Signal-to-Noise Ratio
SoC	System-on-chip
SOM	Self-Organizing Maps
SR	Stepwise Regression
SRAM	Standby RAM
SRAM	Static Random Access Memory
SSB	Single Sideband
SSIM	Smart Sensors and Integrated Microsystem
SSMA	Spread Spectrum Multiple Access
STIM	Smart Transducer Interface Module
STIN	Satellite Terrestrial Integrated Network
STP	Standard Temperature and Pressure
SVD	Singular Value Decomposition
SVM	Support Vector Machine
SVR	Support Vector Regression
SWAMS	Smart Wireless Autonomous Microsystems
TBC	Text Based Classification
TCDMA	Time Division CDMA
TCP	Transmission Control Protocol
TCP/IP	Transmission Control Protocol/Internet Protocol
TCS	Telephony Control Signal
TDD	Time Division Duplexing
TDFH	Time Division Frequency Hopping
TDM	Time Division Multiplexing
TDMA	Time Division Multiple Access
TDNN	Time Division NN
TDOA	Time Difference of Arrival
TEDS	Transducer Electronic Data Sheet
TEM	Traverse Electromagnetic
TFT	Thin Film Transistor
TR	Tobit Regression
TSF	Tree Scatternet Formation
TTL	Transistor-Transistor Logic
TxRx	Transceiver (Transmitter and Receiver)
UART	Universal Asynchronous Receiver Transmitter
UASRT	Universal Synchronous/Asynchronous Receiver/Transmitter
UAV	Unmanned Aerial Vehicle
UBC	User Based Classification
UE	User End
UHF	Ultra High Frequency
UMTS	Universal Mobile Telephone Service
URLLC	Ultra Reliable Low Latency Communication
UV	Ultraviolet
UVLC	Underwater Visible Light Communication
VAE	Variational Auto Encoder

VCO	Voltage-Controlled Oscillators
VCR	Voltage Coefficient
VHF	Very High Frequency
VLC	Visible Light Communication
VLf	Very Low Frequency
VLSI	Very Large Scale Integrated
VPN	Virtual Private Network
VSDM	Variable Slope Delta Modulation
VSWR	Voltage Standing Wave Ratio
www	World Wide Web
WANET	Wireless Ad Hoc Network
WAP	Wireless Application Protocol
WAP	Wireless Access Point
WCDMA	Wideband CDMA
WDSN	Wireless Distributed Sensor Networks
WECA	Wireless Ethernet Capability Alliance
WiMAX	Worldwide Interoperable Multiple Access
WINS	Wireless Integrated Network Sensors
WISP	Wireless Internet Service Providers
WLAN	Wireless Local Area Networks
WMAN	Wireless Metropolitan Area Networks
WPAN	Wireless Personal Area Network
WSN	Wireless Sensor Network
WUR	Wake Up Radio
XAI	Explainable Artificial Intelligence
YIG	Yttrium Iron Garnet
ZC	Zigbee Coordinator
ZED	Zigbee End Device
ZR	Zigbee Router
$\pi/4$ DQPSK	$\pi/4$ Differential QPSK
$\Sigma\Delta$	Sigma-Delta
μC	Microcontroller
μP	Microprocessor
8-QAM	8-Quadrature Amplitude Modulation
16-QAM	16-Quadrature Amplitude Modulation
3GPP	Third Generation Partnership Project

1 Artificial Intelligence

1.1 INTRODUCTION

The recent progress in digital hardware and software has helped artificial neural network (ANN) to form the basis of advanced artificial intelligence (AI) through deep learning (DL) techniques and algorithms. AI requires machine intelligence (MI). But not all MIs is AI as we know today. Particularly over the past four years from 2020, AI has progressed remarkably well such that it has been claimed to have the potential to result in the next industrial revolution. AI is changing our lives in many ways in the economy, entertainment, games, social interactions, healthcare, R&D, transport, automated living, and many others. With its powerful capabilities, its functionalities have been extended to be almost ubiquitous.

AI has advanced considerably due to ANNs, availability of large mostly curated datasets, and implementation of fast processors and advanced algorithms. Advanced graphics processing units (GPUs) operate in parallel structures to perform computations on large and massive datasets.

AI has one advantage that humans do not have. That is, the retaining vast amount of information and being able to make use of that information in a holistic manner. For example, AI in natural language processing (NLP) can learn and store decades of knowledge and information in one place. This can be compared with a person completing degree courses in universities and remembering all of them when demanded. However, the machine-based AI still has a long way to go since it lacks human elements such as feelings, sympathy, empathy, social elements, judgments, survival skills, instincts, and so on. AI is developing human-like behavior such as responding to external stimuli (touch, smell, see, taste, hear), self-thinking, self-decision making, and imagining.

AI has a wide spectrum of applications ranging from technology and medicine to economic and political decision-making. Applications of AI are truly transforming our world and impacting all facets of society. It is also impacting instruments, instrumentation, measurement, and communication systems. In this chapter, we give an overview of applied AI and its usage in wireless communication systems, sensor networks, and instrumentation.

1.2 ARTIFICIAL INTELLIGENCE DEFINITIONS

Intelligence, learning, and training are the words used in describing AI. Yet these words have different meanings depending on the context they are used in. In this section, we will clarify these words.

1.2.1 LEARNING, TEACHING, AND TRAINING

Learning, teaching, and training are often encountered words in AI literature; therefore, it is worth clarifying these points.

Learning is “*acquisition of knowledge or skills through study, experience, or being taught.*” Learning can be done through one’s own experiences or being taught by using prior knowledge. Machine learning (ML) is the ability of machines to learn new things either from human beings or among themselves.

Teaching is “*to give knowledge, to instruct, to train someone.*” Teaching is a well-established discipline as it involves all of us. There are many methods of teaching, and they can be teacher-centric, learner centric, contents focused, technology-based, instruction-based, and so on.

Training is “*the process of learning the skills one need to do a particular job or activity.*”

ML is “*the study of creating computer systems that can improve their own ability to carry out tasks by analyzing new data.*” It is the process of creating or improving a computer representation of a system or process from the available data.”

1.2.2 THINKING AND REASONING

Oxford English dictionary defines thinking as “the process of reasoning about something.” Reasoning is “the process of thinking about things in a logical way.”

Thinking is the activity of the mind to consider something. It involves ideas, opinions, and reasoning about things. Thinking is necessary for reasoning, but reasoning may not be necessary for thinking. Thinking and reasoning have been the province of philosophy for millennia. Today it is within the well-established fields of psychology, cognitive science, and cognitive neuroscience.

1.2.3 INTELLIGENCE

The definition of *intelligence* is a controversial one. Neither has it been fully understood nor agreed upon. Intelligence has been a hot subject of discussion since the dawn of civilization. There are various definitions depending on the social elements of the source and contexts they are described for. Cambridge dictionary defines intelligence as: “*The ability to learn, understand, and make judgements or have opinions that are based on reason,*” whereas Miriam-Webster dictionary defines as “*Intelligence is the ability to learn or understand or deal with new or trying situations.*” Some fundamental differences can be observed between these two definitions.

There are various other definitions of “intelligence” such as the collection of information from rivals. Depending on the discipline one is practicing, other definitions concentrate on various aspect of intelligence, such as emotional intelligence, spatial intelligence, musical intelligence, spiritual intelligence, survival intelligence, and so on.

We will not delve into abstractions and definitions of intelligence in this book except to make some clarifications when necessary for a sound understanding of *intelligence* in the AI technology. Intelligence requires adaptivity, creativity, critical thinking, communications, decision making, forming concepts, intuitive judgments, language usage, logic, planning, problem solving, reasoning, retaining knowledge, self-awareness, and use of knowledge.

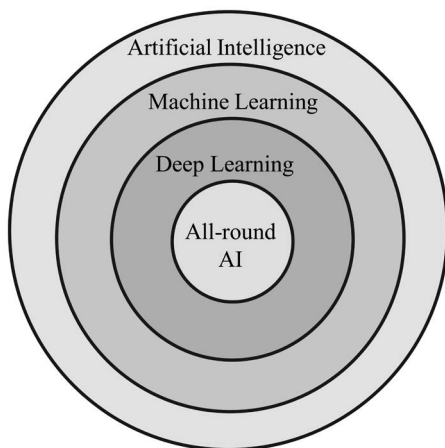


FIGURE 1.1 Artificial intelligence is an umbrella term that encases many different types.

AI is the intelligence in computers, devices, machines, or other human configured systems to make decisions and to provide learned knowledge. Artificial intelligence is an umbrella term it has evolved over many decades. AI has many as shown in [Figure 1.1](#). DL based on ANN is the most successful branch AI outpacing human intelligence in some cases. Today's AI is not close to human intelligence as it still lacks many of human traits such as psychological and social traits, emotions, self-awareness, motivations, ideas, opinions, empathy, and so on. But this does not mean that AI will not reach human intelligence in the future, and that the future can be very near.

AI is developing toward *all-round artificial intelligence* (AAI) and *distributed artificial intelligence* (DAI). AAI focuses on multiple processing capabilities just like human brain does. DAI focuses on the development of agents and systems that can emulate the intellectual, social, interactive abilities of human beings. Distributed AI is based on multiagent system that can operate together to organize their structures, assign tasks, and exchange knowledge as in the way human society operates.

1.3 DATA AND DATA HANDLING

AI makes use of data and datasets. *Data* is the digital information expressed in binary form by having digits of 1s and 0s. For example, binary equivalent of decimal 14 is (1110) or binary equivalent of letter A is (01000001). Computers can process only digital information.

Data can be visual, textual, or numerical which could assume discrete or continuous values. Hence, data describes quantities, characters, shapes, statistics, and symbols that can be processed by digital systems.

Data can be raw data, primary data, metadata, big data, or massive data. *Raw data* is unprocessed data which has not been coded, formatted, or analyzed. *Primary data* is the data collected from the original source in its unprocessed form. *Metadata* is the data that gives information on some other data. *Big data* is large datasets that contain patterns, trends, and multiple information. *Massive data* is often used term

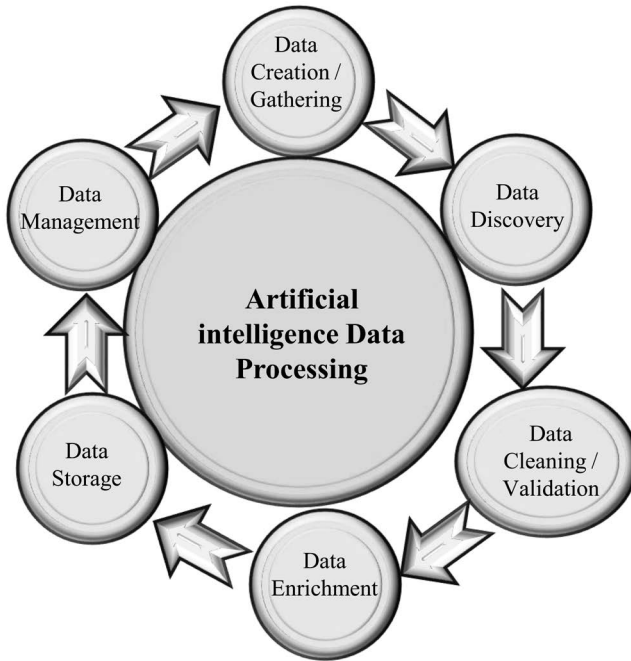


FIGURE 1.3 AI data preparation requirements.

Data processing requires data management, identification, cleaning, validation, enrichment, tracking, indexing, transformation, storing, and reuse of the data as shown in [Figure 1.3](#).

Referring to [Figures 1.2](#) and [1.3](#), the data handling process has two major elements, such as data management and data maintenance.

Data management requires facilitation and/or manipulation of different modalities of data. Management of data requires modification of annotation, labeling, serialization, and storing of the data. When dealing with dynamic and evolving or changing datasets, the large-scale nature of the data can make data management very complex and demanding.

Dataset creation, discovery, and selection can be an arduous and error-prone process due to many factors including some factors within the data itself. Datasets are susceptible to missing data, outliers, adversarial data, and poisoned data. In many cases AI models may have to be continuously updated with new data to avoid degradation and errors in their performances. The lack of ownership, absence of transparency, and incomplete documentation can seriously deteriorate the quality of data (QoD).

Data preprocessing is essential for minimizing data errors, for finding missing data, identifying outliers and adversarial data, and to gather knowledge on biases. The model accuracy may also be affected during the data processing in the data features, missing data, and data splitting strategies.

Data validation, cleaning, and enrichment is necessary since the original data often changes as it evolves over time. AI models are complex black boxes therefore it becomes uncertain if the learned model is still effective in solving the intended problems. Data validation processes need to check for data errors and to monitor those errors constantly to ensure integrity⁽¹⁻⁹⁾.

Once the data is prepared, then the AI trainings models can be implemented. The next steps are the model deployment, model testing, model drift awareness, performance matrices, and model governance. A typical procedure for data implementation in AI is illustrated in [Figure 1.4](#). These points are explained throughout this chapter.

1.3.2 DATASETS, BIG AND MASSIVE DATA

Datasets are a collection of sets of data information in an organized manner. A **database** is a collection of multiple datasets that may be related or unrelated to each other.

Big data is large datasets generated in different ways, such as industrial processes, sensors, social media, R&D, organizational operations, transactional activities, archives holding historical information, clouds held information, health and medical related activities, customer data bases, and many others. For example, large scientific research projects generate massive data expressed in multiple petabytes (10^{15} bytes or 1000 terabytes, or 1 million gigabytes) of data every year. Some

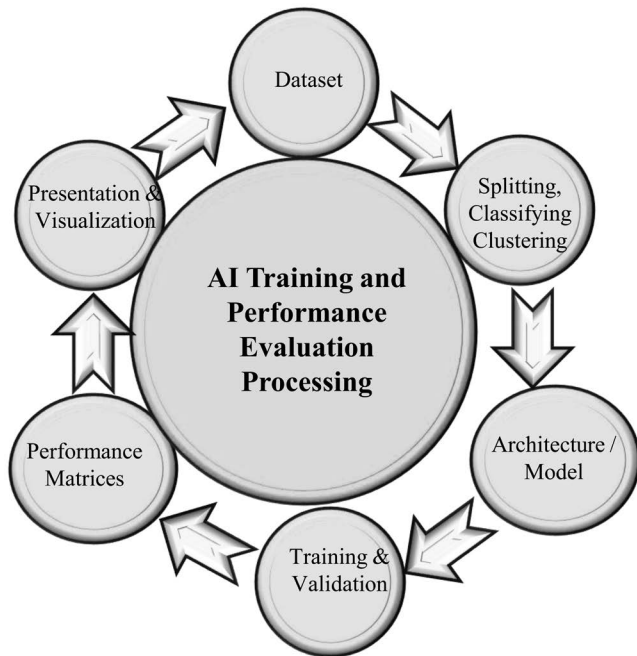


FIGURE 1.4 Implementation of data in AI.

examples of large research projects are Large Synoptic Survey Telescope, Large Hydron Collider, Laser Interferometry Gravitational Wave Observatory, National Ecological Observatory Network (NEON), Square Kilometer Array space observations, and Ocean Observation Initiative (OOI). Just to provide scale of these projects, OOI for example currently serves data from 57 platforms and 31 mobile assets carrying 1227 instruments, providing over 25,000 science datasets. It is backed up with 100,000 scientific and engineering data products.

In research-related applications of big data, the requested information by the researchers is delivered timely and in complete form. The demand can be generated from anywhere in the world thus supported by reliable and efficient communication systems. The equipment, infrastructures, and facilities used for processing the data are more likely to be different. For this, powerful distributed and networked computers form the backbone of the computer infrastructure. Figure 1.5 illustrates a few of the big data sources.

Two examples of big data are cloud data and social data. *Cloud data* which is a major data source where contents are stored on cloud servers. A feature of cloud-based big data is that the users' preferences concerning specific content are often similar and correlated. *Social data* constitutes a large volume of data online social networks. The use of the mobile Internet makes convenient multimedia communications ubiquitously accessible for everyone. The amount of social data has reached an astonishing magnitude and is set to exhibit an increasing trend in the coming years.

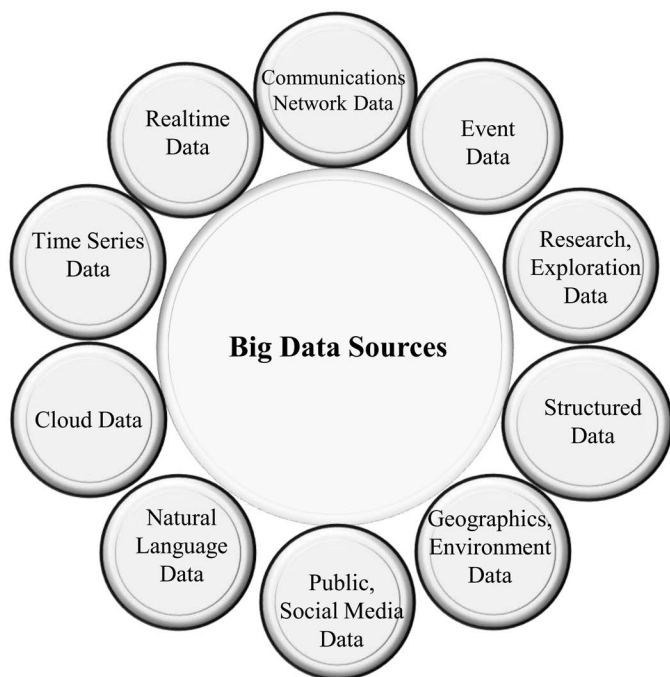


FIGURE 1.5 Typical big data sources.

Recently, AI has advanced considerably because of ANNs, availability of large mostly curated datasets, and implementation of fast computing processors and advanced algorithms mostly backed up by mathematical descriptions. Advanced GPUs are processing assemblies operate in parallel to perform computations on large and massive datasets. This leads to the emergence of new ML hardware and development of diverse range of models in the rapid advancements of the AI.

Big data existed for decades, but the use of it on a large scale began in 2019 by Open AI. The announcement of GPT-2 had a rolling effect in the advancement of the use of big data. GPT-2 is a language model trained on textual datasets using text scraped from the web. GPT-3, released in May 2020, was a larger ML model with 175 billion parameters (weights). GPT-3 is known to include training data of entirety of the Wikipedia, which represented only 3% of its overall training dataset. Other systems followed the trend developing similar applications on a large scale, like Google’s LAMDA system and DALL-Ek, and Imagen1. In Summer 2021, a team at Stanford University came up with ML foundation models that could be trained on general dataset as well as on dataset from specialist applications⁽¹⁰⁻¹⁴⁾.

1.3.3 DATA QUALITY

AI can be model-centric or data-centric. Ideally both should be combined to get good results. In model-centric applications, a benchmark is observed giving a standard and fixed dataset for model accuracy. However, without high-quality training data models the results can exhibit performance discrepancies, reduced accuracy, and persistent fairness. Hence, poor data quality impacts accuracy, reliability, and performance. The steps in data-centric operations are given in Figure 1.6.

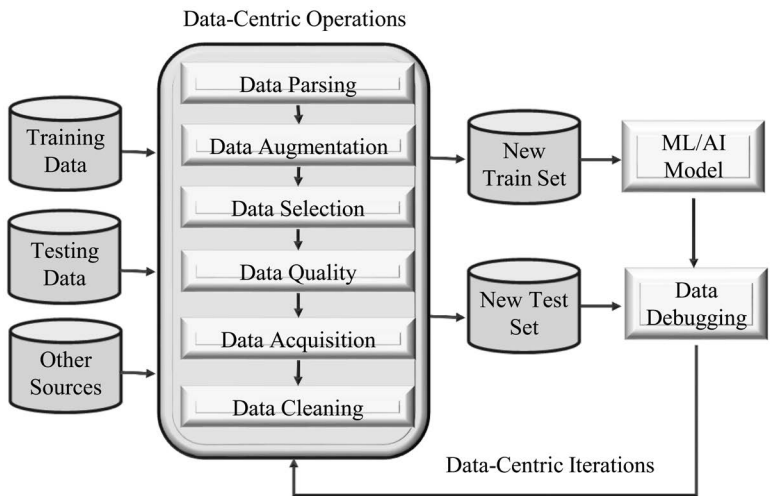


FIGURE 1.6 Data centric operation steps.

The proliferation of data sources and the exponential growth in data volumes make it hard to assess the QoD⁽¹⁵⁻¹⁷⁾. Undoubtedly, high quality data yields better result hence restoring confidence to the users. QoD may be summarized as:

- Accuracy, correctness, validity, precision, and focus,
- Accessibility, availability, and supporting technologies,
- Business and legislative formalities,
- Completeness, pertinence, and relevance,
- Consistency, cohesion, coherence, and integrity,
- Redundancy, minimality, compactness, and conciseness,
- Readability, comprehensibility, clarity, and simplicity,
- Believability, security, and reputation of the data source, and
- Usefulness of data to the end-users.

Good practices in data quality are implemented by data governance, utilizing data quality tools, collaborating with data providers, monitoring data quality with feasible metrics, and using the data quality indicators. *Data quality* indicators are the guides for data preparation, data assessments, and data cleaning. The assessment of QoD requires some basic steps, such as:

- Data quality dimensions and weighting,
- Defining the thresholds for good and bad quality for each data quality dimension,
- Determination the acceptability of data quality,
- Follow-up procedures of the data quality,
- Implementation of corrective actions, and
- Selection of data to be assessed.

1.3.4 DATA-EFFICIENCY

Efficient data reduces efforts and saves computing time and resources. The size of datasets affects training time directly. Data efficiency in AI applications can be divided into three main groups: 1) labeled data, 2) less labeled data, and 3) unlabeled data.

Labeled Data is already known, identified, and tagged data. Some forms of AI are a typical data-driven method that the performance can be correlate with labeled data.

Less Labeled Data is the reduction of amount of labeled data. Less labeled data can build more efficient models. Some advanced technologies such as *active learning*, and *few-shot learning* techniques largely use less labeled data.

Unlabeled Data is much less known data but can be used in various models. These methods are structured to learn from related tasks or totally from unlabeled data. There are two common methods transfer learning, and self-supervised learning. *Transfer learning* aims to solve a given task from labeled data from the previously related ones. It tries to minimize the difference between the source and target domains. *Self-supervised method* learns directly from unlabeled data. It uses either

contrastive learning methods by identifying similar and/or dissimilar data samples. It uses pretext tasks as representative features to learn^(18–21).

1.3.5 DATA CYBERINFRASTRUCTURE

Cyberinfrastructure (CI) consists of multiple computers, data storage systems, advanced instruments, data repositories, multiple visualization environments, high-speed communication networks, and people. CIs are good examples of the use of big and massive data.

A performant and scalable CI can acquire, process, and deliver many sizes of data and related data products. CIs are built to address a range of requirements, including managing the lifecycle of large volumes of complex and heterogeneous data, generating a wide range of data products on demand. It provides services for reliable and pervasive access to data and related products through scalable and easy-to-use data delivery mechanisms ensuring the security, and integrity of data. It supports long-term archiving of data, metadata, and data products.

CI are built on: 1) data types (e.g., CSV, PDF), 2) data acquisition methods (e.g., sensors and instruments), 3) data storage mechanisms (e.g., database, object stores, archival), 4) data delivery arrangements (e.g., graphical interface), 5) access control methods (e.g., open public access, authentication mechanisms), 6) data processing techniques (e.g., generation of data, externally hosted data), 7) cybersecurity arrangements (e.g., firewalls, VPNs, secure socket layer), 8) deployment models (e.g., cloud-based, open source), and 9) operational models (e.g., internal or external support).

To be able to fulfill the functionalities of CI, a set of policies and procedures are put in place, such as information security policy, access control policy, asset management, disaster recovery, password policy, and training^(22–25).

1.4 SEARCH METHODS AND OPTIMIZATION

Search and optimization are the main building blocks of AI. They will be explained in greater detail.

1.4.1 SEARCH METHODS

In AI applications, the relevant information from the dataset(s) is searched first and then conclusions are drawn. There are two fundamental search methods: 1) uninformed/blind search and 2) informed/heuristic search (shown in [Figure 1.7](#)).

1.4.1.1 Uninformed/Blind Search

Uninformed/Blind Search is a method that does not contain information about the goal state. The algorithms explore a problem space without any specific knowledge about the goal state. Search algorithms require information on the initial state to take further actions. There are various types such as uninformed search, Depth First Search (DFS), Breadth First Search, Uniform Cost Search, Bidirectional Search, Iterative Depth Search^(26,27). Some of these methods are explained in this chapter.

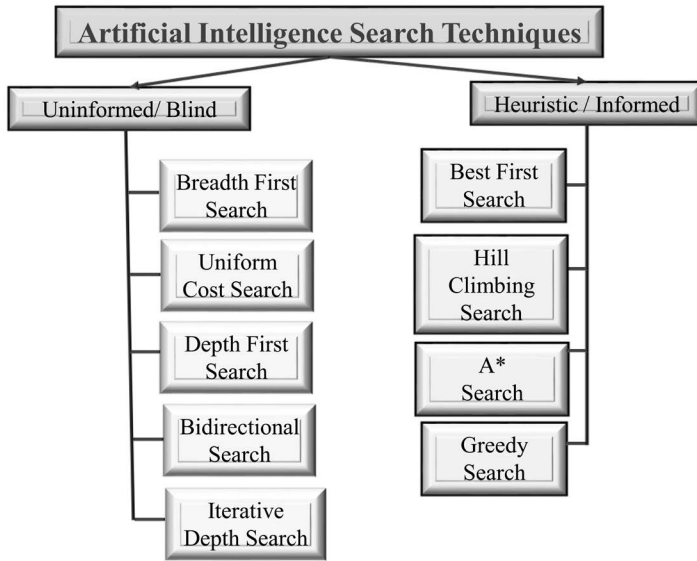


FIGURE 1.7 AI search technique examples.

1.4.1.2 Informed/Heuristic Search

Informed/Heuristic Search contains information about the goal state. Since more information is known, it is more efficient compared to blind search, and it takes less time to find the goal state. Typical examples of informed search are Best First Search, Hill Climbing Search, A* and AO* Search, and Greedy Search^(28,29).

AI search methods can also be classified as *state space search*, *local search*, and *distributed search*.

1.4.1.3 State Space Search

State space search searches to find a goal state using three methods, *simple exhaustive search* or *search spaces* or *adversarial search*. The most used methods in state space search are deep-first search, breadth-first search, hill climbing, simulation annealing, and A* algorithm⁽³⁰⁾.

1.4.1.4 Local Search

Local search uses optimization methods to find numeric solutions on the problem. It begins with a guessed value for the search and then refines the guess incrementally until no more refinements are necessary. These algorithms can be visualized as *blind hill climbing* where search begins at a random point, and then keeps moving by jumps or steps uphill (or downhill) until it reaches the top (or bottom). This local search process is called *stochastic gradient ascent (or gradient descent)*⁽³¹⁾, illustrated in Figure 1.8.

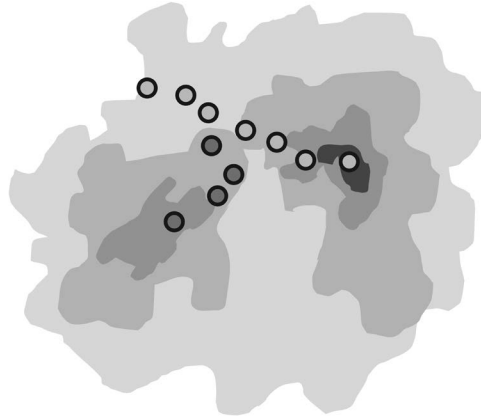


FIGURE 1.8 Gradient ascent method.

1.4.1.5 Distributed Search

Distributed search processes can be implemented by various methods, such as the *swarm* algorithms. Common swarm algorithms are particle swarm optimization, ant colony optimization, and others⁽³²⁾.

1.4.2 OPTIMIZATION

Optimization is minimizing or maximizing some problems for the best outcome satisfying certain constraints and conditions. It employs some rules and measures of deviations from an ideal case. Those measures are usually provided by *performance indices* or *cost functions*. Performance index is a function whose value indicates how well the actual performance of the system matches the desired performance subject to some constraints. Appropriate selection of the performance index is essential since it determines the nature and complexity of the problem. The choice of the performance index depends on data in hand, system specifications, and physical realizability. The choice of an appropriate performance index involves a compromise between a meaningful evaluation of system performance and availability of a feasible mathematical description, which may be linear or non-linear, or stationary, or time varying. In AI applications, the basic rules and mathematical descriptions for optimization are the same as explained in this book.

If the performance index and constraints can be formulated as linear algebraic functions of the variables, then a linear programming may be selected. In most cases, *simplex methods* can provide good solutions to linear programming problems. If the system is non-linear, then the solution involves non-linear techniques or linearization in some sub-regions. This is where AI and learned methods become effective. If the problem involves determination of the largest value of a function with two or more variables, then the *steepest ascent (or descent) methods*, also termed *hill climbing* or *gradient methods* are used.

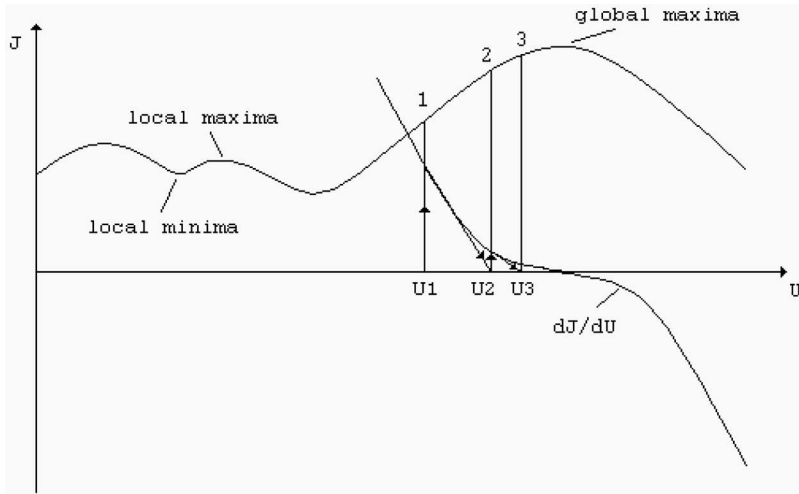


FIGURE 1.9 Hill climbing technique to determine global maxima or minima.

An illustration of hill climbing method in the region of absolute minima is given in Figure 1.9. This figure also illustrates a common pitfall of being trapped in a local minima or maxima. As can be seen in this figure, the sign of the first derivative of the function changes as the derivative passes through the maximum point. This method commonly uses an initial estimated point where the first derivative is considered (guessed) to be zero. This point is then used, in the direction of steepest ascent or descent, for the next point until the absolute maxima or minima is reached without being trapped in local extreme points.

Before starting optimization, it is necessary to formulate the system by understanding the information on parameters and describing equations, constraints, class of allowable control vectors and the selected performance indices. Then the solution can be proceeded by determining the optimal *control vector* $\mathbf{u}(k)$ within the class of allowable vectors. The control vector $\mathbf{u}(k)$ will be dependent on factors such as: the nature of performance index selected, constraints, initial values of state, initial outputs, and the desired state as well as the desired outputs.

In simple cases, errors between the desired and actual responses can be selected as the performance index to be minimized. There is a host of choices for a suitable performance index. As an example, in time-related applications, error function, $e(\tau)$, can be used to determine integral absolute error, integral square error, integral time absolute error, integral time squared error, mean square error, and so on.

$$\text{Integral absolute error (IAE)} = \int_0^T |e(\tau)| d\tau \quad (1.1)$$

$$\text{Integral squared error (ISE)} = \int_0^T e^2(\tau) d\tau \quad (1.2)$$

$$\text{Integral time absolute error (ITAE)} = \int_0^T |e(\tau)| \tau d\tau \quad (1.3)$$

$$\text{Integral time squared error (ITSE)} = \int_0^T e^2(\tau) \tau d\tau \quad (1.4)$$

$$\text{Mean Square Error (MSE)} = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T e^2(\tau) \tau d\tau \quad (1.5)$$

Mean Square Error (MSE) is also commonly used since slight modification leads to inclusion of statistical techniques and analysis of random processes.⁽³³⁾

Tailored optimization algorithms are used at every stage of development and applications of AI including in the AI itself. Optimizations improve the accuracy of the models by lowering the risk of errors and by minimizing their losses. They are particularly useful when machines are trained with limited local data or when training offline is permissible. Ambarella AI optimization methods are: 1) Deterministic Optimization, 2) Stochastic Learning Optimization, and 3) Stochastic Optimization. Figure 1.10 illustrates some of the optimization techniques frequently used in AI.

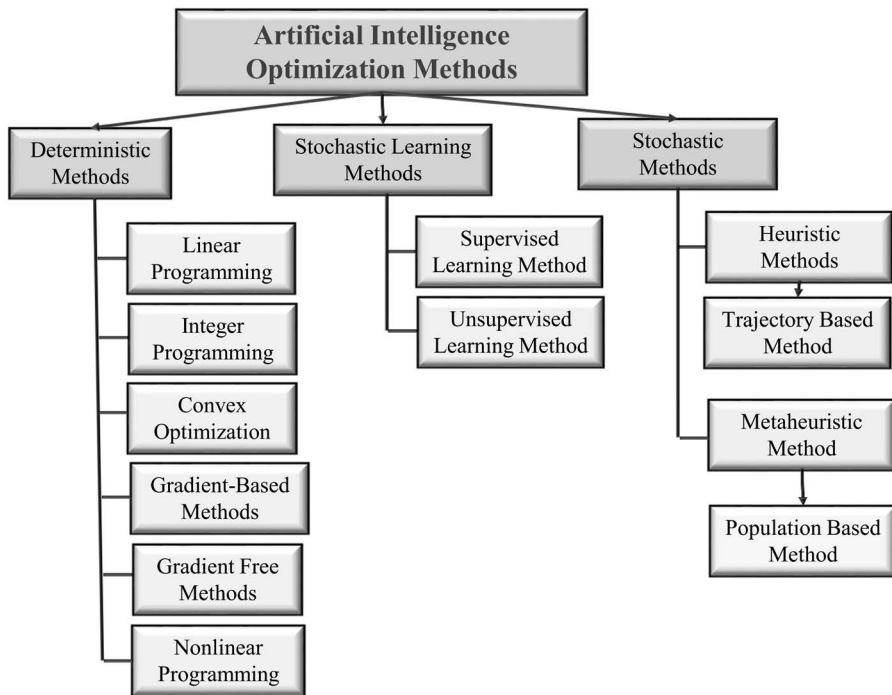


FIGURE 1.10 Some of the typical optimization methods in AI.

1.4.2.1 Deterministic Optimization Method

Deterministic optimization method refers to complete or rigorous classes of algorithms. It explores the cases having clear features, or with cases that can be described by some crisp mathematical models. Deterministic optimization has various techniques, such as the linear programming, non-linear programming, integer programming, convex methods, gradient methods, and non-convex methods^(33,34).

1.4.2.2 Stochastic and Stochastic Learning Optimization Methods

Stochastic Learning Optimization attempts to make the best decision to improve performance. This learning method has randomness as part of the search procedure. This method is based on the information obtained from the observations and analysis of the system. The systems under investigation can be static, or dynamic, or changing over time and space or changing by other affecting parameters. Stochastic learning optimization is employed in 1) supervised learning and 2) unsupervised learning.

Stochastic optimization is employed on stochastic processes to maximize or minimize the objective function. There are quite a few methods, such as heuristic methods, trajectory-based methods, metaheuristic methods, and population-Based methods. Within these methods there are diverse range of algorithms, such as hill climbing, tabu search algorithms search, greedy randomized adaptive search, simulated Annealing, genetic algorithms, differential evolution, particle swarm Optimization, and many others. The two most used algorithms (genetic algorithms, particle swarm optimization), and some newly emerging methods such as the harmony search algorithm (HAS) to seek optimal network weights^(35,36).

1.4.2.3 Genetic Algorithms

Genetic algorithms (GAs) are heuristic (informed) search algorithms that reflect the process of natural selection. They are the most widely explored algorithmic models used by evolution-based computation techniques. In evolution-based computing, a set of candidate solutions are generated and then iteratively updated. These techniques involve algorithms such as genetic programming, evolutionary methods, differential evolution, and cultural evolution.

GAs rely on the use of selection, crossover, and mutation operators to create successive generations of better adapted individuals. In a sense, GA behaves as an intelligent process of stochastic searches within a specified search space to solve a problem. It can also be used to optimize weights of ANNs. The search is guided only by the fitness value associated with every individual in a population, e.g., social behavior of species living in swarms or colonies⁽³⁷⁾. An example of a software flow diagram is given in [Figure 1.11](#). GA suffers from mutation problem that might lead to a premature convergence.

Fitness value of GA is determined by the fitness functions and penalty functions which can be expressed as:

$$f_{\text{Raw}} = \sum_{i=1}^o o_i w_i \text{ with } \sum_{i=1}^o w_i = 1 \quad (1.6)$$

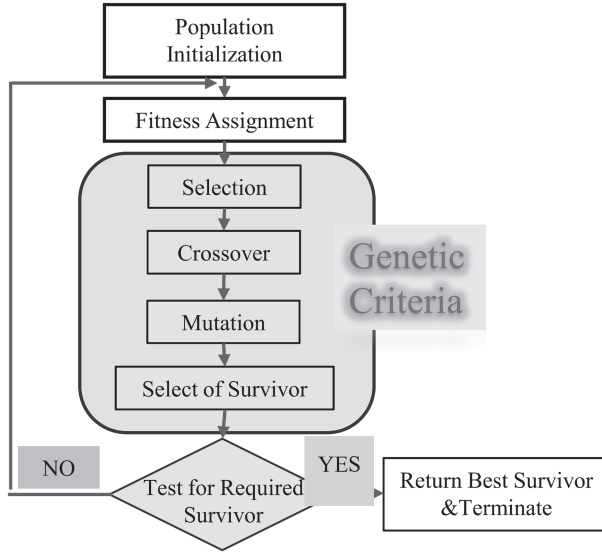


FIGURE 1.11 Genetic algorithm flow diagram.

where f_{Raw} is overall raw fitness

o_i is each objective, and
 w_i is assigned weight for each objective.

$$f_{\text{Final}} = f_{\text{Raw}} \prod_{j=1}^R p f_j(r_j) \quad (1.7)$$

r_j is the restrictions,

R is the violation of restrictions, and

$p f_j(r_j)$ is the return of each restriction with values between zero and 1.

1.4.2.4 Particle Swarm Optimization

Particle swarm optimization (PSO) is a nature inspired search optimization method. It has a few different methods applied frequently, such as the ant colony optimization, cuckoo search algorithms, and bat algorithms. PSO algorithm mimic social behavior of fish schooling or flying birds looking for food sources through collaborative work of the population. The exploration of the search space could be slow since PSO calls evaluation function by the number of its population size in each iteration. It can be trapped in local optimum if improper values are assigned to the parameters.

In many applications, it is common practice to combine two or more techniques to form various hybrids paradigms. The basic idea is to exploit the relative merits of individual techniques while eliminating their weaknesses^(38–40).

1.5 ARTIFICIAL INTELLIGENCE

AI is an umbrella term to describe the intelligence demonstrated by computers or machines. AI is fundamentally ML due to its implementation in machines and/or by machines. AI aims to perform tasks that conventionally require human intelligence. In some cases, it surpasses human capabilities particularly if large amounts of data is involved for analysis, processing, information extraction, and decision making.

AI is based on ML algorithms. There are many algorithms and different methods for creating AI. The most dominant and the most promising method now is the use of ANNs and combination of ANNs with other methods⁽⁴¹⁾.

AI can be classified as capabilities-based or functionalities-based AI (see Figure 1.12).

1.5.1 CAPABILITIES-BASED AI

Capabilities-based AI can be categorized in three main groups^(42,43):

- 1. *Artificial Narrow AI* can perform single tasks, and in many cases faster and better than humans. Capabilities-based AI examples are ChatGPT, IBM Watson, and Amazon Alexa.
- 2. *Artificial General Intelligence (AGI)* can learn skills and apply those skills in different contexts. AGI does not exist yet, but the idea and concept draws considerable research and design (R&D) attention.
- 3. *Super AI* is still in theoretical form and does not exist, but if realized, it will be able to think, reason, learn, make judgments, and demonstrate cognitive abilities just like humans. However, Super AI may be realized sooner than many people think.

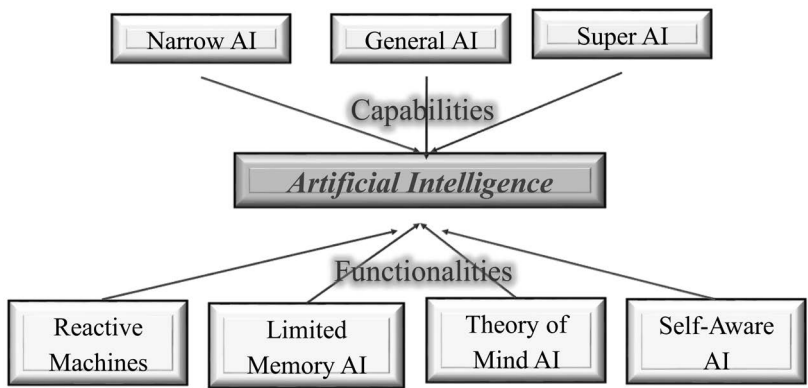


FIGURE 1.12 Capabilities- and functionalities-based AI methods.

1.5.2 FUNCTIONS-BASED AI

Functions-based AI can be categorized in four groups:

- 1. *Reactive Machines* which collect previous outcomes and decisions, and then work on the present data. In functionalities-based AI, the data is analyzed by deterministic or statistical models to make decisions. A typical example of a reactive machine is the IBM Deep Blue.
- 2. *Limited Memory* AI uses past and present data to decide a course of action. Typical examples are ChatGPT and self-driving cars.
- 3. *Theory of Mind* AI can understand the thoughts emotions and other entities. Theory of mind AI does not exist but there are intensive ongoing R&D activities in this area.
- 4. *Self-Aware* AI can exhibit self-awareness and it will be able to interact with humans with emotions, beliefs, gestures, and needs. There are many examples of self-aware AI in robotic applications, but we still are away from reaching the final goal.

A considerable progress in AI can be noted since the 2020s. More complex and large size AI models appear almost daily finding unexpected applications with high degree of efficiency, reliability, and robustness. Examples of applications specific AI developments are illustrated in [Figure 1.13](#).

Many of the AI methods have novel paradigms for current ubiquitous applications in agriculture, automated decision-making, aviation, chemistry, computer vision, defense and military, environmental science, finance, health and medicine, human communications, human-machine interactions, industry, intelligent cities, intelligent sensing, knowledge presentation, knowledge discovery, material science,

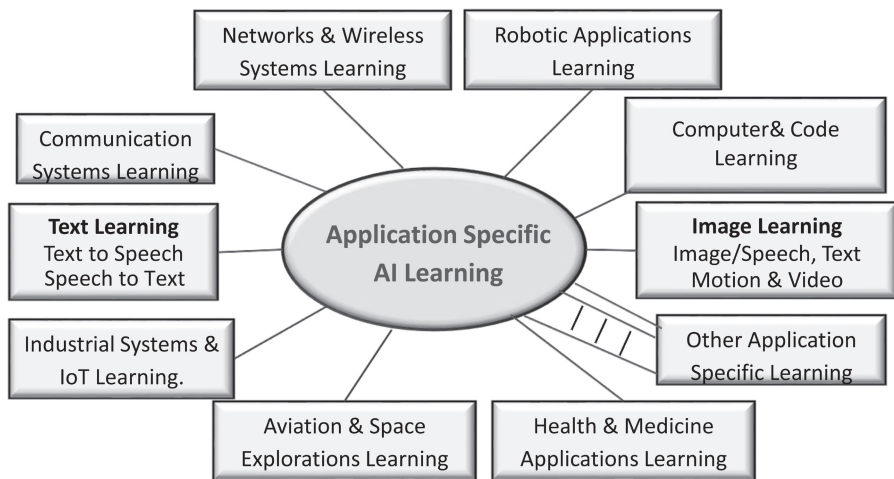


FIGURE 1.13 Typical applications specific implementation of AI.

manufacturing, medical diagnosis, microeconomics and macroeconomics, monitoring conditions of devices, NLP, neuroscience, operation research, physics, robotics, science and engineering, self-driving cars, sensor networks, space explorations, telecommunications, transport systems, and web search engines^(18,44).

1.5.3 BRIEF HISTORY OF AI

In the 1960s, ML and a limited form of AI were based on logical reasoning with symbols and were carried out with relatively slow digital computers. In the 1980s, AI was based on two technologies: rule-based expert systems, and neural networks supported by training algorithms. In the beginning of 2000s, ANN backed up with many related methodologies and algorithms had been developed. In the early 2010s, AI reached new level with the rise of neural networks that were capable of learning from large datasets. The leap forward in AI took place in the beginning 2020s.

Much of the theoretical foundation for the advances in AI can be attributed to 1990s and 2000s. But lack of labeled training data in structured forms and in a machine-readable format impeded AI advances. In addition, the unavailability of AI-specific affordable hardware prohibited possible applications. Advanced computational techniques were used only in a handful of high-end laboratories, government entities, and in some large corporations. However, the foreseen potentials of AI have generated a renewed interest due to confluence of breakthroughs primarily in AI hardware, such as the GPUs and other fast data handling technologies.

The successful applications of deep neural networks (DNNs) have reignited interest in AI because of prominent breakthroughs in machine vision, medical imaging, and language applications. Nowadays, there are numerous specific chips to handle AI training and inference tasks. Training requires accuracy and inference requires flexibility; therefore, different sets of chips are needed to handle these two requirements. The wider applicability and accuracy of AI depends on the specific method applied, in the quality of the dataset, validity of algorithms, computing methods, and hardware used^(45–47). All these points are explained and clarified in this book.

1.5.4 ARTIFICIAL INTELLIGENCE METHODS AND ALGORITHMS

There are multiple methods and algorithms used in different stages of AI. There still are many challenges at each stage. Some AI supporting algorithms and methods serve multiple tasks, some are applicable in various categories, and some belong to a parent method with slight variations. Algorithms are also dependent on the type of computer language selected, such as Python, Java C++, Lisp, R, and Prolog. Some of the methods and algorithms used in AI development are as follows:

- **Machine learning methods:** DL, Knowledge-Enhanced Learning, Physics-informed ML, Reinforcement Learning, Self-supervised Learning, Semi-supervised Learning, Supervised Learning, Transformer Learning, Unsupervised Learning, and others.

- **Machine intelligence:** AGI, artificial narrow intelligence (ANI), artificial super intelligence (ASI), limited memory intelligence (LMI), reactive machine intelligence (RMI), self-aware intelligence (SAI), Theory of Mind™, and others.
- **Data classification:** Audio classification (AC), content-based classification (CBC), image classification (IC), document classification (DC), user-based classification (UBC), text-based classification (TBC), and others.
- **Data classification algorithms:** Decision tree (DT), kernel approximation algorithm (KAA), logistic regression (LR), Naïve bias (NB), K-nearest neighbor (KNN), random forest (RF), stochastic gradient decent (SGD), state vector machine (SVM), and others.
- **Data clustering:** Centroid-based clustering (CBC), constrained-based clustering (CBS), density-based clustering (DBC), distribution-based clustering (DBC), partition-based clustering (PBC), fuzzy clustering (FC), hierarchical-based clustering (HBC), and others.
- **Clustering algorithms:** Agglomerative hierarchical (AH), balanced iterative reduction (BIR), density-based special clustering (DBSC), expectation minimization (EM), fuzzy C-means clustering (FCC), Gaussian mixture models (GMMs), hierarchical clustering (HC), K-means (KM), subtractive clustering (SC), and others.
- **Regression types:** Bayesian linear regression (BLR), decision tree regression (DTR), lasso regression (LR), linear regression (LR), negative binomial regression (NBR), ordinal regression (OR), poisson regression (PR), polynomial regression (PR), principal component regression (PCR), quantile regression (QR), random forest regression (RFR), regularization, ridge regression (RR), stepwise regression (SR), SVR regression, tobit regression (TR), and others.
- **Regression algorithms:** ANN, DT, elastic net (EN), linear regression, polynomial regression, random forest regressor, ridge regression, robust regression, stochastic gradient ascent/descent, SVM, and others.
- **Machine learning networks:** Auto encoder (AE), Bayesian networks (BN), Boltzmann machines (BM), continuous valued neural networks (CVNN), convolution neural networks (CNN), deep belief networks (DBNs), deep Boltzmann machines (DBM), deep feed forward (DFF), DNNs, feed forward networks (FFN), functional link networks (FLN), gated recurrent unit (GRU), generative adversarial network (GAN), hopefield network (HN), long/short term memory (LSTM), perceptron, radial basis networks (RBNs), recurrent neural networks (RNNs), self-organizing maps (SOMs), and others.
- **Deep learning algorithms:** Encoders and autoencoder (EA), convolution neural networks, DBNs, Gaussian Naive Bayes (GNB), GANs, long short-term memory networks (LSTMNs), Markov models (MMs), multinomial Naive Bayes (MNB), multilayer perceptron (MLP), radial basis function networks (RBFNs), RNNs, restricted Boltzmann machines (RBMs), self-organized maps, and others.

- **Search and optimization:** Classic optimization techniques, hill climbing (ascend or descent), ant colony optimization (ACO), genetic optimization (GO), PSO, and others.
- **Probabilistic methods:** Discriminative methods (DM), generative methods (GM), reinforcement methods (RM), and others.

It is not possible to cover all these methods and algorithms in detail in a limited space in this book, but we will provide sufficient information for a comprehensive appreciation of the overall concept. Many of the AI methods and algorithms will be explained clearly for a wide range spectrum of readers from those who have limited knowledge on AI to experts in this field.

1.6 ARTIFICIAL NEURAL NETWORKS

ANN is the backbone of modern AI. A neural network is an interconnected group of nodes known as artificial neurons. Neurons are the processing elements that connect two layers, and the strength of these connections is determined by adjustable weights. Neural networks are emulation of neurons in a biological brain. A human brain contains 86 billion neurons plus 16 billion neurons in cerebral cortex.

Figure 1.14 represents the mechanism of human neurons. At one end of the cell, the nucleus has dendrites that are sensitive to stimuli. Axons are chemical neurotransmitters. When the cell is excited to an external stimulus it sends an electrical signal along the axon which releases neurotransmitter chemicals. This triggers other cells in a stronger and cumulative manner.

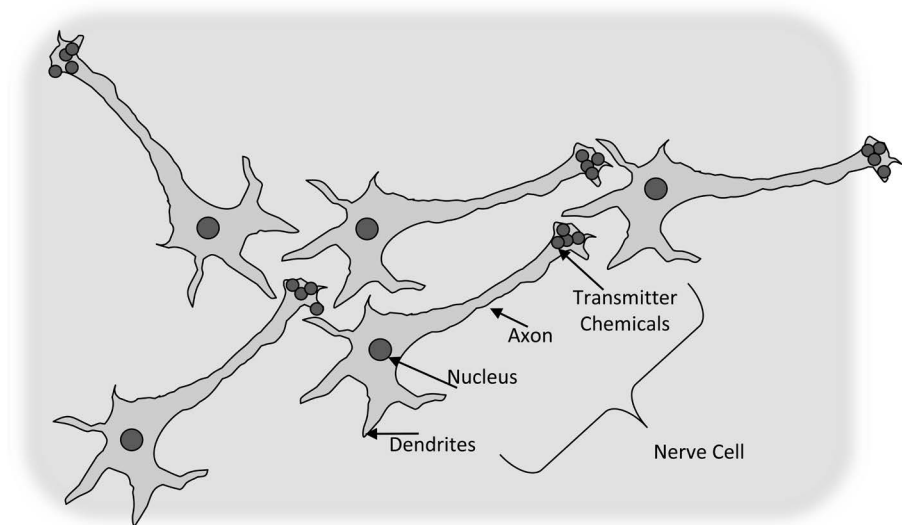


FIGURE 1.14 Representation of mechanism of human neurons.

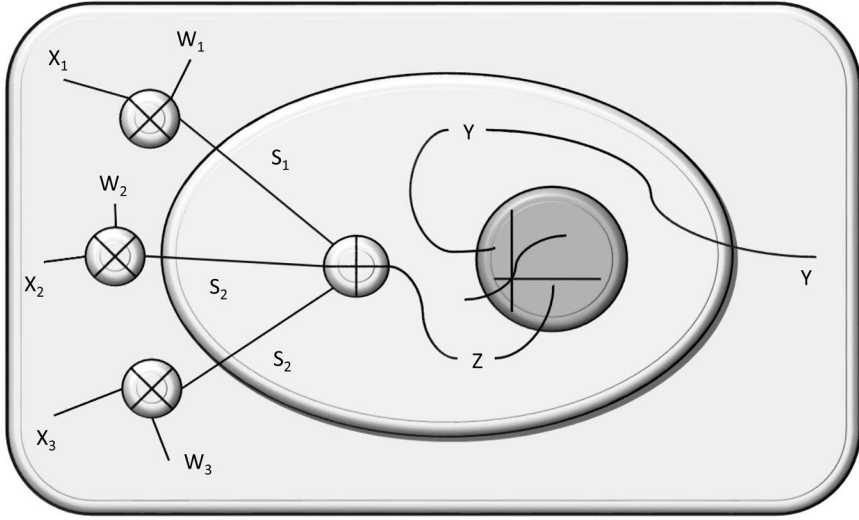


FIGURE 1.15 Mathematical construction of nerve cells (sigmoid).

Signal (x_i) coming from the predecessor nodes (neurons) multiplied by a weight factor (w_i) and then all the inputs are summed as the new state of that neuron as shown in Figure 1.15. which can be expressed by.

$$z = \sum x_i \times w_i \quad (1.8)$$

Extreme values of z is -1 and $+1$ termed as hyperbolic tangent sigmoidal. The hyperbolic tangent sigmoidal is defined as:

$$y = \frac{e^z - e^{-z}}{e^z + e^{-z}} \quad (1.9)$$

A neuron receives information from the predecessor neurons. In general, the following equations describe this operation as:

$$a_j(t+1) = f(a_j(t), p_j(t)), \theta_j \quad (1.10)$$

where $a_j(t)$ the state (activation) of the j th neuron at time t ,

f is the activation function that determines the new state of the j th neuron using previous (already stored) activation function, the new inputs from the predecessor neurons.

$p_j(t)$ input from the previous neurons.

θ_j a fixed threshold unless programmed to change by the learning process.

The output function (termed as identity function) after the activation is:

$$o_j(t) = f_{\text{out}}(t)(a_j(t)) \quad (1.11)$$

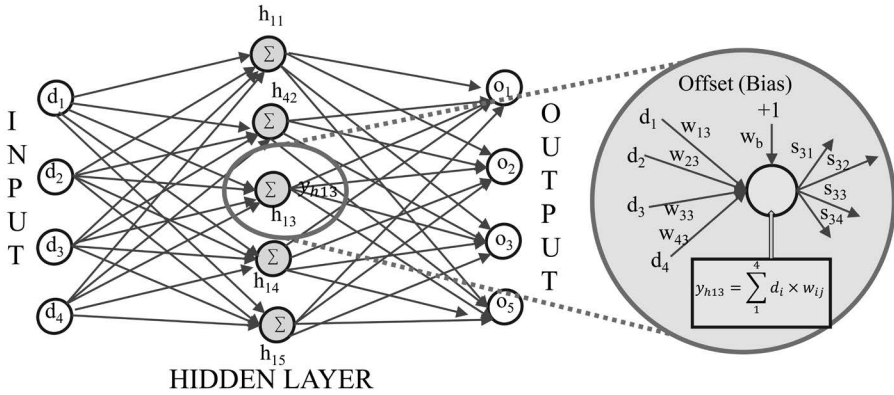


FIGURE 1.16 Typical operational structure of ANN.

The propagation function calculates the input to the neuron from the outputs from the predecessors using the bias w_{0j} as:

$$p_j(t) = \sum_i (o_j(t)) w_{ij} + w_{0j} \quad (1.12)$$

ANN structure mimicking the human neurons is illustrated in [Figure 1.16](#). A neural network has an input layer, one or more hidden layers, and an output layer. Each node applies a default function as an initial weight to be adjusted by the input layer data, known as *training*. Once the *weight* in one hidden layer reaches a specific threshold, the output is transmitted to the next layer that holds the next set of nodes.

ANNs are trained by learning algorithms. Algorithms use *local search* methods (such as Hill climbing, local beam search, simulated annealing, and so on) to choose the weights that will result in the right output for each input during training. Ideally, the trained network should be able to fully estimate the output parameters for all input parameters, but often this is not the case. Weights are adjusted to meet preset estimation requirement within some acceptable level of errors during the training process.

The neural network is trained by calculating the difference between the actual output and the desired output. For this, mathematical optimization is used, and it has many dimensions as there are adjustable weights (parameters) in the network. Training the networks iteratively seeks minimum of this multidimensional loss or cost function. Many training runs are performed incrementally changing the network's state by making small adjustments to the weights. This process is repeated over numerous random selected sets of training samples until the loss function is minimized to provide the best results for any new input.

This process is then repeated for each hidden layer, for example the first neuron in the first layer may need to have its activation increased. Depending on the adjustments of the first layer the neurons' subsequent layers are adjusted, for example, in the second layer activation of third neuron may need to be decreased.

ANN has different methods of operation and learning techniques, some of which are:

- **Backpropagation neural network** where the errors between input and output are compared for retaining purposes. This is a common training technique,
- **Convolution neural networks** strengthen the connection between neurons which are “close” to each other by convolving (rolling or entwining) locally,
- **Deep learning network** uses multiple layers,
- **Feed-forward neural network** where the flow takes place only in one direction,
- **Fine-tuning network** is the base models in which a few final layers on newly tagged data are custom trained without modifying the weights of the other layers,
- **Perception** use only one layer of neurons for training, and
- **Recurrent neural networks** feed the output signal back into the input, which allows short-term memories of previous input events.

The backpropagation technique is used to correct errors between the input and the expected output, shown in [Figure 1.17](#). In training the weights of the network are learned from *labeled training data*. Labeled data is the data already known, for example, data obtained from the image of human face is already known to be a part of human anatomy. In the backpropagation method, the output of each layer is transferred into to the next layer as an input for the weights of this layer to be adjusted. The process is repeated until the final output layer is reached. The generated output

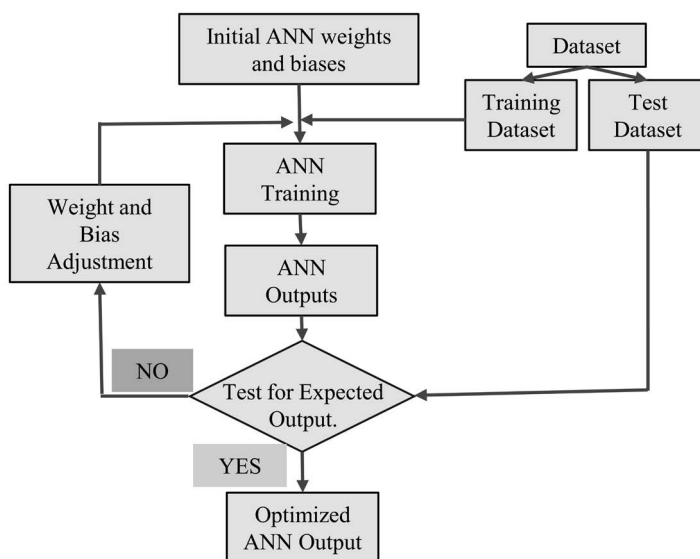


FIGURE 1.17 A flow diagram of back propagation neural networks.

layer values are compared with the expected output. The differences in error between predictions and obtained values are used to update the weights all over again until an error threshold is reached^(48–53).

1.7 MACHINE LEARNING

A machine with AI can solve a wide variety of problems with breadth, diversity, accuracy, and versatility just like human beings. MI can be divided into two main groups: 1) General intelligence aiming the goals of general human intelligence and 2) Social intelligence which can recognize, interpret, process, and simulate human feeling, emotions, and moods. Machine social intelligence does not exist yet, but the concept attracts many researchers and curious investigators⁽⁵⁴⁾.

1.7.1 MACHINE LOGIC

Machine logic has a long history. Machine logic based on hardware and software constitutes a significant part of the operations of computers and digital systems. For example, arithmetic logic units (ALUs) constitute the most essential part of micro-processors and computers.

Machine logic is based on hardware, software, and the combinations of the two. At the basic level, electronic logic components are used for developing sequential and combinational logic circuits. Some of these logic components are OR, AND, and NOR gates. Fuzzy Logic, Bayesian networks, and programmable logic controllers are typical examples of machine logic. AI related machine logic involves a range of algorithms for training, processing, reasoning, and knowledge representation. These algorithms are backed up by mathematical and stochastic methods.

Machine logical deductions (*inferences*) are the process of coming to conclusions from premises (statements) that are already known to be true. A logical Knowledgebase is used to handle queries and assertions by observing inference rules. Inference rules describe valid steps in a proof. The most common inference rule is the resolution in which inference searches to find a path that can lead to conclusions from premises. There may be many steps to reach inference and each step must comply with the setout rules^(55,56).

Machine logic has several types, such as 1) propositional logic and 2) predictive logic. *Propositional logic* operates on statements (premises) that are true or false by using logical connectives such as “AND,” “OR,” and “NOT.” *Predictive logic* operates on objects to predict relations and quantifiers.

1.7.2 MACHINE REASONING

Machine reasoning is based on knowledgebase and knowledge representation. *Knowledgebase* is a body of knowledge representing objects, relations between objects, situations, events, properties, concepts, categories, states, causes, and effects. Basically, a Knowledgebase is the knowledge of the known, domains of knowledge, and default reasoning. *Knowledge representation* supports AI to answer questions intelligently and to draw reasonable deductions about the real-world. Knowledge

representations are used in content-based indexing, retrieval, scene interpretation, decision support, and knowledge discovery.

Reasoning is done by algorithms implementing step-by-step reasoning procedures to solve problems and to make logical deductions. Many algorithms can perform accurate reasoning even if some uncertainties exist in data, incomplete information, and high degree of probabilities⁽⁵⁷⁾.

1.7.3 MACHINE DECISION-MAKING

Many problems in AI include perception, planning, reasoning, and decision making. These can be done using deterministic and/or probabilistic methods as explained above. Some of these methods are Bayesian inference algorithms, learning by means of expectation maximization algorithms, planning by decision networks, and perception dynamic Bayesian networks, probability theory, and some theories in economics and other disciplines.

In some AI applications, the decision making is carried out by *agents* that are capable of perceiving, planning, and acting to achieve a set of goals. The agent has options and preferences and assigns numbers to situations (utilities) for determining the degree of preference so that the *expected utility* can be calculated and acted upon. The utility of all possible outcomes of an action are weighted by probability functions so that the action taken leads to the maximum expected utility. However, in many situations, uncertainties exist due to the unknowns, incompleteness of information, unobservabilities, and uncertainties. Probabilistic guesses are made if the outcomes of each possible action cannot be determined. Then the situations are reassessed to see the effectiveness of these guesses on each action taken.

In many cases, multiple agents may be involved to solve the same problem. This makes the preferences of an agent much more complex and uncertain relative to the possible actions of other agents. In such cases, inverse reinforcement techniques are applied to acquire further information on the case to improve preferences and to make better choices. Information *theory* (such as the Markow decision process) can be used to weigh the value of the experimental or exploratory actions. Markow decision process describes the probability of a particular action influencing the state in a particular way. In many cases, *reward functions* are used to assess the utility of each state and the cost of each action. Depending on the outcome, a policy can be implemented deterministically to calculate the cost of action. The possible outcomes of a policy are determined by *iterations* or *heuristically* (trial and error, or experimentally). Multiple supporting methods are developed to describe the rational behavior of multiple interacting agents as in the case of *game theory*^(58,59).

1.8 CLASSIFICATION, CLUSTERING, AND REGRESSION

Initially, the ML was a tool that originated from automated knowledge processing. It was used to teach computers how to learn from raw data and manage the data more effectively. The data was pre-prepared to make it fit for the learning process. Later, new stages evolved to effective pre-processing, detailed learning, and efficient data evaluation.

The raw data usually comes in an un-structured form containing unwanted noise as well as different and unrelated data elements. Thus, the pre-processing stage converts information into frames by applying cleaning, extracting, thus suitably preparing the raw data. This is done by classification and clustering algorithms as illustrated in [Figure 1.18](#). There are numerous algorithms to fit the purpose^(60–64). A list of most used algorithms is provided here.

- ANN,
- DT, boosted decision tree, and bagged decision tree,
- K-Means clustering classification,
- KNNs,
- Linear discrimination analysis (LDA),
- Linear regression
- Logistic regression,

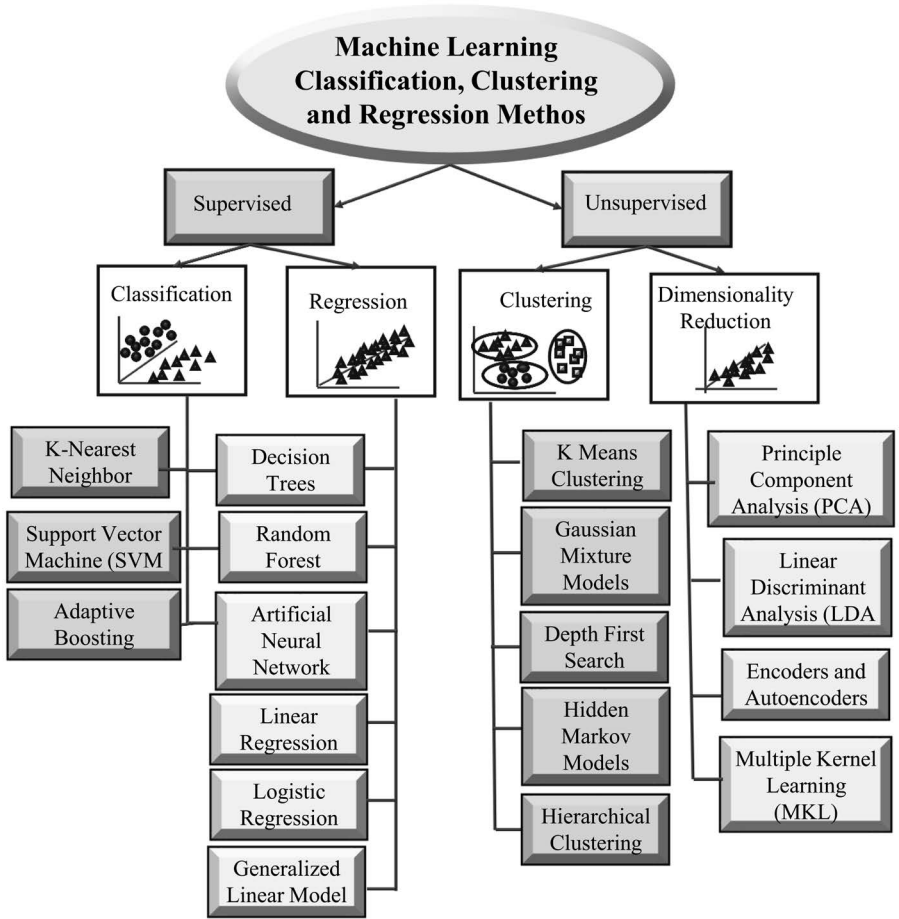


FIGURE 1.18 ML classification and clustering types.

- Naïve Bayes (NB),
- Principal component analysis (PCA),
- Quadratic discrimination analysis (QDA),
- RF,
- Support vector machine (SVM), and
- many the other models and algorithms, such as adaptive neuro-fuzzy learning, fuzzy C-mean clustering, GMM, hidden Markow model, hierarchical clustering, regularization, SOMs, stochastic gradient descent (SGD), stochastic gradient models, and subtractive clustering.

Due to the limited availability of space in this book, only a few highly relevant methods will be explained.

1.9 ARTIFICIAL INTELLIGENCE LEARNING

There are several types of learning methods, shown in [Figure 1.19](#). In this book, we will discuss mostly ANN-based learning methods.

- Supervised Learning and Semi-Supervised Learning,
- Unsupervised Learning and Self-Supervised Learning,
- Reinforcement Learning,
- Active Learning,
- Transfer Learning,
- Collaborative and Distributed Learning,
- Federated Learning,
- DL which is a type of ML that runs inputs through biologically inspired ANNs for learning.

We will expand on some of the most effective learning methods. A subchapter is dedicated to DL as it is the most comprehensive method enjoying diverse applications in AI.

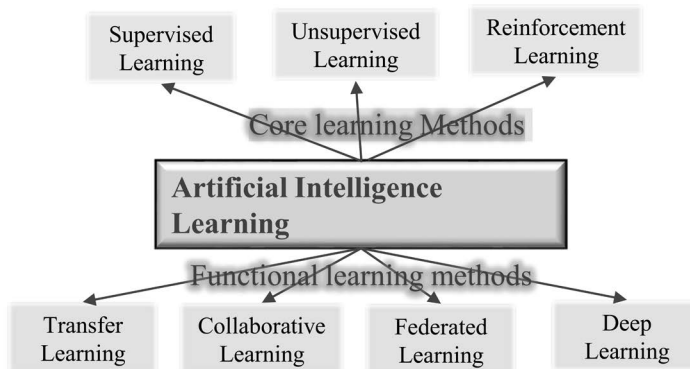


FIGURE 1.19 Core and functional AI learning methods.

Based on its learning capability, we can classify the family of AI learning models into the following subsets:

Batch learning-based algorithms typically train a model after a sufficiently large amount of training data has been collected prior to the learning task. This offline learning procedure has the advantage of allowing for more involved ML based on all the available data, where the model needs updating infrequently. The main advantage of batch learning is that the convergence speed is high. Batch learning may not be suitable for rapidly fluctuating processes subject to stringent requirements.

Online learning algorithms enable a model to learn from a stream of data arriving sequentially. Online learning changes and adapts to its structure and parameters on demand. An advantage of online learning is that it is eminently suitable for real-time and dynamic applications, but its convergence speed may be slow.

AI learning models can be divided into two parts: instance-based learning, and model-based learning.

Instance-based learning algorithms are based on constructing inference structures and predicting unseen instances. This may result in long search times and high memory requirements. However, it does not require prior-model assumptions. It operates best on large datasets for good quality learning.

Model-based learning aims to find the optimal parameters. The algorithms are designed to optimize the objective functions and to maximize the discovery of previously unseen patterns in the test data^(63,65).

1.9.1 SUPERVISED LEARNING

Supervised learning (SL) is a major branch of AI learning, and it has been an extensively researched topic. Large quantities of labeled data must be available for learning the training samples for a desired output. The convergence speed and action quality of SL is high.

SL uses labeled data to build learning models. the process of SL can be divided into two parts, the classification, and the regression. *Classification* assigns a categorical label to every input sample. By classification, the program learns to predict what category the input belongs to. Classification methods include DTs, SVM, KNNs, and others.

In *regression* where the program deduces functions based on the inputs. It predicts continuous values based on the input features. Typical regression analysis contains support vector regression, Gaussian process regression (GPR), and others^(65–69).

Supervised ML requires human interface in the loop. SL input dataset is split into two or more parts as the training datasets, and testing datasets. The training dataset has an estimated or graded output vector. A typical SL structure is illustrated in [Figure 1.20](#).

SL can be implemented by using several methods, some of which are:

- *Artificial Neural Network*: Adaptive boosting, convolutional neural network (CNN), extreme learning machine (ELN), logic learning machine (LLN), long short-term memory (LSTM), RNN, and SOM,
- *Bayesian Networks*: Naïve Bayes analysis,

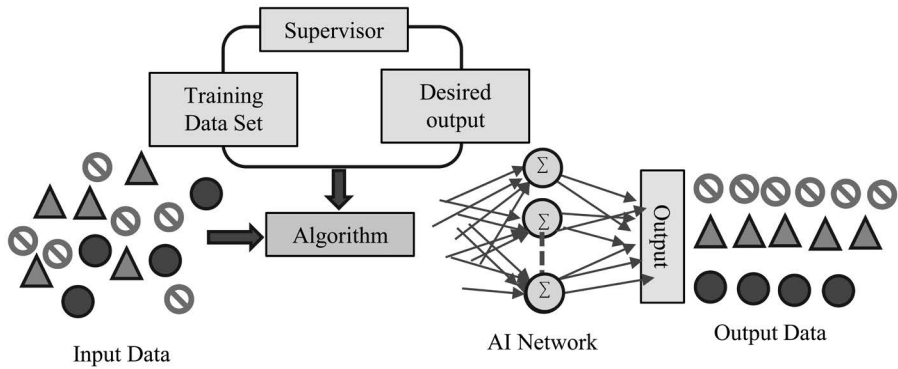


FIGURE 1.20 An illustration of supervised learning.

- *Decision Tree Methods:* Chi-squared Automatic Interaction Detection, Classification and Regression Tree, Conditional Decision Tree, Decision Trees, Decision Stump, Random Forest, and so on,
- *Gaussian Regression,*
- *Inductive Logic Programming,*
- *Fuzzy Networks,*
- *Instance-Based Learning,*
- *K-Nearest Neighbor,*
- *Lazy Learning,*
- *Learning Vector Quantization:* Elastic-net, Lasso, Linear Discriminant Analysis, Generalized Linear Models, Linear Regression, Logistic Regression, Multinomial Logistic Regression, Naïve Bayes Classifier, Ordinary Least Squares, Passive Aggressive Algorithms, Polynomial Regression, Ridge Regression/Classification, SVM,
- *Logistic Model Tree:* Analogical Modeling, and Nearest Neighbor Algorithms,
- *Ordinal Classification,* and
- *Random Forest.*

1.9.2 SUPERVISED LEARNING DATA CLASSIFICATION

Referring to [Figure 1.17](#) and algorithms listed above, the most used SL classification methods are explained below:

1.9.2.1 K-Nearest Neighbor Classification

KNN is a nonparametric algorithm utilized for both classification and regression. In this method, the algorithm aims to classify the given sample when the data is grouped into different classes. The class of an object is determined by a popular vote of its nearest K neighbors. The KNN algorithms are suited to multi-modal classes pertaining to many class labels. The error rate of KNN can be higher compared to other methods since the irrelevant features of data during classification can significantly contribute to the outcome⁽⁷⁰⁾. A typical structure of KNN is shown in [Figure 1.21](#).

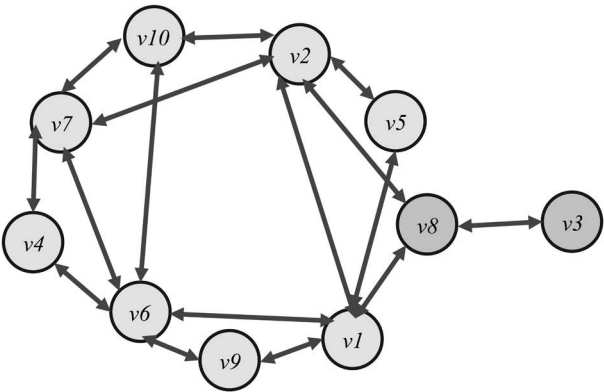


FIGURE 1.21 A typical structure of a KNN method.

KNN uses Euclidian distance formula to find the similarity between the data points. Distance between the data points x and data points y can be determined by:

$$d(x,y)=\sqrt{\sum_1^N(x_i-y_i)^2}=\sqrt{\sum_1^N(x_1-y_1)^2+(x_2-y_2)^2+\cdots+(x_N-y_N)^2} \tag{1.13}$$

Apart from the Euclidian distance formula there are other methods based on multivariate techniques such as Mahalanobis distance calculations.

1.9.2.2 Support Vector Machine

SVMs are classification models with a geometric means of separating and classifying two label classes. An SVM finds the straight line that best separates the two groups of points on a plane. It is used as the basis of building optimal classification and regression models. The training process is complex, but once tuned, state vector machine models provide high accuracy, leading to generalization. The structure of SVM is illustrated in Figure 1.22.

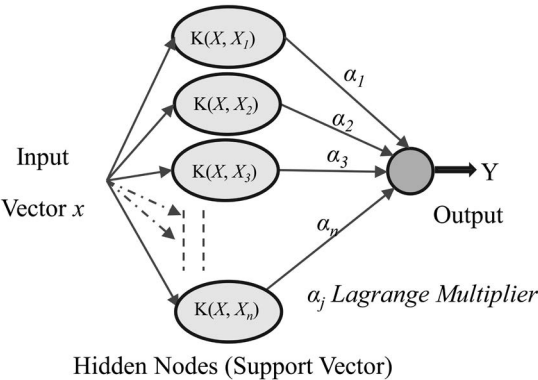


FIGURE 1.22 SVM configuration.

SVM is a sophisticated supervised ML technique. By using training data, the algorithm enhances an optimal hyper-plane, and this describes the decision and classifies new examples. State vector machines support both linear and non-linear classifications. In complicated cases, the points can be projected into a higher-dimensional space to find the plane or hyperplane that best separates the classes. The projection is called a *kernel*, and the process is called the *kernel trick*⁽⁷¹⁾.

There are numerous options for selection of the suitable kernels, some of which are:

Polynomial (homogeneous) kernel $k(\mathbf{x}_i - \mathbf{x}_j) = (\mathbf{x}_i \cdot \mathbf{x}_j)^d$ where $k > 0$ and d is the degree of polynomial and $(\mathbf{x}_i \cdot \mathbf{x}_j)$ is dot product of vectors $\mathbf{x}_i$ and $\mathbf{x}_j$.

Polynomial (inhomogeneous) kernel $k(\mathbf{x}_i - \mathbf{x}_j) = (\mathbf{x}_i \cdot \mathbf{x}_j + r)^d$ where $(\mathbf{x}_i \cdot \mathbf{x}_j)$ is dot product.

Hyperbolic tangent kernel $k(\mathbf{x}_i - \mathbf{x}_j) = \tanh(\kappa \mathbf{x}_i \cdot \mathbf{x}_j + c)$ where $k > 0$ and $c > 0$.

Other kernel functions are Gaussian radial basis function (RBF), Laplace radial function, Bessel function, ANOVA RBF, and linear spline kernel in one dimension.

1.9.2.3 Decision Trees (DTs)

DTs are heuristic structures built upon sequences of choices or comparisons for both non-numerical data and string type data. A tree-like structure is used where decisions are performed at each node at different hierarchical layers (see Figure 1.23). DTs can handle cases with missing data, and they are scalable from linear to non-linear data using the same logic.

Training a DT begins with selecting a metric (e.g., Gini-index, cross-entropy) and a root node. In the implementations, the data is repeatedly split into two or more parts until all the leaf nodes of the tree are reached in all branches in accordance with a stop rule. Aggregation of multiple DTs may be applied to optimize the overall performance, to determine the robustness, and to achieve the generality of the models.

The goal of a DT is to create a model that predicts the value of a target variable by learning simple decision rules inferred from the data features. DTs are easy to

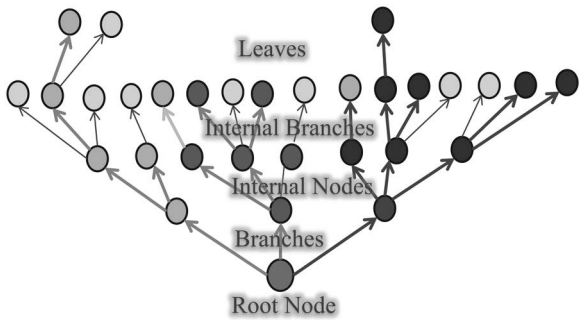


FIGURE 1.23 A decision three with five layers.

interpret and convenient to deploy. They are computationally intensive to train and prone to overfitting. A version of the DT is the *boosting and bagged decision tree*. In this case each tree is dependent on the prior tree. The algorithm learns by fitting the residue of the tree that preceding it. That is the tree boosting starts with a single decision or regression tree, optimizes it, and then builds the next tree from the residuals of the first one^(72,73).

1.9.2.4 Random Forest

RF algorithms are ensemble of sets of DTs. The RF selects the classification that takes the most votes. The RF model produces an *ensemble* of randomized DTs (Figure 1.24) for classification and regression. The aggregated ensemble either combines the votes modally or averages the probabilities from the DTs⁽⁷⁴⁾.

1.9.2.5 Adaptive Boosting

Adaptive Boosting (AdaBoost) is used by combining multiple weak classifiers to form a strong one. It is an ensemble learning technique combining several weak learners to form a single stronger learner. Weak learners are trained sequentially by adjusting their weights to avoid mistakes that their predecessors made. Adaptive Boosting is a powerful technique, and it has become an essential part of AI involving ensemble modeling^(75,76).

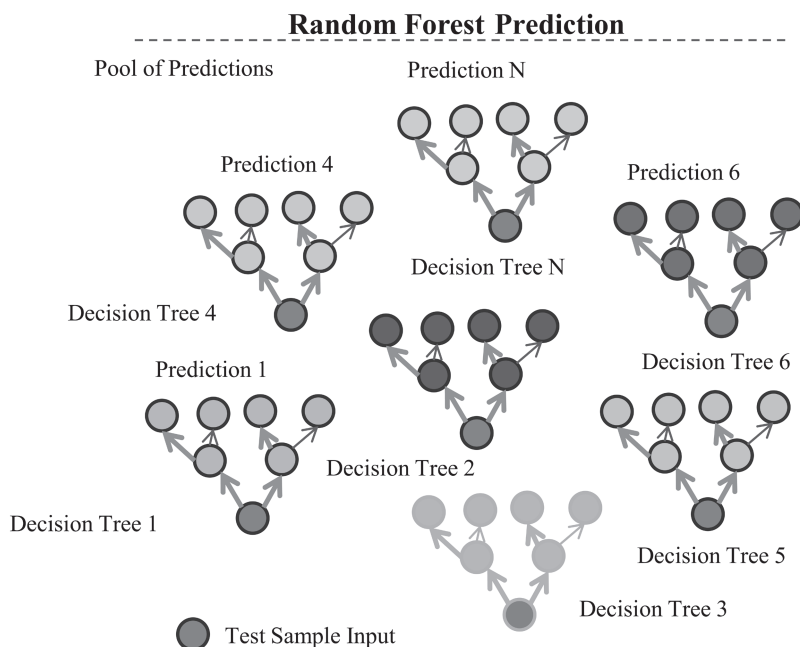


FIGURE 1.24 A random forest structure.

Boosting methods combine multiple models to create a robust learner, thus attracting considerable research and development. There are several different boosting algorithms. *XGBoost* algorithms use DTs for base learners, *Gradient Boosting* algorithms use different loss functions and different base learners, and *Brown Boost* minimizes the exponential loss function. Other less used ones are *Categorical Boost* (CatBoost), *Gradient Boosting Machine* (LightGBM), and so on.

1.9.2.6 Artificial Neural Networks as Classifier

ANNs are used as classifiers or regressors. It classifies the data by class labels based on input features. It regards the dependent variables belonging to discrete classes as the functions of the input. During the classification process, ANN adjusts synaptic weights to get better classification accuracy. They can be configured as *Binary Classifiers* having two neurons in the output layer, and *Multi-Class Classifiers* having several neurons in the output layer, each representing one class⁽⁷⁷⁾.

1.9.3 SUPERVISED LEARNING REGRESSION

Regression is the prediction of continuous values. The fundamental metrics for regression are variance, bias, and error. Algorithms use these fundamental metrics to assess the effectiveness of analysis. The most used algorithms are Linear Regression, Logistic Regression, Bayesian methods (i.e., Naïve Bayes), DTs, RF, and ANN methods.

1.9.3.1 Linear Regression

Linear regression (also called least square regression) predicts the relationships between the different features or variables as a linear objective function (as in [Figure 1.25](#)). It assumes that the objective function is linearly correlated with the independent variables. This assumption may or may not be true for the data in hand.

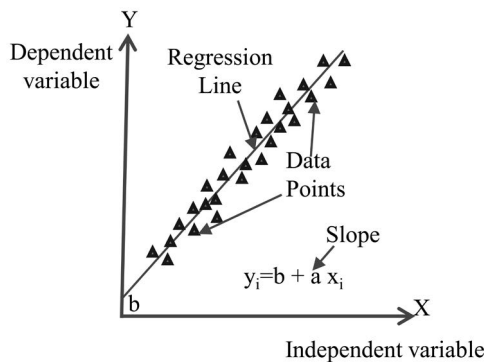


FIGURE 1.25 Linear regression determines the best line describing a given data.

An independent variable is a variable that stands by itself and not dependent on others. Linear regression can be optimized using gradient descent method. In some cases, it is solvable in closed form not requiring an optimizer⁽⁷⁸⁾.

Linear regression finds the linear line that best fits the independent data points x_i and dependent data points y_i which can be expressed as:

$$y = ax + b \quad (1.14)$$

where a is the slope and b is the intercept and calculated from the data points as:

$$a = \frac{[n(\sum y \times \sum x^2) - (\sum x) \times (\sum xy)]}{[n(\sum x^2) - (\sum x)^2]} \quad (1.15)$$

$$b = \frac{[n(\sum x \times y) - (\sum x) \times (\sum y)]}{[n(\sum x^2) - (\sum x)^2]} \quad (1.16)$$

1.9.3.2 Logistic Regression

Logistic regression is a method for solving categorical classification problems that uses linear regression inside a sigmoid or logit function (see [Figure 1.26](#)). It compresses the values to a range of 0 or 1 (Sigmoid function) resulting in a crisp probability. Logistic regression is often used as the first method when categorical predictions are involved⁽⁷⁹⁾.

Equation for logistic regression is the sigmoid function (activation function) which can be expressed as:

$$f(x) = \frac{1}{1 + e^{-x}} \quad (1.17)$$

where e = base of natural logarithms, and
 x = numerical value to be transformed.

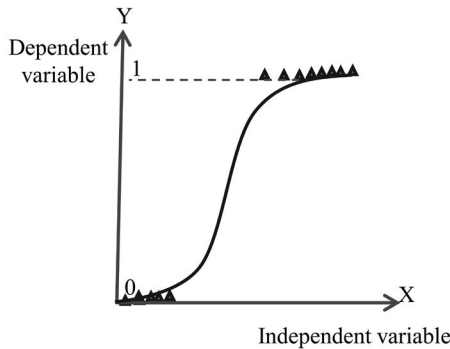


FIGURE 1.26 Logistic regression as a sigmoid function.

1.9.3.3 Naïve Bayes

Naïve Bayes (NB) is based on conditional probability. This technique uses a probability table which is based on feature values. The fundamental assumption is the conditional independence, hence the term *naïve*. It assumes that all features of the input are independent of each other. Naive Bayesian algorithm makes large datasets very easy to handle, and it is a good method for classification. The Bayes theorem provides a basis for later probability calculations⁽⁸⁰⁾. A simple formula used in NB is as follows:

$$P(A / B) = \frac{P(B / A) P(A)}{P(B)} \quad (1.18)$$

where A, B = events

$P(A/B)$ probability of A given B is true,

$P(B/A)$ probability of B given A is true, and

$P(A)$ and $P(B)$ independent probabilities of A and B .

1.9.4 UNSUPERVISED LEARNING

Unsupervised learning models rely on unlabeled data for inferring the underlying information structure without being dependent on external resources or human supervision. The advantage of unsupervised learning is that no prior knowledge is required; however, this comes at the cost of potentially reduction in its accuracy. Another disadvantage is that the automatically discovered data may not always representative of real-world situations. Given its unique features, unsupervised learning is suitable for solving the problems of user groups for hybrid multiple access, attack detection of malicious users, and so on^(60,65).

Unsupervised learning analyzes a stream of data and finds patterns and features from unlabeled data. It makes predictions without additional guidance. Unsupervised learning can also be effective in dimension reduction, clustering, exploring the data patterns and structures, and finding groups of similar objects. It can detect outliers and noise in the data.

Unsupervised learning uses previously learned functionality when new data is introduced for explanation of the data classes. There are numerous methods in unsupervised learning, which can be listed as:

- *Association rule learning*: Apriori, Eclat, and FP-growth algorithms,
- *Auto-encoders*,
- *Cluster analysis*: Conceptual Clustering, Expectation Maximization, Fuzzy Clustering, Hierarchical Clustering, K-means Clustering, K-medians, Mean-shift, and Single-linkage Clustering,
- *Dimensionality reduction*: Canonical Correlation Analysis, Dynamic Mode Decomposition, Factor Analysis, Feature Extraction, Feature Selection,

independent Component Analysis, Linear Discriminant, Multidimensional Scaling, Non-negative Matrix Factorization, Partial Least Squares Regression, PCA, Principal Component Regression, Projection Pursuit, Sammon Mapping, T-distributed Stochastic Neighbor Embedding,

- *Expectation-maximization algorithms,*
- *Information bottleneck methods,*
- *Vector quantization,* and more.

1.9.5 UNSUPERVISED LEARNING DATA CLUSTERING

Unsupervised clustering groups a set of samples into different clusters in accordance with their similarities by using techniques such as hierarchical clustering algorithms. Dimension reduction transforms a high-dimensional data space into a low-dimensional space without losing useful information. It also employs algorithms such as K-means clustering^(60,72).

Among many others, a typical unsupervised ML often uses the following clustering techniques: K-means clustering, GMMs, DFS, hidden Markov models, hierarchical clustering, isometric mapping (ISOMAP), and PCA.

1.9.5.1 K-Means Classification and Clustering

K-Means Classification and Clustering algorithm classifies an object into K-number of objects based on the features and attributes. Grouping is carried out by reducing the number of squares of distances between data and the corresponding centroid cluster, as shown in Figure 1.27. This *technique* uses the Euclidean distance metric to divide n observations into k clusters. Its main objective is to minimize the variance (sum of square distances) between the data and the corresponding centroid of the cluster. K-means clustering is useful for feature learning, and for providing a starting point for other algorithms.

K-means clustering assumes spherical clusters that are separable from each other in a way that the mean value converges toward the center of the cluster. The clusters are expected to be of similar size, so that the assignment to the nearest cluster center can correctly be done^(60,72,81).

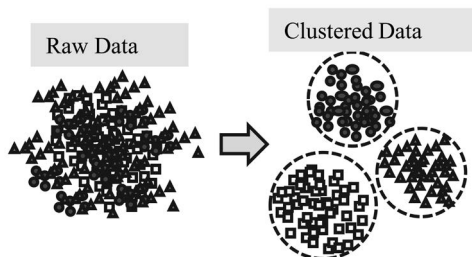


FIGURE 1.27 K-means clustering output.

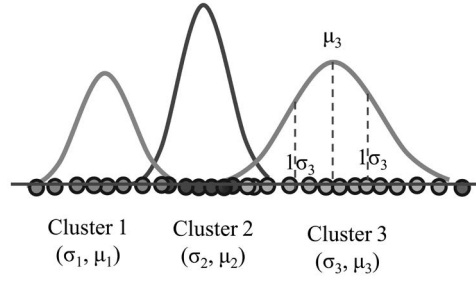


FIGURE 1.28 A GMM classification output.

1.9.5.2 Gaussian Mixture Model

GMM is a stochastic model based on normal distribution technique. A three stage GMM is shown in Figure 1.28. It is a multivariate technique and often used in pattern recognition and other unsupervised, semi-supervised, and SL. It requires specifications and definitions on the number of expected clusters. GMM can perform hard clustering or soft clustering. In hard clustering the model assigns query data points to the multivariate normal components. Soft clustering (Fuzzy Clustering) assigns a data point for each cluster. GMM can work on clusters of different sizes and clusters having different correlation structures⁽⁸²⁾.

The Gaussian probability function in terms of standard deviation and mean values can be expressed as:

$$y = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad (1.19)$$

where σ = standard deviation

μ = mean value

1.9.5.3 Depth First Search

DFS algorithm is recursive algorithm requiring many successive executions by using backtracking. The algorithm starts from a root node which is an arbitrary node and explores deep to the final node (see Figure 1.29). When it exhausts tracking in one

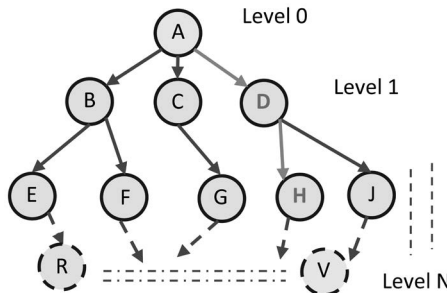


FIGURE 1.29 A DFS structure.

direction, it moves backwards on the same path to find nodes to traverse. All the nodes on the current patch of data are traversed before it goes to the next path. A node is selected, and the next node is visited while all the adjacent nodes are pushed into a stack. The process is repeated until the stack is empty. Stacking avoids visiting the same node multiple times. DFS is employed in traversing or searching for trees or graphing data structures. Another method based on similar principles is the breadth first search algorithms⁽⁸³⁾.

1.9.5.4 Hidden Markow Model

Hidden Markow Model is a method for representing probability distributions over sequences of observations. It is often used in speech, activity, gesture, handwriting, recognitions, and some biological applications. In graphical representation of a typical Markow model is shown in [Figure 1.30](#). In this figure, the state sequence q_1, q_2, q_3 , and q_4 are unobservable (hidden) and x_1, x_2, x_3 , and x_4 the observed variables.

In a hidden Markow Model, there are two distinct stochastic processes: 1) hidden states are variables that are unobservable but are calculated based on observed data and 2) observed states datasets directly measured from random events. This includes Markow assumption which is the probability of reaching the next state depends only on probability of the current state. There are two distinct probabilities: *transition probability* that tracks the transfer of hidden states from one another in a dataset, and *emission probability* is a metric that tracks the possibility of each hidden state certain combination of visible states⁽⁸⁴⁾.

1.9.5.5 Hierarchical Clustering

Hierarchical Clustering is a method that can create homogeneous groups of entities or objects. Each observation is assigned as a single cluster; thus, N observations will have N number of clusters. By observing the similarities and dissimilarities of nearest (or similar) clusters are merged to make a bigger cluster. Similarities and dissimilarities are determined by various methods, such as cosine similarity method, Euclidean distance method, Jaccard similarity coefficients, Manhattan distance method, and Minkowski distance method. The process of merging clusters is repeated until all observations are merged into one cluster.

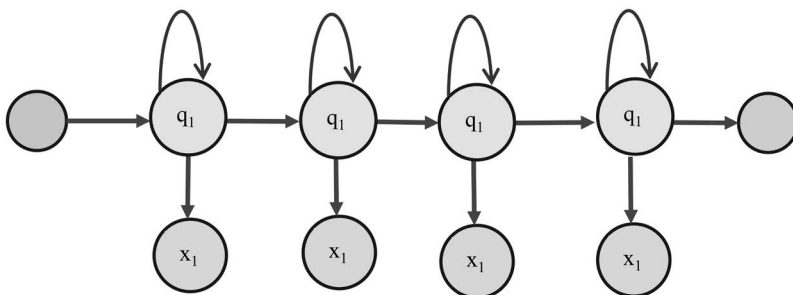


FIGURE 1.30 Operation of a hidden Markow model.

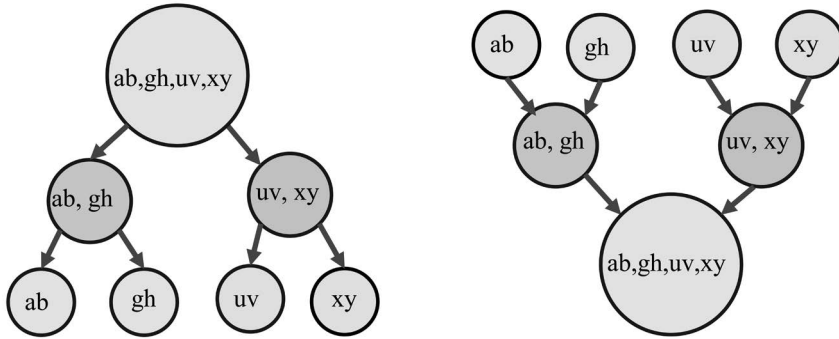


FIGURE 1.31 Hierarchical clustering methods.

There are two types hierarchical clustering (Figure 1.31): 1) divisive clustering and 2) agglomerative clustering. In divisive clustering the entire data is assigned to a single cluster. Then the cluster is split until there is one cluster for each observation. This is a top-down technique. In agglomerative clustering each data or observation is treated as a cluster. Then the pair of clusters are combined until the clusters are merged into one big entity containing all the data⁽⁸⁵⁾.

1.9.6 UNSUPERVISED LEARNING REGRESSION

Unsupervised learning regression finds the hidden structures in unlabeled data. It enables the mapping of raw data into a multidimensional label space. In unsupervised learning, there are five main regression techniques: 1) PCA, 2) apriori algorithms, 3) linear discriminant analysis (LDA), 4) encoders and autoencoders, and 5) multiple kernel learning (MKL).

1.9.6.1 Principal Component Analysis

PCA is a data reduction technique that creates one or more index variables from a larger set. It captures the variance in variables in small sets. The created index variables are called components and are used for building predictive models and in data analysis. The mathematical method employed uses a collection of views of associated variables to transform into a collection of linear variables, known as the principal components. PCA relies on possibly correlated numeric variables into a set of values of linearly uncorrelated variables.

PCA can be accomplished by numerous methods such as the Eigenvalue decomposition of a data covariance or correlation matrix, or singular value decomposition (SVD) of a data matrix. If PCA is used, a normalization step is applied to the initial data. Figure 1.32 illustrates the combining of measured variables into a single variable. All variables contribute to the component variable. Depending on the weights, some Y variables contribute more than others⁽⁸⁶⁾.

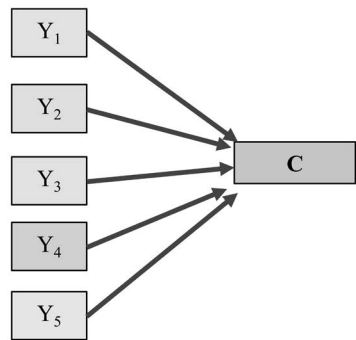


FIGURE 1.32 A PCA structure.

1.9.6.2 Apriori Algorithm Learning

Apriori algorithm learning is based on datamining principles. It seeks relational datasets by applying mining methodologies and association rules, as in Figure 1.33. Apriori algorithms classify various unique items in a dataset. If these items appear sufficiently often, algorithm extends these datasets to larger and more general item sets. The item sets identified by Apriori can be used to discover association-based rules on the itemset which highlight overall trends in the dataset. Apriori algorithms tend to be slow compared to other algorithms. It scans the entire database multiple times to identify the item sets, hence efficiency can be low⁽⁸⁷⁾.

1.9.6.3 Linear Discrimination Analysis

LDA is used in multiclass classification problems in supervised ML. It is based on data dimensionality reduction by separating multiclass datasets having multiple features. LDA is used for dimension reduction as well as for classification and data visualization. It is a simple and robust method and used for preliminary classifications

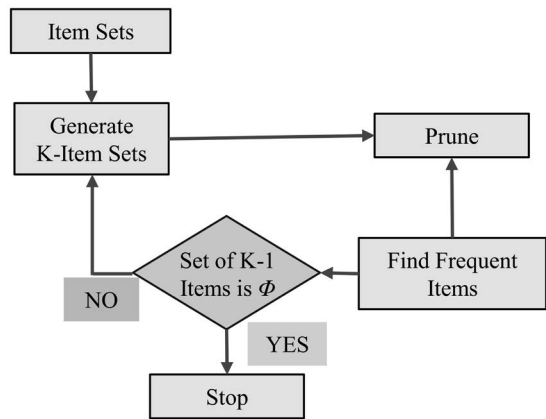


FIGURE 1.33 Operation of an Apriori algorithm.

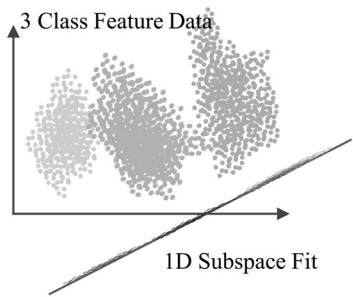


FIGURE 1.34 A typical output of linear discriminant analysis.

before other more rigorous methods are employed⁽⁸⁸⁾. A typical output of LDA is illustrated in [Figure 1.34](#).

1.9.6.4 Autoencoders

Autoencoders are used in supervised and unsupervised learning. They are based on ANN as explained previously. They are useful in efficient coding of unlabeled data mainly for the purpose of dimension reductions.

1.9.6.5 Multiple Kernel Learning

MKL is used to estimate the value of a function at a given point. They are used in regression, classification, and clustering. MKL different kernels are integrated in the same ML model. In the case of clustering applications, multiple kernel clustering (MKC) groups data by integrating information from base kernels ([Figure 1.35](#)). MKC first decomposes the kernels into orthogonal partition matrices to learn from their characteristics⁽⁸⁹⁾.

1.9.7 SEMI-SUPERVISED LEARNING

Semi-supervised learning involves human-in-the-loop (HITL) to review questionable predictions. It applies the SL model from small and tagged datasets to large and

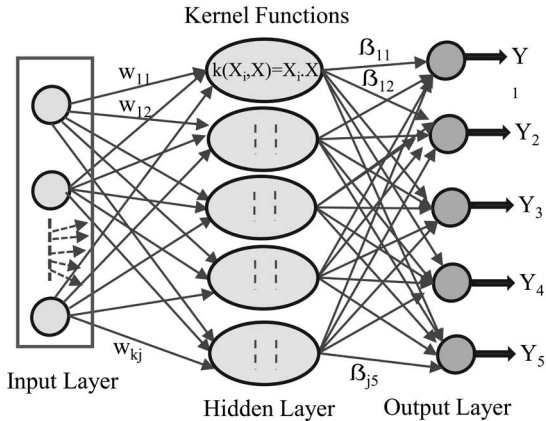


FIGURE 1.35 Multiple kernel learning structure.

untagged datasets. In some applications, the combination of labeled and unlabeled data improves the learning performance. The purpose of semi-supervised learning is to separate the combination of labeled data and unlabeled data.

There are several methods in semi-supervised learning, which can be listed as:

- Active learning,
- Few shot learning,
- Co-training,
- Generative models,
- Graph-based learning,
- Low-density separation, and
- Transduction.

Most of these methods are explained in various places in this book. Here, only the active learning and a few shot learning will be explained.

1.9.7.1 Active Learning

Active Learning uses less labeled data for model training. Active learning intends to find representative samples (e.g., a small portion of large pool of a dataset) to be annotated by an oracle (e.g., human annotator), such that a well SL model can be trained by intelligently on the selected annotated samples. Generally, active learning techniques can be divided into three groups: membership query synthesis, stream-based selective sampling, and pool-based sampling^(18,20,90).

1. *Membership query synthesis method* learners can select any unlabeled samples from instance space or generated samples by the learners for annotation.
2. *Stream-based sampling* is selective sampling sequentially sending the data samples to an active learner. Active learner decides whether to annotate or to discard the sample.
3. *Pool-based sampling* method aims to select samples for annotation from a pool of unlabeled samples. It can select more than one sample at a time. Pool-based sampling is the most widely used technique in active learning.

1.9.7.2 Few-Shot Learning

Few-shot learning is based on meta-learning principles, that is by learning to learn. It consists of teacher and student models, where the teacher learns how to optimize a student's learning while the student learns how to perform downstream tasks.^(18,21,91) Few-shot learning can significantly reduce the amount of labeled data for the model applied.

1.10 REINFORCEMENT LEARNING

Reinforcement learning (RL): Initially designed to discover optimal action spaces through adaption and interactions in uncertain time-varying environments. Today, this model provides an alternative way of learning from unlabeled data as long as either positive or negative feedback can be cleaned during the learning process by

trial and error. RL does not require direct supervision. The interactive learning paradigm is capable of learning to act so that it can prepare itself for achieving an ever-improving performance. The disadvantage of RL is, however, that it relies on large amounts of data and resources. Another drawback is that the resultant high-performance solutions often lack plausible physical interpretations. RL have been applied in unmanned aerial vehicle (UAV) communications, autonomous vehicle driving systems, mobile edge computing (MEC), and wireless caching placements.

RL suitable for supervised and unsupervised learnings by not requiring labeled input and output sets. RL uses agent theory, and each agent learns and makes suitable decisions to maximize a long-term reward. It rewards the desired behavior and punishes the undesired ones (Figure 1.36). RL algorithms include Markov decision process (MDP), Q-learning, policy learning, actor critic (AC), deep reinforcement learning (DRL), and multi-armed bandit (MRB), and few others.

RL applies biological aspects of learning to solve problems without reliance on the availability of labeled data beforehand. There is no separation between the training and application phases, therefore, the system continuously learns as it predicts. RL trains agents to respond to maximize the values by trial-and-error method. Though it is different from supervised or unsupervised learning, it works in a combined manner with them. RL is useful for training computers to play games and training robots to perform tasks^(66,72,92). Some of the algorithms employed in RL are:

- Deterministic policy gradient,
- Learning automata,
- MDP,
- Proximal policy optimization,
- Q-learning,
- State–action–reward–state–action,
- Temporal difference learning,
- Trust region policy Optimization, and others.

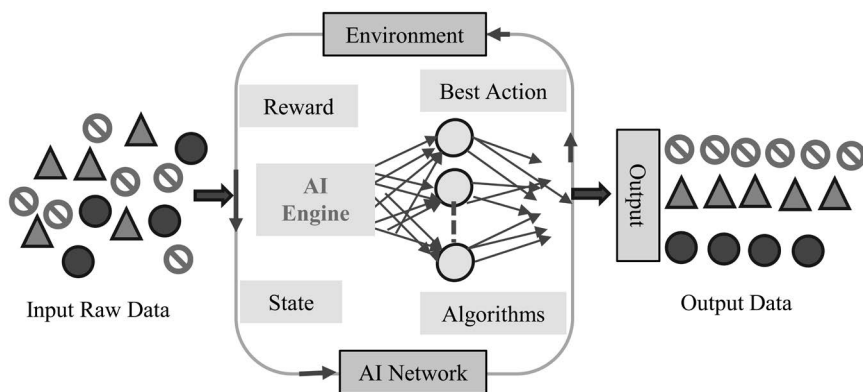


FIGURE 1.36 A typical reinforcement learning methods with feedback loop.

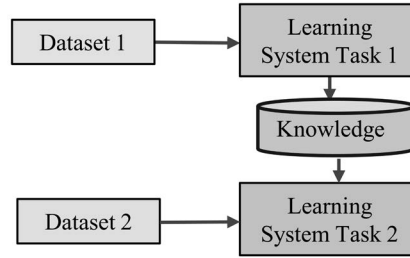


FIGURE 1.37 Transfer learning methods use a knowledgebase.

1.11 TRANSFER LEARNING

Transfer learning (TF) uses the knowledge gained from a solved problem to apply on new ones (Figure 1.37). It aims to solve a given task (target domain) with labeled data from related ones (source domains). It minimizes the differences between the source and target domains. There are two approaches to minimize domain differences: 1) Distance-based methods and 2) Adversarial-based methods.

1. **Distance-based methods** measure and minimize the domain differences such that the gap between the source and target domains is minimal. The source classifier/regressor trained using the labeled source domain data to be used for the target domain classification/regression tasks.
2. **Adversarial-based methods** are inspired by GANs.

TF assumes the availability of a labeled source domain is related to target. For example, when performing an image classification task with no available labeled data, an effective way is to use TF from publicly available datasets.

A form of TF is the *fine-tuning* that assumes the availability of a few labeled data in the target domain. Fine-tuning aims to train few layers of the learnt network from the source domain to train by using few labeled data of the target domain. Source domain can be shared community AI models^(18,93).

1.12 SELF-SUPERVISED LEARNING

Self-supervised Learning is a technique that learns directly from unlabeled data. It has two main approaches: 1) **contrastive learning**, which learns features by identifying similar data samples and dissimilar ones. It learns representations by maximizing the similarity of two different augmented (enlarged) views of one sample and by minimizing the similarity of augmented views of different samples and 2) **pretext tasks** rely on supervised samples for learning representative features.

Self-supervised learning can extract useful information from the data by contrasting between samples with their augmentations or defining pretext tasks. The creation of augmented samples or pretext tasks lead to additional training need. Self-supervised learning requires relatively more training iterations^(18,94).

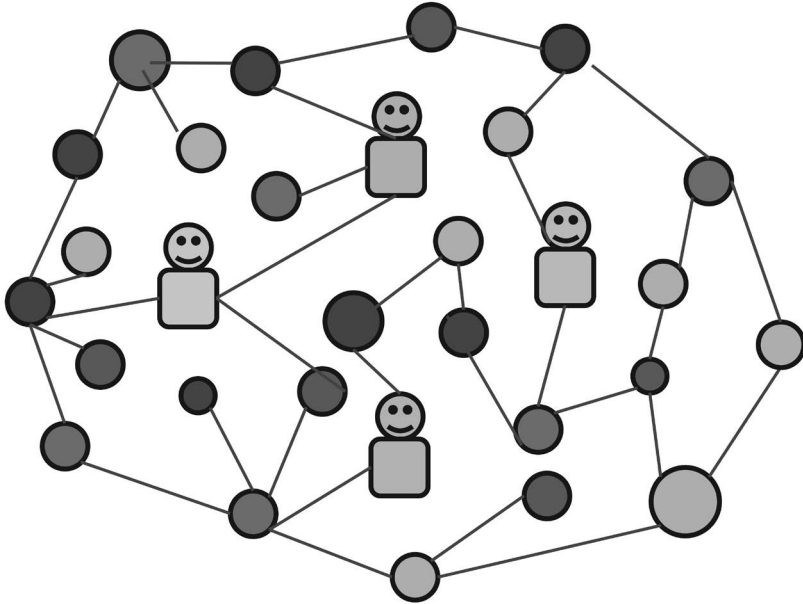


FIGURE 1.38 A typical distributive learning structure involving many devices.

1.13 DISTRIBUTED AND COLLABORATIVE LEARNING

If there are many devices (Figure 1.38) involved AI applications, algorithms can be implemented central devices or/and by local devices to be forwarded to centralized entities (e.g., cloud). Or all the raw data can be forwarded to the central entity for entire processing. In distributed and collaborative learning, the training dataset, labeling, and models can perform with or without any human intervention. Algorithms are opted in accordance with the problem complexity, while taking into consideration time and accuracy requirements. This method is suitable distributed systems such as communication networks, and internet of things (IoT)^(71,95,96).

1.14 FEDERATED LEARNING

Federated learning is a collaborative and distributed learning method. In federated learning, the data is trained in a shared and synchronized format by multiple devices and servers. In some cases, the sharing is organized by a central coordinator (Figure 1.39) thus avoiding the need of sending all the data to a central location. The challenges in FL are the resource allocation, privacy, communication costs, and security. Federated learning is used in large projects such as the deep space surveillance systems (DSSS)⁽⁹⁷⁾.

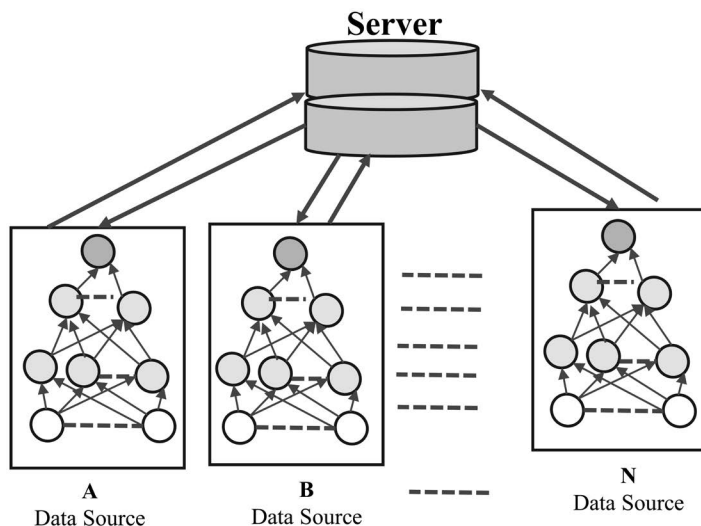


FIGURE 1.39 A federated learning structure.

1.15 DEEP LEARNING

DL is an AI technique that mimics the human brain in understanding and coming to conclusions. DL is based on the ANNs theory. The learning models can be supervised, semi-supervised and unsupervised. There is a repertoire of extremely successful DL algorithms, such as the DNN, CNN, RNN, and LSTM.

DNNs progressed remarkably over the past few years since 2020. DL covers many fields and subfields of AI, and it has introduced a new meaning to AI. It finds a wide range of applications, such as computer vision, speech recognition, NLP, image classification, communications networks, national security, finance, and many others.

DL can combine low level features to form more abstract high-level representations, attribute categories, and novel feature discoveries in the new data. Most DL networks explicitly include feedback loops thus giving them memory capability, and dynamic behavior that allows handling of very large sequences, such as the speech, text, and images.

Unlike the classical ANN models, DL has many folds of hidden layers. Each layer in DL networks contain multiple neurons which are connected to neurons of the next layer. As in the case of ANN, each neuron acts like a summing junction (adder) summing up the values supplied from activation functions. In some models, the difference between the final output and the desired output is calculated to be processed further.

Having multiple hidden layers has many advantages in extracting features and learning capacity. The features learned by the DL models can describe the new data with greater accuracy. The training is done layer-by-layer initialization. DL can be

applied in complicated function approximations by learning deep non-linear network structures and it accepts distributed representation of input data.

DL demonstrates the capacity to learn the essential characteristics of datasets from a few samples by applying certain preset rules. It can extract complex features from a large amount of data to discover hidden patterns and trends. The more hidden layers the neural network has, the deeper and more intense processing required for the data go through. Since the use of more hidden layers entails the need for higher computational and processing capability assisted by powerful GPUs and other advanced hardware devices.

It is worth emphasizing that during the training, if the final output differs from the expected output, the weights in the layers are adjusted and the whole process is repeated, usually starting from the input layer. In each iteration of training, the amount of deviation between the generated output and the expected output is measured to determine discrepancies. The discrepancies are known as the *loss functions*. The goal is to bring the loss function to zero or as close to zero as possible. This makes DL algorithm more accurate in extracting the correct features or patterns from a given dataset. When the loss function is minimized, the network is said to provide the best result it could deliver, and it is ready for a new input.

DL implementations are making revolutionary changes in societies as well as science and technology. The remarkable progress in DL can be attributed to a number of factors: 1) increase in computer power, better computing, and state-of-art hardware organizations (such as the new generation of GPUs, 2) wide availability unexplored curated training data such as the public data on the internet and various social platforms, 3) accumulation of related knowledge over many decades such as optimization methods, 4) availability of highly trained programmers and related expertise, 5) big interest to the extent of a “must win” race between governments and investors, interest of public, seeking military advantages, unexplored aspects in social sciences, scientific research, and space explorations, and 6) large and direct investments by governments and private organizations^(66,98–103).

1.15.1 DEEP LEARNING TECHNIQUES

Most practiced and researched DL techniques include:

- Autoencoders and Deep Autoencoders,
- Boltzmann machine (BM) and RBMs,
- CNNs,
- DBNs,
- GANs,
- LSTM,
- Q-learning,
- RNNs,
- Transformers, and more.

Each of these techniques also has well-established as well as emerging subclasses. Since DL is the game changer in AI, some of these methods will be explained in detail.

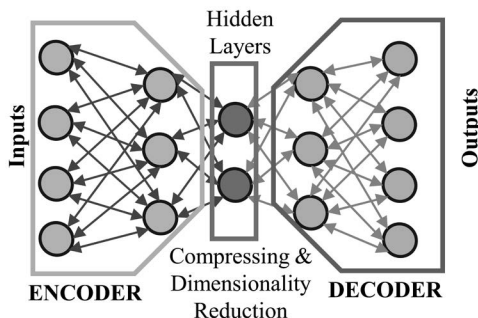


FIGURE 1.40 An autoencoder.

1.15.1.1 Autoencoders and Deep Encoders

Autoencoding compresses information and learns how to reproduce accurate approximation of the original information from the compressed data. Since they are used for data compression, autoencoders are effective in dimension reduction and data denoising. A typical structure of an autoencoder is given in Figure 1.40.

Deep Autoencoder is an algorithm to treat the hidden layer of NN as an encoder and a decoder of the input data. When it reaches the output layer, the output results should be consistent with the input data as far as possible. One advantage of this approach is that the hidden layers capture the features of the input data and remember the features.

Encoder-decoder architectures are used for involving the input data to output data of a different form by machine translation. Encoder processes the input data and decoder uses that data to produce output. This technique is used in RNN-based and transformer-based models making these methods highly efficient.

Encoder processes the input data and compresses the information into a state. This context (state) captures the essence of the of the input data for decoder to process further.

Decoder takes the context from the encoder and generates the output data. The output is sequential, and decoder generates it one element at a time using the context. A generated element decides the next one^(98,101–104).

1.15.1.2 Boltzmann Machine

Boltzmann Machine (BM) is a random neural network because it uses Boltzmann distribution function for the activation. BM is a network where all the neurons are symmetrically connected (Figure 1.41). The neuron-like units randomly decide to turn on or off. It introduces statistical probability that changes the state of neurons. Hence, the equilibrium state of the network obeys Boltzmann distribution. In some cases, BM combines the advantages of multi-layer FNN and discrete Hopfield network in its network structures, learning algorithms, and dynamic operation mechanisms. BM has the capacity to learn large amounts of data. It can find the optimal solution through a simulated annealing process model. The training time is longer than other models, such as the back propagation neural networks^(98,105).

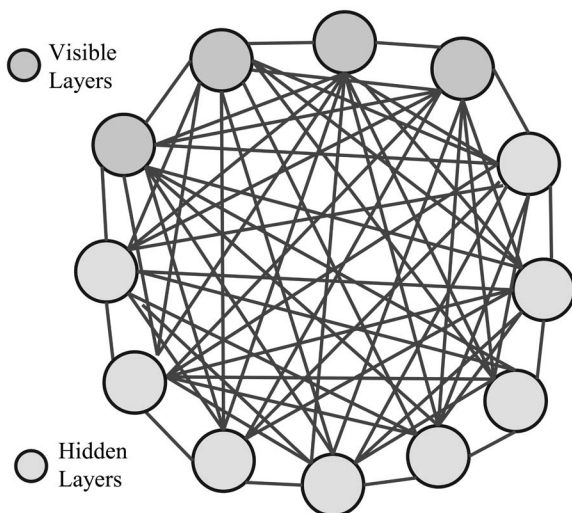


FIGURE 1.41 A Boltzmann machine.

1.15.1.3 Restricted Boltzmann Machine

RBM is a random neural network model with a two-layers structure. Layers are symmetrically connected, and no self-feedback allowed. It has full connections between layers and no connections within layers. The first layer, called the *dominant layer*, consists of the dominant neurons. Dominant neurons are used for training the input data. The second layer, called the *recessive layer*, consists of recessive neurons and acts as a feature detector. The difference between RBM and DNN is that RBM does not distinguish between forward and reverse. The state of the visible layer can be applied to the hidden layer, while the state of the hidden layer can also be applied to the visible layer. The training process of RBM is to find a probability distribution that can best generate training samples^(98,106).

1.15.1.4 Convolution Neural Networks

CNN is basically an ANN acting as a deep forward neural network (DFNN) (Figure 1.42). It has a weight sharing structure which makes it more like biological

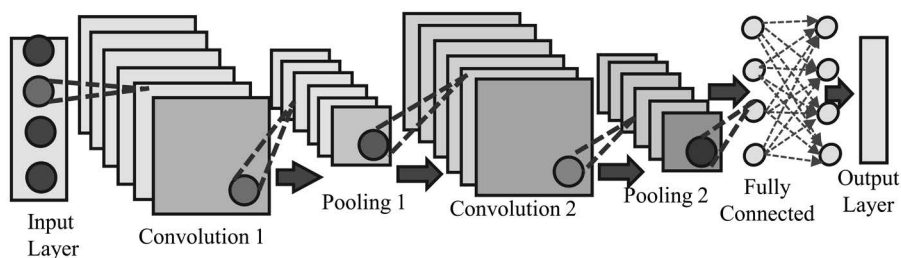


FIGURE 1.42 Operation of a DFNN.

neural networks. Weight sharing simplifies complexity of network model and reduces the number of weights. In the traditional algorithm, different methods are used to extract features, but CNN learns and filters by itself. That means the network is self-sufficient, and it can automatically extract features.

CNNs employ convolutional layers known as subsampling (pooling) layers. Pooling layers serve to reduce the dimensions of the feature maps. This reduces computation needs, minimizes overfitting by retaining the relevant information. The final stage layer is fully connected (Figure 1.43). The fully connected layers execute reasoning based on the learned features leading to classifications and predictions. CNNs use of the convolution mathematical operations in feature extraction processes. Convolution neural networks are used in computer vision, pattern recognition in image processing, and speech recognition, and other applications.

A convolution neural network automatically detects important features without any human intervention. The layers apply a convolution operation to the inputs, passing the result to the next layer. Pooling layer reduces the data dimensions by combining the outputs of the neuron clusters. The fully connected layers compute the class scores, resulting in classifications. The main components of CNN are convolution layer, ReLU layer, pooling layer, and fully connected layer.

Convolution layer is the core building block of CNN. The convolution layer applies several filters in the input. Each filter activates certain features from the

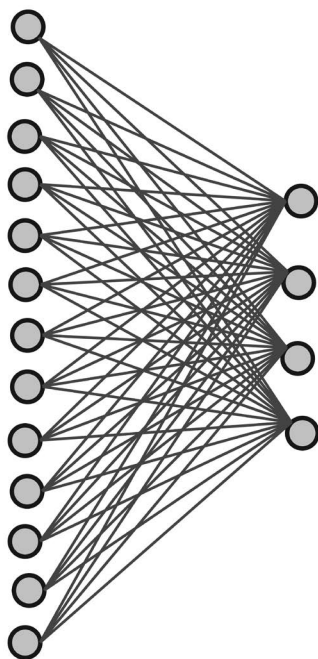


FIGURE 1.43 Fully connected final layers of a CNN.

input, such as the edge of an image. This process is crucial for feature detection and extraction.

ReLU layer (rectified linear unit) is applied to introduce nonlinearity into the model, allowing it to learn more complex patterns.

Pooling layer reduces the spatial size of the representation, decreasing the number of parameters and computations. This way, it controls the overfitting.

Fully connected layer is at the network's end. It maps the learned features to the final output, such as the classes in classification task.

The following equation expresses elements of the output feature that sums an element-wise products of kernel and a part of the input image on which the kernel is positioning. The convolution operation can be expressed as:

$$S(i,j) = (I * K)(i,j) = \sum_k \sum_m I(i+k, j+m) \cdot K(k,m) \quad (1.20)$$

where $S(i,j)$ is the output feature map,

I is the input,

K is the kernel or filter,

i, j are the coordinates on the feature map,

The convolution operation,

k, m are the coordinates in the kernel.

Convolution neural networks employ multiple filters at each convolution layer to capture a wide array of features from the feature map. Multiplicity and the resultant depth are central to the network's success in processing information in a comprehensive and nuanced form. Each filter detects different features in the input, such as colors, edges, textures, and complex shapes^(98,100,107).

1.15.1.5 Deep Belief Networks

DBNs detect the distribution of the input training dataset. DBN is composed of many layers with multiple neurons in each layer. The neurons are divided into dominant neurons and recessive neurons. *Dominant neurons* are used for the input, and *recessive neurons* are used to extract features. The connections between the top two layers are undirected edges and form a federated memory. The other lower layers have directed connections above and below the connections. The bottom layer represents the data vector, and each neuron represents one dimension of the data vector.

Stacking RBMs on top of each other forms *deep belief machines* (DBMs) (Figure 1.44). The process of training DBM is carried out layer by layer. In each layer, the hidden layer is inferred by the data vector, and then the hidden layer is regarded as the data vector for the next layer. Each RBM can be used independently as for clustering. Typical applications of DBMs are speech and object recognition^(97,100,108).

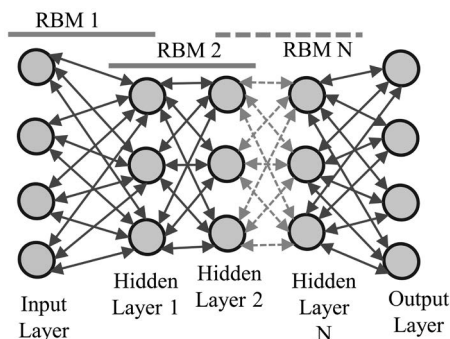


FIGURE 1.44 A Deep Belief Network with multiple Boltzmann layers.

1.15.1.6 Generative Adversarial Networks

GANs adopt an approach for maximum likelihood estimation and employ two neural networks that compete against each other in a zero-sum game. The GAN architecture (Figure 1.45) utilizes a generator model and a discriminator model. The *generator* network takes random noise as input and generates data like the training data. Generator aims to produce data indistinguishable from the real data from the discriminator. *Discriminator* network takes the real and generated data as input and attempts to distinguish between the two. The discriminator is trained to improve the accuracy detecting real vs. generated data while the generator is trained to fool the discriminator. Examples include game development and artificial video generation.

GANs are used in unsupervised ML, implemented by two neural networks competing in a zero-sum game. This enables GANs to generate new data with the same statistics as the training set, such as generating images that look authentic to human observer^(100,102,109).

1.15.1.7 Q-Learning

Q-learning is a model-free, value-based, off-policy algorithm for RL. It finds the best series of actions based on the current state. The “Q” stands for quality. Quality represents how valuable the action is in maximizing future rewards. Q-learning is basically a method that learns by experience. Q-learning is often combined with

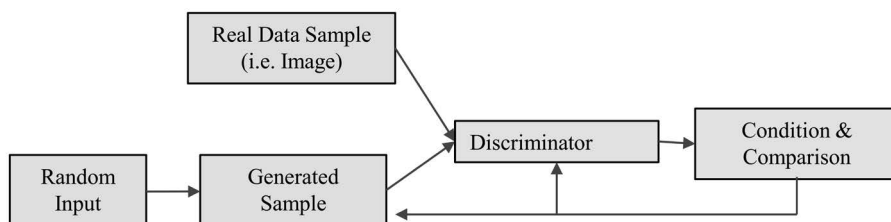


FIGURE 1.45 A GAN structure.

DNNs^(109,110). It is used with CNNs trained to extract features from video frames to play video games or for learning robotic control. Bellman's equations are often used in Q-learning. Bellman's formula is recursive method for optimal decision-making. The following formula helps to calculate the value of a given state and to determine its relative position. The state with the highest value is the optimal state. Bellman's equation is:

$$Q(s,a) = Q(s,a) + \alpha \left(r + \gamma \max_{a'} (Q(s,a')) - Q(s,a) \right) \quad (1.21)$$

where $Q(s,a)$ represents the expected reward for acting in a state,

α is the learning rate,

a' is the next state of a ,

γ is the discount factor,

r actual reward received,

s' , the next state, and

$\max (Q(s,a'))$ highest expected reward for all possible a' in state s .

1.15.1.8 Recurrent Neural Networks

RNNs is designed to describe the relationship between the current output of a sequence and the previous information. In the network structure, RNN remembers the previous information, and then uses it to influence the output of subsequent nodes. RNN can be regarded as the result of the same neural network being copied indefinitely. In theory, RNN can carry information over an indefinite number of steps, but in practice it cannot tolerate too many steps without losing the context. When a greater number of steps are taken the gradient of the network tends to vanish. This interferes with the ability of a gradient-based optimizer, e.g., SGD, to converge. Typical applications of RNNs are speech recognition, language modeling, language translation, robot control, handwriting recognition, time series prediction, anomaly recognition, and image processing^(98,111)

1.15.1.9 Transformers

Transformers have multiple layers of encoders and decoders as shown in [Figure 1.46](#). A transformer model is an unsupervised pre-training algorithm useful particularly for NLP such as ChatGPT/Bing, GPT-3.5 or GPT-4, and Bard based on language model for dialogue applications (LaMDA).

Transformers process data in parallel, thus reducing the training duration. They utilize their attention time to weigh the influence of different words on each other. Transformers can handle data sequences without the need for sequential processing. This makes them highly effective in NLP tasks including translation, text summarization, and sentiment analysis. The main components of transformer are attention mechanism, encoder layers, and decoder layers⁽¹¹²⁾.

Encoder layer process the input data in parallel, applying self-attention and position-wise fully connected layers to each input part.

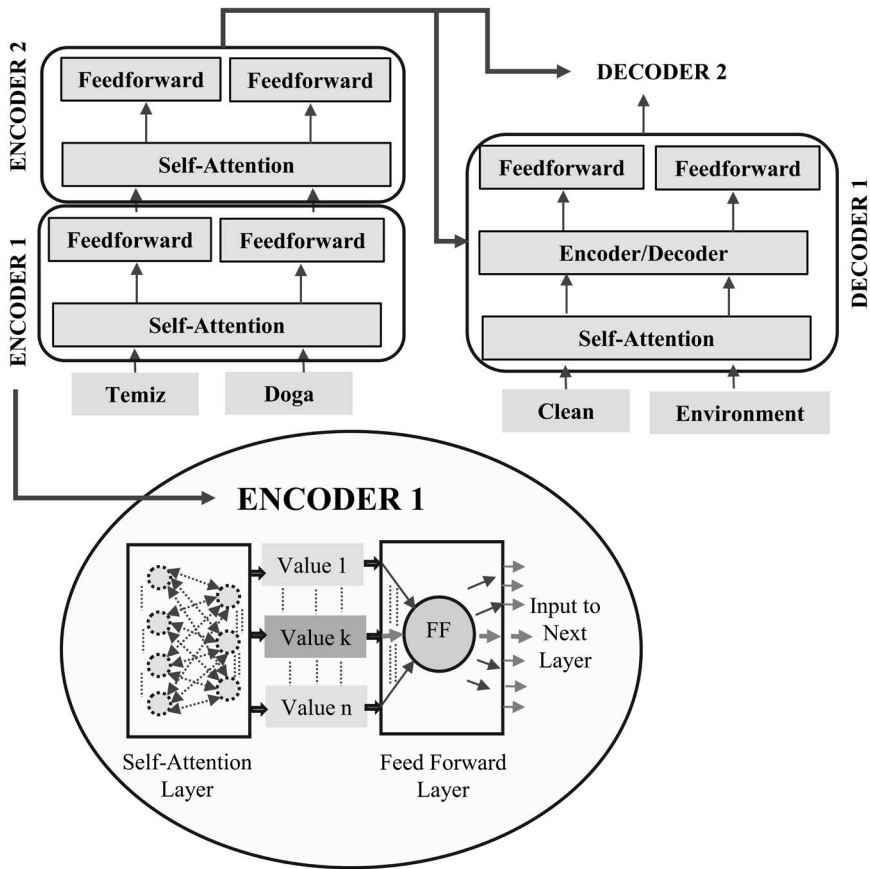


FIGURE 1.46 A transformer structure.

Decoder layer uses the output from the encoder to produce an output. It also applies self-attention in the way that prevents position from attending to the next position to maintain causality.

1.15.1.10 Long Short-Term Memory

LSTM is a version of recursive neural network model. To solve the problem of long-term dependence, different interval of useful information and different length. LSTM is a special network structure (Figure 1.47) with three “gates.” It relies on these gates to make information selectivity affecting the state of sequence in the neural network. The “gate” structure is a combination of a sigmoid and a bitwise multiplication operation. It is called a gate because of having a fully connected NN layer. The sigmoid is an activation function that outputs a value between 0 and 1 describing how much information of the current input can pass through this structure. LSTMs were explicitly designed to avoid vanishing gradient problems. Although LSTM introduces more complexity in training it works well for long sequences⁽¹¹³⁾.

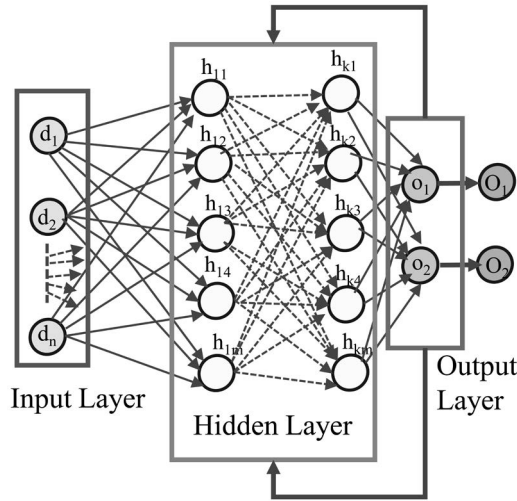


FIGURE 1.47 An LSTM with gates and hidden layers.

1.16 GENERATIVE ARTIFICIAL INTELLIGENCE

AI can be considered to have two basic types: 1) predictive AI (PAI) and 2) generative AI (GAI). Predictive AI aims to generate accurate predictions and forecasts about the future events. It represents a well-established and relatively old technology. *Generative AI* uses most AI techniques. It focuses on creating new and original content, which basically gives it a degree of intelligence. GAI attracts much attention due to its flexible applicability in many areas.

GAI finds wide applications in generating images, texts, audio, music, computer codes, videos, and so on. It can produce a variety of novel contents without repeating the trained data. GAI can be constructed by using supervised, unsupervised, and self-supervised ML methods. GAI can be unimodal trained with one type of input, or multimodal that can take multiple forms of datasets such as images, text, and audio simultaneously.

Types of GAIs are:

- Autoregressive models,
- Diffusion models,
- GANs,
- Generative pretrained transformer (GPT),
- RNNs,
- Reinforcement learning models,
- Transformer base models, and
- Variational auto encoders (VAEs).

GANs, RNNs, Reinforcement Learning Models, and Transformer-Based Models are explained above. The remaining models: Autoregressive Models, Diffusion Models, GPT, VAEs will be explained next.

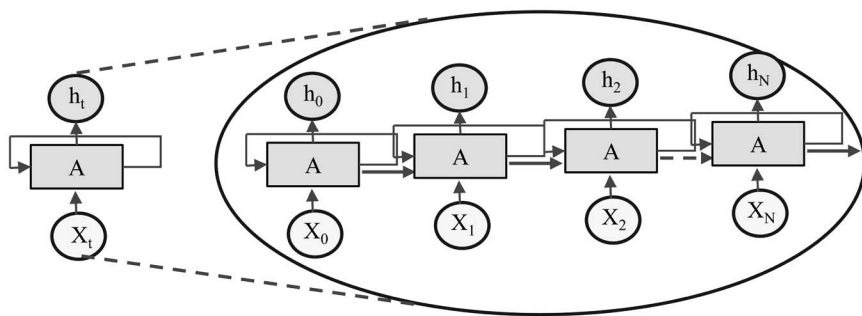


FIGURE 1.48 An autoregressive AI model.

1.16.1 AUTOREGRESSIVE MODELS

Autoregressive models (ARs) are statistical models for understanding and predicting the future values from the past values. AR is a feedforward model (Figure 1.48) that learns from its own past experiences to predict the future. It is mostly applicable in time series-based situations and time varying processes, e.g., the speech and image recognition and generation. There are several AR methods such as the Pixel CNN, PixelCNN++, Wavenet, and Transformer⁽¹¹⁴⁾.

1.16.2 DIFFUSION MODELS

Diffusion models use probabilistic data generation. It generates new data based on the data it is trained on. In the process, sequences of transformations are applied to random noise. The model scales and centers data so that ideally the data is converted to a distribution with a mean zero and a variance of one. The data is basically destroyed by adding gradual noise (i.e., Gaussian noise) and then reversed to recover the data from the learning process. Initially a forward distribution method is used to diffuse the sample step-by-step. The forward diffusion generates complex data distribution then the reverse diffusion is applied to map it back to simple distribution through a sequence of inverse transformation. DMs are used for text recognition and video generations. There are various variations of this method, such as denoising diffusion probabilistic models (DDPMs), score-based generative models (SGMs), and stochastic differential equations (SDEs)⁽¹¹⁵⁾. A block diagram of a diffusion model is shown in Figure 1.49.



FIGURE 1.49 A diffusion model.

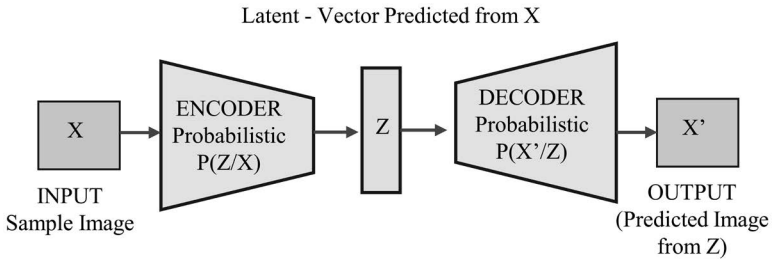


FIGURE 1.50 A variational autoencoder model.

1.16.3 GENERATIVE PRETRAINED TRANSFORMER

GPT uses a neural network that contains self-attention features to process input sequences. Transformer architecture is used for parallel processing of various length inputs. A block diagram of a typica; GPT is illustrated in Figure 1.46.

In this figure, *encoders* understand the input (i.e., text). GPTs are composed of multiple layers of self-attention FNN. Decoders generate the outputs, and they also have self-attention and Feed-Forward NNs. They also have *Language Model Head* for speech and text interpretation. GPT predicts the next token in the sequence from the previous tokens. *Dialogue Generation Heads* generate the response to given prompts in the context of dialogue. The Language Model Head and the Dialogue Generation Heads have linear layers with trained weights learned during the pre-training⁽¹¹⁶⁾.

1.16.4 VARIATIONAL AUTO ENCODER

VAEs are probabilistic graphical models. They are regenerative models that require neural networks for encoding and decoding as part of their overall structure. In this model, encoders produce multiple different samples as results of the same distribution. Decoders map the latent space into input space⁽¹¹⁷⁾. They are applied for generation of realistic sounds, images, and texts (Figure 1.50).

1.17 NATURAL LANGUAGE PROCESSING

NLP enables computers to perform a wide range of natural language related tasks ranging from parsing and part-of-speech tagging to machine translations and human dialogues. NLP uses various techniques such as CNNs, RNNs, LSTM, and transformers. These models can be constructed on multiple natural languages to translate from one language to another. NLPs can analyze grammar, summarize text, express sentiment, and generate non repeated texts. Language models are usually trained on very large language datasets.

ML approaches for language processing had been based on SVM and logistic regression for training. Neural networks based on dense vector representations

proved to be superior in NLP processing. This trend is sparked applications of DL methods. DL enables multi-level automatic feature representation learning^(118,119).

1.17.1 WORD EMBEDDING

Word embeddings are the first stage of data processing. AI networks are pre-trained by optimizing an auxiliary objective in a large unlabeled corpus. The learned word vectors can capture general syntactic and semantic information. Word representations, once compiled into sentence representations of joint probability of the word sequences. The context words are assumed to be located symmetrically to the target words within a distance equal to the window size in both directions. This method is called the *window approach*. As the embedding dimension increases, the accuracy of prediction also increases until it converges at some point. This is the optimal embedding dimension as it is the shortest distance without compromising the accuracy.

There are several other methods for word embedding, such as the “*count-based*” model. In this model, the word co-occurrence count matrix is preprocessed by normalizing the counts and log-smoothing them. This matrix is then factorized to get lower dimensional representations by minimizing a reconstruction loss^(118–120).

1.17.2 CNN IN NATURAL LANGUAGE PROCESSING

CNNs in NLP tasks use a window approach, which assumes that the tag of a word primarily depends on its neighboring words. For each word, a fixed-size window surrounding itself is assumed and the sub-sentence are considered ranging within that window. The window approach is coupled with a time-delay neural network (TDNN) in which the convolutions are performed across all windows throughout the sequence. These convolutions are generally constrained by defining a kernel having a certain width. TDNN considers all windows of words in the sentence at the same time. The layers are stacked like CNN architectures to extract local features in lower layers and global features in higher layers⁽¹²¹⁾.

1.17.3 RNN IN NATURAL LANGUAGE PROCESSING

RNNs process sequential information. They perform the same computation (recurrent) over each token of the sequence and each step is dependent on the previous computations. A fixed-size vector is produced to represent a sequence by feeding tokens one by one to a recurrent unit. That enables RNNs to have “memory” over previous computations and use this information in current processing. The memory capabilities of RNNs are used in other applications such as machine translation, speech recognition, and image captioning.

Languages exhibit a natural recursive structure, where words and sub-phrases combine into phrases in a hierarchical manner which can be represented by a constituency parsing tree. Tree-structured models have been used to make use of syntactic

interpretations of sentence structure. In an RNN, the representation of nodes in a parsing tree is determined by the representations of all its children.

RNN language generators are trained as agents by maximizing the likelihood of each token in the ground-truth sequence given the current hidden state and the previous tokens. Ground-truth tokens are then replaced by a token generated by the model itself. This training scheme (teacher forcing) provides the real sequence prefix to the generator during each generation (loss evaluation) step. The discrepancy between training and inference, known as the *exposure bias* and it can yield errors that can accumulate quickly along the generated sequence. Some mechanism stores a series of hidden vectors of the encoder, which the decoder is allowed to access during the generation of each token. The hidden vectors of the encoder represent *internal memory*^(118,122).

1.17.4 GENERATIVE PRE-TRAINED TRANSFORMER

Generative pre-trained transformer (GPT) is a large language model based on the semantic relationships between words in sentences in a natural language. Text-based GPT models are pre-trained on a large corpus of text. The pre-training of GPT uses *tokens* to predict the next token. Tokens are a group of characters such as words, or sub-words, punctuations, emojis, or symbols^(98,122,123). ChatGPT 3 has 499 billion tokens and 175 billion parameters (weights) in its network. It is estimated that ChatGPT 4 has 1.7 trillion parameters. ChatGPT was trained using thousands of GPUs operating in parallel. It is estimated that the training took continuous 90–100 days.

ChatGPT has several types of parameters, such as the position parameters to understand the order of words, learned parameters that allow the model to make accurate predictions, hyper parameters which are set before the training for batch size, and model configuration parameters. Model configuration parameters define the architecture of the model such as the number layers in the model. ChatGPT has 96 layers, the dimensions of hidden layers are 12,288, and the sequence length is 2,048. ChatGPT is trained on very large datasets 60% of the data comes from web-pages. Metadata extracts the text from “common crawl” web crawling information of which 22% comes from Reddit posts, 15% non-fiction and academic articles, and 3% from the entire English version of Wikipedia.

GPT models accumulate knowledge about the world and can then generate human-like text by repeatedly predicting the next token. Typically, subsequent training phases make the model more truthful, useful, and harmless, usually using reinforcement learning from human feedback (RLHF). Current GPT models are still prone to generate falsehoods called *hallucinations*, although this can be reduced with RLHF and training on quality data. *Chatbots* allow users to ask a question or request a task in simple short texts.

Large language models are trained using data available in the worldwide web. However, worldwide web contains toxic, misinformation, misused and misleading contents which can make models toxic too. This is called *toxic training*.

Current GPT models and services include *Bard*, *ChatGPT*, *Claude*, *Copilot*, and *LLaMA*. *Multimodal GPT*. Most models process different types of data or modalities such as images, videos, sounds, and texts simultaneously by the same algorithms.

1.18 QUALITY IN MACHINE LEARNING

ML and AI comes in three categories: 1) predictors, 2) optimizers, and 3) generators. In *predictor* models, algorithms function as a predictor for designed quality of result (QoR). The *Optimizer* algorithms include continuous algorithms such as quadratic optimization, non-linear optimization, or combinatorial optimization such as the minimum spanning trees and mixed integer programming. There are a range of heuristics and greedy algorithms for synthesis, routing, buffering, and sizing. *The generator* produces new data that resembles the data it is trained on.

Algorithms have different strengths and weaknesses. The strength of algorithms is that they are suited well in solving known problems efficiently by learning from complex data. The weakness is they usually rely on the well representation and description of the problems. Often several algorithms are integrated to take advantages of strengths of each, while preventing their weaknesses.

AI systems have unique quality concerns during development due to their data-dependent behavior, being sensitive to drifts, handling style of biases, and timely capturing all necessary elements in the training and retraining. Consequently, several steps need to be taken for training and learning to achieve good quality results. These steps are model training, hyper-parameter optimization (HPO), model governance, model monitoring, model testing, model drift, and model deployment^(1,124–127).

Model training requires setting up the infrastructure to automate the training pipeline and monitor the model performance. Retraining may be necessary when the new data becomes available.

Hyper-parameter optimization (HPO) finds optimal solutions by trial-and-error methods or by expert knowledge. Desired model performance, efficiency, and rate of convergence are all dependent on HPO. Wrong choice of parameters directly influences the learnability and the rate of coverage. HPO can be implemented by several methods, such as Bayesian optimization, meta-learning, and neural architecture search (NAS).

Model testing is necessary due to the stochastic nature of AI. In the real world, the inputs and the expected output can change over time.

Model deployment is the transition from test or prototype for the production. A model may be deployed and integrated with other models.

Model drift is caused by many factors, such as the drift in data, changing and evolving data sources, fluctuation in data collections, model staleness, and degradation in performance over time.

Model monitoring and maintainability is necessary due to factors like evolving input data, prediction biases, and lack of understanding of the data characteristics.

Model governance is required when cross-disciplinary knowledge is involved.

1.19 AI HARDWARE

The success of creating and applying AI relies on 1) advanced AI hardware, 2) relevant data, 3) human talent and AI related R&D, 4) well-designed algorithms, 5) business investments, and 6) governmental, commercial, and public interests. In

recent years, the remarkable developments of AI have been due to the availability of advanced hardware. In recent years since 2020, AI attracted considerable interest from the public, researchers, governments, and private companies. Trillions of dollars of investments, by private organizations and by governments, are taking place to gain advantages. There seems to be a race going on between the countries, and companies to develop a better and comprehensive AI^(45,46).

1.19.1 AI HARDWARE TYPES

AI hardware refers to commercial off-the-shelf or custom designed integrated circuits on which algorithms can operate. *Data* refers to structured or unstructured digital information on which models can be trained to make inferences and draw conclusions based on the evidence. *Talent* refers to human know-how to develop *algorithms* and synthesize the hardware, produce the supporting software, and the integration of data into an AI system to create a value.

Since the 1980s, the development of scientific, technological, and social components of AI has occurred at different stages with talent and algorithms leading the way. The availability of appropriate hardware and data had been lagging far behind. At the same time, suitable and unused data had been accumulating over the years, but processing of data had not yet been fully developed. Today, powerful computers and specialized AI specific hardware have found their way in the marketplace. Advance algorithms supported by hardware is helping an exponential growth of AI finding applications in diverse range disciplines.

AI hardware refers to semiconductors chips that ubiquitously power all types of information-related technologies. Running computation-intensive AI applications requires data processing by high-end chips or fast computers.

Apart from the fully equipped computer systems, there are four types of chips suitable to handle AI workloads: 1) central processor units (CPUs), 2) GPUs, 3) field-programmable gate arrays (FPGAs), and 4) application-specific integrated circuits (ASICs). These chips have their own specialized benefits and tradeoffs for speed, efficiency, accuracy, and flexibility. At moment, the GPUs are the winners and leading the way in AI training and applications^(45,47,128–130).

1.19.1.1 CPUs

CPUs manage the system as well as carry out additional computing tasks. Most CPUs have limited computing capability and small memory. Therefore, they do not meet the needs of AI training and implementation algorithms. Still, CPUs are used in AI training and inference, but only in simple AI computations⁽⁷²⁾.

1.19.1.2 GPUs

GPUs are mainly used for graphics processing and for games. Currently, they are the most suitable and flexible chips available for AI. GPUs with thousands of computational cores can achieve 10~100× computing power compared to CPUs.

In AI, GPUs are used for parallel model training particularly in DL. Depending on the complexity of the task in hand, they can be clustered in vast numbers from few GPUs to thousands of GPUs. A typical supercluster contains more than 25,000

GPUs. Parallel operations can reduce DL training time from years of continuous operations to a couple of days. Parallel DL uses two methods: 1) data parallelism, where large dataset is distributed across multiple GPUs and 2) model parallelism, where a DL model too large to fit in a single GPU is distributed across multiple devices⁽⁷²⁾.

1.19.1.3 FPGAs

Unlike CPUs and GPUs, FPGAs can be reprogrammed by the end user with supporting software suitable for the specifications of applications. Their functionalities can be changed dynamically for the program requirements. Thus, the programming flexibility makes FPGAs adaptable to most AI applications. FPGAs come with millions of parallel system logic cells making it flexible and suitable for inference, optimizing, implementing algorithms, and neural network training. FPGAs are also used extensively in non-intensive AI applications. FPGA reduces the power consumption by performing approximate computing and optimizing the algorithms on the hardware level^(72,131,132).

1.19.1.4 ASICs

ASICs chips are popular form of AI chips due to their optimization features to accelerate computing. In these chips, many parallel calculations can be carried out opposed to sequential calculations of CPUs. Memory access can be made inside the chip or nearby memory devices. On-chip memory capability increases the capacity of ASICs thus allowing faster running of algorithms. Compared with GPUs and FPGAs, ASIC chips have a longer development cycle and less flexibility. They have high efficiency and low power consumption. Some manufacturers have introduced different ASIC-based processors intended for various computation-intensive AI tasks requiring high efficiency and high performance.

GPUs, FPGAs, and ASICs are all designed to handle different AI workloads. GPUs are particularly suitable for image-processing and gaming systems as they can process parallel computations efficiently and accurately. Hence, they are preferred for AI training with labeled data. FPGAs offer greater flexibility for customization on specific workload. FPGAs can handle inference tasks requiring high efficiency and high customization. ASICs can handle training or inference workloads depending on the choices of the designer. Also, ASIC circuitry can be customized by the user to handle specific algorithms^(45,72,133,134).

1.19.2 HARDWARE MANUFACTURERS

The semiconductor supply chain is based on design, fabrication, and outsourced semiconductor assembly and test (OSAT). The main intellectual property (IP) is licensed mostly to chip designers. OSATs rely on specialized components that take finished chips and prepare them for the markets such as cell phones.

Once AI found wider applications, the ASIC, GPU, and FPGA manufacturers started to concentrate on the production of AI-related chips. Companies such as Alphabet, AM, Apple, Arm, Cerebras, IBM, Intel, Nvidia, Qualcomm, and TSMC, and many others offer a wide variety of devices and services. At the moments Nvidia

and TSMC are the leaders in the AI hardware market. Others are intensely concentrating on improving their products. TSMC is a contract manufacturer producing chips on demand from other firms. TSMC is based in Taiwan, which is a politically sensitive area.

Manufacturers are concentrating on all-purpose AI chips of various sizes, efficiency, speed, and accuracy. Many applications require specific hardware to satisfy their needs. For example, autonomous vehicles require powerful AI chips to perform inference at network edge at nanoseconds level latency. Wearable devices chips process AI tasks under strict constraints of power consumption, accuracy, and privacy^(41,45).

1.19.3 AI MEMORY AND MEMORY ACCESSORIES

AI has achieved revolutionary performance in providing intelligent solutions in smart applications in some cases surpassed human capabilities. But it requires suitable hardware, large memory, computation power, and adequate bandwidth for high performance outcomes. The accuracy of the outcomes predicted by the algorithm relies on the number of data points and data capacity since more data leads to better precision. On the other hand, too much data may result in irrelevant information and biases. Therefore, it has been challenging to figure out the right amount of data to train AI for an optimum performance in a particular application.

AI implementation can be hardware-centric or software-centric. In many applications, the hardware-centric implementation is preferable over software for enhancing the AI inference. Nevertheless, hardware suffers from a bottleneck called the “memory wall.” Memory wall is the limited capacity and bandwidth of memory transfer^(135–139).

Non-von Neumann architectures, in-memory and near-memory computing is a preferred option to enhance memory capabilities. Processing-in-memory (PIM) partly bridges the separation between memory and processor by implementing processing units inside the memory chip itself. Nevertheless, the PIM processing elements can achieve only limited functionalities due to the fabrication limitations of the memory chips. To address memory problems, accelerators are used in a wide variety of applications. Traditional accelerators are deployed on the underlying platforms of ASICs or FPGAs. Newer accelerators utilize approximate computing to reduce resources and power consumption, while other types of accelerators are deployed on neuromorphic hardware that is based on emerging memory technology.

In some cases, the network architecture can be optimized by reducing the model size along with the hardware optimization. In the other cases, model compression/pruning are the preferred techniques involving large and complex networks. Some methods involve the removal of ineffectual neurons, compressing the weights to a binary number, or reducing the weight precision. Such methods are shown to have reduce the network size and complexity leading to improved performances^(140–143,145,146).

There are two basic types of AI accelerators: 1) hardware accelerators and 2) software accelerators. These two types of accelerators will be explained next.

1.19.4 HARDWARE ACCELERATORS

Hardware and accelerators have improved remarkably on the underlying hardware platforms, computation paradigms, and compiler frameworks. Different hardware designs use different frameworks, different evaluation platforms, and different design paradigms for the viable techniques.

From the hardware perspective most of the traditional methods use multipliers targeting approximation of the mantissa multiplier to improve the overall resource consumption of the floating or fixed-point multipliers. This is done by reducing the multiplication by truncating the least significant bit (LSB) precision scaling.

The DNNs and convolution neural networks tend to be resilient against computation errors making them suitable for approximate computation (AC). Approximate computing is done by software-level approximation concentrating on programming. The hardware-level approximation focuses on the physical system. Such techniques lead higher performance, and faster inference^(133,135,147). There are numerous accelerators, such as DeepBurning, heterogeneous multicore accelerators, and many others, some of which will be explained next:

1.19.4.1 DeepBurning

DeepBurning is a form of hardware accelerator using register transfer-level (RTL-level). In this case, a coordinated compiler generates the control flow and data layout for various AI topology implementations. RTL library holds scripts to perform logical, arithmetic, approximate lookup table (LUT) operations utilized for the implementation of the accelerators^(137,148).

1.19.4.2 Heterogeneous Multicore Acceleration

Heterogeneous multicore acceleration uses shared memory to assist acceleration. Many interactive AI and user specific I/O devices require large computational power. Heterogeneous multicore systems provide parallel computing and exhibit high performance at low latency with improved reliability^(137,149,150).

1.19.4.3 Approximate Computing on FPGA

Approximate computing on FPGA is based on selective violation of some specifications and allowing selective approximations result for efficiency. An effective approximate computing requires a careful selection of suitable data and an approximation strategy. There are various approximation strategies such as precision scaling, skipping memory accesses, performing inexact tasks, loop perforation, approximate neural networks, pruning, and weight sharing. Deploying these strategies requires tradeoffs between the performance loss and design optimization. For example, the neural approximate acceleration has been shown to yield in considerable improvement in terms of performance and efficiency when used in CPUs^(137,149,150).

1.19.4.4 Approximate-Computing-Based In-Memory AI Accelerators

Approximate-computing-based in-memory AI accelerators are primarily designed to alleviate the data communication overheads between logic and memory elements. Computations are carried out inside the memory to reduce the bottleneck, latency,

and energy waste and improve performance. The main form of approximate computing in processing in memory (PIM) architectures which improves the energy efficiency and maintains the reliability of the system. Approximating memories can also give significant reductions in the hardware^(137,151). Several PIM architectures have been proposed, such as DrAcc, SCOPE, and xnor-POP.

1.19.4.5 Neuromorphic Accelerators

Neuromorphic accelerators-based on emerging memory is inspired from biological neural networks based on the underlying biological intelligence. The number of weights in neural network grows linearly or quadratically with the number of activation units^(137,152).

1.19.4.6 Non-Volatile Memory Crossbars Acceleration

Non-volatile memory (NVM) crossbars acceleration is the most common way to integrate into neuromorphic architectures. Another method is the *Spiking neuron* circuits which is a key component of neuromorphic systems as the circuitry emulates neuronal behavior. Neuromorphic neuron circuits are implemented in CMOS, memristors, and biristors. They have varying degrees of biological realism. The threshold logic gates based on high-gain CMOS logic provide the simplest level of modeling for first-generation neural networks. These circuits facilitate the implementation of binary activation functions.

Training NVM-based neuromorphic accelerators are based on neuromorphic DNN accelerators. These accelerators are designed for inference functionality or training plus inference. Neuromorphic accelerators are programmed by training a model of the neuromorphic hardware offline using standard CPU/GPU resources. Some of the software frameworks are Tensorflow and PyTorch. The parameters of the offline-trained model are transferred to the neuromorphic hardware by programming the conductance states of individual NVM elements. The offline model contains sufficient detail about the behavior of the neuromorphic hardware such as the number of neurons/synapses, maximum neuron fan-in and fan-out, and number of conductance states per NVM element. The training process is required via on-chip training or hardware-in-the-loop training to avoid accuracy degradation^(137,153).

1.19.4.7 Computing-in-Memory AI Engines

Computing-in-memory (CIM) AI Engines can alleviate the “memory wall,” problems. There are various types of memory that can be used in CIM AI engines, like SRAM, RRAM, ferroelectric field-effect transistors (FeFETs), and flash memory. In SRAM applications for example, there are three steps in implementing: 1) the digital inputs are converted into multilevel voltages and are stored on the parasitic capacitors of the SRAM bit line (BL), 2) the charge on all capacitors is shared to realize matrix multiplication in the analog domain, and 3) the voltages are converted into digital outputs by ADCs^(154,155).

1.20 AI SOFTWARE

AI model training requires significant amounts of data with sufficient quality and composition. *Quality* refers to the information the data contains about the targeted application and the ease of extracting this information. *Composition* refers to the representation of required information in the dataset. If a dataset is sufficiently good

or not for the target application is difficult to predict but can be identified during model development and testing.

There are many different commercial software available for AI training, offered by a diverse range of companies. An example of a popular framework is TensorFlow software suite tailored for neural networks applications. This software suite includes several tools that target different steps including model training, optimization, and model conversion. Other frameworks are Eloquent, Emlearn Scikit-Learn, Amazon Lex, Vortex, and many others. Semiconductor companies develop software platforms such as TinyML Cube.AI, Silicon Labs' Machine Learning Toolkit, ARM's CMSIS-NN library, and so on⁽¹⁵⁶⁾.

1.20.1 SOFTWARE ACCELERATORS

Software accelerators focus on the orchestration AI model designs, model optimizations, computing, and effective resource management. There are various software acceleration methods, such as training acceleration, inference acceleration, all-digital acceleration, plug-and-play acceleration.

1.20.1.1 Training Acceleration

Training Acceleration addresses training shortfalls of AI models. AI training can be time-consuming, computationally intensive due to limited memory, and computing power. There are software methods to accelerate the training processes. For example, in TF, already learned knowledge on previous models is transferred to new similar application scenarios. In federated learning, global models from a server are uploaded to user-end devices to make use of already trained models^(41,157).

1.20.1.2 Inference Acceleration

Inference Acceleration uses compression and pruning. In *compressions*, fewer input parameters are used for model optimizations by narrowing the searching space and region-of-interest (RoI) encoding. *Pruning* can be applied in many ways. In some cases, only the essential connections are kept and unimportant parameters having connections with fewer weights are deleted. Pruning is also used in other cases: 1) matrix factorization by decomposing a matrix into multiple small-size matrices, 2) knowledge distillation and parameter quantizing, 3) elimination of redundant parameters, and 4) lowering floating-point numbers in the expense lower precision in some precision tolerant applications^(41,158).

1.20.1.3 All-Digital AI Acceleration

All-Digital AI Acceleration is based on hardware efficient algorithms. In many digital accelerators, multiple access control (MAC) arrays can be reconfigured for data-flow and stationary schemes. MAC engines with specific dataflows reuse input the activations and the weights. In most cases, precision is supported for both activations and weights for AI inference⁽¹⁵⁹⁾.

1.20.1.4 Plug and Play Accelerators

Plug and play accelerators are AI platforms already developed by third parties. PnPs can offer solutions in many applications to save time of training if an

excessive amount of time is needed for learning techniques to be implemented and tested. PnP can be implemented by either hardware or software or the combination of the two⁽¹⁶⁰⁾.

1.21 AI-COMPUTING

With the evolution of high-speed digital computers, hardware such as GPUs, and the use of computational intelligence (CI) techniques have significantly advanced the development, and proliferation of AI. AI-related computing platforms can be listed as cloud computing, edge computing, fog computing, and quantum computing,

1.21.1 CLOUD COMPUTING

Cloud computing is a platform that connects many end devices used by consumers, research activities, industry, and business activities distributed all around the world. These interconnected devices generate large amounts of data that require transmission, analysis, and storage. To ensure the integration between distributed devices and cloud computing different interconnection structures and architectures have been configured.

Cloud computing is a paradigm that is defined by the National Institute of Standards and Technology (NIST) as “a model that allows ubiquitous, convenient and on demand access to shared networks.” It supports configurable computing resources (such as networks, servers, storage, applications, and services) that can be provisioned and released with minimal administration. Cloud computing is characterized by unlimited storage capacities and computing power that can be accessed from anywhere.

The case for cloud computing infrastructures AI is widely established. The ability to offload computationally intensive tasks on remote data centers is growing as a preferable option in many applications. Some examples of everyday use of cloud computing are in speech recognition, pattern recognition, object identification, and language translations^(72,152,154,161–164).

End devices architectures in a cloud environment has the following layers: 1) connected devices layer, 2) communication layer between connected devices (e.g., Bluetooth, ZigBee), 3) access network layer, 4) cloud layer, and 5) application layer.

1.21.2 EDGE COMPUTING

Enhanced data rates for GSM evolution (EDGE) computing brings information storage and computing capabilities closer to the device producing that information. There are many architectures proposed or adopted in practice to reduce latency and to have fast response times. Some examples are Fog Computing, MEC, Cloud-Edge-Beneath, Cloudlet, FemtoCloud, Nebula, Mobile-Edge Offloading, and Foraging.

AI is applied in serverless edge computing in runtime mechanisms for scaling, placement, and routing. Typical methods are analytical, heuristic-based approaches. Analytical solutions are not resource intensive, but they are short of capturing details. Heuristic-based platforms rely on optimization processes, and they capture

important details and perform optimization. AI driven platforms are based on various methods, such as RL, to continuously learn and optimize the function deployment^(1,154,165,166).

1.21.3 FOG COMPUTING

Fog computing advocates a user-centric architecture to utilize the context and resources locally. User-driven shift in computation and storage enables many applications requiring less interaction with remote services (or data centers) to perform efficiently^(1,154,167).

1.21.4 QUANTUM COMPUTING

Quantum computers make use of quantum mechanics phenomenon of atomic and subatomic particles. They find applications in solving complex optimization problems much faster than classic computers. Quantum computing is still in development stages, but they have been tried in the embedding stage to learn and offer efficient solutions to AI problems^(168–171).

1.21.5 COMPUTING EFFICIENCY

We need to clarify that computing efficiency is a requirement for all types of computing, not in only AI applications. One way of increasing computing efficiency is sharing the computer networks. Sharing trained AI models can significantly reduce repeated model training efforts. Compressed models can be shared on resource-limited devices (e.g., embedded systems) thus requiring less memory and low computational power.

In computation- and data-efficient learning, a technical challenge is to reduce the sizes of AI models and the amount of training data. For instance, Compressing AI models into smaller sizes without significantly compromising their performance can save time, and can reduce the need for resources during the deployment stage. For data-efficient learning, the main objective is to reduce/alleviate the use of large, labeled datasets for model training, as the collection of large, labeled datasets requires large resources⁽¹⁸⁾.

Computing efficiency using software for AI are achieved by 1) compressed AI modules, 2) pruning, 3) quantization, 4) knowledge distillation, 5) matrix decomposition, and 6) early exit of inference (EEoI).

1.21.5.1 Compressed Modules

Compressed Modules are effective ways to acquire computation-efficiency. Compressed AI models are particularly useful in applications where models run for routine and long-term tasks, as in the case of wireless communication systems. Compressions can be lossless compressions or lossy compressions. Generally, *lossless compression* reduces the data size for all the data, but it cannot achieve high compression. The *lossy compressions* take place at the expense of losing some data. The original data cannot be reconstructed fully but they still are applied for efficiency

in the expense of some losses. The compression ratio (CR) provides a direct measure of the compressor's performance, which can be expressed as:

$$CR = \frac{\text{input raw data size}}{\text{compressed data size}} \quad (1.22)$$

The reconstruction quality in terms of sample-to-sample error can be determined by percent-root-mean-square difference (PRD) and percentage root-mean-square difference normalized (PRDN) as:

$$PRD = 100 \times \sqrt{\frac{\sum (x[k] - x_r[k])^2}{\sum x^2[k]}} \quad \text{and} \quad PRDN = 100 \times \sqrt{\frac{\sum (x[k] - x_r[k])^2}{\sum (x[k] - x)^2}} \quad (1.23)$$

where $x[k]$ is the raw data, and

$x_r[k]$ is the compressed data.

The error estimates are expressed in terms of MSE, normalized mean square error (NMSE), root-mean-square error (RMSE), and normalized root-mean-square error. As an example, NMSE can be determined from:

$$\frac{\sqrt{\sum (x[n] - x_r[n])^2}}{\sqrt{\sum x[n]^2}} \quad (1.24)$$

where for a length of dataset N , $x[n]$ = raw sample and $x_r[n]$ = reconstructed sample^(171,172).

1.21.5.2 Pruning

Pruning is used when AI models are over-parameterized. The network pruning serves to remove redundant weights or neurons that have the least impact on accuracy, resulting in significant reduction in memory storage and computational cost. Pruning is one of the most effective methods to compress AI models. [Figure 1.51](#) shows the process of pruning for a typical neural network-based AI model.

One of the simplest heuristics is to estimate importance in terms of the absolute value of a weight, *called magnitude-based pruning*. In this method, zero valued weights or weights within a certain threshold are removed. A regularization term is often adopted during training to increase weight sparsity. However, magnitude-based pruning methods may result in unstructured network organizations which are difficult to compressed or accelerated without specialized hardware support^(18,173).

1.21.5.3 Quantization

Quantization is another approach for computation-efficient AI. In quantization floating-point parameters are quantized into lower numerical precision. For example,

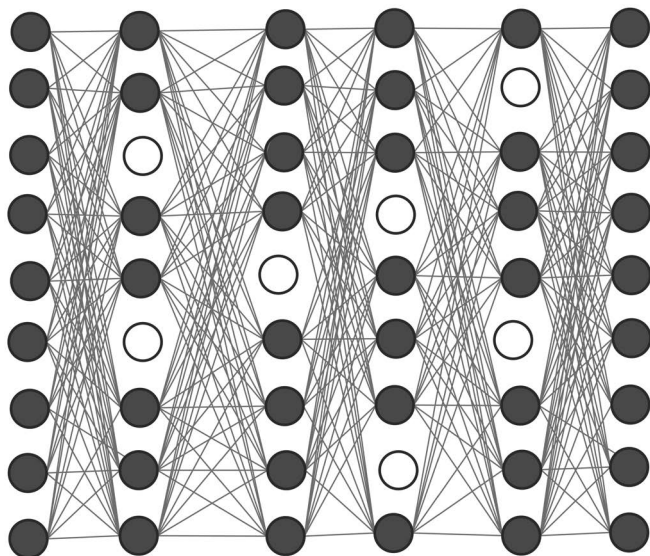


FIGURE 1.51 Pruning eliminates ineffective neurons that have zero (or near zero) values.

32-bit is reduced to 16-bit, 8-bit or even 1-bit. Typically, a 32-bit single-precision floating-point format is quantized to 16-bit half-precision float, or 8-bit mini float.

Using low-bit numerical representations not only reduces model storage cost but also results in significant speed-up during inference. The floating point multiply-accumulate operations are minimized or even avoided. Another method is the incremental network quantization (INQ). INQ converts full-precision weights into power-of-two values with lower precision. It allows the replacement of multiplication operations with bit-shift operations which are more efficient for hardware such as the FPGAs.

Although quantization is generally applied to weights and activations, it can also be extended to gradient computations. Instead of quantizing a single weight, vector quantization approaches focused on factorizing the weight matrix by employing weight clustering and weight sharing.

Combination of pruning or quantization for network compression leads to higher compression rates. Both pruning and quantization are effective ways to reduce model size for efficient learning. One limitation is that they can only deal with models that share a common network architecture, which necessitates compression of cumbersome network to an efficient one with less weights (or neurons) in expense of lower precision^(18,174).

1.21.5.4 Knowledge Distillation

Knowledge distillation (KD) uses the information from large models (called teachers) to train small models (called students). The student model mimics the teacher

model to obtain a competitive or even a superior performance using significantly smaller sizes. Knowledge distillation can be categorized as: logits-based, feature-based, and relation-based.

Generally, knowledge distillation methods are more flexible since AI has already been trained. All the students need to do is, to learn from the existing structures to implement on new cases. KDs can have homogeneous or heterogeneous architectures, which are significantly different from pruning and quantization. However, in most cases, KD methods require more training efforts, as they involve two sessions: 1) train the teacher first and 2) train the students to learn from the teacher to use the acquired knowledge^(18,175).

1.21.5.5 Matrix Decomposition

Matrix decomposition relies on modeling of parameters of AI models as matrices, e.g., describing the weights connecting two layers of an FNN. The parameter matrices form lower ranks can be decomposed into smaller matrices with an overall reduction on the number of parameters^(71,175).

1.21.5.6 Early Exit of Inference

EEoI introduces outputs at different points across a neural network. During the inference process, when an early inference can be performed with high confidence, EEoI allows obtaining a reliable result without reaching the last layers. This method typically incorporates additional trainable layers or a supplementary network, indicating the layers that may be skipped during inference^(171,176).

1.22 AI SAFETY

DL-based AI systems have grown remarkably due to their impressive performances. They have been expanded into critical applications, such as self-driving vehicles where the safety of such systems is paramount, health and medical systems, and others. Notably, some motor accidents have already been connected to self-driving systems and indirectly to AI, highlighting the importance of AI safety. Apart from implications on human safety, AI needs to build trust in society pivotal for the acceptance and usage of it. Recent research has shown that an adversary can undermine the safety of the system by targeting either the AI's inference or training phase^(18,171,178).

1.22.1 DATA POISONING

Modern DL-based AI models rely heavily on large amounts of training data, typically mined from public sources, such as the Internet or crowdsourcing platforms. This exposes the models to the threat of data poisoning where an attacker can degrade a model's performance by corrupting subsets of training data. The data corruption typically involves a combination of altering the data labels and making edits to the original training samples. A sophisticated variant of data poisoning, called *backdoor poisoning*, allows an adversary to control a model's prediction through poison signatures in the input data while eluding detection as the model performs seemingly well on clean inputs.

Several defenses can counter this threat under certain conditions. The first type of defense works by filtering poisoned samples that contain spectral signatures where their hidden states would have different statistics compared to most uncorrupted samples. Initial iterations of such defense either require prior knowledge, such as the ratio of poisoned samples, size of the poison patterns or only work for small poison patterns.

The defenses and attacks on AI systems is likely to see a protracted race especially in high-stake applications. In addition to safety against adversaries, much work has also been conducted on improving the reliability of the models in the face of high valid and invalid data ratios. Exposing AI models to training samples augmented with a wide diversity of corruptions has shown to improve the performance of models.

The deployed AI system might be attacked by specific deception algorithms, known as adversarial attacks, which can generate subtle perturbations imperceptible to human observers but may greatly mislead state-of-the-art AI systems. Algorithms that can resist such attacks are named adversarial defenses^(18,65,66,179).

1.22.2 ADVERSARIAL ATTACKS

Adversarial attacks can be realized by adversarial algorithm fed directly into a ML classifier or end device as an input. Although AI models have achieved remarkable results in identifying many adversarial attacks. State-of-the-art ML algorithms are vulnerable to adversarial attacks in particularly image classification tasks. Adversarial data can be generated by simply adding subtle perturbations to original inputs. Although these adversarial examples may look the same as the original clean data, the experimental result shows that DNN can misclassify adversarial examples into an incorrect category with high confidence^(18,65,66,179,180).

1.22.3 ADVERSARIAL DEFENSE

Adversarial defense for adversarial attacks can be grouped into two categories, the first of which is improved training schemes. A natural adversarial defense is to extend the training set with these generated adversarial examples and train the DNNs with mixed clean and adversarial samples. This scheme is known as adversarial training. The second category is improved network architectures to acquire immunity from adversarial attacks^(18,65,66,180,181).

1.23 AI AND SOCIETY

AI is changing our lives in many ways in the economy, entertainment, games, social interactions, healthcare, R&D, transport, automated living, and so on. With the powerful capabilities of AI, its functionalities have been extended to be almost ubiquitous. However, AI systems are black boxes, where it is difficult to explain how they work, especially to the public. The users may not trust AI models if they cannot understand their decision-making processes particularly in critical applications like finance, medicine, healthcare, and self-driving cars^(18,81,182,183).

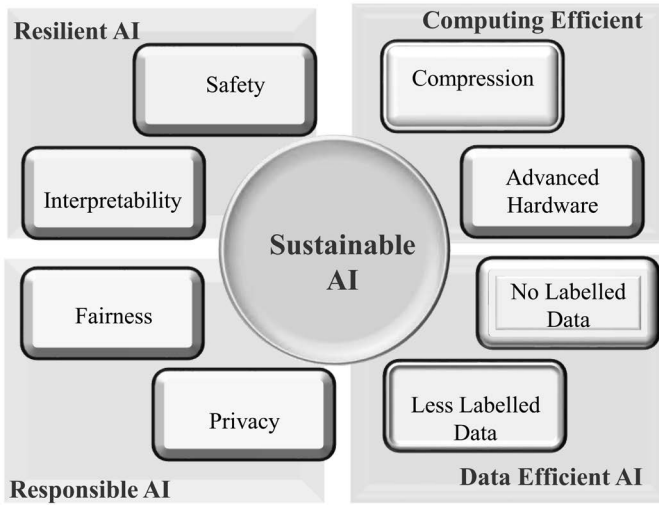


FIGURE 1.52 Components of sustainable AI.

1.23.1 AI SUSTAINABILITY AND RESPONSIBILITY

Wider acceptance and use of AI depends on its sustainability (Figure 1.52). Sustainable AI can be achieved by its: 1) resilience, 2) computing efficiency, 3) data efficiency, and 4) responsibility. *Resilience* is provided by safety, and interoperable AI. *Computing efficiency* is realized by many methods such as compression, pruning, hardware as explained above. *Data efficiency* depends on the quality and the nature of the data. *Responsible* AI observes fairness and privacy of users.

Recently, more and more attention has been focused on explainability of AI. Explainable AI (XAI) techniques can be divided into three categories: 1) model-based explainability, 2) model-agnostic explainability, and 3) example-based explainability^(18,183–185).

1.23.1.1 Model-Based Methods

Model-based methods explore AI models that can be explained, Examples are linear models like linear regression, logistic regression, and tree-based methods like DT, and decision rules. The *if-then* rules in a decision rule algorithm indicate the relationship between inputs and outputs of the model^(18,186).

1.23.1.2 Model-Agnostic Methods

Model-agnostic methods are more general in explainable often used in many AI models. A typical method is based on *partial dependence plot* (PDP) which shows the marginal effect of features on prediction results. Perturb in the inputs can lead to observable changes in outputs, which can explain the relationship between inputs and outputs^(18,65,66,186,187).

1.23.1.3 Example-Based Methods

Example-based methods use a set of examples from a dataset to explain the underlying behaviors of AI models. Counterfactual explanations describe how an instance can change and influence the prediction. By designing counterfactual samples, it is possible to show how the model produces certain predictions and how to explain individual predictions. Another example-based method leverages prototypes and criticisms in which prototypes select representative influential samples from data that are not well represented by prototypes. The way of selecting influential samples is to identify training instances that are most critical to the model or predictions.

It is important for explainability in AI not just to be considered for decisions and predictions but also for generative and creative algorithms too. For example, interpretable and XAI focuses on decisions and predictions particularly for generative/creative algorithms or the ideas and results they produce. Interpretability for generative and creative models is critical as they are becoming widely used in many applications, such as in science, engineering, design, arts, health, and medicine.

Beyond human interpretability, explainability further requirements are reproducibility and replicability. Although this holds directly in settings of AI-based decision-making and predictions, there is some subtlety to this point in generative/creative settings. The purpose of creativity is to generate ideas or artifacts that have never been imagined before. Many algorithms are developed with the intention of producing novel ideas for applicability. These algorithms are influenced by specific prompts, seeds, or conditioning variables but contains various forms of randomness^(18,187–191).

1.24 APPLICATIONS OF AI

AI is applied extensively in many areas. Detailed applications relevant to sensors, instruments, and instrumentation are given throughout this book. A brief overview of the applications is given in [Chapter 4](#). Some of the applications areas can be mentioned as: agriculture, archeology, astronomy, automation, autonomous vehicles, biomonitoring, Chatbots, climate monitoring, climate change, communications systems, computer vision, control systems, cyber security, education, entertainment, environmental monitoring, earth monitoring and earthquakes, facial and image recognition, financial services, governments, healthcare, image processing, industrial processes, instrumentation and measurements, lifestyle, marketing and sales, medicine, navigation, R&D, robotics, smart cities, smart homes, social media, space explorations, trade and commerce, text mining, travel and transportation, video and gaming, wireless communication networks, and many more.

1.25 CONCLUSIONS

In this chapter, we have shown how AI evolved from humble beginnings to a highly sophisticated technology. AI is basically an ML backed up with state-of-art hardware and software. As shown in this book, AI is based on well-established mathematical methods and uses search and optimization techniques. Consequently, there is not

only one type of AI and there are many different ones. Different AIs find different applications that it fits best. Nevertheless, deep ANNs has led to training of massive datasets. The use of massive datasets resulted in ubiquitous applications in many scientific and social fields. We can witness that AI is replacing conventional methods of describing, modeling, and controlling physical and social entities. AI is making remarkable changes in the design and use of instruments, instrumentation systems, sensors, sensor networks, and wireless communication systems.

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2 Wireless Sensors and Networks

2.1 INTRODUCTION

The term *communication* refers to the transfer of information from one point to another for the purpose of exchanging information. Wireless communication signals carry the information in the form of electromagnetic, or optical, or sonic energies. In radio communications, the information is generated electronically at the source and transmitted to the sink wirelessly through electromagnetic radiation in space.

Multiple classes of wireless technologies have been developed in recent years. Some examples are the near-field communications, wireless wide-area networks (WWAN), long range (LoRa), advanced wireless local area networks (WLANs), and sensor networks such as the ZigBee. These developments together with advance computing enabled real-time connections of billions of devices dispersed all around the world.

Communication of instruments takes place via electromagnetic transmission of signals by minimum of two devices, the source, and the sink. The originator of the information is called the *source*, and the receiving end is called the *sink*. The task of the transmitter is to process the communication signal into a suitable form for successful transmission along a selected channel. The *communication channel* is the medium that connects the transmitter to the receiver. The *receiver* extracts the signal coming through the communication channel and processes it such that the information can be recovered.

This chapter concentrates on the principles of wireless communication techniques using electromagnetic waves particularly at radiofrequencies (RF) and microwaves. This is because the technology for such communication of instruments and instrumentation system has become a common practice, well established, and related devices are easily available.

Chapter starts by introducing the basic principles of electromagnetic wave propagation, which is common to all types of electromagnetic communication systems. Important elements of electromagnetic radiation, losses, fading, reflections, refractions, and attenuations are explained. The components necessary for a successful wireless communications are introduced. Modern communication methods, modulation and multiplexing techniques, frequency spreading, and multiple access methods relevant to construction and operation of wireless sensors and instruments have been explained and examples are provided.

Communication systems are modeled and described by mathematical equations. In addition to those descriptions, artificial intelligence has been applied at different

stages of communication networks. It has been shown time and time again that AI enhances communication capabilities and its applications. This book illustrates the applications of AI in communications systems and networks.

2.2 WIRELESS COMMUNICATION PRINCIPLES

Wireless communications cover a wide spectrum of applications: radio and TV broadcasting, broadband satellite communications, cellular systems (GSM/GPRS/EDGE/CDMA/3GPP), wireless cables, WLANs (Bluetooth, Wi-Fi), broadband wireless access (WiMAX), global position system (GPS), phased array RF systems, RFID, and smart handheld devices. Wireless communication systems empower us to conveniently communicate voice, exchange messages, data, images, and videos.

Parameters for wireless communication systems include the carrier frequency, channel bandwidth, modulation schemes, coding algorithms, power consumption, size, mobility, production techniques, and low cost. The devices and technologies include primarily CMOS and bipolar IC circuits, a range of compound semiconductor devices and related technologies.

In wireless communication systems, information from the source to sink is transmitted through a communication channel as shown in [Figure 2.1](#). The information is carried by messages. A *message* contains features of information plus overheads such as encryptions and protocols. The messages can be in different formats and in a variety of forms. The source and sink must be able to synchronize and understand each other to enable an effective communication.

There are several types of wireless communication methods: sonic, ultrasonic, optical, visual, and electromagnetic. In the case of electromagnetics, communication takes place by electromagnetic waves propagating in free space or in a guided medium. Electromagnetic waves are generated by antennas transforming electric power into electromagnetic energy. A receiving antenna located at a distance picks up the radiated electromagnetic energy and converts it back to the original signals. The transmitted and received energy are expressed in terms of *decibels*, which will be explained next.

Decibel (dB) is a measure of ratio of relative power between the output and input of a communication system. Since it is relative ratio, decibel does not indicate the absolute signal levels at some point in time and space. The decibel is defined as:

$$10 \log \frac{P_{\text{out}}}{P_{\text{in}}} \text{dB} \quad (2.1)$$

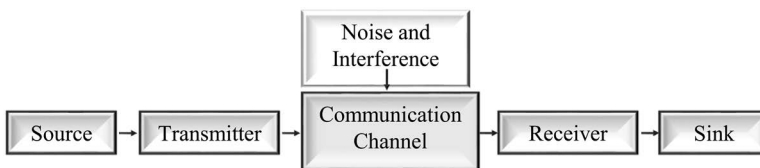


FIGURE 2.1 Block diagram of a simple communication system.

As an example, in an amplifier has input of 2 mW and the output of 5 mW

$$10 \log \frac{P_{\text{out}}}{P_{\text{in}}} = 10 \log \frac{5 \text{ mW}}{2 \text{ mW}} \cong 4 \text{ dB} \quad (2.2)$$

Similarly, the input is 10 mW and output is 100 mW the power ration will be 10, which can be as 10 dB. An output to input power ratio of 100 gives 20 dB, ratio of 1,000 gives 30 dB, and so on.

If the power output is less than the input the gain of the circuit assumes negative decibels. For an output of 0.1 mW for and input of 2 mW the ratio is 0.5 or -3 dB. The output to input ratio of 0.1 is -10 dB, ratio of 0.01 is -20 dB, and so on.

There are several meanings for 0 dB; for example, 0 dBj = 1 millivolt, 0 dBk = 1 kilowatt, 0 dBm = 1 milliwatt at 600 ohms, 0 dBv = 1 volt, and many more. The decibel referred to one milliwatt of input is known as the dBm; therefore, 1 mW output for 1 mW input is represented by 0 dBm. In dBm, the input is understood to be 1 mW; hence it is expressed relative to an input of 1 mW. Following this philosophy, an output of 10^{-2} mW is -20 dBm; similarly, an output of 300 mW is 24 dBm and the input power is assumed to be 1 mW. In actual operations, if the input power assumes any value other than 1 mW then the output power changes proportionally hence the ratio in dBm would remain the same.

Similarly, decibel referred to 1 W is known as dBW, hence 1 W output is 0 dBW. An output of 1 mW will be -30 dBW; an output of 20 W would be 13 dBW.

Expression of currents and voltages in decibels is slightly different due to the following reason. Assuming that the resistances of the input and output are same, using the definition of decibels we get.

$$10 \log \frac{P_{\text{out}}}{P_{\text{in}}} = 10 \log \frac{(E_{\text{out}})^2 / R}{(E_{\text{in}})^2 / R} = 20 \log \frac{E_{\text{out}}}{E_{\text{in}}} \quad (2.3)$$

In radio engineering circuits are often quoted in dBmV, or dBuV. In the case, of dBuV decibels is relative to a reference signal level of 1 μ V. Therefore, a signal level of 20 dBuV is 10 μ V, similarly 0.5 μ V is -3 dBuV. Similar reasoning is applicable for dBmV and others.

2.3 ELECTROMAGNETIC WAVE PROPOGATION

In RF and microwave communications, an antenna transforms electric power generated by the circuits into a radiating electromagnetic energy. Electromagnetic energy propagates through the media as traverse electromagnetic (TEM) waves. That is, the wave has a traverse electric (E) field and a traverse magnetic (H) field propagating at the same time at right angles to each other and both being at right angles to direction of propagation. The electric field strength, E , is expressed in volts per meter (V/m) and the magnetic field strength, H , is in ampere per meter (A/m). The propagation of a TEM wave is illustrated in [Figure 2.2](#).

The wavelength of a propagating wave is the distance that it travels in one complete cycle. The distance traveled in one cycle depends on the speed of the wave

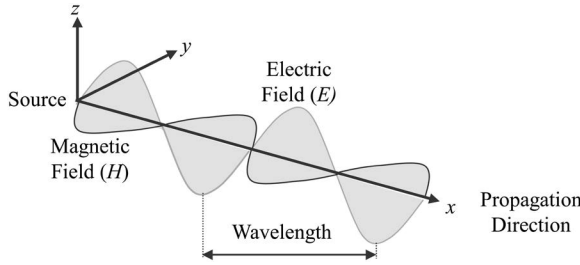


FIGURE 2.2 Propagation of electromagnetic waves.

in the medium. In free space the speed of RF electromagnetic wave is equal to the speed of light (approximately 3×10^8 m/s), which could be calculated by

$$\lambda = \frac{c}{f} \quad (2.4)$$

where λ is the wavelength in meters, c is the speed of light, and f is the frequency in Hertz.

The radiated electromagnetic waves are characterized by wavelength, frequency, intensity or power density, and the direction of polarization. The *wavelength* is the distance that an electromagnetic wave travels in one complete cycle. The *frequency* is the number of cycles per second. The *intensity* is the electric field strength of the wave expressed in volts per unit distance (e.g., V/m). The *power density* of the wave indicates the power carried, in units of watts per unit area (e.g., W/m²).

Waves can be longitudinal or transverse waves. In a *longitudinal wave* the vibration is parallel to the direction of the propagation. In *transverse wave* the oscillations of wave are perpendicular to the direction of the propagation. *Polarization* is the restriction of transverse waves in one direction.

In free space electromagnetic propagation, the direction of polarization is the direction the electric (E) field. If the direction of polarization is horizontal with respect to the surface of the Earth, the wave is said to have horizontal polarization. Other polarizations are possible, such as elliptic polarization or circular polarization. Circular polarization, for example, may be assumed to be composed of two linearly polarized waves that have orthogonal (right angles or perpendicular) polarization properties changing in time. In such cases, the polarization vectors rotate in either left or in right directions as the wave propagates in space. The rotation depends on the phase shift between the two linearly polarized components as $+90^\circ$ or -90° .

2.3.1 NEAR-FIELD AND FAR-FIELD

An electromagnetic propagation disseminated by a source (antenna) can be viewed to take place in two separable forms: the *near-field* and the *far-field*. [Figure 2.3](#) shows wave behaviors in the near-field and far-field areas.

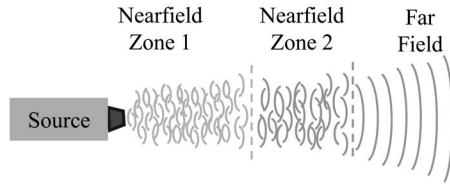


FIGURE 2.3 Shows wave near-field and far-field wave.

2.3.1.1 Near-Field

Near-field propagation takes place in distances less than about $1/6$ wavelength from the transmitting antenna. In this region, propagation waves are still dependent on the electrical characteristics of the source and the circuit impedance. The reactive energy dominates which can be treated in two separate components, the electric field (E) and the magnetic field (H).

The properties of near-field are exploited for short distance communication purposes. Near-fields communication (NFC) is available in some mobile devices with communication capabilities of about 40 mm or less.

2.3.1.2 Far-Field

Far-field is the propagation of the waves at distances greater than $1/6$ wavelength from the source antenna. Once the propagation takes place, it becomes independent of the characteristics of the source. Far-field propagation assumes constant wave impedance irrespective of the space and the location of the propagation. The impedance is defined as the ratio of electric field intensity to magnetic field intensity, which is $377 \, \Omega$ in free space. This basically represents the ratio of the voltage and current ($R = V/I$).

Since electromagnetic propagation differ in characteristics, the techniques of measurements differ too. In the near-field regions, both E - and H -fields must be measured separately. The far-field strengths are normally measured in terms of E -fields or in terms of *power density*, expressed in watts per square meter. The power density denotes the amount of radiated power passing through per square meter of a surface perpendicular to the direction away from the source. The *peak-power density* is known to be equal to $E \times H$, and, for a sinusoidal source, the *average power density* is half of this value. In the case of non-sinusoidal propagation, each frequency component must be considered separately. The total average power can be determined as the sum of the average powers for all frequencies. It follows that the total average power is $P = E^2/377 \, \text{W}$.

During the measurements of the field strength and intensity of radio communication signals, large variations intensity of the field can be encountered over distances with a minimum scale length of half the wavelength, λ . At the smallest scale, in the near-field region, the variations are sensitive to the frequency of the waves and the presence of objects on the path of propagation. Variations depend on the time as well as shapes and positions of the objects, such as movement of people in the vicinity. With the use of moving receivers rapid near-field spatial variations (also

known as *fast fading*) can be monitored. Variations can be explained in terms of the existence of multiple paths as the electromagnetic energy can propagate away from the antenna. Variations over the distance greater the half wavelength show only a gradual variation with frequency, termed as *slow fading*.

2.4 RF PROPAGATION LOSSES

The electromagnetic waves propagating in free space are subject to reflection, refraction, diffraction, absorption, polarization, and scattering; collectively called the *excess path losses*⁽¹⁾.

2.4.1 RF PROPAGATION LOSS EXAMPLES

RF propagation losses have been investigated for decades. Characteristics of losses dependent on conditions of the media of propagation. There are numerous losses in the RF propagation, some of which will be visited next.

2.4.1.1 Atmospheric Losses

As in the case of communication satellites and spacecrafts, when the generated electromagnetic waves pass through the atmosphere, they attenuate due to atmospheric absorption. At the lower frequencies, the atmospheric attenuation is small hence the propagation losses are close to that of free-space propagation losses. But, at high frequencies, 10 GHz and above, atmospheric absorption loss becomes excessive. In addition to the normal atmospheric losses, factors such as rain, snow, and fog cause losses. These additional losses become effective in frequencies 1.3 GHz and above.

2.4.1.2 Free Space Path Losses

Free space path loss (also known as *free-space loss*) refers to attenuation of signal strength between two antennas. The system design calculations are based on the assumptions that two antennas are in line-of-sight (LOS) through the air and the path is obstruction free. Free space loss increases with the square distance between the antennas. It decreases with the square of the wavelength. Path loss represents signal attenuation as a positive quantity and is measured in dBs.

2.4.1.3 Reflection of Electromagnetic Waves

Reflections of electromagnetic waves take place from the Earth's surface and other objects. The reflected wave may interfere with the direct waves. Depending on the relative phase angles of two interacting signals the interference can be constructive or destructive. The interference is constructive if the reflected waves are additive to produce larger waves at the receiver antenna. Or the reflected waves may negatively be added causing smaller waves than the original one. The vector addition of a direct signal and a reflected signal is referred to as the *multipath*. Generally multipath is not desirable, but in some cases, it can be put in useful form like in some indoor wireless communication applications.

Multipath losses can be significant in communication systems. For example, in cellular systems, multipath can be a propagation problem that needs careful

attention as there may be several propagation paths between the transmitter and the receiver. Even if the path between transmitter and receiver LOS, multiple signals will be received due to reflections from ground and other objects on the path. In cellular systems, in some cases the reflected signals may be constructive thus resulting in signals greater than a direct line signal, which may improve the quality of reception.

2.4.1.4 Refraction Losses

Refraction losses can be an important effect of the atmosphere on the electromagnetic wave propagation. The atmosphere does not have a uniform density everywhere and its chemical composition and ionization effects change with respect to the altitude and time of the day. An implication of this is that the propagation velocity of the electromagnetic wave changes as it travels through the atmosphere. The transmitted waves do not propagate in straight lines but follow curved paths thus slightly changing the velocity as it passes through different atmospheric conditions.

2.4.1.5 Diffraction Losses

Diffraction losses of electromagnetic waves refer to bending of electromagnetic waves by materials located on the propagation path. Particularly, large objects such as buildings, houses, warehouses, trees, etc. cause diffraction in open spaces. Objects such as machinery, columns, walls, and other structures cause diffraction in closed spaces such as plant floors in industrial applications. The diffraction effect may be useful as a fraction of energy in the wave bent around an object (e.g., buildings) may be received by a receiver that does not have a direct line of sight from the transmitter. Although there is a loss associated in the diffraction process, the signals received may still be strong enough for a desirable signal reception.

2.4.1.6 Indoor Propagation Losses

Indoor propagation losses are very important in the operations of wireless instruments and wireless networks applications in closed spaces such as large buildings, factories, and plants. The signal transmitted from a fixed base station in an indoor environment reaches the receiver via one or more paths, all slightly delayed in time. The received waves may consist of the intended LOS a signal and several other signals reflected or scattered by structures such as the outer walls, and columns.

2.4.1.7 Multipath Fading

Multipath fading can seriously degrade the performance of an indoor communication system, but the effect can be minimized by careful design of transmitters and receivers, and by selecting good locations. In indoor radio communication, a detailed characterization of the spatial properties of signal propagation is necessary for the successful operation of the system.

2.4.1.8 Impedance Matching and Standing Waves

Impedance matching and standing waves between components is significant if there is an impedance mismatch in the system, some energy transmitted intended for transmission reflects to the source causing standing waves. The amount of energy

that reflects depends on the frequency of the signal and the degree of mismatch. Standing waves are expressed in terms of voltage standing wave ratio (VSWR). The VSWR is a unitless ratio ranging from 1 to infinity, mainly showing the amount of reflected energy that exists in the device. A value of one for the VSWR indicates that all the energy passes through without any reflection, while higher values indicate an increase in the degree of reflection.

2.5 FREQUENCY ALLOCATION

Electromagnetic radiation has different properties at different frequency bands. Therefore, for practicality, the electromagnetic radiation has been divided into several groups as shown in Table 2.1. Some of these frequency bands are extensively used for some specific communications purposes. The wireless instruments and associated networks are primarily supported by RF electromagnetic radiation. However, as technology progressed, the meaning of RF has changed over time. In this book, we will use the RF at frequencies from low frequencies (LF) to infrared (IR) region that are used for communication purposes. This includes the microwave bands as well.

2.5.1 FREQUENCY ALLOCATION BANDS

Traditionally, based on the frequency dependent characteristics and the communication traffic, frequency bands are broadly allocated as in Table 2.1. For communication purposes, as the technology progresses, these frequency bands are reclassified,

TABLE 2.1
Frequency Bands of Electromagnetic Radiation

Frequency	Wavelength	Frequency Band	Applications in Communications
30–30 kHz	$10^5\text{--}10^4$ m	VLF	
30–300 kHz	$10^4\text{--}10^3$ m	LF	<i>Navigation, time standards</i>
0.3–3 MHz	$10^3\text{--}10^2$ m	MF	<i>AM broadcasting, marine/aircraft navigation</i>
3–30 MHz	$10^2\text{--}10$ m	HF	<i>AM broadcasting, shortwave broadcasting</i>
30–300 MHz	10–1 m	VHF	<i>FM/TV broadcasting, amateur radio</i>
0.3–3 GHz	1–0.1 m	UHF	<i>Cellular phones, mobile radio, wireless LAN, PAN</i>
3–30 GHz	10–1 cm	SHF	<i>Satellite, Radar, TV, WLAN, 5G cellular</i>
30–300 GHz	1–0.1 cm	EHF	<i>Satellite, Radar, experimental, 5G cellular, upcoming 6G networks</i>
0.3–3 THz	1–0.1 mm	Band 12	<i>R&D, experimental</i>
1–417 THz	300–0.72 mm	Infrared	<i>IR communications</i>
417–789 THz	0.72–0.38 mm	Visible light	<i>Li-Fi</i>
$789\text{--}5 \times 10^6$	$0.38\text{--}6 \times 10^{-5}$ mm	Ultraviolet	
$3 \times 10^4\text{--}3 \times 10^8$ THz	$100\text{--}1 \times 10^{-2}$ A	X-rays	
$>3 \times 10^7$	<0.1 A	Gamma rays	

merged, or separated in accordance on the demand of usage and availability of technology.

2.5.1.1 Very-Low Frequency and Low Frequency

Very low frequency (VLF) and low frequency (LF) have large wavelengths requiring large antennas. The bandwidths of VLF and LF are also small thus their applications are limited. These frequency bands are inefficient to be used in communication systems, but they find applications in submarine and maritime, underwater communications, and radio navigation systems.

2.5.1.2 Medium Frequency

Medium frequency (MF) band is used in communication systems involving moderate distances. This band is used in commercial radio broadcasting, maritime and aeronautical communications, and navigation systems. Amateur radio broadcasting is also allowed in this band.

2.5.1.3 High Frequency

High frequency (HF) band provides good propagation properties over long distances. This band is used in some fixed and mobile services.

2.5.1.4 Very High Frequency and Ultra High Frequency

Very high frequency (VHF) and ultra-high frequency (UHF) is commonly used in communication systems that require large bandwidths. They can be transmitted and received by small directional antennas thus adding to the efficiency of these bands. VHF and UHF bands are used in fixed communication services, ground-to-air communications, television broadcasting, and some mobile services such as cell phones. These bands are particularly efficient in applications where line of sight operations is possible.

2.5.1.5 Super High Frequency and Extremely High Frequency

Super high frequency (SHF) and extremely high frequency (EHF) has very short wavelengths therefore they require small antennas. The distance of communication is short therefore long-distance propagation requires a series of repeaters and relay stations. These bands are used for television broadcasting and high-speed data transmission, and mobile devices such as cell phones.

Frequency bands above EHF are also used for communication purposes using fiber optics. Infrared frequencies are used in WLANs and controlling devices such as TVs.

The frequency allocations for the different communications services are provided on a worldwide basis by the International Telecommunication Union (ITU). For example, the allocation for region 2, for North and South America, has been adopted by the US Federal Communications Commission (FCC). The FCC is responsible for RF spectrum management within the United States. Individual frequency assignments must be requested by the users for approval before any transmission at the requested frequencies can take place. Other regulations must be followed, such as the maximum transmitted power, out-of-band harmonics, spurious signal levels, and so on.

2.6 DIGITAL TECHNOLOGY IN COMMUNICATIONS

Modern wireless communications rely mostly on digital technology and the integrated circuits (ICs) constitute the key components. RF ICs contain analog and digital circuits for signal amplification, channel selection, signal filtering, analog-to-digital and digital-to-analog conversion, frequency synthesis, and power management. In the era of mass communications, the RF demands highly integrated low-cost system solutions, such as the single-chip radios, single-package radios, and single-module radios.

All wireless communication systems contain in their ICs analog and digital circuits. Functions of these IC can be viewed in five major blocks: 1) analog mixed signal (AM), 2) power amplifier (PA), 3) RF transceiver (TxRx), 4) power management (PM), and 5) digital signal processor (DSP). IC constructions are supported by various types of materials and technologies, such as Si, SiGe, GaAs, InP, SiC, GaN, and so on. The device structures can be bipolar, heterojunction bipolar transistor (HBT), high electron mobility transistor (HEMT), lateral-diffused metal-oxide semiconductor (LDMOS), metal semiconductor FET (MESFET), MOSFET, pseudomorph high electron mobility transistor (PHEMT), and on-chip passive components⁽¹⁻⁴⁾.

Total radio approach creates challenges in terms of system design, manufacturing yields, and system signal isolation. Various technologies are used in total radio approach, such as GaAs-based systems, RF CMOS (complementary metal-oxide semiconductor) or bipolar CMOS (BiMOS) systems. Silicon solutions offer smaller module areas and lower costs making them suitable for next generation of fast and HF applications.

CMOS has adequate RF performance for 2.4 GHz ISM WLAN applications and offers the advantage of adding further RF functions on the same chip. However, InGaP HBTs have significant performance advantages at 5.2 GHz and above over both CMOS and SiGe. In addition, SiGe outperforms CMOS at 5.2 GHz in PAs.

SiGe bipolar devices are not well developed as CMOS devices, but they have some advantages and a degree of freedom since the SiGe epitaxy enabled the use of bandgap engineering technology. The alloying of Ge to create an electric field in the base increases the speed of carriers moving through the base. The addition of carbon doping helps to freeze the doping profile in place so that it is not smeared by subsequent processing.

System-on-Chip (SoC) wireless devices are available for almost all communication standards like Bluetooth, GSM, GPS, and WLAN. CMOS technology allows integration of analog and digital components on the same chip. In addition to RF components, SoC integrates an application microcontroller, audio driver, microphone amplifier with A/D and D/A converter, loudspeaker driver, DC/DC converter, power supply regulator, charger, supervision circuit and SIM card interface.

2.7 RF COMPONENTS

The front end of wireless communications ICs consists of numerous analog circuits. The physical layer of wireless transmission protocols is more efficient using analog coding techniques than digital techniques. Therefore, both analog and digital RF components will be explained in this book.

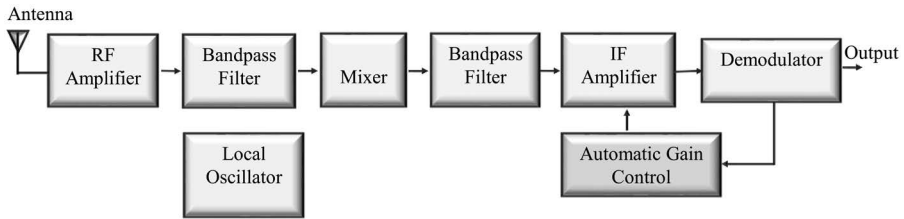


FIGURE 2.4 Components of a typical receiver system.

Depending on the functionalities, the architecture and components of wireless communications systems vary from one another. However, there are several essential components performing similar functions in most RF systems. The typical common components are amplifiers, attenuators, filters, oscillators, multipliers, mixers, modulators, demodulators, multiplexers, and antennas. Other important components are phase detectors, phase shifters, delay lines, duplexers, and diplexers, isolators, circulators, up-converters, down-converters, power dividers, power combiners, transformers, EMI filters, RFI filters, gain controllers, transceivers, and wireless modems. As an example, the block diagram of the components of a typical receiver system is illustrated in [Figure 2.4](#).

2.7.1 ANTENNAS

Antennas are devices used to radiate or receive electromagnetic energy in free space. They can be viewed as transducers that convert electrical energy into electromagnetic energy or electromagnetic energy into electrical energy. The transmitting antenna converts the electrical energy supplied by electronic devices into electromagnetic form. The receiving antenna intercepts the electromagnetic energy and transforms into electrical form.

If the electromagnetic wave from an antenna is uniform in all directions, it is called an *isotropic antenna*. An ideal isotropic antenna does not exist in practice. But it is conventional to refer to the isotropic antennas as a benchmark to determine the comparable operational characteristics of other practical antennas. For example, the gain of an antenna in a direction is a measure of how the power radiated in that direction compares had it been an isotropic antenna. On the other hand, directional antennas radiate power in a particular direction. If the directional antenna radiates electromagnetic energy in one direction, it is said to be a *unidirectional antenna*. If it radiates in two directions it is called the *bidirectional antenna*. A pair of antennas is said to be *bilateral* if the gain is the same in both the receiving and transmitting antennas.

2.7.2 ANTENNA TYPES

Antennas are essential component of wireless communications, e.g., mobile devices, cell phones, transceivers, repeaters, base transmitting stations, satellites, ground stations, local area networks (LANs), wide area networks (WANs), and so on. Hence,

there are many different types of antennas to radiate or receive electromagnetic energy. Antenna characteristics depend on their structures, shapes, constructions, and materials used⁽⁴⁾. Some of these antennas are briefly discussed below.

2.7.2.1 Thin-Wire Monopole Antennas

Thin-wire monopole antennas are commonly used in RF systems. Monopole antennas are constructed in various forms as wideband monopoles, conical monopoles, large-size monopoles, and electrically small monopole antennas.

2.7.2.2 Whip Antennas

Whip antennas resemble monopole antennas, but they are made from flexible rods. They are capacitive antennas or electric doublets that are carefully insulated from the structure and supported on a base insulator. The lengths of whip antennas are usually between 1/10 and 5/8 wavelengths of the radiated electromagnetic wave. Whip antennas are the most widely used in mobile communication systems. *Helix antennas* constitute slight variations from whip antennas.

2.7.2.3 Microstrip Antennas

Microstrip antennas are popular for their low profile and ease of fabrication on printed circuit boards (Figure 2.5). They can assume an unlimited number of shapes and sizes and can be made to conform to odd surface profiles. They are constructed using special dielectric materials that fit in small packages. Microstrip antennas are known to have low efficiency but higher convenience in their construction and use.

2.7.2.4 On-Chip Antennas

On-chip antennas are used in ICs for their small footprints to enable wireless communications. Although they have limited range, they are suitable for cell phones, tablets, headsets, global positioning system (GPS) devices, WLAN routers, Wi-Fi, Bluetooth, and the likes. They are mostly based on CMOS and BiMOS technologies. On-chip antennas differ in their sizes and shapes, such as on-chip 3D Yagi antennas, ring antennas, slot ring antennas, ring antennas with silicon lens, and many more. Examples of on-chip antennas are illustrated in Figures 2.5 and 2.6. On chip antennas are researched heavily as mass communications are moving toward mmWave and THz band systems. THz bands are envisioned to provide unprecedented improvements in handling Terabit-per-second data rates^(5,6).

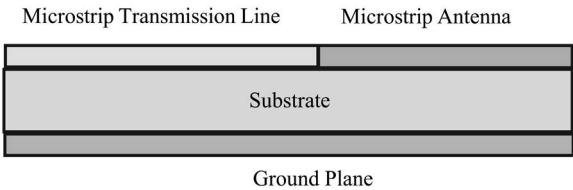


FIGURE 2.5 A microstrip antenna.

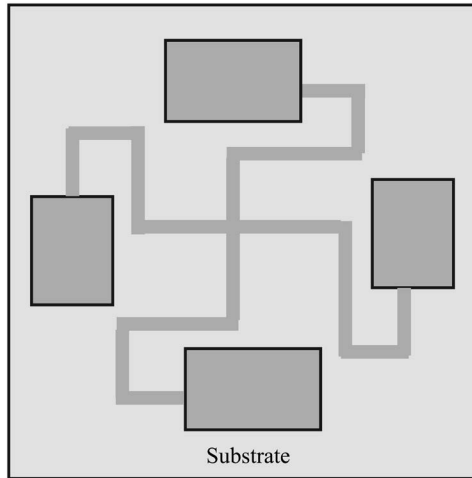


FIGURE 2.6 A microstrip antenna with four elements.

2.7.2.5 Thin-Wire Dipole Antennas

Thin-wire dipole antennas are fed from two-wire lines. Thin-wire dipole antennas is the most used types in communication systems. There are many variation of thin-wire dipole antennas such as fat dipole antennas, folded dipole antennas, helical antennas, horn antennas, lens antennas, log-periodic antennas, loop antennas, notch antennas, reflector type of antennas, sleeve antennas, small dipole antennas, slot antennas, spiral antennas, turnstile antennas, and Yagi antennas.

2.7.2.6 Array Antennas

Array antennas are a collection of various antenna elements such as dipoles, slots, loops, and microstrips (see [Figure 2.6](#)). Two of the common antennas applied in arrays are Yagi-Uda and log periodic antennas. Yagi antenna arrays consist of three or more half-wave antenna elements, called *driven*, *reflector*, and *director*. Yagi arrays offer high directivity, and good gain. Another type of antenna arrays is the log periodic arrays. They are multi-element, unidirectional, narrow-beam antennas that have impedance and radiation characteristics functioning as a logarithmic function of the excitation frequency. Consequently, the length and spacing of the elements of log periodic antennas increase logarithmically from one end to the other. Log periodic antennas are often used in broadband applications^(5,6).

2.7.2.7 Panel (Sector) Antennas

Panel (sector) antennas are like multiple array antennas but differ in their arrangements. They have a multitude of elements to radiate or receive signals. Panel or sector antennas usually have flat panel shapes and are enclosed within a plastic cover to protect their elements from each other or touching other metals, and for protection from nearby noisy electronic components.

In communication systems, at frequencies below 3 GHz, different types of antennas can be opted for, like simple monopole, dipoles, and array antennas. The most common type of antenna at microwave frequencies is the dish antenna particularly frequency of 3 GHz and above. The dish antennas are applied in long distance microwave transmissions, satellite communications, and other high bandwidth microwave systems. Other commonly available antennas include omnidirectional, directional, array, microstrip, panel, and monopole antennas.

2.7.3 RF AMPLIFIERS

Amplifiers and attenuators are one of the most essential building blocks of the RF communication systems⁽¹⁾.

2.7.3.1 Amplifiers Characteristics

Amplifiers are electronic devices that produce output signals with larger amplitudes compared to the input signals. In an ideal amplifier, the output signal should resemble the wave shape and the frequency of the input signal, and only the amplitude increased by a *gain*.

In radio engineering, many different types of amplifiers are configured and used, including low noise amplifier, PA, pulse amplifier, bi-directional amplifier, multi-carrier amplifier, buffer amplifier, and limiting amplifier. The input signal of an amplifier can be in current or voltage. The output signal is usually of the same nature as the input signal. But in some cases, converters may be necessary to change current-to-voltage or voltage-to-current.

Specifications for RF amplifiers include three main characteristics: the frequency range, the gain, and the output power. In voltage amplifiers, the *gain* is the ratio of the output voltage to the input voltage. In current amplifiers, the gain is the ratio of output current to the input current. Similarly, in PAs, the ratio of output power to input power yields to power gain. Normally, gains are expressed in decibels (dB), e.g., $G_{dB} = 20 \times \log (V_o/V_i)$ for voltage gain or $G_{dB} = 10 \times \log (P_o/P_i)$ for power gain. The output power is determined by the signal power at the output of the amplifier under specific conditions such as temperature, load, VSWR, and the supply voltage. The output power is expressed in dBm, or in milli watts (mW), or in watts (W).

Electrical characteristics of RF amplifiers include nominal operating voltages, currents, and impedances. Additional performance specifications may include the noise figure, and the input VSWR. The noise figure (NF) is the ratio (in dB) of the signal-to-noise ratio at the input and the signal-to-noise ratio measured at the output. Therefore, noise figures indicate a measure of the amount of noise added to the signal under the normal operation conditions. Clearly, a low noise figure indicates better performance. The noise figure value sets the lower limit of the dynamic range of an amplifier.

RF amplifiers find applications as: transmitter and receiver amplifiers, intermediate-frequency (IF) amplifiers, low-frequency (LF) amplifiers, PAs, buffer amplifiers, and so on.

2.7.3.2 Transmitter and Receiver RF Amplifiers

Transmitter and receiver RF amplifiers are used in all RF applications, including VHF and higher frequency transmitters and receivers. The IC technologies supporting these amplifiers are largely based on bipolar junction transistors (BJTs) and field-effect transistors (FETs). Depending on the application, amplifiers can be designed and constructed in different types such as Class-A, Class-B, Class-C, Class-E, and push-pull amplifiers.

Class A RF amplifiers are generally used for the low-noise amplifications of small signals. They have good linearity, low distortion, broad bandwidth, good noise figure, low band pass ripple for all output levels, medium output power capability, good stability, and good gain. But they have poor efficiency, more heat dissipation, and tend to be larger in size. There are many different design approaches to Class A RF amplifiers.

Class B and Class AB amplifiers are used for large-scale and high-power amplifications.

Class C amplifiers are used in some cases, but they are known to have non-linearity that may not be suitable in some applications.

In the design of amplifiers, impedance matching between follow up circuits is critical. Impedance matching is often realized by lumped capacitive-inductive circuits known as LC-circuits. In ideal cases, the signal input of an RF amplifier should result in an increase in amplitude without changing the frequency components or introduction of any other form of distortion. Nevertheless, in practice, there will be some distortions in the output signals due to non-linear operation of transistors and other components in the ICs.

2.7.3.3 IF Amplifiers

IF amplifiers are used for providing the necessary gains for receiver circuits at intermediate frequencies. They are preferred amplifiers because it is easier to operate with narrowband and passband filters at IF frequencies. These amplifiers usually include automatic gain controllers (AGC).

2.7.3.4 LF Amplifiers

LF amplifiers are often used after modulators and transmitters, and before de-modulators and receivers. In many applications, RC or transformer coupled Class A amplifiers are preferred as the LF amplifiers.

2.7.3.5 Power Amplifiers

PAs deliver transmission power to transmitting antenna with a high degree of linearity. Their linearity strongly depends on the operating class of the amplifiers. Linearity and performance of a given PA is also controlled by the modulation and access schemes employed. That is, different communication standards have different PA needs and transistor circuits. [Figure 2.7](#) shows a block diagram of an RF PA.

PAs must carefully be designed or selected not to violate the rules set by the FCC or other related authorities. The power must not exceed a certain maximum power levels that allowed at the selected frequency. PAs handle large signals; therefore, they have stringent requirements in ruggedness, construction, and breakdown voltages.

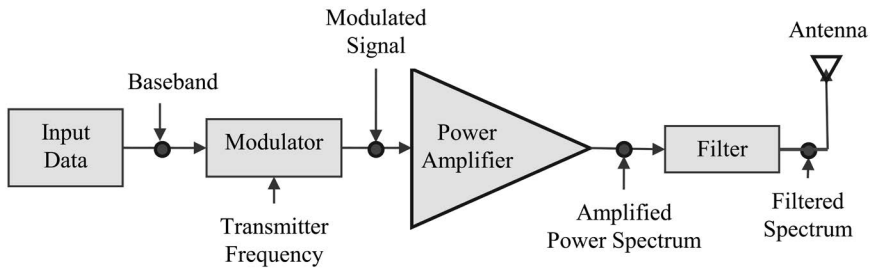


FIGURE 2.7 Block diagram of an RF power amplifier.

IC-based technologies developed to support RF PAs are diverse. Some examples are GaAs, InP HBT, MESFETs and pHEMTs, SiC MESFETs, Si CMOS, and RF-LDMOS. Lateral diffused MOS (LDMOS) are the most used devices due to their superior performance, reliability, and low prices. High gain PAs are used for frequencies from 1 MHz to 3.5 GHz. The RF-LDMOS RFICs are manufactured to handle powers exceeding 100 W⁽²⁾.

2.7.3.6 Low Noise Amplifiers

Low noise amplifiers (LNAs) deliver undistorted signal to the follow up circuits down the stream. LNAs are used as the first stage amplifier if the input signal is too low. Many LNA amplifiers can operate in various carrier frequency bands, such as global system for mobile communications (GSM), enhanced or extended GSM (EGSM), code division multiple access (CDMA), wideband CDMA (WCDMA), global position system (GPS), Bluetooth, WLAN, and the like.

2.7.3.7 Power Management Circuits

Power management (PM) circuits control the power supply to/from PAs to avoid spurious tone generation in transmit/receive channels. PM circuits manage the power supply to all other circuit blocks and regulate battery power to provide an efficient distribution. High voltage operation, low series resistance, and efficient voltage conversion are the key considerations in PM technologies. PM modules integrate the matching networks, transmission switches, pre-filtering, and band selection functions in a single package.

2.7.4 RF ATTENUATORS

Attenuators are circuits to reduce the power level of a signal by a determined amount. Ideally, attenuators should not cause any frequency change, distortion, or reflection but they do. An attenuator reduces the output signal with respect to the input signal. The attenuators can be fixed or variable types. A *fixed attenuator* reduces the power level of a signal by a predetermined amount. They are designed with respect to the characteristics of the input and output signals. A *variable attenuator* has a specified attenuation range that makes it adaptable to different load demands.

2.7.5 RF FILTERS

Filters are one of the most significant components of the RF circuits. The RF filters can be constructed as lowpass, high pass, bandpass, band stop, or notch filters. A *lowpass* filter passes low frequencies and rejects high frequencies. A *high pass* filter passes high frequencies while rejecting the low ones. A *bandpass* filter allows the signals to pass within the desired frequency band. Figure 2.8 shows block diagram of bandpass filter. A *band stop* filter blocks the frequencies of a specified range while passing frequencies outside that band. *Notch* filters are basically narrowband band-pass, or band stop filters operating in a limited range of frequencies⁽¹⁾.

Performance specifications of filters are described by rated voltages, rated currents, cutoff frequencies, bandwidths, ripples, insertion losses, VSWRs, and operational temperatures. For lowpass filters, there is one specified frequency, which is the *cutoff frequency* above which the frequencies are blocked. For high pass filters, the specified frequency refers to the *cut-on frequency*, above which the frequencies are allowed to pass. For bandpass and band stop filters, an important specified frequency is the center frequency in the middle of the *bandwidth*. Specified lower and upper frequencies determine the range of the bandwidth.

Two other important terms in filters are the ripples and insertion losses. The *ripple* is the peak-to-peak variation in the passband response. The ripple is expressed in decibels, and it is a function of the degree of matching of the devices in the system. *Insertion loss* is the total RF power transmission loss resulting from the insertion of a device in a circuit. Insertion loss is defined as the ratio of signal power at the output of the inserted device to the input signal power at the input of the inserted device.

Filters are necessary to offset unwanted electromagnetic interference (EMI) and radio-frequency-interference (RFI) signals. The EMI refers to the interference of electromagnetic energy at all possible frequencies, while RFI refers exclusively to the RF range. Both EMI and RFI can be eliminated or minimized using appropriate filters. EMI and RFI filters consist of circuits or devices containing series of inductive load bearing and series of parallel capacitive non-load-bearing components. Non-load-bearing components provide low impedance paths around the protected circuit for HF noise. Filters attenuate impulses since impulses contain high-frequency components. HF components can be verified by Fourier's Analysis.

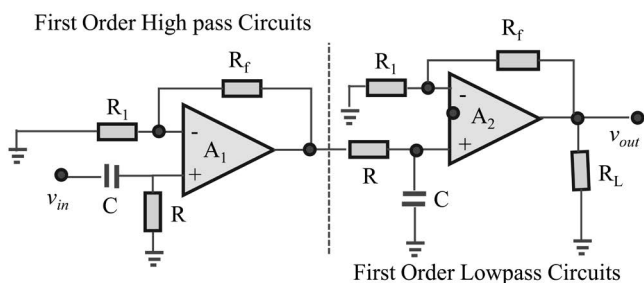


FIGURE 2.8 Block diagram of bandpass filter.

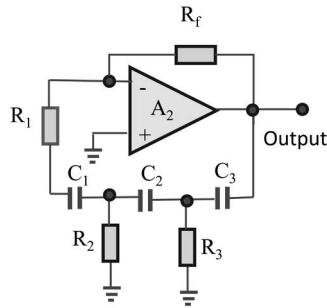


FIGURE 2.9 An op-amp-based oscillator.

2.7.6 RF OSCILLATORS

Oscillators generate required frequency signals in both transmitters and receivers. The signals at different frequencies are generated by using suitable components known as *resonators*. The frequencies of oscillations are determined by the resonant circuits usually located at the feedback loop. There are many types of oscillators suitable for RF applications, some of which are LC oscillators, crystal oscillators, dielectric resonator oscillators (DROs), yttrium iron garnet (YIG) resonator oscillators, ceramic resonators, and voltage-controlled oscillators (VCOs). A few of these oscillators will be explained below. [Figure 2.9](#) shows an op-amp-based oscillator.

2.7.6.1 LC Oscillators

LC oscillators are frequently used in RF communication systems. There are many different types such as Colpitts oscillators, Clapp oscillators, tuned-input tuned-output JFET oscillators, differential-pair oscillators, and a few others.

2.7.6.2 Quartz Cristal Oscillators

Quartz crystal oscillators are one of the most common oscillators used in RF systems. Quartz oscillators provide steady frequencies with high accuracy. They can be tuned to precise frequencies by using external components such as the trimming capacitors.

2.7.6.3 Dielectric Resonator Oscillators

DROs make use of properties of high-Q dielectric materials as the resonator. They give stable frequency and exhibit good phase noise properties. The YIG resonator oscillators are used in UHF and microwave applications. They constitute essential parts of RF mixers and frequency converters.

2.7.6.4 Voltage-Controlled Oscillators

VCOs are the key part of a phase-locked loop (PLL) that synchronizes communication between an IC and the outside world. It is essential to minimize the timing jitter which is the spectral density of phase-noise power. Jitter is measured at a frequency

offset from the carrier frequency of the output waveform when the total power consumed. Most practical VCOs are IC-based. But IC VCOs present serious challenges in integration with standard CMOS technologies. This is due to stringent requirements in VCO such as accurate frequency tuning, low power consumption, and low phase noise. Frequency tuning, phase noise, and power consumption are primarily determined by the passive components^(1,2).

2.7.7 RF FREQUENCY MULTIPLIERS

Frequency multipliers produce larger frequency output signals than the input signal frequency by a predetermined amount. Although, they are nonlinear devices, most frequency multipliers can eliminate undesired frequencies from the output signal thus acting like a filter. There are two types of frequency multipliers: active multipliers, and passive multipliers. A typical frequency multiplier is depicted in Figure 2.10.

2.7.7.1 Active Multipliers

Active multipliers produce output signals with a higher power level than the input signal power level. The difference in power level (expressed in dB) between the output signal and the input signal is called *Conversion Gain*. The conversion gain is a positive number.

2.7.7.2 Passive Multipliers

Passive multipliers produce output signals with a power level smaller than the input signal power level. The difference in power level between the output signal and the input signal is called the *conversion loss*.

2.7.8 RF MIXERS

Mixers are simple devices. They have two input signals at different frequencies and two outputs. Their functionality is to convert RF power of one frequency to the power of the other frequency. The signal of which the power to be shifted from is

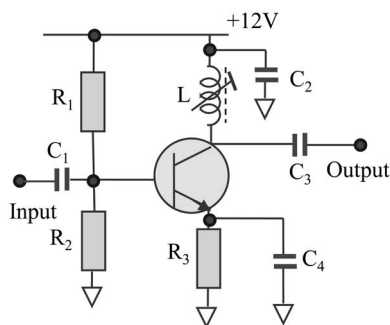


FIGURE 2.10 A frequency multiplier.

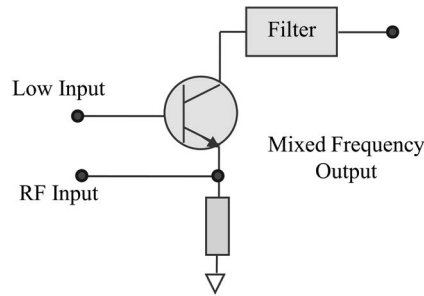


FIGURE 2.11 A frequency mixer.

applied at one of the RF inputs. The signal required to carry the power, obtained from a local oscillator (LO), is applied into the LO port. The frequency sum and the frequency difference of the input two signals are available at the intermediate frequency (IF) ports. A mixer is illustrated in [Figure 2.11](#).

Performance specifications of mixers include RF frequency range, LO frequency range, IF frequency range, conversion loss, and LO power. *RF frequency range* is the range of input frequencies that the mixer can handle without errors. *LO frequency range* is the frequency range of the LO. *IF frequency range* represents the selection of signals that will be subjected to further signal processing for transmission in the case of transmitters or signal extraction in the case of receivers. *Conversion loss* is defined as the loss of power between the RF input signal and the IF output signal during the frequency transition. The conversion loss is expressed in dB as the ratio of the IF output power to the RF input power.

There are many common commercially available mixers, or they can be constructed on circuit boards. Some of these mixers are single balanced mixers, double balanced mixers, triple balanced mixers, quadrature mixers, and image reject mixers. The majority of mixers are made from diodes. There are also different types of diode mixers used in the RF systems, such as the single-ended diode mixer, balanced diode mixer, double-balanced diode mixer, and triple-balanced diode mixer. Transistor mixers are generally made either from BJTs or FETs. Transistor mixers also come in different ranges as single-ended mixers, balanced mixers, and double-balanced mixers. One important advantage of transistor mixers is that they can provide conversion gain rather than conversion loss. FET mixers are known to produce less intermodulation and cross-modulation distortions hence they are preferred in HF operations.

In the application of RF mixers, isolation is a priority. *Isolation* may be defined as the ratio, in dB, of the power level applied at one port of the mixer to the resulting power level at the same frequency appearing at another port. Isolation is a measure of the internal circuit balance for the two inputs of the mixer, and it indicates the level of attenuation of a signal when supplied into a specific port, but its effect is observed at a different port. Clearly, when the isolation is high, the amount of “leakage” or “feed thru” between the mixer ports will be small.

2.7.9 RF MODULATORS AND DEMODULATORS

Shifting or translation of a signal from one frequency to another frequency band is achieved by the process of *modulation*. In modulations, a base band signal (voice, video, etc.) is impressed on a carrier signal at a higher frequency thus altering the frequency and amplitude characteristics of the carrier. Modulation permits the transmission of a message in an efficient manner. In addition, multiplexing, such as the frequency division multiplexing (FDM), can be used to allow multitude of messages to be transmitted simultaneously on the same system. A *demodulator*, on the other hand, takes the received signal at the receiver end and extracts the useful information from it.

2.7.10 RF PASSIVE COMPONENTS

Passive components like resistors, capacitors, and inductors are fundamental circuit elements used in analog and digital circuits. A wide range of passive components are available to use in external circuits. However, passive components in the IC are the topic of interest as an RF IC would contain both large number of active and passive devices as the circuit elements inside the chip. There are three approaches here: 1) SoC, 2) system-in-package (SiP), and 3) system-on-package (SoP). SoC is a method of embedding components in the same substrate or different substrates in the same chip, but the style of integration of components differentiates them from each other^(7,8).

The surface areas in IC constructions are dominated by the size of passive devices. Examples of passive devices occupy the most surface areas include: 1) resistors, 2) inductors, 3) capacitors, 4) matching passive circuits, and 5) stability of the circuit such as biasing components, damping of resonances, rejection of power supply ripples.

Multilayer ceramics and multilayer polymer technology for embedding passive components provide efficiency in high-Q inductors, capacitors, matching networks, low-pass and band-pass filters, baluns, combiners, and antennas. Also, a 3D design approach using multilayer topologies leads to high quality, compact components to support multiple bands and standards and wider bandwidth in a compact form factor and at low cost.

RF ICs such as VCOs, mixers, PAs, isolators, circulators, delay lines, and filters rely on mainly passive components that includes capacitors, varactors, resistors, inductors, and transformers. To realize low-cost RF CMOS technology, the performance of these passive components is a tradeoff between the cost of incorporating them on the chip and their usefulness. BiCMOS technologies have been demonstrated to provide good integration capability.

A common problem in mixed circuits is the RF power reflections. These reflections can interfere with neighboring blocks and with nearby active components. Mitigation of RF power reflection can be accomplished by using RF isolators inserted between active elements and the reflective components. Another common method employed is the use of reflectionless and absorptive materials and components^(2,9-11).

2.7.10.1 RF Capacitors

Capacitors are used in filters and for ac coupling, RF bypass, filters, impedance matching, and resonance circuits. Their characteristics include parasitic capacitance to the substrate, quality factor Q , voltage coefficient VCC, temperature coefficient, and leakage currents.

The capacitors can be formed by MOS gate oxide, polysilicon-insulator-polysilicon (PIP), or metal-insulator metal (MIM) technologies. Capacitor technologies have various advantages and disadvantages depending on the requirements of the modules. For example, MOS capacitors achieve high capacitance densities, but they suffer from high parasitic losses. The polysilicon-based capacitors with oxide or nitride as a dielectric have lower parasitic capacitances and smaller VCC compared to MOS capacitors. Metal-insulator-metal capacitors are formed between two metal layers away from the substrate later in the processing. Depending on which dielectrics are used, they may have low parasitic capacitance to substrate, low VCC, high Q , and relatively high capacitance density.

2.7.10.2 RF Varactors

Varactors are voltage-dependent capacitors. They are used in LC-tanks for VCOs, tuned filters, and tuned matching networks. They are the vital elements in VCO to adjust the frequency with a controlling voltage. The key parameters of varactors include frequency tuning range, quality factor Q , and linearity. Varactors are categorized into two types: junction (diode) varactors and MOS varactors. *Junction varactors* are implemented as a P/N well junction in a standard CMOS process. *MOS varactors* are obtained depending on the polarity of source at the drain region.

2.7.10.3 RF Inductors

Inductors in RF circuits are used in filters, chokes, impedance matching, resonance circuits, baluns, and transformers. They are manufactured by using aluminum on medium resistive silicon or aluminum on high resistive silicon. Alternatively, copper on high resistive silicon is used. At high frequencies greater than 3 GHz using high resistive substrates provide ease and advantages in construction of inductors^(2,9).

2.7.10.4 RF Resistors

Resistors are usually used for biasing circuitry, signal isolation, and impedance matching. In RF ICs, their characteristics include resistivity, temperature coefficient, voltage coefficient (VCR), parasitic capacitance properties, inductance properties, and resistance sensitivity to process variation. There are various methods of obtaining resistors, the polysilicon resistors usually have low parasitic capacitance useful in highly sensitive parasitic operations. Thin film resistors are special resistors built between upper metal levels. Thin film metal resistors also have much lower noise than polysilicon resistors and better linearity and good matching properties. Some of the representative materials used in IC construction of resistors include TaN, SiCr (sichrome), and NiCr (nichrome). TaN is especially attractive for RFCMOS application^(2,10).

2.7.10.5 RF Transformers

RF transformers are devices that transfer electrical energy from one circuit to another by electromagnetic induction. RF transformers are used to increase or to decrease the voltage depending on the turns ratio of the secondary and primary windings. If the turns ratio is greater than one, the device acts as a step-up transformer indicating that the amplitude of the output voltage is larger than the input voltage. If the turns ratio is less than one, the device acts as a step-down transformer. RF transformers find applications primarily in low-power circuits for impedance matching purposes to achieve maximum power transfer. Transformer provides dc isolation between the input and output circuits while maintaining ac continuity.

2.7.11 RF ACTIVE COMPONENTS

Active components constitute the main bulk of RF circuits.

2.7.11.1 RF Duplexers

Duplexers are devices used in transceivers with single antenna to receive and transmit information. Thus, transmitter operating on one frequency uses the same antenna as the receiver operating on a different frequency. The task of the duplexer is coordination of the time sharing of the transmission and reception activities with a minimum of interaction and degradation of the incoming and outgoing RF signals.

2.7.11.2 RF Duplexers

Duplexers are three-port frequency-dependent devices that are used as a separator or a combiner of different signals. A typical duplexer consists of two fixed tuned band-pass filters that share a common port. Signals applied to common ports are separated in accordance with their passband frequencies of the filters. A typical duplexer is shown in [Figure 2.12](#).

IC-based RF duplexers and duplexers include surface mount, flat pack, through hole, plug-in, and waveguide assembly techniques in their constructions. RF duplexers and duplexers, performance specifications are signal rejection of the receiver over transmission signal and vice versa, insertions loss, and VSWR.

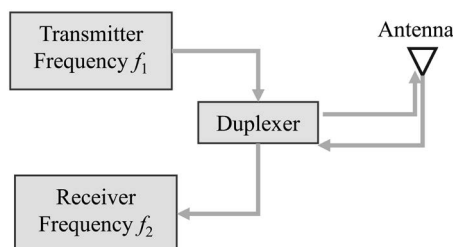


FIGURE 2.12 A typical duplexer.

2.7.11.3 RF Delay Lines

Delay lines are devices used to slow down a signal by a preset time interval in a network. There are two basic types: active delay lines and passive delay lines. *Active delay* lines are built primarily with digital components and applied for delaying digital signals. *Passive delay* lines are built with analog components, but they can be used for both analog and digital signals.

2.7.11.4 RF Isolators and Circulators

Isolators and circulators are the two passive devices that are used for controlling propagation of an RF signal. That is, an *isolator* permits the signal to pass in one direction while providing high isolation against the reflected signal coming in the reverse direction. A *circulator* consists of three or more ports allowing the signal entering a port to be transferred to the port adjacent to it in either clockwise or counterclockwise direction.

Signal isolation is essential RF ICs particularly in SoC devices and SIP devices. Noise coupling from the digital parts to the analog parts on the same chip is an issue since the signal cross-coupling prevents complete integration of various RF functions. But there are many emerging technologies such as the non-classical CMOS such as double gate or 3D vertical transistors, and non-CMOS devices such as resonant tunneling devices, and carbon nanotubes.

2.7.11.5 RF Phase Detectors

Phase detectors are circuits that provide a dc output voltage proportional to the phase difference between two same frequency RF signals depending on the phase difference between the two. A simple way obtaining phase detectors is the use of double-balanced mixers. These mixers can be configured as phase detectors by supplying the two RF signals to the LO and RF input terminals. The DC output of the phase detector (mixer) can be observed at the IF output terminal.

2.7.11.6 RF Phase Shifters

Phase shifters are two-port devices that alter the phase of a signal in response to an external signal. There are two basic types: the analog phase shifters and the digital phase shifters. Although there are many different types of phase shifters in analog and digital categories, the variable phase shifters are the most used ones, in which the phase of the signal can be changed by a variable control signal. A basic phase shifter is illustrated in [Figure 2.13](#).

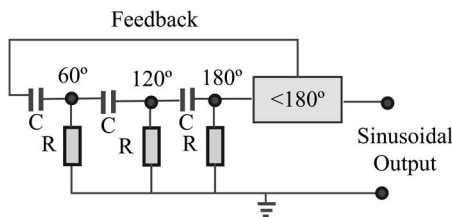


FIGURE 2.13 A typical phase shifter.

2.7.11.7 RF Phase Lock Loops

Phase lock loops (PLLs) handle incoming and outgoing signals in transceivers. The PLL is responsible for relating the phase and frequency to the reference signal from the sender. They are used as for phase locking the baseband processing units. An important sub-circuit in the PLL is the VCO which generates a sinusoidal signal by voltage in the feedback loop. This sinusoidal signal is used for the down- and up-conversions of the signals modulated onto the carrier^(2,9).

2.7.11.8 RF Power Providers

Power dividers deliver multiple outputs, with equal in phases and amplitudes, from a single input signal. Power dividers are reciprocal devices that can be used as *power combiners* by simply using their output ports as input ports. Power dividers divide the input signal in 2-way, 3-way, 4-way, 5-way, 6-way, 7-way, and 8-way forms.

Performance specifications for power dividers include frequency range, insertion loss, isolation, and RF connector type. The *insertion loss*, in dB, is defined as the measured loss through the device excluding the power division factor. The power division factor is calculated as the ratio of power output to power input. The *isolation* is defined as the isolation, in dB, between any set of output ports. It is calculated as the ratio of the power of one output port to the power at any other output port, with matched terminations of all other ports. Common package types for power dividers include surface mount, flat pack, through hole or plug-in, and waveguide assembly technologies.

2.8 RF TRANSCEIVERS

Transceivers are electronic devices that are used for transmitting and receiving RF signals. For transmissions, a transceiver performs all necessary RF signal processing functions such as modulation, up conversion, and power amplification before supplying the signal to the antenna. The receiver can operate on different techniques such as heterodyne, homodyne, image-reject, digital IF, sub sampling principles, and so on.

Heterodyne receivers translate the frequency band of the received signal to much lower frequencies to relax the Q -factor required by channel select filter. The translation is carried out by a mixer often referred to as the down conversion. *Homodyne receivers* convert the incoming signal to carrier frequency. The channel selection requires only low-pass filters with relatively sharp cutoff characteristics. *Digital IF receivers* operate on dual-IF heterodyne principles at LF. They need additional mixer and filter circuits. *Subsampling receivers* sample RF input at lower rates since the narrowband signals exhibit only a small change from one carrier cycle to the next. Sampling receivers may suffer from aliasing and noise problems.

Performance specifications for a transceiver are the frequency range, sensitivity, output power, and data rate. The *frequency range* is the operating frequency of the transceiver. The *sensitivity* in a transceiver is normally taken as the minimum input signal required to produce a specified output signal within a specified

Signal-to-noise (S/N) ratio. The *output power* is the maximum signal power that a transceiver can transmit. The *data rate* is the amount of information that can be sent by the transceiver per unit time.

For duplex operations, transceivers must be used in matching pairs. They may be selected from amplitude modulation, frequency modulation, on-off keying (OOK), amplitude-shift keying (ASK), frequency-shift keying (FSK), or phase-shift keying (PSK) modes of operations.

The technologies used for RF transceivers include Si, bipolar transistors, SiGe, HBT, RF CMOS, and integrated passive components. Si bipolar, SiGe and HBT technologies have dominated the RF transceiver applications with carrier frequency less than 10 GHz because of their favorable gain and noise performances. RFCMOS transceivers find applications in low cost and less performance demanding situations, e.g., Bluetooth and WLAN. High quality and low parasitic on-chip passive components, such as capacitors, varactors, resistors, and inductors are essential elements of RF transceiver building blocks to improve circuit configurations and provide cost effective integrated solutions.

2.9 DIGITAL RF COMPONENTS

Modern communication systems are largely based on digital technology. The major functions and components of a typical transmitter for a digital communication system is given in Figure 2.14. Most of the wireless systems discussed in this book have similar architectures as in Figure 2.14. in their constructions and operations.

The functionalities of components of a digital transmitter are as follows: The *source* converts the real-world signal (e.g., voice, images) into electrical signals. The output signals of the source are compressed through *source coding* since

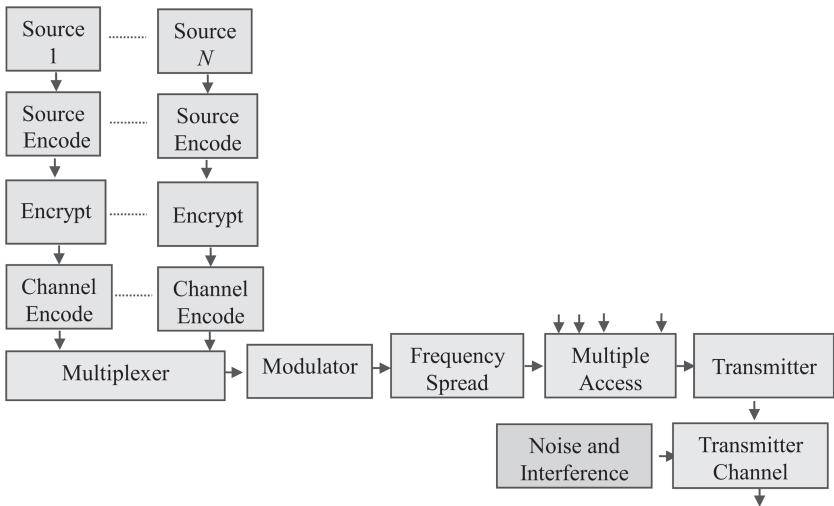


FIGURE 2.14 Components of a typical wireless digital transmitter.

there is usually redundancy in the original signal. Source coding also shapes the signal to make it suitable for transmission within the bandwidth of the communication system. *Encryption* encodes the signal into complex form so that interception and interpretation of the signals by third parties or unauthorized recipients can be avoided. *Channel encoding* reformats the signals to prepare for any anticipated noise and distortions that might take place in the selected channel during transmissions. Once information on the characteristics of the channel encoding of the transmitter is known by the receiver, it can take corrective action errors that might have occurred during the transmission. When more than one signal is transmitted, *multiplexers* are used to merge several signals together to be transmitted by the same channel. Multiplexers increase the efficiency of the system since channel bandwidth is normally greater than bandwidth of any one of the signals transmitted. Multiplexed signals are then *modulated* and *frequency spread*. Frequency spreading has many advantages, for example the signature of the transmitted signal is reduced so that probability of interception and interpretation becomes low. After the frequency spreading, *multiple access* methods used by sharing the channel with the other users. Once the signals are modulated, they are then ready for transmission.

Some of the common digital components will be explained next.

2.9.1 RECEIVERS

Receivers have components that extract the intended information by de-spreading, demodulating, and demultiplexing the received signals coming from the transmitter. The major functions and components of a typical receiver are illustrated in Figure 2.15.

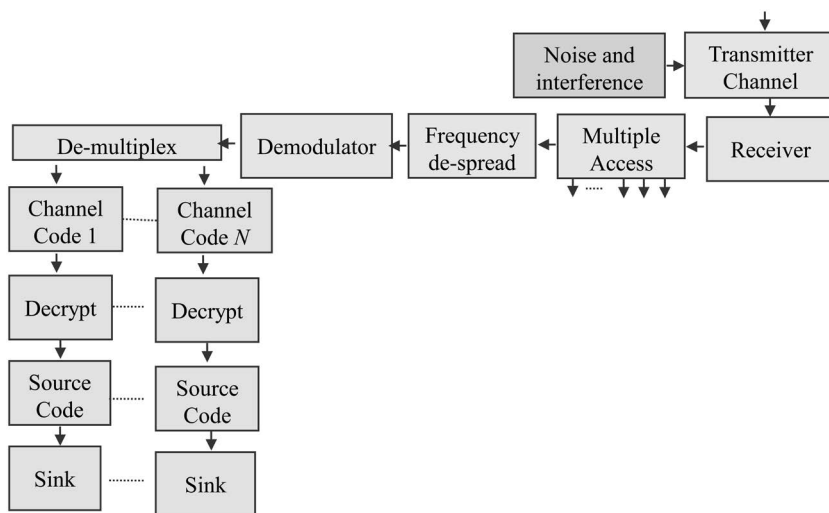


FIGURE 2.15 Components of a typical wireless digital receiver.

The input signal at the carrier frequency is received by the antenna, amplified by a LNA, and then filtered by a bandpass circuit. Next, the signal is down converted to a LF by a mixer circuit, again filtered, and then amplified to an optimal input level for the digital-to-analog converter (D/A). The rest of the receiver side signal processing is usually done in the analog domain^(2,11).

2.9.2 MIXED-SIGNAL DEVICES

Mixed-signal devices like A/D converters and D/A converters are the primary components of all digital communication systems as they interface the interaction between the real world and machine world. Common technologies supporting mixed signal devices are CMOS ICs, and bipolar devices. CMOS devices provide higher voltages, better precision, and good matching with a reasonably adequate speed. Bipolar devices are commonly used for good gain and noise properties, precision, linearity, and matching.

2.10 MODULATION AND DEMODULATION

Modulation can be made by analog or digital methods. Analog modulation represents relatively older but well-established technology still finding many applications. Digital modulation, on the other hand, is relatively new but there are some similarities with the analog modulations at the fundamental levels. Also, analog modulation techniques are used particularly in mixed signal modules. The analog modulation is easy to understand and forms the basis of digital systems, therefore greater detail will be provided in this book.

In the analog modulation process, one or more properties of the carrier signal, either the amplitude, or the frequency, or the phase can be used. There are three major types of modulation techniques:

1. Amplitude modulation
2. Frequency modulation
3. Phase modulation

2.10.1 AMPLITUDE MODULATION

Amplitude modulation (AM) allows the amplitude of the carrier to vary proportional to the amplitude of the base band signal. The amplitude modulated waveform is known as modulation envelope, as illustrated in [Figure 2.16](#). In this figure, the carrier frequency is modulated by a sinusoidal waveform.

When a base band signal, in the frequency range of f_l to f_h , modulates the carrier signal having a fixed frequency f_c , the resulting signal assumes a new frequency spectrum and a new bandwidth.

If the carrier signal is $V_c \sin(\omega_c t)$ and the modulating signal is $v_m(t)$, ignoring the phases, the amplitude modulated signal can be expressed by

$$v(t) = [V_c + v_m(t)] \sin(\omega_c t) \quad (2.5)$$

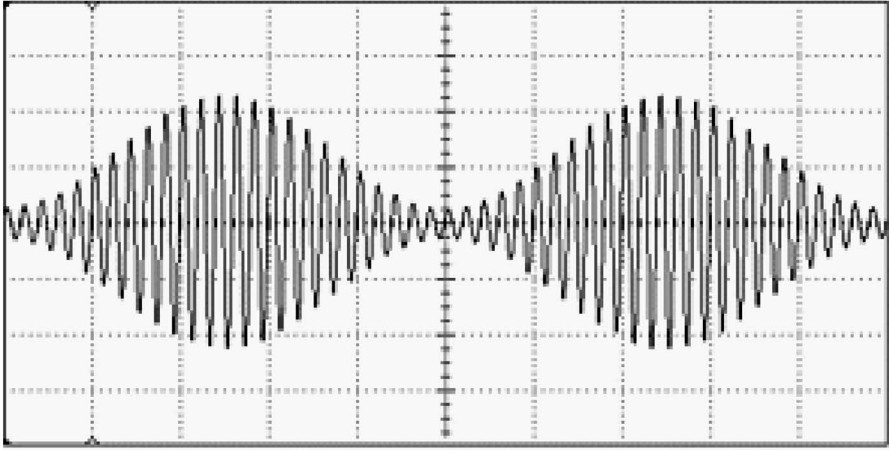


FIGURE 2.16 An AM waveform.

If modulating signal (e.g., the base band signal) is $v_m(t) = V_m \sin(\omega_m t)$, then

$$v(t) = [V_c + V_m \sin(\omega_m t)] \sin(\omega_c t) \quad (2.6)$$

Expanding this equation and using trigonometric relationships results in

$$v(t) = V_c \sin(\omega_c t) + \frac{V_m}{2} \cos(\omega_c - \omega_m)t - \frac{V_m}{2} \cos(\omega_c + \omega_m)t \quad (2.7)$$

The envelope has a maximum amplitude ($V_c + V_m$) and minimum amplitude ($V_c - V_m$). It also contains three different frequencies (f_c), a lower side frequency ($f_c - f_m$), and an upper side frequency ($f_c + f_m$).

Although carrier signal frequency is constant, the frequency of the modulated signal will have a frequency band between the higher frequency (upper frequency) and the lower frequency, f_h and f_l respectively. Hence, the modulated signal produces lower sideband and upper sideband in the frequency spectrum, as illustrated in [Figure 2.17](#).

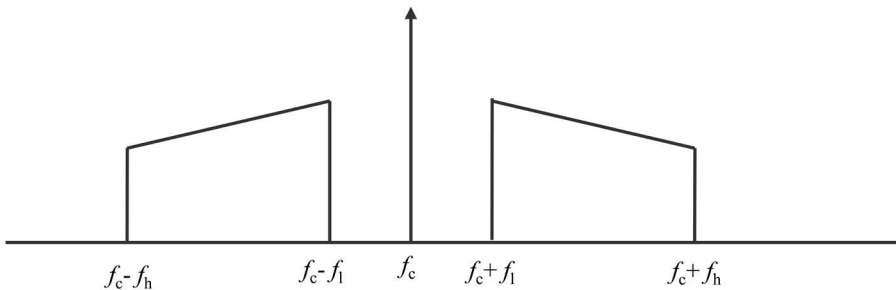


FIGURE 2.17 The frequency spectrum of an AM waveform.

The bandwidth (BW) occupied by the modulated signal depends upon the modulation scheme used. The required bandwidth for a conventional AM is typically twice the highest modulating frequency. For single sideband (SSB) the bandwidth is nominally equal to base band bandwidth.

The *modulation index* of an AM signal is defined as the ratio of the peak base band signal amplitude to the peak carrier signal amplitude. The modulation index is often expressed in percentage values hence it is termed as the *percentage modulation*. Importance of modulation index is that the relative power in the sidebands and the carrier signal depends on the percentage of modulation.

AM modulators are commonly made from Class C amplifiers using plate-modulated or grid-modulated or collector-modulated circuits. There are many other types of conventional amplitude modulators, including FET gate-biased AM systems, FET push-pull modulators, and so on. Also, various forms of side-band modulators are available in the form of single-sideband modulators, double-sideband modulators, Vestigial-sideband modulators, etc.

2.10.2 FREQUENCY MODULATION

Frequency modulation (FM) alters the RF frequency of the carrier proportional to the amplitude of the modulating signal while keeping the amplitude of the carrier signal constant. [Figure 2.18](#) illustrates a binary modulated FM signal.

The carrier wave has the highest frequency when the modulating waveform is at its maximum amplitude. In between the maximum and minimum amplitudes of the modulating signal the frequency of the carrier changes to maximum and minimum above its resting frequency. This process is termed as the *frequency deviation*. For a given system, a maximum frequency deviation, called the *rated system deviation*, ($\Delta f_{\max}$) must be defined due to limitations in the operational bandwidths.

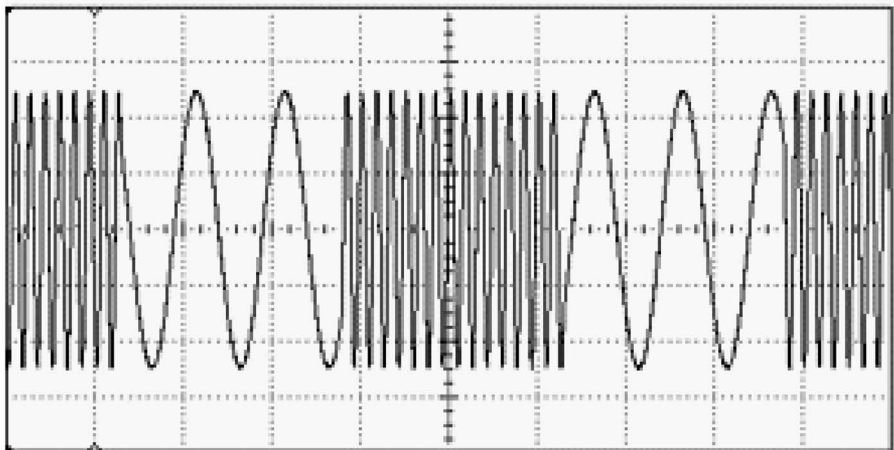


FIGURE 2.18 The FM waveform.

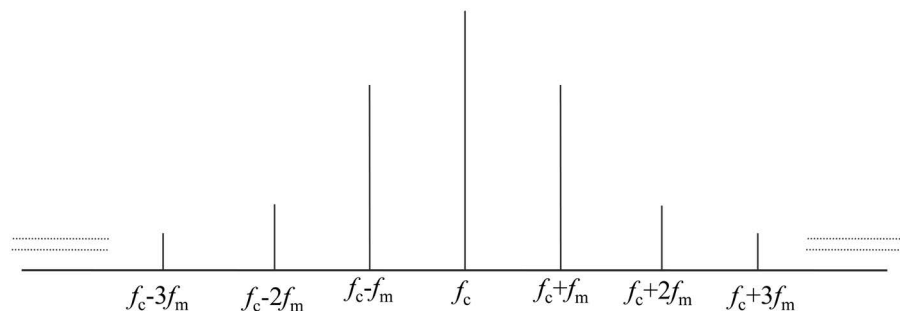


FIGURE 2.19 A typical frequency spectrum of an FM wave.

FM produces an infinite number of side frequencies spaced equally on both sides of the carrier frequency f_c as shown in [Figure 2.19](#).

FM modulators can use many signals in parallel by using different frequencies for each channel. Frequency-modulated signals are often produced by amplifier chains (cascaded multiple amplifiers). The modulation can be accomplished either a direct variation of the frequency of an oscillator or an indirectly by phase modulation. Hence, there are two basic methods of generating an FM signal: the direct method and the indirect method.

In the *direct method*, the carrier frequency is directly varied in accordance with the input modulating signal. VCO is used to vary the frequency of the carrier signal in accordance with the variations in the baseband signal amplitude.

In the *indirect method*, a narrowband FM signal is generated using a balanced modulator and frequency multiplication is used to increase both the frequency deviation and the carrier frequency to the specified levels.

[Figure 2.20](#) illustrates a simplified block diagram of a FDM obtained using FM modulation for each channel. In this system, the signals are sent using suppressed-carrier

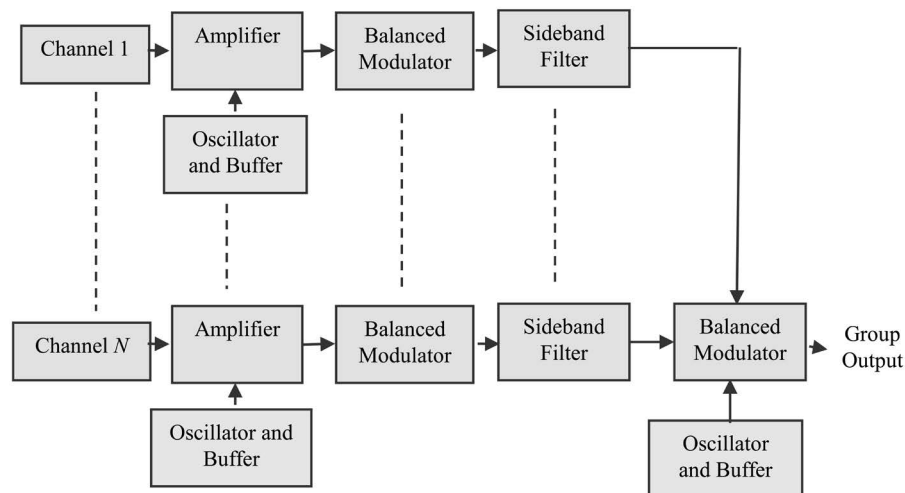


FIGURE 2.20 A basic N channel of an FDM.

modulation and each channel has its own amplifier, oscillator, balanced modulator, and sideband filter. Channels are grouped together by adders and group filters.

2.10.3 PHASE MODULATION

Phase modulation (PM) is like FM. The instantaneous phase of the carrier signal is varied between the values of $(\Phi + \Delta\Phi)$ and $(\Phi - \Delta\Phi)$ at a rate proportional to the modulating frequency and by an amount equal to the amplitude of the modulating signal. The equation for the PM can be expressed by

$$v(t) = V_c \sin(\omega_c t + \Phi_m \sin \omega_m t) \quad (2.8)$$

where Φ_m is the modulating index.

Despite the similarities between PM and FM, PM has a significant difference, that is, the maximum deviation in phase depends only on the amplitude of the modulating waveform and is independent of the frequency.

2.11 DIGITAL MODULATION AND DEMODULATION

Digital RF modulators are based on the carrier signal being modulated by a digital baseband signal. It offers many advantages over analog modulation hence it forms the backbone of modern wireless communication systems. The digital counterparts of analog modulators for AM, PM, and FM are called ASK, PSK, and FSK respectively.

2.11.1 AMPLITUDE-SHIFT KEYING

ASK is a simple form of AM and can be achieved by OOK of the carrier frequency, where *on* represents binary 1 and *off* represents binary 0. Although, simple, OOK method is not normally used in digital data transmission due to significant variations in received the signal strength at the receiver.

The ASK is not a desired form of transmitting data. If used it is accompanied with some complex codes such as the Murray code. OOK presents significant noise problems at the receiver, and it introduces difficulties in determining transmitted binary 1s and 0s, particularly in noisy environments. In some simple cases, two-state or four state AMs are used in combination with pulse PM techniques to increase the number of bits per baud or bits per pulse being transmitted.

2.11.2 FREQUENCY-SHIFT KEYING MODULATORS

FSK is the binary equivalent of analog FM. In this case, binary 0 is transmitted as frequency f_0 and binary 1 is transmitted as frequency f_1 while keeping the amplitude of the RF waveform constant as illustrated in [Figure 2.21](#). Hence the binary signal effectively modulates the carrier in a less complicated manner than a continuous analog base band signal. In practice, the frequency shift is only a few kilohertz. The receiver is required simply to differentiate between the two frequencies f_0 and f_1 for a successful reception. This is particularly important in mobile communication systems where there is an overriding need to minimize the transmitted bandwidth.

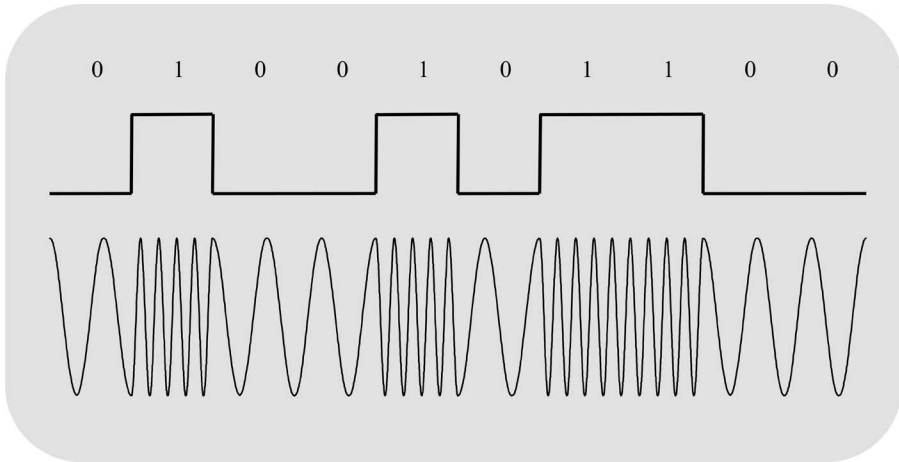


FIGURE 2.21 An FSK waveform.

To produce the smallest error probability, the carriers f_0 and f_1 , must be orthogonal. That is, they must have correlation coefficient of 0. To minimize the bandwidth of the transmitted signal it is necessary to determine the minimum difference between frequencies f_0 and f_1 , which produces orthogonal signals. This method is called *minimum-shift keying* (MSK).

FSK is the main method of transmitting data over older radio communication links since analog FM modulation can easily be replaced by FSK. Also, existing analog data carrying capable FM receivers can be used. Nevertheless, the radio links specifically designed for data transmission largely employ phase-shift modulation technique, and it will be explained next.

2.11.3 PHASE-SHIFT KEYING MODULATORS

PSK was developed for wireless communication systems for early deep space explorations, but, today, it is extensively used in civilian, industrial, and military, communications. In this method, digital information is transmitted by shifting the phase of the carrier among several discrete values. The performance of binary PSK is like FSK. Nevertheless, there is one important difference between the two, that is, the bandwidth required by the transmitted signal in PSK is much less than FSK since transmission can be done only on one frequency.

There are many variations of PSK, these are: binary phase-shift keying (BPSK), differential phase-shift keying (DPSK), quaternary phase-shift keying (QPSK), offset QPSK (OQPSK), differential QPSK (DQPSK), $\pi/4$ DQPSK, and octonary phase-shift keying (OPSK). These modulation techniques form the basis of today's digital communication systems therefore will be explained in detail next. There may be subtle differences between some of the methods depending on the age of technology involved.

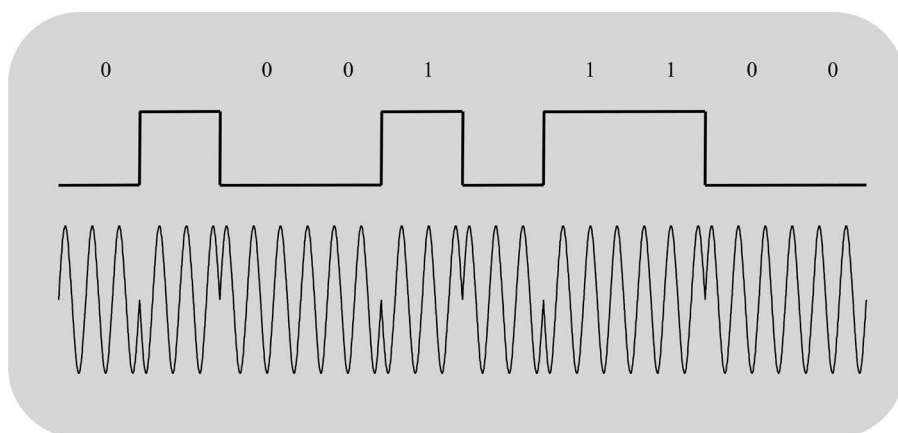


FIGURE 2.22 Amplitude-shift keying.

2.11.3.1 Binary Phase-Shift Keying

BPSK is equivalent to analog PM as it is another form of two-state modulation technique. In BPSK, both frequency and amplitude of the carrier remain constant, and only the phase of the signal changes. The binary symbols are transmitted as a phase shift of zero or π radians, hence the two binary states are 180° apart in phase. A typical BPSK waveform and the amplitude spectrum are illustrated in [Figure 2.22](#).

In practical implementations, the reference signal is the signal supplied from an oscillator. Similar reference signals are used for both transmitter and receiver.

2.11.3.2 Quaternary Phase-Shift Keying

It is also possible to have two state phase-shift keying to four, eight, or sixteen states. [Figure 2.23](#) illustrates a typical four state QPSK waveform. This type of modulation provides two or three bits of information for each pulse sent. The phase shift keyed waveform can be represented by the expression.

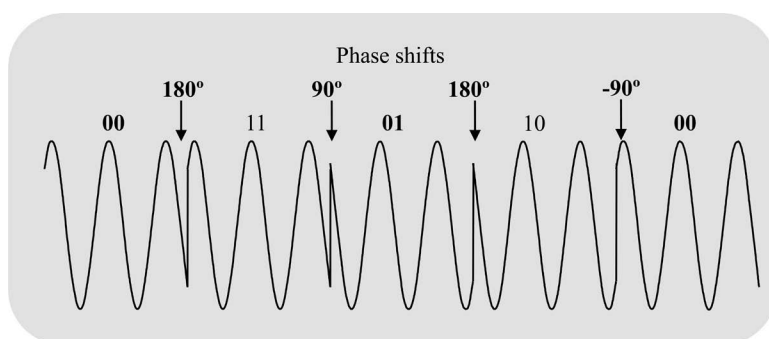


FIGURE 2.23 A typical four state QPSK waveform.

TABLE 2.2
Phase Shift in QPSK

Digital Bits	Phase Shift	In Phase Amplitude	Quadrature Amplitude
00	$\pi/4$	$+1/\sqrt{2}$	$+1/\sqrt{2}$
01	$3\pi/4$	$-1/\sqrt{2}$	$+1/\sqrt{2}$
11	$-3\pi/4$	$-1/\sqrt{2}$	$-1/\sqrt{2}$
10	$-\pi/4$	$+1/\sqrt{2}$	$-1/\sqrt{2}$

QPSK involves four phase states each state separated by 90°. In four-state modulation, QPSK, Φ_k , takes on two possible values for phase separated by 90°. QPSK is equivalent to PM by a symbol with four possible levels, as shown in Table 2.2.

$$v_c(t) = V_c \cos(\omega_c t + \phi_k) \tag{2.9}$$

Each pulse represents two bits of information (00, 01, 10, or 11). An important version variation of QPSK is $\pi/4$ -DGPSK modulation. This is a differential four-state PM technique in which the phase reference is derived from the previous RF pulse. Information is sent by the changes in phase from pulse to pulse rather than in absolute phases. The $\pi/4$ indicates that the minimum phase shift with respect to the reference of 45°.

In QPSK, the input data stream is grouped into *dibits*. Each group of digits produces a unique phase shift of the carrier, separated by interval of $\pi/2$. The resulting QPSK can be represented on diagram as shown in Figure 2.24.

On the receiving side, the data carrying signal is recovered from the QPSK waveform by using two coherent detectors supplying oscillators which generate in-phase and quadrature components of the carrier wave. This process produces an in-phase data waveform and a quadrature data waveform separately. A drawback of this method is that a single error made in one of the digits in the detection process affects two digits in the reproduced data waveform.

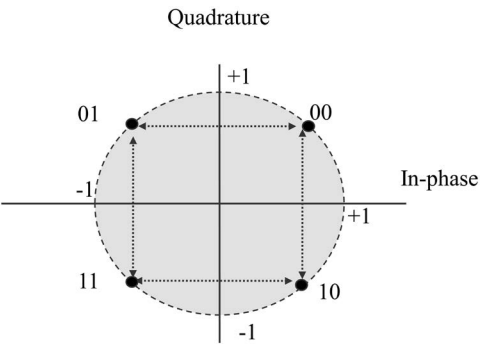


FIGURE 2.24 I/Q diagram for QPSK.

2.11.3.3 Eight-State Phase-Shift Keying

Eight-state Phase-Shift Keying (8-PSK) involves the use of eight phase states with the phase states separated by 45° . Each pulse represents three bits of information (001, 001, 010, 011, 100, 101, 110, or 111).

2.11.3.4 16-Quadrature Phase-Shift Keying

16-quadrature phase-shift keying (16-PSK) provides 4-bit pulse capability. An improved version of 16-PSK is 16-QAM (16-quadrature amplitude modulation). 16-QAM uses 4 states and 4 possible amplitudes. This 4×4 combination results in a 16 possible state modulation.

In 16 phase-state keying (16-PSK), each phase step is separated by 22.5° , thus allowing the transfer of four bits of information per pulse. An advanced version of 16-PSK is the 16-amplitude phase keying, which is often used in high-capacity time division multiplexing (TDM) systems.

2.11.3.5 $\pi/4$ DQPSK

$\pi/4$ DQPSK is an improved form of QPSK. In this case each dibit combination produces a specific phase transition multiple of $\pi/4$. The phase transition is independent of the current carrier phase and has maximum value of $3\pi/4$, which is less than the maximum phase shift produced QPSK. Although there are eight separate carrier phases, there are only four possible phase transition 00, 01, 10, and 11.

2.11.3.6 Differential Phase-Shift Keying

Differential phase-shift keying (DPSK) is another form of BPSK. In DPSK, the information sent is determined by a change in the phases between the successive pulses. By using the phase of the carrier signal in the previous digit interval as the reference the phase of the present digit interval can be determined. To make this possible, binary 0 is transmitted to be the same phase as the previous digit, and binary 1 is transmitted as a change of the phase. On the receiver side, the receiver constantly compares the phase of the current digit with the phase of the previous one. No change indicates binary 0. If there is a phase change the digit is interpreted as a binary 1.

2.11.3.7 Offset QPSK

OQPSK delays the data for the quadrature channel by a pulse duration T . This results in maximum phase duration of the carrier signal, thus restricting it to 50% of the phase transition of a standard QPSK. As a result, a much smaller envelope variation takes place after bandpass filtering. Hence, the phase transition, which pass through the origin are removed by the delay of T , thus restricting the maximum phase shift to $\pi/2$.

2.11.4 DIGITAL DEMODULATION

Demodulators are used on the receiver side and are constructed to decode information coming from transmitters. The receiver demodulator and the transmitter modulator

must a matching pair for successful transmission and reception. Depending on the type of modulators, the demodulators can be amplitude demodulators, frequency demodulators, or phase demodulators.

2.11.4.1 Amplitude Demodulators

Amplitude demodulators convert an amplitude modulated RF signal into audio, video, or pulse signals of the original form as the modulated signals. An important type of AM demodulator is called envelope detector. Improved versions of envelope detectors include capability for volume control, and automatic gain control.

A different form of amplitude demodulator is the product detector, which consists of a balanced demodulator or balanced mixer circuit and filters. Product detectors are used to demodulate *single sideband* (SSB) or *double sideband* (DSB) signals. They are made either from diodes or transistors.

2.11.4.2 FM Demodulators

FM demodulators, also known as *FM detectors*, use diodes, transformers, discriminators, rectifiers, limiters, and filters to demodulate the received signals.

2.11.4.3 Phase Detectors

Phase detectors use filters, power splitters, couplers, and mixers. The structures of phase detectors are different depending on types of signals they handle, e.g., BPSK or QPSK. In the case of BPSK detectors, double balanced diode mixers are often preferred. In this case, one of the inputs of the mixer is the phase-modulated carrier frequency from an IF amplifier and the other input is the reference carrier frequency from an oscillator. Consequently, the frequency of the output signal of the balanced mixer is either $f_{\text{out}} = f_{\text{in}} - f_c$ or $f_{\text{out}} = f_c - f_{\text{in}}$. The correct frequency and phase angle can be obtained by a suitable synchronization technique.

2.11.4.4 QPSK Demodulation

QPSK demodulation provides two-bits per pulse. *QPSK* demodulators are made from filters, three-way power splitters, balanced mixers, quadrature-phase couplers, adders, and timing circuits. In some cases, additional power splitters are used in the output stages to recover the carrier frequency. The timing circuits are used to determine the binary states for the signal as one or zero. Timing circuits also provide necessary time delays to convert the parallel signals into serial forms.

In the demodulation process, if the carrier of the received signal and the oscillator of the receiver requires phase synchronization. This method is called *coherent detection* also known as *correlator*. Correlators provide a high degree of signal-to-noise ratio and hence probability of error becomes low. Coherent detectors require that the phase of the LO in the receiver must have a consistent relation with the carrier of the received signal. Some modulated waveforms can be demodulated using *non-coherent detection*. Non-coherent detection is based on a series of filters to determine received frequencies and to eliminate interference. Non-coherent detectors find wider applications due to their simplicity.

TABLE 2.3
Comparison of FDM and TDM

Function	FDM	TDM
Signal type	Analog – channels carry original waveform	Digital channels carry waveform samples
Speed	Slow – depends on bandwidth availability	High – depends on channel and modem
Transmission	Parallel	Serial
Network type	Widespread	Clustered

2.12 DIGITAL MULTIPLEXERS

In traditional communications systems, many different signals (e.g., data, image, voice) are transmitted on a single communication link by three generic forms of multiplexing: FDM, TDM, and code division multiplexing (CDM).

Although, in digital transmission both FDM and TDM techniques are used, FDM is preferred for slow data flow between terminals spread over wide geographical areas. The TDM is generally applied in high data rate terminals clustered in one location or in situations where long-distance digital signal transmission is required. [Table 2.3](#) compares FDM and TDM use in signal transmission.

2.12.1 FREQUENCY DIVISION MULTIPLEXING

FDM is illustrated in a simplified block diagram in [Figure 2.25](#). As seen in this figure, many different inputs are fed to each sub-channel for modulation. Oscillators

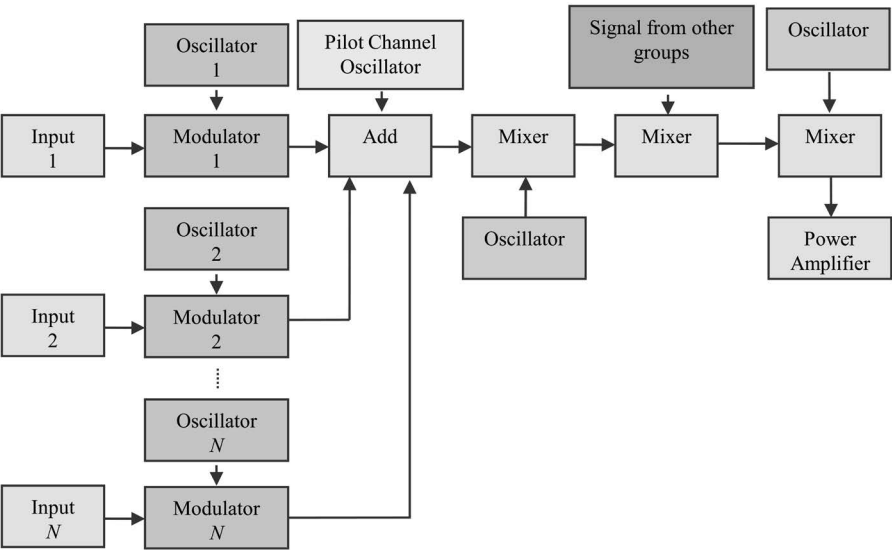


FIGURE 2.25 A simplified FDM system block diagram.

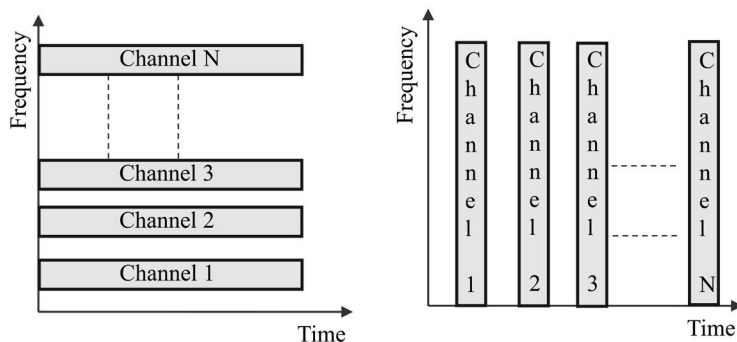


FIGURE 2.26 (a) Frequency division multiplexing, and (b) Time division multiplexing.

are used to provide selected frequencies for each channel. The output of each modulator within a particular group is added to pilot channel oscillator signals by using RF add circuits. The combined signal feeds a mixer circuit that acts as a frequency up-converter.

An FDM band limits each channel to a maximum frequency, as illustrated in Figure 2.26 (a). Vacant frequency bands, called the guard bands, are allocated between channels to prevent *crosstalk* through overlapping of spectra of adjacent channels. This helps effective filtering and frequency selection on the receiver side. Each channel waveform is modulated onto a sinusoidal *sub-carrier* to translate the channel to its allotted position of frequency. In such systems, many sub channels are used in parallel in the main allocated channel hence they can transmit simultaneously over a single physical circuit.

TDMs use a single wideband channel for many input channels by assigning time slots for each channel data. In TDM, at any instant of time only one signal can be transmitted. As illustrated in Figure 2.26 (b), Channel 1 occupies the entire channel during the first time slot. Channel 2 occupies the entire time slot during the second, and so on.

In the next stage, the outputs from all the different groups of modulators are combined. Since the mixers associated with each group use different frequencies, the signal from each group is distinguishable from one another. This helps easy recovery of individual signals on the receiving side. A PA chain is used to increase the power level of the combined signal to the desired output level for transmission. The PA connects the overall signal to antenna of the transmitter.

2.12.2 TIME DIVISION MULTIPLEXING

TDM is illustrated in Figure 2.27. In a typical TDM, analog inputs are fed to sampling circuits to be converted into digital forms. The task of delay lines is to space data in time. Delay line data is added to be reorganized again in the output stages. The output stage contains other components such as mixers, oscillators, PAs, and transmitter antenna.

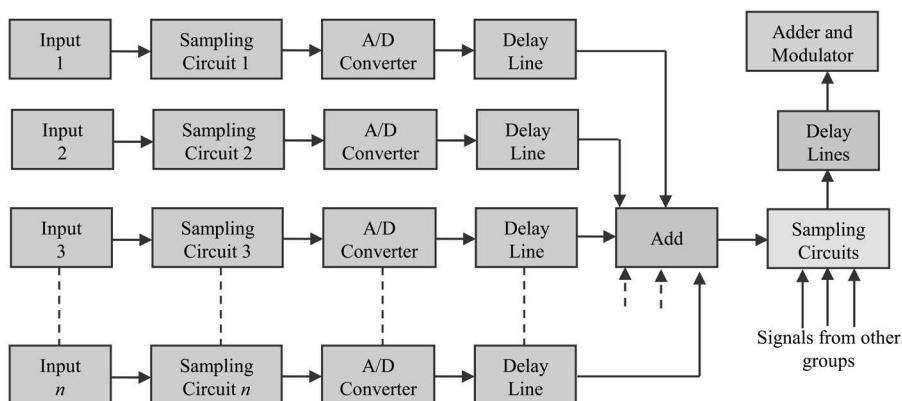


FIGURE 2.27 A typical TDM system.

The concept of time sharing of a channel is shown in Figure 2.28. In this case, there are 36 major time slots and 12 time slots in each major time slot. As a result, there are 432 time slots available for data transmission.

2.12.3 CODE DIVISION MULTIPLEXING

In CDM, multiple data signals are combined for simultaneous transmission in a common frequency band. This allows multiple users to share the same channel. This technology is called CDMA.

In code division multiple access, CDMA systems, the narrowband message signal is multiplied by a very large bandwidth signal called the *spreading signal*. The spreading signal is a pseudo noise code sequence that has a chip rate which is greater by order of magnitude than the data rate of the message. As seen in Figure 2.29, all users in a CDMA system use the same carrier frequency and may transmit simultaneously. Each user has its own pseudorandom codeword which is approximately orthogonal to all other codewords. The receiver performs a time correlation operation to detect only the specific desired codeword. All other codewords appear as noise due to decorrelation. For the detection of the message signal, the receiver needs to know the codeword used by the transmitter. Each user operates independently with no knowledge of the other users.

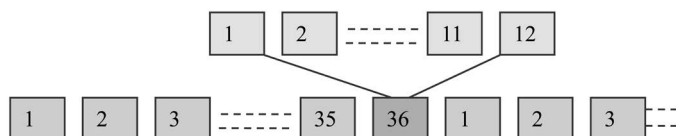


FIGURE 2.28 Time slot assignment in TDM.

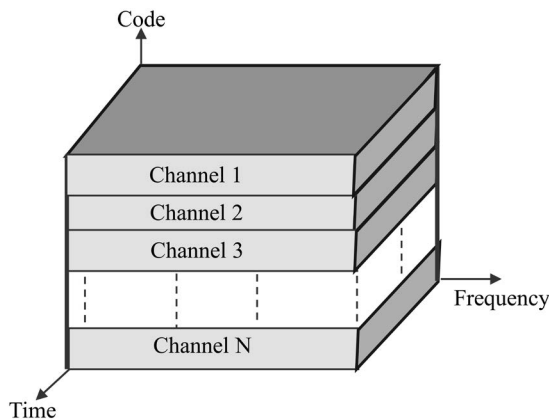


FIGURE 2.29 Spread spectrum multiple access CDMA.

In CDMA, the power of multiple users at a receiver determines the noise floor after decorrelation. If the power of each user within a cell is not controlled such that they do not appear equal at the base station receiver, then the *near-far* problem occurs. To combat the near-far problem, a *power control* scheme is used in most CDMA implementations. The power control is provided by each base station and ensures that each unit provides the same signal level to the receiver in a coverage area. The power control is implemented by rapidly sampled signal indicator (RSSI) levels of each unit and sensing a power channel over the forward radio link.

2.13 DIGITAL DUPLEXERS

There are two types of digital duplexers: 1) frequency division duplexing (FDD) and 2) time division duplexing (TDD).

2.13.1 FREQUENCY DIVISION DUPLEXING

FDD provides two distinct bands of frequency one transmission channels for the transmitter and one for reception channel for the receiver (Figure 2.30 (a)). In FDD,

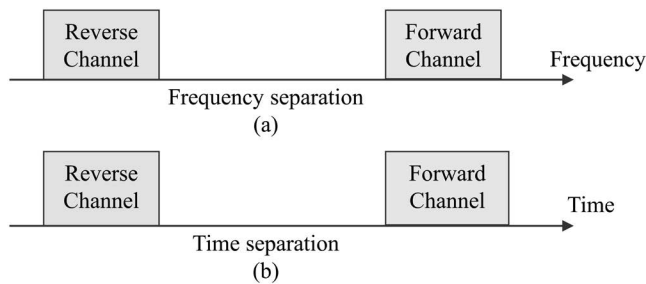


FIGURE 2.30 (a) Frequency division duplexing and (b) Time division duplexing.

separate transmit and receive antennas are used to accommodate the two separate channels. However, in some cases, a single antenna is used backed up by duplexers for simultaneous transmission and reception. To facilitate FDD, it is necessary to separate transmit and receive frequencies by about 5% of the nominal RF frequency, so that duplexer can provide sufficient isolation.

In FDD, a pair of simplex channels with a fixed known frequency separation is used to define a specific radio channel in the system termed as *forward channel* for transmission and *reverse channel* for reception. In FDD, any duplex channel consists of two simplex channels and the frequency separation between forward and reverse channels is constant.

2.13.2 TIME DIVISION DUPLEXING

TDD uses time instead of frequency to provide both a forward and reverse links. In TDD, multiple users share a single radio channel by taking turns in time domain (Figure 2.30 (b)). As the forward and reverse links share a single radio channel in time, a portion of time is used to transmit, and the remaining time is used to receive from the other side within the allocated *time slots*. Each duplex channel has both a forward time slot and a reverse time slot to facilitate bidirectional communications. If the data transmission rate in the channel is much greater than the end user's data rate, it is possible to store information bursts and provide the appearance of full duplex operation even though there are not two simultaneous radio transmissions at any instant. TDD is only possible with digital transmission systems and with digital modulation, and it is sensitive to timing. TDD finds applications for indoor and small area wireless networks where physical coverage distances are relatively small.

There are several tradeoffs between TDD and FDD. TDD enables each transceiver to operate either as a transmitter or as a receiver on the same frequency. This eliminates the need for separate forward and reverse frequency bands. There is a time latency created by TDD because communication is not full duplex in its truest sense, and this latency can create inherent sensitivity for propagation delays for the individual users. Because of the rigid timing required for time slotting, TDD is generally limited to short range access. Clearly, TDD is effective for fixed wireless access when all users are stationary so that propagation delays do not vary in time among the users. FDD, on the other hand, is geared toward a radio communication system that allocates individual RF for each user. Because each transceiver simultaneously transmits and receives RF signals, the frequency allocation in forward and reverse channels are coordinated within the system users as well as with out-of-band users that occupy the same spectrum.

2.14 FREQUENCY SPREADING

In the RF systems, the bandwidth is a limited resource, one of the primary design objectives of modulation techniques is to minimize transmission bandwidth. Spread spectrum techniques require transmission bandwidths with several orders of magnitude greater than the minimum required signal bandwidth. Clearly, this system is

bandwidth inefficient. But frequency spreading offers advantages in allowing many users to use the same bandwidth without significantly interfering with one another.

Apart from occupying a very large bandwidth, spread spectrum signals are pseudorandom signals that have noise-like properties when compared to digital information data. The spreading of the waveform is controlled by a *pseudo-noise sequence* or *pseudo-noise code*, which is essentially binary sequence that appears random but can be reproduced in a deterministic manner by the intended receivers.

Spread spectrum has an inherent interference rejection capability. Each user is assigned a unique pseudo-noise code which is approximately orthogonal to the codes of the other users. The receiver can separate each user, based on the information on their allocated codes even though they occupy the same spectrum. Resistance to multipath fading is another useful property of the spread spectrum technique. Since spread spectrum signals have uniform energy over a very large bandwidth at any given time only a small portion of the spectrum undergoes fading.

There are two main versions of spread spectrum techniques: the direct sequence spread spectrum (DSSS) and the frequency hopped spread spectrum (FHSS).

2.14.1 DIRECT SEQUENCE SPREAD SPECTRUM

A DSSS system spreads the baseband data by directly multiplying the data pulses with a pseudo-noise sequence. The pseudo-noise sequence is produced by an internal code generator. A single pulse or symbol of the PN waveform is called a chip. Synchronized data symbols, information bits or binary channel code symbols, are added to the chip before being phase modulated.

Receiver demodulates the transmitted signal in several different forms by using either coherent phase-shift-keying demodulators or differentially coherent PSK demodulators. Figure 2.31 (a) shows the received spectra of the desired spread spectrum signal and the interference at the output of the receiver wideband filter. Multiplication by the spreading waveform produces the spectra, see Figure 2.31 (b), at the input of the demodulator. The signal bandwidth is reduced to while interference

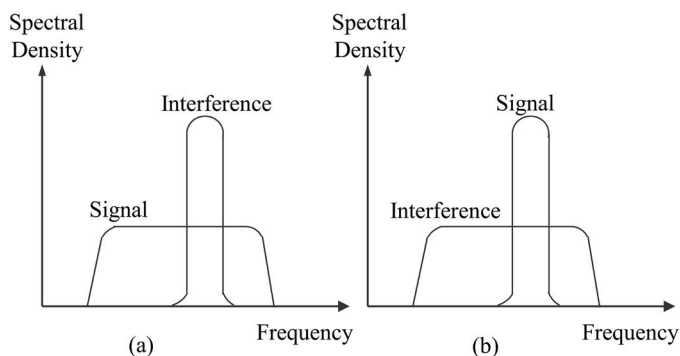


FIGURE 2.31 Spectra of received signal: (a) Wideband filter output and (b) Correlator output.

energy is spread over an RF bandwidth. The filtering action of the demodulator removes most of the interference spectrum. This process is very important as the interference energy is eliminated by spreading minimally affecting the desired receiver signal.

2.14.2 FREQUENCY HOPPED SPREAD SPECTRUM

Frequency hopping involves periodic change of transmission frequency within an allocated frequency band. A frequency hopping signal may be regarded as a sequence of modulated data bursts varying in time at pseudorandom carrier frequencies. Hopping occurs over a frequency band that includes several sets channels each operating at an allocated frequency. A set of possible carrier frequencies is called *hopset*. Each channel is defined as a spectral region with a central frequency in the hopset and a bandwidth large enough to include most of the power in a narrow modulation burst (e.g., FSK) within the corresponding carrier frequency. The bandwidth of a channel used in the hopset is called the *instantaneous bandwidth*. The bandwidth of the spectrum over which the hopping occurs is called the *total hopping bandwidth*. Data is sent by hopping the transmitter carrier to seemingly random channels which are known only by the intended receiver. On each channel, small bursts of data are sent using conventional narrowband modulations before the transmitter hops again. When only a single carrier frequency is used in each hop, digital data modulation is called *single channel modulation*. The time duration between hops is called *hop duration* or *hopping period*.

On the receiver side, when the frequency hopping is removed from the received signal, the resulting signal is said to be *dehopped*. Before demodulation, the dehopped signal is applied to a conventional receiver. The frequency pattern produced by the receiver synthesizer is synchronized with the frequency pattern of the received signal, the mixer output is a dehopped signal at a fixed difference frequency. Whenever an undesired signal occupies a particular hopping channel in the input signal, the noise and interference in that channel are translated in frequency so that they enter the demodulator. Thus, in this process, it is possible to have collision in a frequency hopping system where an undesired user transmits in the same channel at the same time as the desired user.

If binary frequency-shift-keying, FSK, is used, the pair of possible instantaneous frequencies changes with each hop. The frequency channel occupied by a transmitted signal is called *transmission channel*. The channel that would be occupied if an alternative symbol were transmitted is called the *complementary channel*. The frequency hop rate of an FHSS system is determined by the frequency agility of receiver synthesizer, the type of information being transmitted, the amount or redundancy used to code against collision, and the distance to the nearest potential interferer.

FHSS has an advantage over DSSS in that it is not as susceptible to the near-far problem. The near-far problem occurs when close by and stronger signals prevents the recognition of weaker signals coming from far transmitters. Because signals are generally not utilizing the same frequency simultaneously, the relative power levels of signals are not as critical as in DSSS. The near-far problem cannot totally avoid, however, since there will be some interference caused by stronger signals bleeding

into weaker signals due to imperfect filtering of adjacent channels. To combat the occasional hits, error correction coding is required on all transmissions. By applying appropriate (e.g., Reed-Solomon) error correcting codes performance can be increased significantly.

2.15 MULTIPLE ACCESS TECHNIQUES

Multiple access schemes are used to allow many users to simultaneously share a finite amount of radio spectrum. The sharing of spectrum is required to achieve high-capacity usage of the available bandwidth. In wireless communication systems, it is desirable to allow users to send information to a station while receiving from the station at the same time. This can be achieved by duplexing, which can be done using frequency or time domain techniques.

Multiple access techniques can be grouped into two broad terms as *narrowband* multiple access and *wideband* multiple access. They both depend on how the available bandwidth is allocated to a user. The Both narrow band and wideband systems are supported by duplexing techniques such as the FDD and the TDD.

Narrowband system is a term used to relate the bandwidth of a single channel to the expected coherence bandwidth of the channel. In a narrowband multiple access system, the available radio spectrum is divided into many narrowband channels, usually operated using FDD. To minimize interference between forward and reverse links on each channel, the frequency separation is made as large as possible within the frequency spectrum, while still allowing duplexers and transceivers in each unit.

Wideband system has transmission bandwidth of a single channel that is much larger than the coherence bandwidth of the channel. Thus, multipath fading does not greatly vary the received signal power within a wideband channel, and the frequency selective fades occur in only a small fraction of the single bandwidth at any instance of time. In wideband multiple access systems, many transmitters are allowed to transmit using the same channel.

There are few different types of multiple access techniques, these are:

- Frequency division multiple access,
- Time division multiple access,
- Spread spectrum multiple access,
- Code division multiple access,
- Frequency hopped multiple access,
- Hybrid spread spectrum multiple access,
- Space division multiple access,
- Carrier sense multiple access, and
- Data sense multiple access.

2.15.1 FREQUENCY DIVISION MULTIPLE ACCESS

Frequency division multiple access (FDMA) assigns individual channels to individual users. These channels are assigned on demand to the user who requests service. Upon the request, the users are assigned a channel as a pair of frequencies for

forward channel and reverse channel. Once allocated, during the transmission no other user can share the same channel.

FDMA requires tight RF filtering to minimize adjacent channel interference. Also, if a channel is not used it is said to be an *idle channel* and cannot be used by others to increase or share the capacity. To increase capacity techniques such as FDD can be used. Then the system is called FDMA/FDD.

2.15.2 TIME DIVISION MULTIPLE ACCESS

Time division multiple access (TDMA) allows each user to occupy a cyclically repeating time slot. It may be thought of having frames that comprises N reoccurring time slots. TDMA systems transmit data in a *buffer-and-burst* method therefore the transmission for any user is non-continuous. Transmission sequence from various users is interlaced in a repeating frame structure as it illustrated in Figure 2.32. Each frame is made up of several sections: a preamble, an information message, and the tail bits. *Preamble* contains the address and synchronization information that both the base station and units to be able identify each other. *Guard times* are utilized to allow synchronization of the receivers between different slots and frames. Different TDMA wireless standards have different frame structures.

In TDMA systems, the number of time slots per frame depends on the modulation techniques and the bandwidth available. The adaptive equalization is usually necessary since the transmission rates are generally higher as compared to FDMA channels. Also, high synchronization overhead is required because of burst transmissions. TDMA transmissions are slotted, and this requires the receiver to be synchronized for each data burst. In addition, guard slots are necessary to separate users. Guard times should be minimized but too much shortening guard times can cause interference to adjacent channels.

TDMA is applied in narrowband as well as wideband systems. Narrowband TDMA allows users to share the same radio channel but allocates a unique time slot to each user in a cyclic fashion on the channel. For narrowband TDMA system, there are large number of radio channels allocated using either FDD or TDD, and channels are shared using TDMA. Such systems are called TDMA/FDD or TDMA/TDD access systems.

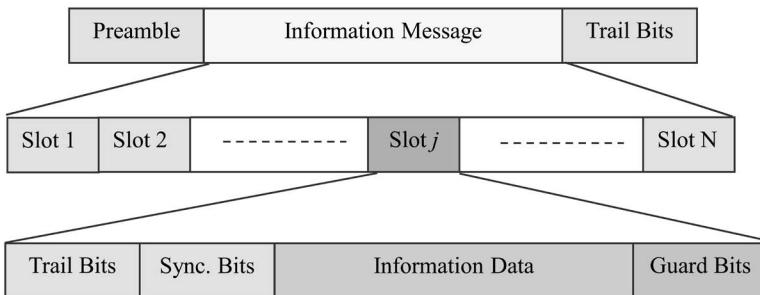


FIGURE 2.32 TDMA frame structure.

In wideband systems, TDMA allocates time slots to the many transmitters on the same channel and allows only one transmitter to access the channel at any instant of time. This can be compared to CDMA where spread spectrum allows all the transmitters to access the channel at the same time.

2.15.3 SPREAD SPECTRUM MULTIPLE ACCESS

Spread spectrum multiple access (SSMA) uses signals which have transmission bandwidth that is several orders of magnitude greater than the minimum required RF bandwidth. A pseudo-noise sequence converts the narrowband signal to a wideband noise-like signal before transmission. SSMA also provides immunity to multipath interference and robust multiple access stability. Since many users can share the same system SSMA is known to be bandwidth efficient and attracts much attention of the wireless communication designers.

2.15.4 CODE DIVISION MULTIPLE ACCESS

CDMA multiplies the narrowband message signal by a very large *spreading signal*. The spreading signal is a pseudo noise code sequence that has a chip rate which is greater by orders of magnitude than the data rate of the message.

2.15.5 FREQUENCY HOPED MULTIPLE ACCESS

Frequency hopped multiple access (FHMA) is a digital multiple access system in which the carrier frequencies of the individual users are varied in a pseudorandom fashion within a wideband channel. FHMA allows multiple users to simultaneously occupy the same spectrum at the same time, where each user dwells at a specific narrowband channel at a particular instance of time, based on the pseudo-noise (PN) code of the user. The digital data of each user is broken into uniform sized bursts which are transmitted on different channels within the allocated spectrum band. The instantaneous bandwidth of any one transmission burst is much smaller than the total spread bandwidth. The pseudorandom changes of the channel frequencies of the user randomize the occupancy of the specific channel at any given time, thereby allowing for multiple access over a wide range of frequencies. On the receiver side, a locally generated pseudo-noise code is used to synchronize the receiver's instantaneous frequency with that of the transmitter. At any given point in time, a frequency hopped signal only occupies a single, relatively narrow channel since narrowband FM or FSK is used.

The difference between FHMA and a traditional FDMA is that the frequency hopped signal changes channels at rapid intervals. If the rate of change of the carrier frequency is greater than the symbol rate, then the system is referred to the *fast frequency hopping system*. If the channel changes at a rate less than or equal to the symbol rate, it is called *slow frequency hopping*. A fast frequency hopper may thus be thought of as an FDMA system which employs frequency diversity. FHMA system often employs energy efficient constant envelope modulations.

Frequency hopped system provides a level of security, especially when many channels are used, since an unintended or interception receiver does not know the pseudorandom sequence of frequency slots must retune rapidly to search for the signal it wishes to intercept. In addition, the FH signal is immune to fading, since error control coding and interleaving can be used to protect the frequency hopped signal against deep fades which may occasionally occur during the hopping sequence. Error control coding and interleaving can also be combined to guard against erasures which can occur when two or more users transmit on the same channel at the same time. Bluetooth and HomeRF wireless technologies have adopted FHMA for power efficiency and low-cost implementation.

2.15.6 HYBRID SPREAD SPECTRUM MULTIPLE ACCESS

Hybrid SSMA techniques provide advantages. There are several versions of it.

Hybrid FDMA/CDMA (FSDMA) technique is used as an alternative to direct sequence (DS-CDMA). The available bandwidth spectrum is divided into several subspectra. Each of the sub-channels becomes a narrowband CDMA system having a processing gain lower than the original CDMA. This hybrid system has an advantage that different users can be allocated different subspectra depending on their requirements.

Hybrid DS/FHMA technique consist of direct sequence modulated signal whose central frequency is made to hop periodically in a pseudorandom fashion. Direct sequence, frequency hopped systems have an advantage in that they avoid the near-far effect. However, frequency hopped CDMA systems are not easy to synchronize frequency hopped station receivers to the multiple hopped signals.

Time division CDMA (TCDMA) or **TDMA/CDMA** system uses different spreading codes for different cells. Within each cell, only one user per cell is allotted to a particular time slot. Thus, at any time, only one CDMA user can transmit in each cell. It avoids near-far problem since only one user transmit within a cell.

Time Division frequency Hopping (TDFH) is a multiple access technique that the spectrum hops to a new frequency at the start of new TDMA frames. It is used when severe multipath problems or severe co-channel interferences occur.

2.15.7 SPACE DIVISION MULTIPLE ACCESS

Space division multiple access (SDMA) controls the radiated energy for each user. Different areas covered by the different antenna beam may be served by the same frequency as TDMA or CDMA or different frequencies as in FDMA. Extensive studies are concentrated on adaptive antennas to simultaneously steer energy in the direction of many users. Adaptive antennas used at the base station promise to mitigate some of the problems on the reverse link. In the limiting case of infinitesimal beam width and infinitely fast-tracking ability, adaptive antennas implement optimal SDMA, thereby providing unique channel that is free from the interference of all other users in the cell. With SDMA, all users within the system would be able to communicate at the same time using the same channel. Perfect adaptive antennas do not exist at this moment in time but reasonable sized arrays with moderate directivities exist.

2.15.8 CARRIER SENSE MULTIPLE ACCESS

Carrier sense multiple access (CSMA) protocols assume that each terminal on the network is able to monitor the status of each channel before transmitting information. If the channel is idle, then the user is allowed to transmit a packet based on a particular algorithm which is common to all transmitters on the network.

In CSMA protocols, detection delay and propagation delay are two important parameters. *Detection delay* is a function of the receiver hardware and is the time required for a terminal to sense whether the channel is idle. *Propagation delay* is a relative measure of how fast it takes to travel from a base station to a mobile terminal. With small detection time, a terminal detects a free channel quite rapidly, and small propagation means that a packet is transmitted through the channel in a small interval of time relative to the packet duration. There are several variations of the CSMA strategy.

Persistent CSMA – The terminal listens to the channel and waits for transmission until it finds the idle channel. As soon as the idle channel is identified, the transmitter transmits with probability of one.

Non-persistent CSMA – after receiving not-acknowledged (NACK) the terminal waits a random time before retransmission. This is a popular technique in wireless LAN applications.

P-persistent CSMA – is applied to slotted channels. When the channel is found to be idle, the packet is transmitted in the first available time slot.

CSMA/CD – In CSMA with collision detection (CD), a user monitors its transmission for collision. If two or more terminals start a transmission at the same time, the detected transmission is immediately aborted.

2.15.9 DATA SENSE MULTIPLE ACCESS

Data sense multiple access (DSMA) is a special type of CSMA. Transmitter demodulates the forward control channel before broadcasting data in the reverse channel. Each user attempts to detect a *busy-idle* message. When the busy-idle message indicates no user is transmitting in the reverse channel, then a user is free to send a packet.

2.16 WIRELESS MODEMS

Modems (MODulator-DEModulator) are RF transceivers to transmit and/or receive signals from another matching radio modems. In many applications, several modems are connected to an *access point* that enable wireless network connectivity. Specifications for wireless modems are form factor, modem speed, and network type.

Form factor describes the basic types of modems, such as internal modems that are computer cards installed in a slot of the motherboard; external modems that are modules that can be attached to communications ports of computers. *Modem speed* is the maximum data transfer rate at which the modem can deliver data, expressed in bits/second. *Network type* for wireless modems include dialups, Ethernet, GSM, ISDN, PAN, GPRS, and so on.

Modems are also identified by their radio link specifications, which include the frequency bands and the radio technique used. *Frequency band* on wireless modems can be 900 MHz, 2.4 GHz, 5 GHz, 23 GHz, VHF, and UHF. Choices for *radio technique* include DSSS, and frequency hopping spread (FHS) spectrum. Spread spectrum reduces the impact of localized frequency interferences by using more bandwidth than the system may need.

2.17 BRIDGES, ROUTERS, GATEWAYS, AND REPEATERS

In data transmission, if transmitter and receivers are out of range then the wireless bridges, routers, gateways, and repeaters can be used. They are fundamental components of wireless networks.

2.17.1 WIRELESS BRIDGES

Wireless bridges provide long-range point-to-point or point-to-multipoint links. Some bridges use DSSS while other use FHSS or some other techniques explained above. Wireless bridges comply with standards such as the IEEE-802.11b connecting one or more remote sites to a central server or the Internet.

2.17.2 WIRELESS ROUTERS

Wireless routers are for indoor and outdoor applications. They use narrowband or broadband technology such as the 802.11g. Broadband wireless router supports more bandwidth allowing more communicating devices to be networked.

2.17.3 WIRELESS GATEWAYS

Wireless gateways are used primarily in centralized WLAN management and control. They have a wide range of applications such as managing traffic in voice-based systems, connecting remote locations to central hubs, and providing local and rural connectivity over GSM networks.

2.17.4 WIRELESS REPEATERS

Wireless repeaters are employed to simply regenerate a network signal to extend the range of the existing wireless network infrastructure. For example, a WLAN receives radio signals from access points or end-user devices, or from repeaters.

2.18 PACKET RADIO

In a *packet radio* (PR) access technique, many subscribers are allowed to attempt to access a single channel in an uncoordinated manner. Transmission from each user is done in bursts of data. Simultaneous transmission of the multiple transmitters is detected at the base station receiver. The base station coordinates the operations by issuing acknowledge (ACK) and NACK signals to the

requesting transmitters. The ACK signal indicates an acknowledgment of the received burst from a particular station, and NACK indicates that the previous burst was not received correctly by the base station. By using ACK and NACK signals, a packet radio system employs effective feedback, even though traffic delay due to collisions may occur.

Packet radio multiple access is easy to implement but has low spectral efficiency and may result in considerable time delays. The subscribers use contention techniques to transmit on a common channel. There are many protocols used in traffic coordination of PR systems. For instance, ALOHAnet (Areal Locations of Hazardous Atmospheres network) protocols, developed for early satellite systems, are good examples of managing contention and collision. ALOHAnet allows computers (or subscribers) to transmit packets over a shared channel. The transmitter listens to the acknowledgment feedback to determine if transmission has been successful or not. If a collision occurs, the transmitter waits for a random amount of time and then retransmits the packet again. The advantage of packet contention technique is the ability to serve many users with virtually no overheads.

2.19 WIRELESS NETWORKS

Wireless technologies and wireless networks are applied ubiquitously, including in the following:

- Advanced wireless Internet access for data flow,
- Connection to the Internet of mobile devices or users,
- Connecting people-to-people, people-to-machine, machine-to-machine,
- Delivery of network access into isolated areas,
- Environmental monitoring such as air pollution, soil degradation, water quality,
- Exchange of information between fixed and mobile device,
- Health, medicine, and patient monitoring,
- Home or office networks for wireless connections,
- Infrastructure health monitoring, machine health and diagnostics,
- Networking of instruments in production, plants, and factories,
- Secure communications such as corporate LANs,
- Smart cities,
- Transport systems,
- Wireless instrumentation systems and control, and
- Wireless networking at which layout change such as wildlife and habitat monitoring.

Wireless networks are deployed in 1) centralized, 2) converged, and 3) cloud-based forms. They can be grouped as: 1) private wireless networks and 2) publicly wireless networks.

Private wireless networks are configured in special situations such as the campuses, hospitals, defense, security, tactical networks, maritime, border security, etc.

It is estimated that there are more than 1,000 private networks installed worldwide. These networks are installed and operated by the owners. Most of the private networks are based on cellular networks such as the 5G.

Public wireless networks are configured for the use of public and organizations. There are 6 types of public networks:

1. Ad hoc networks,
2. Cellular networks,
3. Global area networks (GANs),
4. Light-based and infrared networks,
5. LANs,
6. Metropolitan area networks (MANs),
7. Personal area networks (PANs),
8. Satellite networks,
9. Space networks, and
10. WANs.

2.19.1 WIRELESS AD HOC NETWORKS

Wireless ad hoc networks (WANETs) evolved from Packet Radio systems. They are also known as mobile ad hoc networks (MANETs). WANETs are decentralized networks without relying on any existing wireless networks. The structure organized in such a way that each node acts as an access point to receive data from others and forward data to the requesting targets. This decentralized and self-organizing nature of WANET gives them advantages over structured networks as they do not need any pre-existing infrastructure.

They are used many applications such as the air and navy defense systems, sensors networks, hospital communication systems, disaster response teams, smart phone systems, environmental monitoring, transport systems, and distributed sensor data collection. For example, the Airforce uses unmanned aerial vehicles (UAVs), Army and Navy tactical units use WANETs, transport systems use vehicular ad hoc networks (VANETs). Others common applications are smart phone ad hoc networks (SPANs), and mobile wireless sensor networks (MWSNs).

2.19.2 WIRELESS CELLULAR NETWORKS

Cellular networks are made from many cells distributed over a wide geographical area (Figure 2.33 (a) and (b)). Cells send and receive data from end nodes such as mobile telephones for transmission of voice, data, image, and other forms of data. Cells are also wirelessly connected to base stations known as base transceiver stations. Cell structure enables coverage of wider geographical areas. End devices, like mobile phones, tablets, laptops, machine-to machine devices can exchange data via cells. Worldwide distribution of cellular networks together with public telephone networks and satellite communications systems enables communication all end devices and provide internet access.

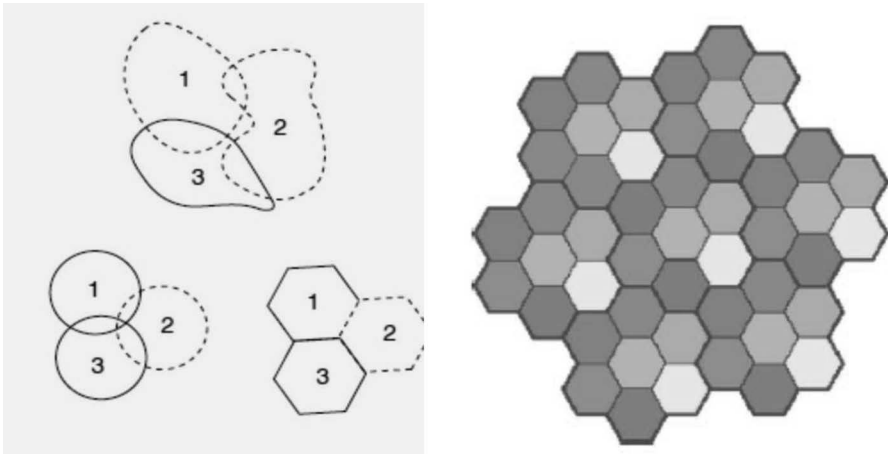


FIGURE 2.33 (a) Localized vs actual cell coverage and (b) Cellular frequency reuse pattern (Source: C. W. Wendt ⁽¹²⁾).

2.19.3 WIRELESS GLOBAL AREA NETWORKS

GAN is made up by interconnection of many networks for worldwide coverage. It works seamlessly enabling communication from one local coverage to another one. Broadband GAN (BGAN) involves satellite communication systems for coverage to remote areas.

There are various types of GANS: GPS, Satellite television networks (i.e., DIRECT TV), and the Internet that partially uses wireless networks.

2.19.4 LIGHT-BASED AND INFRARED NETWORKS

Infrared wireless transmitters use light beam at infrared frequencies to send communication signals from transmitter to receiver. Infrared transmitters generate strong signals to prevent interference from other light sources. In many applications, infrared communication systems work well mainly because of its high bandwidth. These systems can deliver data from 10 Mbps to 100 Mbps. There are four primary types of infrared systems:

1. *LOS systems* require an unobstructed view, between transmitters and receivers.
2. *Reflective infrared systems* signal from optical transceivers are received by an intermittent device called the central hub. Central hub then forwards messages to the intended recipients.
3. *Scatter infrared systems* operate on the principles of bouncing the transmitted signals from walls or other solid objects. The bounced signal is then picked up by the receiver. This approach limits the distance of transmission

to typically 30 meters or less depending on the strength of the transmitted signal, sensitivity of receiver, and presence of interference from other sources.

4. *Broadband optical telepoint systems* provide broadband services. In some applications, they can match capabilities of most modern wired communication systems.

Laser-based transmission also requires a clear line of sight between sender and receiver. In many applications, laser technology-based communication systems are subject to limited use since excessive radiation can affect human vision and deteriorate health.

2.19.5 LOCAL AREA NETWORKS

LANs provide connectivity between mobile or fixed devices. LAN coverage can be enhanced by *Extended LANs* which are certain kinds of wireless network equipment beyond their normal distance limitations of LANs. Devices, such as bridges and repeaters, provide connectivity outside the area of coverage.

For example, a typical wireless bridge can connect networks up to 4–5 km. Such bridges enable the linking of locations within or outside the line of site. LAN bridges rout dedicated digital communications lines from one site to another through communication carrier. Spread spectrum radio, infrared and laser-based equipment used as bridges are readily available on the commercial market. Longer-range wireless bridges are also available, including spread spectrum that work with Ethernet, or token ring, or others, ranging over distances of up to 40 km.

2.19.6 METROPOLITAN AREA NETWORKS

MAN is a collection of computers connecting LAN networks in a city area. In a sense, it is WAN within a metropolitan area. MANs are particularly useful for providing internet connections.

2.19.7 PERSONAL AREA NETWORKS

PANs are connection of devices withing an individual's workspace. The devices can be computers, smart phones, tablets, wireless keyboards, headphones, printers, or other personal used devices.

The meaning of PAN has been expanding due to availability of diverse range of wireless devices individuals use. *Wearable devices* such as smart watches, fitness devices, health monitoring devices are included in PAN. *IoT* devices are used smart homes to link cameras, home security devices, gaming and entertainment, and other control equipment.

PANs enable different technology devices to seamlessly communicate within short range. There are to prominent technologies: Wi-Fi Direct and Bluetooth PAN. *Wi-Fi Direct* offers high data transfer features on peer-to-peer connected devices

without the need traditional access points. Bluetooth PAN offers simplicity, low power consumption and wider compatibility with various devices.

2.19.8 SATELLITE NETWORKS

Satellite networks are communication systems that use satellites, ground-based stations, transceivers, and networks of ground stations. They provide point to point transmission by relaying information from one place to another on global scale. There are two types: of satellite networks low earth orbit (LEO) and geostationary orbit (GEO) networks. LEO are realized by multiple low orbiting satellites (Figure 2.34). GEOs are high altitude satellites and much fewer in number. Satellite networks are applied telecommunications, cellular communications, navigation systems, television and radio broadcasting, border security, and military systems.

Satellite communications are employed to provide global Internet services exploiting large coverage of satellites. The beam size of GEO satellites is typically 200–3,500 km. Low orbit and LEO's are miniaturized in size and their beam size is 100–1,000 km. Although LEO satellites coverage is small compared to GEO, they have low latency and high throughput due to their low altitudes.

Satellite communication occurs within four allocated bands of frequencies: the L, C, Ku, and Ka bands. *L band* communication occur within the spectrum of frequencies between 1.53 GHz and 2.7 GHz. This range of frequencies is a “long” wave and capable of penetrating many structures and materials. *C band* is traditionally used for the direct broadcast of television market. C band uses large diameter antennas to capture weak signals. *Ku band* operates in the range of frequencies between 11.7 GHz and 12.7 GHz for downlink and 14.0 GHz and 17.8 GHz for uplink. This is a medium wave transmission that has good penetrating power. It can carry large data due to the large bandwidth per channel. *Ka Band* operates between 18.0 GHz and 31 GHz millimetric wavelengths. This band is subject to fading due to atmospheric conditions. It requires powerful transceivers for reliable communications.

Satellite-terrestrial integrated networks (STIN) integrates terrestrial network systems, aerial platforms, and satellites in massive machine type communications (MTCs). STIN architecture uses licensed and unlicensed bands to interconnect

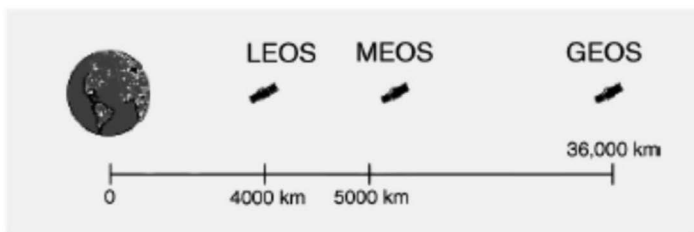


FIGURE 2.34 Comparison of satellite orbits (Source: C. W. Wendt ⁽¹²⁾).

terminal devices such as Bluetooth, Wi-Fi, 5G, a backbone network. STIN consists of perception, cognition, and intelligence layers with self-organizing capabilities. The use of Markov decision algorithms, reinforcement learning, multi-agent deep reinforcement learning algorithms are some of the techniques being investigated.

Futuristically, LEO satellite constellation together with 6G networks can provide global coverage, particularly for underserved and isolated areas. To provide seamless global coverage and connectivity, LEO satellites serve as relays and have inter- and intra-plane connections via intersatellite links. AI devices together with cloud and edge computing address many network problems. Federated learning in LEO communication systems is investigated for AI parameter exchange rather than data exchanges between the devices^(13–15).

2.19.9 DEEP SPACE NETWORKS

Deep space network (DSN) is a worldwide communication network configured for spacecraft missions and ground facilities. Ground facilities are in the United States, Spain, and Australia. It also facilitates radio and radar communications in solar system and universe explorations. Existing ground facilities shielded by shape of terrain from radio interferences. They are located strategically such that at least one ground station is in sight with spacecraft aligned with the rotation of Earth. Continuous data delivery and transmission are realized by steerable parabolic reflector antennas.

DSN, a project of NASA, allows to acquire data, transmit commands, upload software, track position and velocity of spacecraft, and gather scientific data from the space observations. DSN collaborates with other country's deep space facilities. Various independent DSNs are run by other countries, Russia, China, India, Japan, and Europe, mainly for safety and to reduce the risks associated with space explorations.

2.19.10 WIDE AREA NETWORKS

WANs are telecommunication networks that spread over a large geographical area (Figure 2.35), several countries, or even around the world. They are mostly leased communication systems and private networks for wide coverage of business operations of large companies. They are made from multiple computer systems making use of LAN networks, and Ethernet. They are often linked across the internet via virtual private networks (VPNs) and IP tunnels. WANs use a diverse range of communications facilities and methods such as cable modems, cellular networks, dial-up internet, fiber optic facilities, Wi-Fi, Wi-Max, digital subscriber lines, and so on.

2.19.11 BLUETOOTH AND BLE

Bluetooth is the most widely used technology in shortrange and personal area communication systems. Bluetooth low energy (BLE) transmission power varies from 0.01 mW (−29 dBm) to 10 mW (+10 dBm). This power transmission is 10–20 times less than conventional Bluetooth. BLE enters deep sleep to save energy.

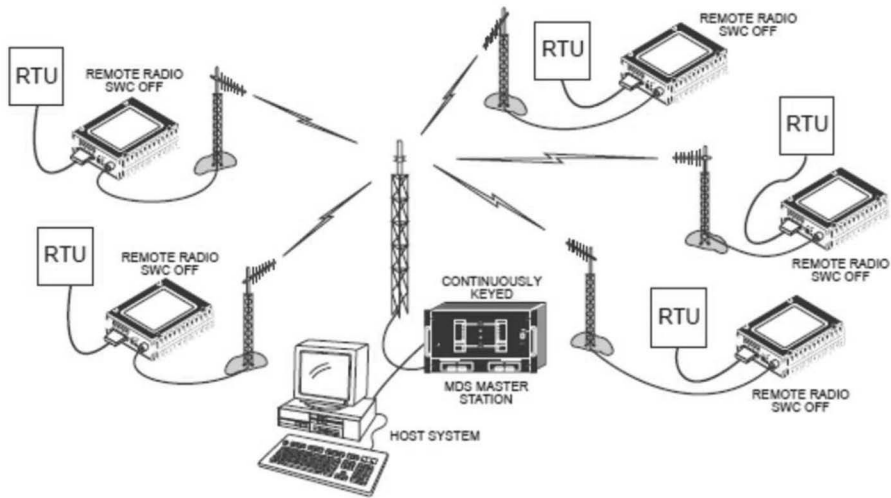


FIGURE 2.35 Typical wide area network (Source: C. W. Wendt (¹²)).

Some of the common appliances that use Bluetooth are cell phones, PCs, computer mice, keyboards, joysticks, cameras, printers, audio handsets, speakers, and musical instruments. It has an *ad-hoc* structure enabling various devices to communicate with one another with a nominal distance of 10 m. This range can be increased to 100 meters by using amplifiers.

Bluetooth operates in the unlicensed 2.4 GHz ISM (Industrial, Scientific, and Medical) band. In most countries, there are 79 channels available for use; however, some countries allow only 23 channels. The nominal bandwidth for each channel is 1 MHz. Bluetooth uses the frequency hopping technique at a hopping rate of 1,600 times per second with a residence time of 625 μ s. Transmissions are performed within the 625 μ s slots with a single packet transmitted over a single slot. For long data transmissions, a particular user may occupy multiple slots using the same transmission frequency, thus slowing the instantaneous hopping rate to below 1,600 hops per second. In an inquiry or page mode it hops 3,200 times per second with a residence time of 312.5 μ s. The frequency hopping scheme of each Bluetooth is determined from a cyclic code of length $10^{27}-1$, and each user has channel symbol rate of 1 Mbps using GFSK modulation.

Bluetooth specification uses TDD and TDMA. All communications between devices take place using master and slave relationship in TDD. The master polls each active slave to see if they have data to transmit. If the slave has data, it must send it immediately soon after it has been polled. The master transmits only in even numbered time slots, while slaves transmit only on odd numbered time slots. In each time slot, a different frequency channel is used for the hopping sequence.

Specification divides Bluetooth protocol stack into three logical groups: transport, middleware, and application. Bluetooth core protocol groups are illustrated in Figure 2.36.

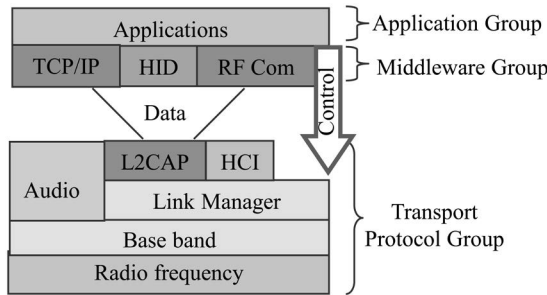


FIGURE 2.36 Bluetooth protocol groups.

Transport group protocols (equivalent to Data link and Physical Layers of OSI model) allow Bluetooth devices to locate each other and manage the physical and logical links with higher layer protocols and applications. The radio, base band, link manager, local link control, and adaptation (L2CAP) layer, and host controller interface (HCI) are included in this group.

Radio layer is concerned with design of transceivers. *Base band* layer defines how Bluetooth devices search for other devices and how they establish connections. The master and slave arrangements, typical of Bluetooth, are made at this level. The RF communication is realized by frequency-hopping sequences and TDD and synchronous or asynchronous transmission determined at this level. Packet transmission and retransmission, packet processing procedures and strategies for error detection and correction, signal scrambling, encryption, are arranged at this layer. The *Link Manager* layer implements link management protocol (LMP), which manages the properties of air interface link between devices. The bandwidth allocation for general data, the bandwidth reservation for audio traffic, authentication, trust relations between devices, encryption of data and internal and RF power usage are under the control of LMP protocol. *L2CAP* layer provides interface between higher level and lower-level transport protocols. Link manager is also responsible for multiplexing of higher-level protocols, packet segmentation and reassembly, and maintenance of services between devices. The HCL enables interoperability and defines a standard interface for upper-level applications to access the lower layers of the stack. All the protocols in the group are required to support communications between Bluetooth devices.

Middleware protocol group includes third-party and industry-standard protocols as well as Bluetooth special interest group (SIG) developed protocols. These protocols allow already available and new applications to operate over Bluetooth links. Industry-standard protocols include point-to-point protocol (PPP) internet protocol, IP; transmission control protocol (TCP) wireless application protocol, WAP; and object exchange (OBEX), protocols adapted from Infrared Data Association (IrDA).

Application group depends on the application in hand. Bluetooth SIG developed protocols include a serial port emulator (RFCOMM) that enables legacy applications to operate seamlessly over Bluetooth transport protocols. Other protocols are the

service discovery protocol (SDP) allows devices to obtain information about each other's available services and the packet-based *telephony control signaling* protocol (TCS) manages telephony operations.

Different countries have allocated various channels for Bluetooth operations. In the United States and most of European countries, the FHSS 2.4 GHz ISM band is available. The wireless PAN based on Bluetooth specifications is now within the IEEE 802.15 standards. The IEEE 802.15 aims to interconnect pocket PCs, personal digital assistance, PDAs, cellphones, and other devices.

2.19.11.1 Networking Bluetooth

Networking Bluetooth devices is relatively easy for the user as they can group themselves for the purpose of communications. Bluetooth devices can be interconnected to form small scale or large-scale networks. In small scale applications, a typical Bluetooth wireless personal area network (BT-WPAN) consists of predefined *piconet* structures. Each piconet is a cluster of up to eight Bluetooth devices. One of the devices in the cluster is designated as the master, and the others assume the slave position. Two piconets can be connected through a common Bluetooth device. This common device serves as a gateway or bridge.

2.19.11.2 Piconets

Piconet may be defined as an ad-hoc and spontaneous clustering of Bluetooth devices. The formation a piconet is as follows: when two Bluetooth devices enter the communication range of each other, they communicate to determine the master-slave hierarchy, which is formed automatically. One of the devices assumes the master position, the device selects the frequency, the frequency hopping sequence, the timing of hops, and the polling order for the slaves. The piconet then communicate within the assigned parameters⁽¹⁾.

Any Bluetooth device can function as a master, or a slave, or a bridge on temporary basis. In a piconet cluster, a maximum of seven devices, can be active slaves at any given time. If there are more than seven devices, the rest of the slaves cannot be active, and they are said to be *parked*. The maximum number of parked slaves can be 255 per piconet. Each parked slave is assigned with a dedicated direct parked slave addresses as defined by the SIG. However, indirect addressing of parked slaves by their specific Bluetooth addresses is also permitted thus allowing any number of parked slaves within the range.

A Bluetooth device can assume many other tasks such as: standby, inquiry, page, connected, hold, park, or sniff. A *standby* device is powered on but has not yet joined piconet cluster. In *inquiry*, the device is seeking if there are other devices that it can collaborate with to form a network. In *paging*, the device is already in a master position and actively seeking other potential devices prompting them to join in the piconet. A *connected* device is accepted in the piconet, and all the necessary communication links have been established. A slave assumes *transmit* position when sending data to the master. At the end of transmission, it returns to *connected* state until it has another data to send, or it has been requested by the master to respond. The *sniff* state is a low-power consumption state in which the slave sleeps for a predetermined time. In the *hold* state slave is not active for a predetermined time and it

does not communicate until the time lapses. In the *park* state, the slave has no data to receive or transmit. In the parked state, the master advises the slave to relinquish its active address and device will be inactive. The active relinquished address can then be used for other devices that may require parking.

2.19.11.3 Bluetooth Scatternet

Bluetooth scatternet is an ad-hoc network consisting of two or more piconets connected by bridge nodes. Piconets coexist in the same area by using different hopping sequences. The construction of ad-hoc networks is generally complex, as a result, several models for constructing Bluetooth scatternet have been developed. Most prominent of these models are: BlueTrees, TSFs, and Bluestars.

2.19.11.4 BlueTrees

BlueTrees protocol is based on the formation of Tree network topology simply called *trees*, shown in Figure 2.37. There are many different types of BlueTree protocols. In One type, in initial formation of a tree, a node is designated as the *root* of the tree to be formed onto.

Once the root is decided, this node acquires its direct, one hop neighbors as slaves. Each slave then pages its one-hop neighbors that are still unconnected and attempts to acquire them as slaves. This process continues until the entire devices in the vicinity are acquired as masters and slave arrangement and a tree is built. The tree is then optimized by limiting each master to have a maximum of seven slaves. In another type of BlueTree protocol, several initial roots are formed and each root acts independently in the way that explained above. Once several trees are formed, it then trees merge form a single tree.

Formation and maintenance of BlueTrees require considerable overheads. For the information to flow from bottom of the tree to the top or vice-versa, a considerable amount of routing process and timing may be required. There are several modifications of BlueTrees, including the ones that are based on determining positions using GPS information⁽¹⁾.

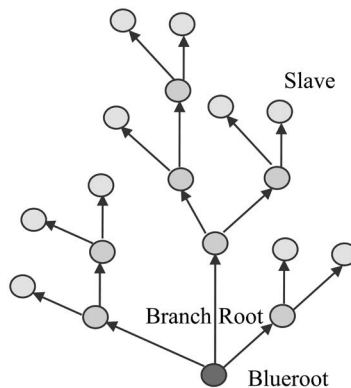


FIGURE 2.37 An example of rooted BlueTree.

power consumption, and long-range communication capabilities. ZigBee is mainly used in industrial control systems, embedded sensors, and healthcare applications. These applications benefit from energy saving sleep modes. ZigBee is promoted by ZigBee Alliance, an international community supported by more than 100 companies (Motorola, Mitsubishi, Phillips, Samsung, Honeywell, Siemens, etc.).

There are three types of devices playing different roles in the network, these devices are ZigBee coordinators, ZigBee routers, and ZigBee end devices (ZEDs). The *ZigBee coordinator* (ZC) oversees the overall control of the network. There must be one coordinator per network. *ZigBee router* (ZR) connects separate devices in the network topology, and it also provides user application layer. ZED transmit information to coordinators or routers.

There are three main topologies for ZigBee networks, star topology, mesh topology, and tree topology. In *star topology* (shown in Figure 2.39) communication is established between a central device and the other ZigBee devices. This central device is called a PAN coordinator, and it is unique in the network. *Mesh topology* is like star topology. The difference between the two is that any device can connect with another device, not just only with the PAN coordinator. *Cluster tree topology* connects ZigBee compliant full-function devices (FFDs) and reduced-function devices (RFDs) to the network. One of the FFD is established as the PAN coordinator, the others can start communicating data.

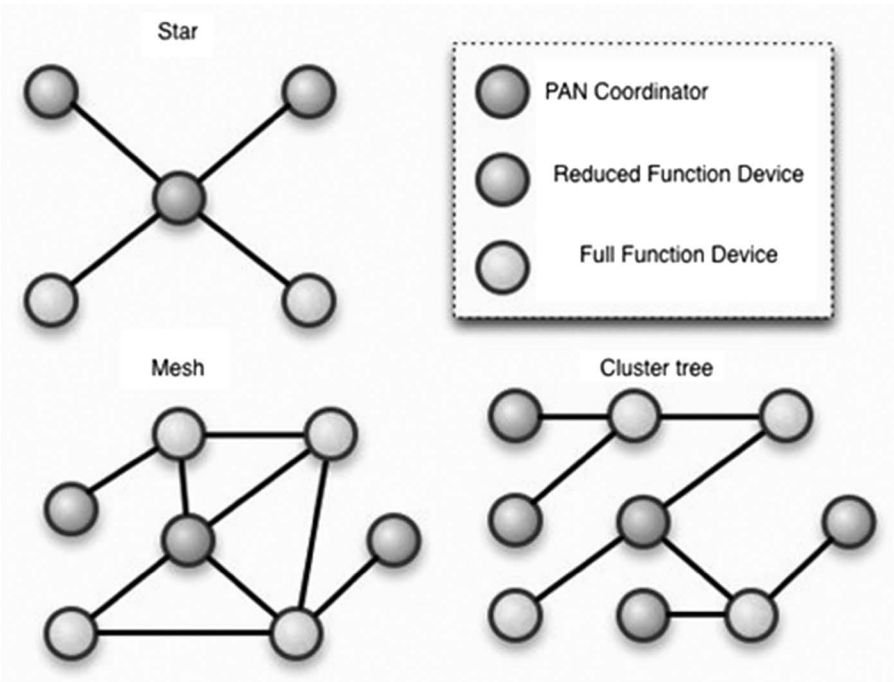


FIGURE 2.39 ZigBee topologies (Source: E. Dorrnoro et al. (17)).

2.21 THE WIFI

Wi-Fi networks connect multiple devices via wireless routers. Wi-Fi complies with IEEE 802.11 standards and there are progressive versions of it developed over a few decades. Recently, the IEEE 802.11be has been amended to cover Wi-Fi 7 in the 2.4 GHz, 5 GHz, and 6 GHz frequency bands. The Wi-Fi 7 has 4,096 constellation points operating on QAM scheme. The Wi-Fi 6 has 1,024 constellation points operating on QAM.

Wi-Fi 6 is based on the IEEE 802.11ax amendment intending to improve the spectral efficiency. This is achieved by many methods, including orthogonal frequency division multiple access (OFDMA) particularly in MIMO applications^(18,19).

2.22 WIRELESS MACHINE TYPE COMMUNICATION NETWORKS

MTC devices include home appliances, sensors, wearable devices, vehicles, low-altitude AUVs. New wireless technologies are emerging to serve these devices, such as Bluetooth 5, Wi-Fi HaLow, NB-IOT and WUR. *Bluetooth 5* operates in 2.4 GHz ISM band with improved channel selection algorithms by employing pseudo random hopping sequences. Communication range extends to several hundred meters though the maximum transmit power is the same as conventional Bluetooth. It can support 32,000 devices in Mesh network mode. *Wi-Fi HaLow* operates at sub-1 GHz ISM band with an extended transmission range of 1 km. It has mandatory 1 MHz and 2 MHz, and optional 4 MHz, 8 MHz and 16 MHz channel bandwidths. *NB-IoT* is designed to handle a massive number of devices distributed in a wide coverage area. It opts for small and infrequent packet transmission. *Wake-up radio* (WUR) operates at low power consumptions complying IEEE 802.11ba standards. It is designed for low latency data reception in fixed, portable, and mobile devices. Other emerging technologies are LTE-M, vehicle-based and unmanned aerial vehicle networks, LoRa, and so on^(20–23).

2.23 IEEE-802.11 WIRELESS LAN STANDARDS

IEEE-802.11 and 802.15.4 have been established for low-rate wireless PANs. These standards are extensively applied in wireless sensors, RFID, Bluetooth, ZigBee, and Wi-fi therefore they will be revisited briefly.

In IEEE-802.11 Wireless LAN standards, frequency hopping and direct sequence approaches are used. The DSSS IEEE-802.11b standard has been named *Wi-Fi* by the *wireless ethernet capability alliance* (WECA), a group that promotes adoption of 802.11b DSSS WLAN equipment and interoperability between vendors. IEEE-802.11g is developing *complimentary code keying orthogonal frequency division multiplexing* (CCK-OFDM) standards in both the 2.4 GHz (802.11b) and 5 GHz (802.1a) bands and supports roaming capabilities and dual band use for public WLAN networks.

Figure 2.40 illustrates WLAN channels specified in the IEEE-802.11b standard at the 2.400–2.4835 GHz band. 802.11b and 802.11ds operate DSSS, DPQSK/DBPSK modulation schemes with a channel bandwidth of 22 MHz. 802.11fh

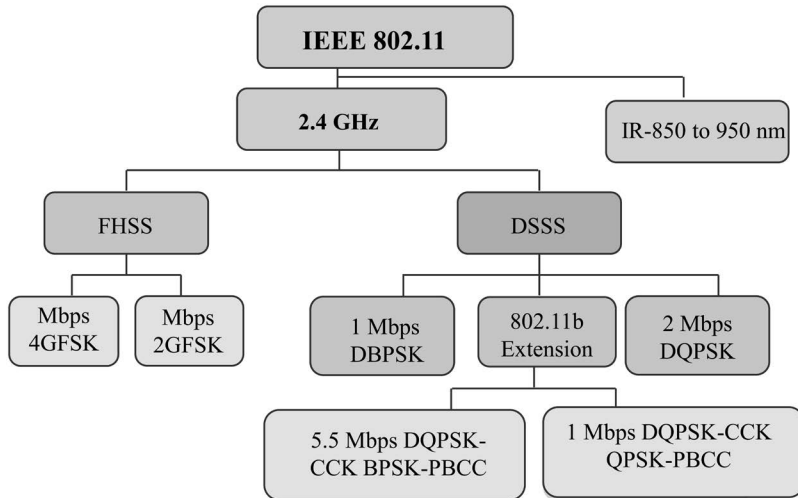


FIGURE 2.40 IEEE 802.11 wireless LAN standards.

operates DPQSK/DBPSK modulation schemes with a channel bandwidth of 1 MHz. WLANs 802.11g operates DSSS, DPQSK/DBPSK, OFDM, CCK, QAM/QPSK/BPSK modulation schemes with a channel bandwidth of 16.25–22 MHz operating on any of the specified channels are assigned to a particular channel by the network operator during the installation. Even though WLAN networks are designed to work in an interference-rich environment by specific installation of access points can provide significant improvement in end-user data throughputs. The channelization scheme used by the network operator becomes very important for a high-density WLAN network for avoiding interference and maintaining good performance^(23,24).

There is a range of software available for the deployment of WLAN. With the use of relevant software, a WLAN can be set up quickly. Figure 2.41 shows a 2.4 GHz channelization scheme for IEEE 802.11.

IEEE 802.11ah is a wireless communications PHY and MAC layer standard that operates in unlicensed bands. It combines CSMA.CA and TMA into a dynamic adaptable MAC scheduler. 802.11ah application categories include smart city, smart power, hot spot, monitoring, industry and backhaul. Other standards such as 802.15.4g is used in smart utility, smart city, smart building/home, monitoring, and industry, 802.15.4w is used in smart city, monitoring and manufacturing. LoRa is a proprietary radio communication method based on spread spectrum modulation technique. LoRa finds applications in agriculture, smart cities, industrial IoT, smart homes and buildings. SigFox is a wireless network for connecting low power objects such as water meters to smart wearables. It finds applications in retail, smart city projects, equipment monitoring, and industrial operations.

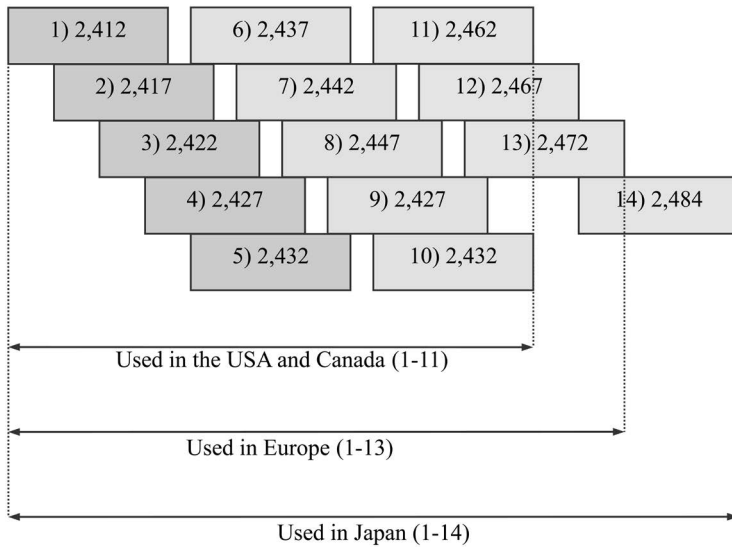


FIGURE 2.41 2,400 MHz channelization scheme for IEEE 802.11.

2.24 THE HIPERLAN STANDARDS

The *high-performance radio local area network* (HIPERLAN) standard was developed in Europe to provide similar capability as the IEEE 802.11 standards. There are two specifications: HiperLAN/1 and HiperLAN/2. The European Telecommunications Standards Institute (ETSI) has adopted both.

HiperLAN/1 provides communications at up to 20 Mbps in the 5-GHz range of the RF spectrum. **HiperLAN/2** operates at up to 54 Mbps in the same RF band. HiperLAN/2 is compatible with WLAN systems for sending and receiving data, images, and voice communications. HiperLAN/2 has the potential for implementation worldwide in conjunction with similar systems in the 5-GHz RF band.

2.25 AI IN WIRELESS SYSTEMS

AI are used in three levels in wireless systems:

1. Physical layer design and implementation,
2. Intelligent and cognitive radios, and
3. Intelligent wireless networks and spectrum sharing.

2.25.1 AI IN WIRELESS COMMUNICATIONS PHYSICAL LAYERS

As explained early in this chapter, in communication systems, the transceiver consists of many individual blocks to fulfill different functions, such as amplification, filtering, signal generation, modulation, coding, channel estimation. AI is applied in

physical layers on one or more blocks to enhance the performance of transceivers. AI is also applied transmitters and receivers achieve a joint optimization. This is called a *DL-based end-to-end transceiver*.

In physical layers implementation, the AI techniques enable automatically learn to make decisions for self-configuration, self-optimization, self-organization, and self-healing, power control, spectrum access, and routing management. Decision-making is an important cognitive task that enables agents to intelligently reason, plan, and choose the most suitable decisions to meet the high-quality service requirements. The decision-making is achieved by using AI techniques such as MDP, game theory, RL, and optimization theory.

AI is particularly effective on physical design of multiple-input multiple-output communication systems. Deep neural network (DNN) implementations have been shown to be effective in reducing model mismatches. DNN uses multiple models for model free communications capabilities. It is also effective in managing non-linearities, A/D and D/A operation shortfalls, PA saturations, and antenna modulations^(21,25–27).

2.25.2 AI INTELLIGENT AND COGNITIVE RADIO

Cognitive radio (CR) contains intelligent elements that mitigate spectrum sparsity by improving spectrum utilization. CR techniques attracted interest by the new regulation of the FCC allowing the usage of licensed frequency bands by unlicensed users. Unlicensed users radios sense if a particular spectrum is occupied by a licensed user to make decisions whether to access the spectrum for data transmission^(28–32).

Cognitive Radio CR (shown in Figure 2.42) is an opportunistic communication method to use an idle spectrum. Spectrum sensing is realized by cognitive information from the authorized spectrum occupancy by the primary user in specific geographic area. Modulation recognition (MR) and frequency hopping recognition (FHR) are the main methods in CR signal detection and classification^(34–37).

Cognitive radio uses AI techniques to perceive the spectrum occupancy by learning and adapting to the changes. Learning and understanding also enables adjustments of the internal communication mechanisms. It changes the operation

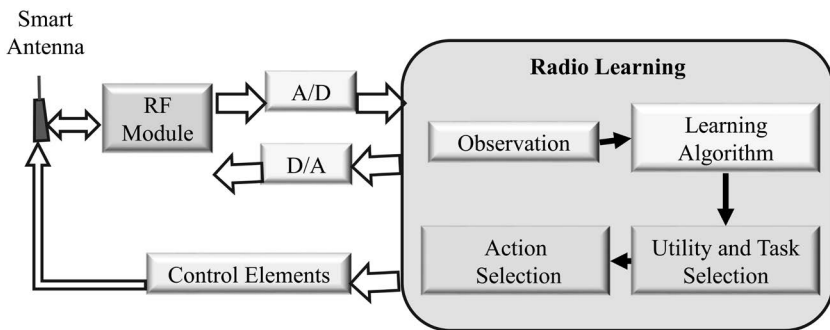


FIGURE 2.42 Block diagram of a cognitive radio.

parameters, power, carrier modulation, and coding of the unlicensed user's system on a real-time basis. Dynamic spectrum sharing (DSS) improves the efficiency of spectrum utilization as it allocates spectrum resources for different modes of communication technologies in the same frequency band dynamically and efficiently.

In advanced CR communication systems, such as the 5G networks, Blockchain technology is used. Blockchain makes use of cryptography principle and consensus mechanism to ensure fairness, credibility, construction, and sharing of networks. Blockchain authentication algorithms prevent malicious nodes from accessing spectrum which are not complying with authentication processes. Integrated AI with Blockchain improves performance by learning more abstract concepts^(28–30,33,38,39).

2.26 AI IN WIRELESS RADIO NETWORKS

AI harnesses potentials of supervised and unsupervised learning methods for effective network layouts and improved services. AI yields faster and more accurate optimization in radio access networks (RANs) and open radio access networks (O-RANs). AI helps wireless systems continuously learn to update the related parameters. AI is also applied in CR, edge computing (EC), channel measurement (CM), and end-to-end encoders/decoders (EEDs)^(40,41,42).

Wide usage communication networks and internet of everything (IoE) introduces challenges in low latency transmission, efficient computing power, higher throughput, spectrum sensing, and spectrum efficiency. Intelligent wireless communication network with self-organization, self-configuration and self-recovery functions addresses the needs of most of those challenges. AI in communication systems learns the hidden rules from the big data, and then uses these rules to make the necessary prediction or decision. There are three main application scenarios: 1) enhanced mobile broadband (eMBB), 2) ultra reliable low latency communication (URLLC), and 3) mass machine type communication (MMTC).

In network applications, it is necessary to reduce data dimension of the raw data, to filter abnormalities, and to achieve a more reasonable dataset. AI-based data mining, such as PCA and ISOMAP, are common AI algorithms to help networks to transform higher-dimensional data into a lower-dimensional subspace. This significantly decreases computing time, storage space, and model complexity. For example, the collected channel information, traffic flows, images and videos from sensors, devices or social media are high-dimensional data, which can be compressed into a small set of useful variables of the raw data.

As discussed earlier in this book, all wireless communication systems encounter problems such as path loss, multipath delays, power fading, Doppler spread, scattering, diffraction, shadow, and fading. AI helps to overcome some of these problems well; particularly in channel estimation, channel prediction, and channel state information (CSI) feedback compared to traditional methods.

AI is used in various stages of wireless networks. Wireless networks need to be managed efficiently to improve resource allocation, spectral efficiency, and enhance user quality of service (QoS). For example, load balancing relies on proactive wireless resource allocations for efficiency transmitted power, selection of frequency, and adjusting the direction of base stations depending on the user demand. Resources

include signal transmission power, time slots, frequency bands, and spatial beams for systems with multiple antennas. AI methods, such as game theory and reinforcement learning (RL), play important roles in spectrum management^(40,42–44).

Specific AI applications we will discuss in this book are spectrum sharing, modulation recognition, channel estimation, and channel measurements.

2.26.1 INTELLIGENT SPECTRUM SHARING

Intelligent spectrum sensing uses AI for reasoning, availability, and usage of a spectrum other than primary users. This method seeks unoccupied spectrums (spectrum holes) that are not used by primary users. Spectrum hole changes over time, frequency, and location, therefore a certain level of intelligence is required. Other indicators of spectrum holes are energy levels, statistical characteristics, cyclostationary properties, and covariance primary signals. Synthetic data is generated to fulfill intelligence requirements. There are various methods of synthetic data generation by generative adversarial networking (GAN), cooperative spectrum sensing, compressive spectrum sensing, deep reinforcement learning, and neural networks^(2,4).

2.26.1.1 Synthetic Data Gathering with GAN

Synthetic Data Generation with GAN is used as a classifier when a set of large and labeled data is available. GANs are suitable to generate synthetic data to train the classifier together with the real data. Once the classifier is trained, GANs can perform real-time spectrum sensing and data adaptation for the new spectrum environment^(34,45,46).

2.26.1.2 Intelligent Cooperative Spectrum Sensing

Supervised and unsupervised learning classification techniques are used for spectrum sensing. In cooperative spectrum sensing, each secondary user reports its sensed average energy of a frequency band to the fusion center or to a base station or to a secondary user. Based on the energy vectors from all involved secondary users the base station decides to use the available spectrum.

Intelligent Cooperative Spectrum Sensing classifiers are designed by using convolution neural networks (CNNs), K-means clustering, Gaussian mixture model (GMM), Bayesian learning, support vector machine (SVM), or weighted K-nearest-neighbor (KNN). K-means clustering and GMM are unsupervised ML, while SVM and KNN are supervised ML. Giving few examples, CNN-based cooperative sensing improves sensing accuracy with low complexity, the combination of K-means clustering and SVM achieves real-time sensing by training low-dimensional input samples, and Bayesian learning addresses large-scale heterogeneous sensing problems by tackling heterogeneous data fusion^(34,45,47,48).

2.26.1.3 Intelligent Compressive Spectrum Sensing

Intelligent Compressive Spectrum Sensing is based on accurate sensing of sparsity level of blocks in the spectrum. The sparsity level is an indication of spectrum occupancy which is usually unknown before performing sensing and it varies with time. The sparsity level at each block is sensed based on primary user's activities statistics,

secondary user's neighbors, previous sparsity levels, and current spectrum measurements, and number of occupied bands^(34,45).

2.26.1.4 Deep Reinforcement Learning Spectrum Sensing

Deep reinforcement learning spectrum sensing has been investigated on the device-to-device (D2D) communications. The idea is that if the users in a cellular network are close enough, they can communicate directly without going through the base station as in the case of vehicular networks.

2.26.1.5 Neural Network Spectrum Sensing

Neural networks are implemented by various algorithms (i.e., Levenberg Marquardt algorithm). The NN architecture based on spectrum includes three layers: spectrum sensing layer, data processing layer, and situation analysis layer.

Unused spectrums are utilized by secondary users opening possibilities of unwanted and opportunistic entries into the system. AI proves to be effective needed to filter out the legitimate secondary entries from the intruders. The spectrum sensing data falsification (SSDF) attacks and eavesdropping opportunistic attacks are intensely researched areas^(34,45,49–51).

2.27 INTELLIGENT MODULATION RECOGNITION

Modulation recognition (MR) dynamically selects the modulation mode from the optional modulation types suitable for a transmission requirement. Modulation recognition is particularly effective if the receiver is uncertain about the modulation mode of the incoming signal. MR proved to be useful in software defined communication systems as technology moves toward universal systems.

MR can be divided into 1) feature-based and 2) likelihood ratio-based recognitions. In *feature-based* recognition, AI is used as a classifier with input as a signal feature and output as a modulation mode. In *likelihood ratio-based* systems, deep learning is preferred for finding high-dimensional mapping relationship from input to output data^(34,52–54).

2.28 INTELLIGENT CHANNEL ESTIMATION

Channel estimation is the process of using prior information to set the channel state from the received data. There are various channel estimation algorithms such as pilot frequency-based channel estimation, blind channel estimation, and semi-blind channel estimation. These algorithms include the least square estimation methods, minimum mean square error (MMSE), linear minimum mean square error (LMMSE), and singular value decomposition (SVD). There are other AI methods investigated, such as the Rayleigh fading channel estimator based on NN.

2.29 INTELLIGENT CHANNEL MEASUREMENTS

Intelligent channel measurements (ICMs) algorithms are designed to minimize the estimation error. Channel measurements are done by deep learning algorithms for channel estimation, channel prediction, and CSI. The channel prediction helps to

improve the performance at the transmitter or receiver, adaptive encoding and modulation, decoding processing, channel equalization, and antenna beam forming. CSI feedback makes the communication system adapt to the current channel conditions, thus providing a guarantee for reliability and high-speed communications^(34,55).

2.30 INTELLIGENT FREQUENCY HOPPING RECOGNITION

FHR enhances the communication systems by providing strong anti-fading and jamming ability, confidentiality, low probability of interception, and good networking ability. Therefore, frequency hopping is widely used in military and civilian communication systems. Despite these advantages it can introduce problems for non-cooperative receptions in military communications, radios, and reconnaissance applications.

The frequency hopping MR system based on CNN addresses the difficulty of feature extraction. There are numerous in-depth learning types applied in FHR. Also, there are various varieties of CNN such as LeNet, AlexNet, and ResNet^(33,34,52,56).

2.31 AI 5G/6G NETWORKS

Due to shortage in the available bands below 6 GHz, the necessity to use millimeter-wave (mmW) in the bands of 26 GHz, 28 GHz, and 37–40 GHz for 5G wireless communications is becoming inevitable. 5G are designed to achieve high speed, high efficiency, low latency, and low power consumption for mass communications, massive IoT, M2M, tactile internet and robotics. There is intensive research and design centering on mW development and applications at chip level, complex network level, and network management level.

5G networks are expected to accomplish a transition from network entities into network functions that are configured according to the target applications. This requires effective resource orchestration, network planning, and firm network management. For these requirements AI is investigated using genetic algorithm (GA). SVM, long short-term memory (LSTM) and recurrent neural network (RNN). Network slicing provides applications of different algorithms at different stages of the network operations.

Graph neural networks (GNNs) have been investigated in 5G and 6G network applications. GNNs conduct graph convolution for structured data to obtain the weighted sum over the vertices according to its graph structure. GNNs consist of multiple layers which can be divided into two parts aggregator and updater. An aggregator collects the data as the weighted sum of the neighbor vertices. The updater is a fully connected layer that maps and aggregates the features of the next layer with the activation functions like neural network.

Since 5G wireless networks are expected to support various communications connectivity and services for massive number of wireless smart intelligent devices and machines. AI assists in intelligent learning, autonomously optimizes the networks and makes intelligent decisions with minimum human intervention with optimum performance. AI observes the network environment, analyzes data, and makes decisions. In supervised learning, SVM and KNN are used for classification. After the

classification, regression analysis aims to estimate or predict continuous quantities by using techniques such as support vector regression (SVR) and Gaussian process regression (GPR). In unsupervised learning, dimension reduction techniques such as principal component analysis PCA and isometric mapping (IOMAP). Markov decision process (MDP), Q-learning, actor critic (AC), and deep reinforcement learning methods are used for decision-making. The new trend is exploring applications of multi-armed bandits (MAB) as new signal processing tools. They are useful in resource allocation problems carefully proportioning the resources in channel conditions or some other environmental parameters.

Artificial intelligence has been a paradigm for the design and optimization of 6G networks as in the case of 5G networks. AI in 6G networks is investigated for intelligent knowledge discovery, smart resource management, automatic network adjustment, and intelligent service provisioning. 6G networks are expected to support high-quality services compared to 5G networks for the new emerging applications, virtual and augmented reality, remote surgery, holographic projection, and many other areas with an unlimited connectivity for the massive number of users.

Supported by AI technology in intelligent service provisioning, the 6G networks are expected to meet ultrahigh data rates, a peak data rate of near 1 Tb/s, ultralow latency, low end-to-end delay, high reliability, high energy efficiency, high mobility, seamless connectivity, bulk connection, large frequency bands in the region of 1 THz–3 THz.

Intelligent service provisioning in 6G networks with a high level of intelligence consists of four layers: sensing layer, data mining and analytics layer, control layer, and application layer. This architecture is capable of intelligently extracting valuable information from massive data, learning, supporting different functions for self-configuration, self-optimization, and self-healing.^(56–59)

2.32 HANDLING BIG DATA IN WIRELESS NETWORKS

The new generation wireless networks aim to provide quality communication services that rely on high throughput, massive connectivity, and low delays. Current wireless networks allow for conventional high-speed cellular communication to coexist with machine-to-machine services and internet of things (IoTs), with an emphasize on coverage, advanced, and dense connectivity allowing more than one million sensor connections within a square kilometer area. Just two services, the IoT networks and health monitoring generate massive amounts of sensing data. In addition, autonomous connected vehicles of the near future will need to support millions of high velocity devices that will significantly increase the data traffic^(60,61,63,65,66).

The communication-aware AI techniques reduce the network requirements, while maintaining a high performance. To reduce network requirements for massive data handling methods like model splitting, model downloading, and distributed/federated learning are deployed.

2.32.1 MODEL SPLITTING

Model splitting relies on the execution of trained AI model being divided into two parts: fraction of the model is executed at the user end (UE), while the remaining

part runs at a network endpoint to communicate the result to the user end. Depending on the splitting point, the uplink traffic is limited compared to the user end transmitting the entire local data to the network^(26,60–62,65).

2.32.2 MODEL DOWNLOADING

Model downloading is used when user end requests a fresh model from the network endpoint to execute a new task and/or operate under new conditions. For this, a large collection of AI models is available at the network end point to cater multiplicity of tasks thus resulting in accuracy and alleviation of memory requirements of the user ends^(64,65).

2.32.3 FEDERATED/DISTRIBUTED LEARNING

Federated/Distributed learning enables user end to receive a global model from the network endpoint. After training with local data, the user end transmits their updated models back to the network endpoint, which then updates the global model by federated averaging and sharing it with the user end itself and with other user ends^(60,65–67).

2.32.4 BIG DATA-AWARE INTELLIGENT WIRELESS NETWORKS

Big data-aware intelligent wireless networks provide efficient operation by sensing the big data characteristics. The ability to acquire, analyze, and exploit traffic characteristics can be accomplished by learning units (LUs) at both base stations and in central processing units (CPUs). Common algorithms for wireless traffic analysis are stochastic modeling, data mining, machine learning, and large-scale data analytics.

Examples of stochastic model are K-Markow model, hidden Markow model, time series, and so on. Data mining focuses on implicit structures in data sets. It uses prediction methods and clustering. Machine learning establishes a functional relationship between the input data and output action thus providing auto-processing capability for unseen patterns of data inputs. It employs classification and regression with techniques like reinforcement learning, Q-learning, Active learning, and deep learning^(60,65–70).

2.33 CONCLUSIONS

This chapter summarizes modern wireless communication systems and wireless networks. The principles of modern communication systems are explained. It is pointed out that when the electromagnetic waves propagate in space, they encounter losses fading, reflection, refraction, and attenuation. The components necessary for RF communications have been introduced and the fundamentals of digital communications technology were discussed in detail. Modern communication methods, modulation and multiplexing techniques, frequency spreading, multiple access techniques were detailed. Applications of artificial intelligence at various stages of communication systems and networks have been explained.

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3 Instrument and Sensor Networks

3.1 INTRODUCTION

Instruments are essential in industrial operations, environmental applications, research and development, transportation, military, border security, health and medicine, space explorations, and in our daily lives. *Instrumentation* is a collection of instruments networked to communicate directly with each other or via some intermediate devices such as computers and microprocessors. The wireless machine-to-machine (M2M) communication of instruments is well established.

If the behavior of a physical variable is known, its performance can be monitored and assessed by means of suitable methods of sensing the variable, conditioning, and displaying the signals. Applications of instruments range from laboratories, mobile telephones, to arduous environments such as inside nuclear reactors or remote locations such as satellite systems, deep space, and planets. Majority of instruments have an extensive degree flexibility in their range of use, types of displays, and methods of communications with other devices hence they have a high degree of complexity.

All instruments can be classified to be analog, digital, combination of the two, and virtual instruments. Nowadays, most instruments are digital because of the advantages that they offer. However, the front end of many instruments is still analog; that is, most sensors and transducers generate analog signals. Initially, the signals of such instruments are conditioned by analog circuits before they are converted into digital forms for further processing.

Digital systems are based on digital processors, which provide signal processing and data handling which can simply be achieved by software and hardware implementations. Another important advantage of digital systems is that data generated by instruments can be managed by computers and other digital platforms. This provides a wide range of possibilities for complex data processing, ease in communications and data transfer, multiplexing, data storage, visual displaying, efficient calibrations, enabling broadcasting of information, application of internet of things (IoTs), data mining, and artificial intelligence (AI).

AI and wireless communications are applied expensively, ranging from industrial operations and military to file exchanges and mass communications of people. This book explained how and where the AI and wireless networks applied in measurements and instrumentation.

3.2 FUNDAMENTALS OF MEASUREMENTS

Instruments are man-made devices that are designed to maintain prescribed relationships between the parameters being measured and the physical variables. The physical parameter under investigation is known as the *measurand*. Sensors and

transducers are the most basic and primary elements responding to physical variations. The relation between sensor signal and physical variations can be expressed in the form of *transfer functions*. The transfer function may be linear, non-linear, probabilistic, or stochastic. The non-linear transfer functions, for example, can be logarithmic exponential or in other forms. The signal processing of each transfer function may require different mathematical and signal processing approaches some of which will be explained in this book.

In modern instruments, the original data acquired from the physical variables are usually in the analog form. Analog signals are converted to digital counterparts before they are processed further. For conversions, analog-to-digital (A/D) converters are used together with appropriate sample-and-hold devices. In addition, analog multiplexers enable the connection of multiple sensors and transducers to the same signal-processing media. The typical components of a digital instrument are illustrated in Figure 3.1. The digital systems are particularly useful in performing mathematical operations, applying AI, storing, and transmitting data.

A digital instrument consists of three main subsystems – the general-purpose microprocessor or computer core with appropriate display, mass storage, communication capabilities, application specific input/output signal conditioning hardware and supporting software, and power supply. All the components may not be present in some simpler instruments. However, in all cases, the analog sensor signals are amplified or attenuated, filtered by the signal conditioning circuitry, and converted to their digital representations by (A/D) converter. Digital signals are then processed in accordance with the measurement algorithms and the results are displayed, stored, or send to processing or storage media. Some signals may be sufficiently high quality to be interfaced directly to a digital system as in the case of optical measurements of rotation, and some linear motion measurements.

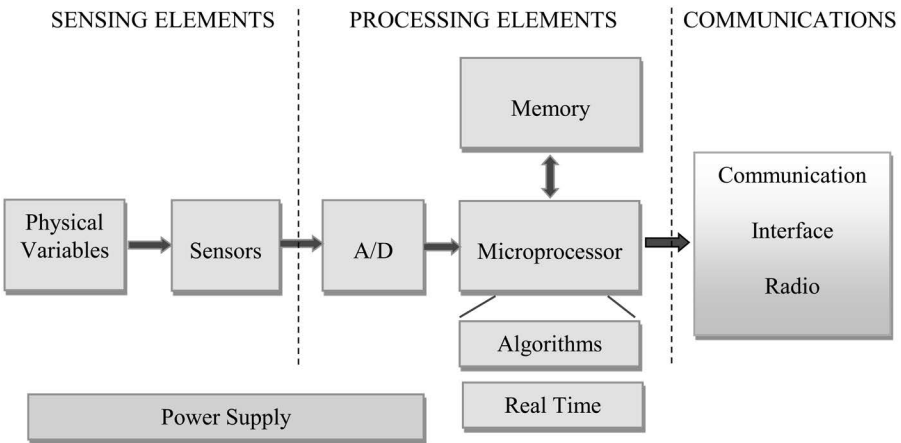


FIGURE 3.1 A general structure of a digital instrument.

A/D conversion involves three stages: sampling, quantization, and encoding. In all digital instruments, the Nyquist sampling theorem must be observed; that is, “the number of samples per second must be at least twice of the highest frequency present in the continuous signal.” As a rule of thumb, depending on the significance of the high frequencies, the sampling must be about five to ten times of the highest frequency component in the signal. The next stage is the quantization, which determines the resolution of the sampled signals. Quantization involves errors that decrease as the number of bits increase. In the encoding stage, the quantized values are converted to binary numbers for processing digitally.

Many modern digital instruments are equipped with effective communication capabilities, which can be achieved in wired or wireless media. In these instruments, integration of fast central processor units (CPUs), vast memories, and advanced wireless communications make them flexible and easy to use, multifunctional, and networkable.

3.3 ERRORS AND UNCERTAINTIES

There are many sources of errors in instruments; therefore, it is important to identify these sources and draw up an error budget. In the error budget, there may be many factors, such as 1) imperfections in electronic components (e.g., high tolerances and noise or offset voltages), 2) drifts, shift in gains, changes in chemical compositions and structure, and aging, 3) external and ambient influences in temperature, pressure, and humidity, 4) improper hardware and software support, and 5) inherent physical fundamental laws such as thermal and electrical noises, Brownian motion in materials, and radiation.

Controlling errors is an essential part of instruments and instrumentation systems. Various techniques are available to achieve this objective. The error control begins in the design stages by choosing the appropriate components, filtering, and bandwidth selection, and by reducing noise. Minimizing the errors generated by the individual subunits of the complete system will result lowering the errors of the instrument. In a good design, the errors of subunits may be compensated adequately by the ensuing subunits.

3.3.1 TYPES OF ERRORS

Errors can broadly be classified as systematic errors, random errors, or gross errors.

3.3.1.1 Systematic Errors

Systematic errors remain constant with repeated measurements. They can be divided into two basic groups: instrument errors and environmental errors. Instrument errors are inherent within the instrument itself, arising because of the mechanical structures, electronic designs, improper adjustments, wrong applications, and so on. Environmental errors are caused by factors such as radiation, temperature, and humidity.

3.3.1.2 Random Errors

Random errors appear as result of noise and interference, backlash, and ambient influences. In applications, random errors may vary by small amounts around a mean value. Therefore, the future value of any individual measurement may not be predicted deterministically. Stochastic approaches are adapted in the analysis and compensation of these errors. This can be realized by using the laws of probability. Depending on the system, random error analysis may be made by applying different probability distribution models such as normal distribution, χ^2 distribution, t distribution, F distribution, etc. But most instrumentation systems obey the normal distribution laws; therefore, the Gaussian model can be applied thus enabling the determination of the mean values, standard deviations, and confidence intervals.

3.3.1.3 Gross Errors

Gross errors are the result of human mistakes, equipment fault, and the like. Human errors may occur in the application stages or during the recording and interpretation of experimental results.

3.3.1.4 Uncertainties

Uncertainties differs from error in that it involves such as human judgmental factors in estimating the possible values of errors. For example, tolerance values of components lead to uncertainties and necessitate some applicable analysis. This is because the absolute error of each component may not be known but the possibility of overall error may be expressed within an error band.

3.4 CALIBRATION OF INSTRUMENTS

Calibration is another point of importance and is essential for checking instrument performances against known standards. This provides consistency in readings and reducing the errors, thus universally validating of the measurements. The calibration procedure involves comparison of the instrument against primary or secondary standards. In some cases, it may be sufficient to calibrate a device against another if the accuracy is known. After calibrations of instruments, the future operations are considered to be error-bound for a given time for similar conditions of operations.

Calibrations may be made under static or dynamic conditions. During calibration, the input is varied in increments in increasing and decreasing directions over a specified range. The observed output then becomes a function of that single input. For better results, this procedure may be repeated by varying the input thus developing a family of relationships between the inputs and outputs. The input/output relationship usually demonstrates statistical characteristics. From these characteristics, appropriate calibration curves can be obtained, and statistical techniques can be applied for analysis.

There are many methods for calibrations including self-calibrations, remote calibrations, opting for specialized calibration experts and agencies, and using AI methods.

3.5 SIGNALS AND TYPES OF SIGNALS

A *Signal* is defined as “any physical quantity that varies in time, space or any other independent variable or variables.” Mathematically, a signal can be described as a function of one or more independent variables. Signals generated by sensors in response to physical variables can be classified in several ways, such as multi-channel and multidimensional, continuous time or discrete, deterministic, or random (stationary, non-stationary), or transient signals. A good understanding and successful treatment of signals determine the quality of the instruments. AI has been applied extensively on signal identifications, filtering, and processing.

3.5.1 TYPES OF SIGNALS

Signals picked up by sensors are classified in numerous ways as will be explained next.

3.5.1.1 Multichannel and Multidimensional Signals

Multichannel and multidimensional signals are functions of one or more independent variables. The independent variable can be real-valued scalar or complex valued quantity.

3.5.1.2 Continuous Signals

Continuous signals are in analog forms. They can be periodic or non-periodic. Periodic signals repeat themselves and can be represented by a sinusoidal, triangular, sawtooth, rectangular waveforms, and so on. Signals can be represented in time domain or in frequency domain.

3.5.1.3 Discrete-Time Signals

Discrete-time signals are defined only at discrete values of time. Discrete values may not be equidistant, but in practice, in many cases for computational convenience, they can be treated equally spaced.

3.5.1.4 Deterministic Signals

Deterministic signals can be expressed by an explicit deterministic mathematical expression, or by some well-defined rules. That is, the past, present, and future values of the signals are known precisely without uncertainty.

3.5.1.5 Random Signals

Random signals evolve *randomly* in time in an unpredictable manner. They cannot be deterministically described to any reasonable degree of accuracy or description becomes too complicated for practical use. There are two broad categories of random signals: 1) the *Stationary random signals* have statistical properties and do not vary in time, and 2) *non-stationary random signals* have statistical properties but vary in time. In these cases, methods, such as time averaging, and other statistical techniques can be employed.

3.5.1.6 Transient Signals

Transient signals are signals of short duration and sudden occurrences. For analysis, statistical methods and Fourier transforms can be used.

3.5.1.7 Single-Ended Signals

Single-ended signal generated by a source that has one of its two output terminals at a constant voltage. Usually, one of the terminals is grounded. A source of a *differential signal* on the other hand has two output terminals whose voltages change simultaneously by the same magnitudes but in opposite directions. The output voltages of bridge circuits are typical examples of differential signals.

3.5.1.8 Broadband Signals

Broadband signals have a large frequency range relative to their central frequency. The value of central frequency is important depending on the application. For example, a signal from 1 Hz to 10 Hz may be considered a broadband in some applications, whereas, say 20 kHz sideband signals around 2 MHz may be regarded as narrowband. *Narrowband signals* have small frequency range relative to its central frequency.

3.5.1.9 Low or High Output Impedance Signals

Low or high output impedance signals determine the requirements of the input impedance of the conditioning circuits to avoid loading effects. This is commonly known as the *impedance matching*.

3.5.1.10 Digital Signals

Digital signals are represented in 1s and 0s, where 1 being a voltage level and 0 being another (e.g., 5 V and 0 V). These signals can be generated digitally or obtained from analog signals by applications of appropriate signal processing and analog-to-digital signal converters (D/A). Since digital signals have only two voltage levels, they are easy to generate, convenient to store, to multiplex, and to transmit. They are relatively immune to noise; error corrections can be carried out easily. Security issues in data handling, encryptions, decryptions can be addressed conveniently. More information can be found in [Chapter 1](#) of this book.

3.6 NOISE AND INTERFERENCE

Noise can be described as the unwanted signals either inherently generated by a device or imposed externally, known as the *interference*. There are many different types inherent and interference noise.

Inherent noise is generated within the device. They can appear as such as thermal noise, shot noise, excess noise, burst noise, partition noise, recombination noise, and spot noise.

Interference signals are generated by external sources. There are several classifications of interference noise depending on it affects on the output signals and how it enters to the electronic circuits. Interference noise can be magnetic, capacitive, or electromagnetic. They can be *additive* or *multiplicative* with respect to the output signals.

Spectral density of a noise signal is defined as the mean-square value of noise per unit bandwidth. Spectral density is independent of the frequency, for example, thermal noise is known to have a flat or uniform distribution therefore is called white noise.

3.6.1 NOISE IN DIGITAL SYSTEMS

Most noise in digital systems are created by the devices themselves. Major noise sources include power supply noise, signal return coupling, crosstalk, inter-symbol interference, and parameter offsets. Lower-level noise sources including alpha particles, thermal noise, shot noise, and flicker noise. Some of the important noise generated due to digital operation is the quantization noise, and the phase jitter noise.

Quantization noise is the result of the difference between the signal and its equivalent quantized value.

Phase jitter noise is due to variations in the phase of the carrier as it passes through a communication network. The variations in the phase of the signal cause a smearing in the zero-crossing of the received signals. As a result, severe jitters may result in pulses moving into time slots allocated for neighboring pulses.

3.7 DIGITAL INSTRUMENTS

Digital systems are based on digital components such as microprocessor, microcontrollers, memory devices, digital inputs/outputs, and digital communication components. The operations of all digital instruments are governed by common mathematical theories, such as the sampling, and digital signal-conditioning.

Microprocessors and microcontrollers constitute the heart of almost all types of modern digital instruments and instrumentation. They play a vital role in data acquisitions, data processing, and data control. The simplified form is shown in [Figure 3.2](#) highlighting the functionality of the digital control devices microprocessors, microcomputers, or computers.

Microprocessors and microcontrollers in instruments fulfill multiple functions, as:

1. *Data handling* includes data acquisition, processing, information extraction, compression, interpretation, recording, storage, and communication,
2. *Instrument and instrumentation control* includes sensors, actuators, system resources, and system control,
3. *Human-machine interface* provides a meaningful and flexible user interface to the operators,



FIGURE 3.2 A simplified block diagram of a digital instrument.

4. *Experimentation and procedural development* consist of commissioning, testing, and general prototyping of the targeted system under investigation,
5. *Application of AI*, includes system check, calibrations, and instrument health, and
6. *Sensors and instrument integration* includes device communications, clustering, and coordination.

Hardware for data processing in instrumentation can be summarized as⁽¹⁾:

- General purpose microprocessors (μ P).
- Microcontroller (μ C),
- Digital signal processor (DSPs), and
- Application specific integrated circuit (ASIC), and
- Field programmable gate array (FPGA).

In selecting an appropriate processor, one needs to consider the followings:

- Ease of use, reuse, upgrading, or modification,
- Execution speed,
- Memory and external supporting circuit requirements,
- Non-recurring engineering (NRE) cost,
- Power consumption,
- Software complexities, and
- Weight and size.

For many smaller instruments, a microcontroller or several microcontrollers will suffice to meet the data processing requirements. Most microcontrollers meet the application requirements as they integrate memory and multiple functions onto a single chip to reduce the parts count and to increase the reliability.

3.7.1 MICROPROCESSORS AND MICROCONTROLLERS

Microprocessors and microcontrollers provide the basic functionality to modern digital instruments. *Microprocessors* are integrated circuits (ICs) that handle and process data. *Microcontrollers* can be defined as dedicated single-chip devices, which have memory for storing and processing information. They can control read/write functions. The performance of a microcontroller is classified by its size, i.e., the number of bits that the CPU can handle at a time. There are many types and families of microcontrollers offered by a diverse range of manufacturers.

Fabrications of microprocessors and microcontroller technologies are complementary metal oxide semiconductor (CMOS) and the post metal programming (PMP). CMOS is the most common method, and it requires less power hence permits longer battery operation. The design architectures can be achieved by several methods such as

- A. *Von-Neumann architecture* – Microcontrollers built upon this architecture has single data bus, which is used to fetch instructions and data from a

common main memory. When the microcontroller addresses the memory two separate fetches occur, i.e., first the instructions are fetched and then the data is fetched.

- B. *Harvard architecture* – There are two separate buses, i.e., data and instructions buses in a microcontroller. Therefore, the fetching occurs in parallel, which is faster than the Von-Neuman but the complexity of the hardware architecture increases.
- C. *Complex instruction set computer (CISC)* – In a typical CISC microcontroller have inbuilt instructions within the structure to perform specific control tasks. These instructions include addressing registers and recognizing addressing modes.
- D. *Reduced instruction set computer (RISC)* – This type of architecture commonly applied and reduces the instructions within the structure. Some of the features of a typical RISC processor are separate instruction sets, separate data buses, ability pipelining of instructions, and execution of orthogonal (symmetrical) instruction sets.

3.7.1.1 Digital Signal Processors

DSPs are dedicated devices for signal processing. They fall in between microcontrollers and microprocessors. DSPs are designed for mathematic intensive and iterative types of calculations, filtering, and image processing. DSPs are useful in signal processing therefore will be explained further in the signal processing hardware section.

3.7.1.2 Field Programmable Gate Arrays

FPGAs are configurable chips that can be programmed to implement digital hardware circuits. They consist of an array of different programmable blocks, logic gates, and IOs. They can be interconnected by using router tracks with programmable switches between them. The configuration of FPGAs is realized by using static random access memory (SRAM) cells. In FPGAs, there are millions of SRAM cells that each can assume user-defined functionality, set by a hardware description language (HDL). High-level languages, such as C or OpenCL, are used to translate to HDL. HDL is compiled by using computer-aided design (CAD) packages for bit-level configuration of SRAM cells. The bit level configuration ability of FPGAs leads to implementation of exact hardware needed in specific and specialized applications. This gives advantages on fixed architecture devices such as general-purpose processors, ASICs, and graphic processors units (GPUs). FPGAs enjoy a wide range of applications in instruments and instrumentation systems⁽²⁻⁵⁾.

3.7.1.3 Embedded Controllers

Embedded controllers are essentially microprocessors or microcontrollers with added components to perform functions within an instrument. Once embedded controllers are implemented in instruments may have limited flexibility for improvements and making changes when programs are written and burnt into the ROM or EEPROM, the system will be inflexible for further alterations.

3.7.1.4 Dedicated Computers

Dedicated computers are often used in automatic instruments where there are many sensors and transducers involved as in the case of some mobile air-pollution monitoring systems, self-driving cars. Such instruments consist of a laptop or palm computers like devices with displays. Dedicated computers are designed to perform a set of specific tasks rather than for general purpose computing.

3.8 SIGNAL PROCESSING HARDWARE

DSPs fulfill fast operational needs for signal processing in many applications. Some applications are telecommunication systems, audio signal processing, speech recognition, text and image recognition, radar, lidar, sonar, and vision sensing applications, HDTVs, mobile phones and consumer electronics, drones, robotics, and automotive.

The DSPs incorporate special hardware features that can speed up calculations for digital filtering, fast Fourier transforms (FFTs), and other repetitive and handling numerical tasks. The distinction between the DSP processor and microprocessors is in the degree of specialization. For example, for the implementation of signal processing, say finite impulse response (FIR) filters, an *X*-memory facilities can be used to store the samples of input signals, and a *Y*-memory facilities to store the impulse responses.

General-purpose DSPs: A general-purpose DSP is fundamentally deterministic in terms of code execution timing. These processors have straightforward architectures and are supported by tools that help the programmer determine code execution times. Some general-purpose processors have extremely complex architectures, for example, superscalar architectures that dynamically select instructions for parallel executions of instructions.

Floating-point DSPs: A floating-point DSP can support multiple data types, such as 40-bit floating-point. Some of these processors are based on modified Harvard memory architecture with two memory spaces: program memory (PM) and data memory (DM), each with its own address and data buses.

Fixed-point DSPs: Fixed-point DSPs provide multiple distinct arithmetic units: a multiplier-accumulator, a shifter, and an arithmetic logic unit (ALU). Inputs to all arithmetic operations from the register file, and results of all arithmetic operations are delivered back to the register file. Fixed-point DSPs implement a load-store architecture in which the source operands for arithmetic operations are loaded into registers before computation.

Data handling of DSPs depends on the measurement characteristics and computations that must be accomplished in a fixed time. The term *real-time computing* is used to describe a computing environment that sets an upper limit on the execution time. Digital signal processing functions often must be done in real-time, so that processing tasks should be completed before next data arrives.

3.9 SIGNAL HANDLING

Analog signals need to be conditioned first by means of analog circuits before they are converted into digital forms. Analog signal conditioning has two main goals before the A/D conversion: the first is the amplification of small signals or

attenuation of large ones, and the second goal is the removal of unwanted frequencies by suitable filters.

Small signal handling is necessary in many instruments and instrumentation applications. The A/D converters can achieve full accuracy and resolution only when A/D input signal optimally matches the dynamic range of the A/D.

3.9.1 AMPLIFIERS, FILTERS, CONVERTERS, AND DATA ACQUISITION

Essential components of instruments are amplifiers, filters, converters, and data acquisition electronic circuits.

3.9.1.1 Amplifiers

Amplifiers typically provide gains under the condition of low internal noise and immunity to external interference. *Operational amplifiers* and *instrumentation amplifiers* are the most common devices used for amplification of signals. They can be configured as *inverting amplifiers* or *non-inverting amplifiers*.

Charge amplifiers are used in current or charge generating devices such as active capacitive sensors, piezoelectric sensors, and pyroelectric sensors. Charge amplifiers are designed to have extremely low bias currents.

3.9.1.2 Filters

Filters are used to modify the frequency spectrum of signals generated by sensors to extract the useful information from the sensors and to eliminate external and internal noise. There are two basic types of filters: active filters and passive filters. *Active filters* contain active elements such as operational amplifiers in addition to passive components. *Passive filters* are made from only the passive components, such as the resistors, capacitors, and inductors. The filters can be configured as: lowpass filters, high pass filters, bandpass, and band stop filters, and notch filters. Most popular filters are Butterworth filters, Chebyshev filters, elliptic filters, Bessel filters, and raised-cosine filters. They can be constructed as first-order or higher-order filters. Second-order and higher-order filters are realized by common first-order filter circuits, such as the Sallen and Key lowpass filter integrated with multiple feedback bandpass filters.

Butterworth filters can be realized with simple resistor-capacitor networks around op amps. They are maximally smooth filters with linear rolloff in the stop band with no ripples. *Chebyshev filters* have sharper cutoffs and steeper rolloffs than Butterworth filters. The pass band for Chebyshev filters has significant ripple. *Elliptic filters* have the sharpest cutoffs and steepest rolloffs over Butterworth and Chebyshev filters, and the pass and stop bands have significant ripples. *Bessel filters* have maximally smooth delays in the pass band. They have a linear phase response. *Raised-cosine filters* provide the maximum information content for the allotted frequency bandwidth.

Digital filters have significant advantages over analog counterparts; they are programmable, use less space on a circuit board, and are not subject to variations in temperature and other physical parameters. There are various types of digital filters as explained below.

Finite impulse response (FIR) filters (or non-recursive filters) are simple linear combination of a finite number of input samples. They produce less ripple and narrower filter bandwidth.

Infinite impulse response (IIR) filters (or recursive filters) use feedback of the output in linear combination with a finite number of input samples. Their primary advantage over FIR filters is that IIR filters require one half the number of taps of equivalent FIR filters, but due to feedback these filters can be unstable.

Discrete Fourier Transform and Fast Fourier Transform Filters operate on the data point-by-point. They provide spectral analysis to determine the energy at different frequencies within a signal.

3.9.1.3 Converters

Sample-and-hold (S/H) circuits acquire the value of the input analog signal (*sample*) and holds it throughout the conversion process until the conversion is complete. Thus, a fix level of voltage is provided to the converter during the conversion process and the uncertainty provoked by the variation of the input voltage is minimized or eliminated to some acceptable levels.

Analog-to-digital (A/D) and *digital-to-analog (D/A)* converters provide the interface between the analog environment (the real world) and the digital environment for computation and data processing purposes. Sensing applications require different ADC architectures, therefore there are many different types to choose from. Some of the A/D converters will be explained next.

Counter-ramp converters are one of the simplest forms of direct A/D conversion. An improvement to the counter ramp A/D converter is the *tracking converter* that uses up-down counters.

Successive approximation A/D converter successively compares with the voltage generated by a D/A converter.

Flash (Parallel) A/D converters contains as many comparators as there are digital word length except for one. They continuously compare the analog input signal with a reference signal.

Delta-sigma (Δ/Σ) A/D converters are based on an inside feedback loop with an analog lowpass filter, and a D/A converter. The block diagram of a Δ - Σ ADC is shown in [Figure 3.3](#). In this figure, cooperative working of analog and digital components can be seen. The analog modulator contains an integrator, a comparator, and a single bit digital-to-analog converter. The comparator provides single-bit output signal, called *bit stream*. The operations of analog modulator and a digital filter are synchronized by a high-frequency clock.

Dual slope A/D converters use integration techniques to produce digital outputs for analog inputs. It consists of integrator, comparator, clock, signal generator, control logic, and counter blocks.

Pipelined A/D converters are suitable for signals requiring less than 100 Msps (mega samples per second) for CCD cameras, imaging, ultrasonic medical imaging, digital videos, cable modems, and Ethernet. As an example, the operation of a pipelined A/D converter is as follows: pipelined A/D converter quantize three bits

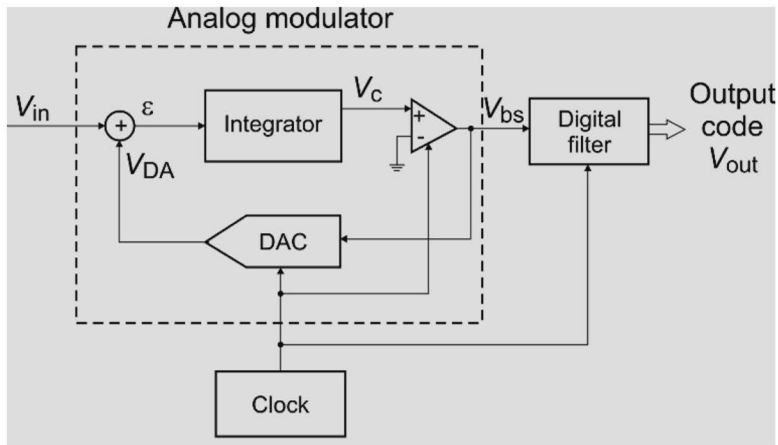


FIGURE 3.3 Delta-sigma (Δ/Σ) A/D converters (Source: A. Ferrero ⁽⁶⁾).

to be fed into a 3-bit D/C, then the output is subtracted from the input. The residue is gained up by a factor of 4 and fed into the next stage. This continues through the pipeline providing 3 bits per stage until it reaches 4-bit ADC to resolve the last 4 LSB.

Digital-to-analog converters (DACs or D/A) are used to convert binary words into analog voltages. A typical D/A converter has three major blocs: 1) resistive or capacitive networks, 2) switching circuitry, operational amplifiers, latches, and 3) a reference voltage. A popular D/A converter is the weighted current converter.

Weighted current D/A converters generate a weighted current for each bit containing logic ONE. This kind of D/A converter is useful when an analog current needs to be implemented directly. When voltage is needed, the output voltage can simply be determined by a known load resistor that carries the current.

Data-acquisition systems are compact devices to handle multiple analog and digital inputs and outputs on real-time bases. They are usually produced as add-on devices.

3.9.1.4 Oscillators and Modifiers

Oscillator circuits are the generators of electrical signals, and they are essential in many instruments and instrumentation applications. Oscillators are basically unstable circuits that comprise of some gain, non-linear features, and positive feedback. There are three types: *RC* oscillators, *LC* oscillators, and *crystal oscillators*. In the case of *RC* and *LC* oscillators, the operating frequencies are determined by the values of resistive, inductive, and capacitive components. In crystal oscillators, the frequency is defined by mechanical resonant properties of quartz crystals.

There are many different types of oscillator circuits that can be found in literature based on operational amplifiers and digital systems some examples are provided in the [Chapter 2](#) of this book. Some oscillators use operational amplifiers, while others use *npn*-transistors as amplifiers and *LC* networks to obtain variable frequencies. Also, various types of multivibrators together with the logic circuits constitute the basics of many oscillators.

3.9.1.5 Data Acquisition Components

Passive devices such as bridges are commonly used to measure basic electrical quantities such as resistance, capacitance, inductance, impedance, and admittance. There are two groups of bridges, the AC bridge, and the DC bridge. Also, in each type there are many different types such as Wheatstone, and Kelvin DC bridges, and Schering, Maxwell, Hay, and Owen bridge.

Single-channel data-acquisition system primarily handles only one signal. They can handle sensor signals that may not be suitable for A/D converters due to low amplitudes or signals mixed with heavy noise.

Multichannel systems have several data acquisition channels coming from various sensors. They have *multiplexers* to stream the signals for conversions. The multiplexing can be done by analog or digital multiplexers. In analog devices, the multiplexing function is carried out on the analog signals before D/A conversion takes place. *Digital multiplexers* coordinate the timing of A/D converted multiple inputs on the internal system bus.

Data-acquisition boards are plug-in devices that allow the processing of analog signals inserted in digitals systems or computer slots.

Pulse width modulators and linearizers are often used to process the sensor output as amplitude-modulated signals. Typical examples of such sensors are the capacitive vibrational and displacement sensors. The output of the sensor may be pulse width modulated to directly be interfaced with digital circuits.

Voltage followers are often used for interfacing sensors to loads. They act as buffers by converting high-level to low level impedances.

3.10 DIGITAL DATA PROCESSING

Signals from sensors need to be analyzed and processed for extracting the useful information. Among many others, the *Fourier transforms*, and *Fourier series* are common methods that can be used in signal analysis. The signal representation involves decomposition in terms of sinusoidal or complex exponential components. The decomposition is determined by *frequency analysis tools* that provide a mathematical and pictorial representation of frequency components, which is referred to as the *frequency spectrum*. The frequency spectrum can be determined by various *frequency* or *spectral analysis* techniques.

AI is deployed in data processing of sensors and instruments. Data from devices can be fused, combined, reduced, filtered, or correlated. The data processing can also involve self-calibration, multiple channel filtering, data fusion between different sensors, and decision-making.

The basic instrument data processing requirements using AI are.

- *Signal filtering*: noise removal, noise cancellation, and averaging,
- *Signal analysis and feature extraction*: detection of changes, frequency spectrum analysis,
- *Synthesis*: generating complex signals, encoding, data encryption,
- *System identification*: numerical analysis to describe system properties, and
- *Compression*: reducing bandwidth and memory of signals,

3.11 VIRTUAL INSTRUMENTATION

Traditional instrumentation systems are made from multiple stand-alone or interconnected instruments. Nowadays, cell phones, palm computers, and laptop computers are extended by appropriate hardware and software to function as instruments. Instrumentation systems configured in this way are referred to as *virtual instrumentation*.

The major advantage of virtual instrumentation is the *flexibility*. The changes in the functionality of an instrument can easily be done with the use of convenient software tools. Virtual instruments also offer analysis function, such as the Fourier analysis, mathematical manipulations, graphical displays, and so on by using the corresponding libraries.

3.12 WIRELESS INSTRUMENTS

Wireless sensors and instruments have largely been supported by improvement in digital components. RF circuit fabrication methods, advances in signal processing theory, emergence of large-scale wireless ICs, advanced networks that allow easy data flow, use of AI at various stages of the process, convenience of the internet, and advanced computing techniques such as the edge and cloud computing. Modern IC technology makes radio equipment smaller, cheaper, and more reliable. In parallel to technical advantages, the consumer acceptance and the need for such products permit widespread deployment of wireless instruments and supporting communications systems. Trends are likely to continue at a greater pace in the coming years.

Wireless sensors contain sensing elements, signal processing circuits, and wireless communication components on the same chip. Several miniaturization techniques are available, including system-on-chips, MEMS, and ASICs. In wireless sensors, there are five main components that need to be integrated for a complete system: the sensor(s), signal processing circuitry, radio circuits, power circuitry, and the package. The radio, sensors, and signal processing circuitry can be reduced in size through hybrid circuits, MEMS, or mixed signal ASIC designs.

Wireless instruments consist of five main components – sensors and signal conditioners, programmable digital hardware, memory and storage, input/output, and communication components. A typical wireless instrument is illustrated in [Figure 3.4](#). Instruments differ from each other in the way of sensing, handling data, transmitting and displaying information.

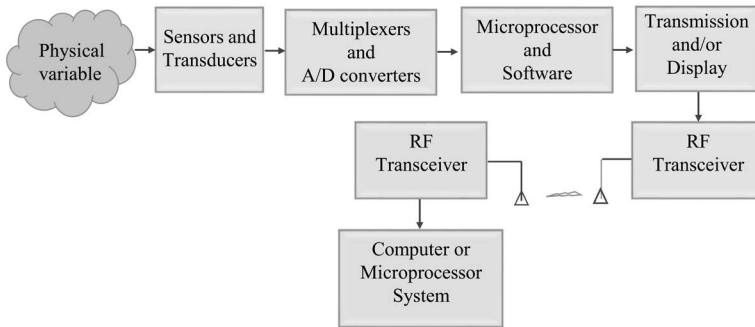


FIGURE 3.4 Components of a wireless instrument.

3.13 INSTRUMENT SOFTWARE

Electronic instruments are applications specific devices that perform specific tasks. Therefore, there is no universal software available that can address the needs for all kinds of instruments. The software for instruments is mainly determined by the type of microprocessor and microcontroller used for the internal operations. The software in the basic instrument level assists signal handling, signal conversion, data gathering, diagnostic, troubleshooting, displays, parameter adjustments, instrument settings, and so on.

Once the data is transferred externally from an instrument for further processing, different software is required using various software languages. The major software languages are C++, C, EC++, Java, Python, Ada, Basic, and Forth.

3.14 INSTRUMENTS COMMUNICATIONS AND NETWORKS

The capabilities of electronic instruments and instrumentation systems extraordinarily improved due to the new generation of digital processors, microcomputers, digital communication systems, computers and supporting algorithms, databases, AI, and IoTs. Computers, microprocessors, microcontrollers, and DSPs provide flexibility in the design, and computational power to process signals as well as the related data. Computational power has been growing constantly, and this allows the development of much capable and more challenging applications.

Wireless communication give flexibility to instruments thus making any instrument behave like a portable instrument in handling data. That is an instrument may be fixed but the wireless transmission of data generated by it gives them mobile instrument behavior. Today, instruments can easily be networked independent of their geographic locations albeit fixed or mobile. In many measurement systems, instruments operate autonomously but in a coordinated manner. Information generated by instruments can instantly be communicated between external devices, instruments themselves, local controllers, or other devices such as recorders, routers, base stations, and host computers.

In complex measurement systems, instruments become much flexible 1) for their networking capabilities by means of remote communication methods such as radio frequencies, microwaves, Internet, and optical techniques, 2) for their internal data handling capabilities, e.g., data compression, and 3) for availability of on-board large memory capabilities for internal data handling and storage. In most cases, the

transmission of data between digital devices is carried out by using wireless transmission techniques. However, as the measurement system becomes large communication can become very complex this is where the applications of AI is a preferred option.

Various techniques are used in wireless instrument communications, these are: optical and infrared methods, radiofrequency methods, and sonic methods. Application of radiofrequency methods yields to large networked systems. Since the main theme of this book is wireless instruments and networks, most used RF methods are explained in detail.

3.14.1 RF INSTRUMENT COMMUNICATIONS

RF system design in instrumentation applications is a multidisciplinary field and requires a good understanding and knowledge of many areas of disciplines including modern IC design and implementation. In the design process, RF component architectures are planned according to the available off-the-shelf components. The ICs components may have many options in construction, capabilities, and architectures, thus leading to a great deal of redundancy at both system and circuit levels.

When data is transmitted over long distances, appropriate hardware and software must be used such as modems, and other RF devices. On the receiving end, appropriate hardware and software interprets the received signals and extracts the intended information transmitted. In such cases, various modulation techniques are used to convert signals to suitable formats. For example, most modems, with medium-speed asynchronous data transmission, use frequency shift keyed (FSK) modulation. The digital interface with modems uses various protocols such as MIL-STD-188C to transmit signals in simplex, half-duplex, or full duplex forms depending on the directions of the data flow. The simplex interface transmits data in one direction, whereas full duplex transmits it in two directions simultaneously. Figure 3.5 illustrates the components of transmitter and the receiving base station.

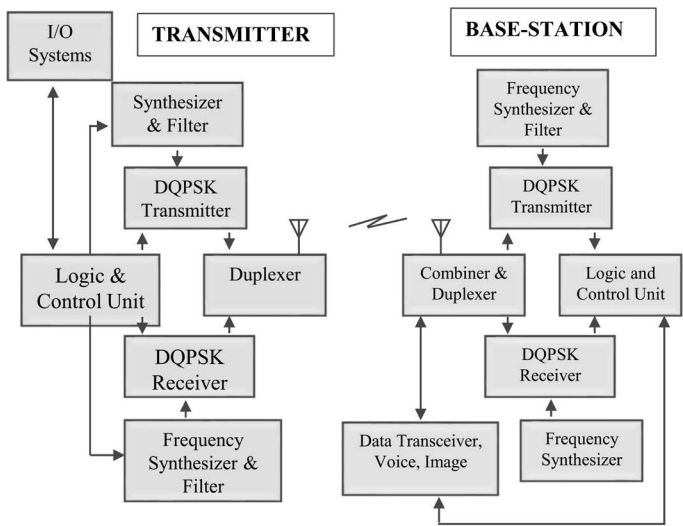


FIGURE 3.5 Components of transmitter and base station.

3.15 AI IN MEASUREMENTS

Applications of AI in instrumentation and measurements cover detections, tracking, monitoring, characterizations, identifications, estimations, diagnosis, predictions, sensor design, calibrations, uncertainty predictions, measurement data interpolations, software, indirect measurements, fault detections, and system identifications. AI is practical when the precise and accurate mathematical modeling of the measurement system or instrument at hand is highly complex, non-linear, dynamic, or impossible to model due to lack of knowledge on the system except for limited number known parameters. Measurement prediction is another useful feature of AI, which can be applied when actual measurement is either costly or impractical. AI has been shown to predict measurements more accurately and orders of magnitude faster than non-AI techniques. AI algorithms automate the process of extracting discriminative features in datasets.

Technological developments and AI have gone through changing and improving the way measurements are being performed. Starting from entirely mechanical designs, today's measurement instruments are electronic, computerized and, in many cases, connected. This has enabled a largely automated collection of physical quantities with high resolution and reliability. The recorded data are used as the basis for decision-making⁽⁷⁻⁹⁾.

Measurement systems are becoming much more complex involving a multitude of sensors and instruments. The construction and the validation of the model underlying the measurement process is based on priori knowledge. The complex measuring systems lead to consideration of perceptive agents aligned for the AI operations. The cooperative multiagent systems or agencies extensively studied in distributed instruments applications and robotics. AI paves the way to complex and articulated intelligent measurement systems which are capable of cooperation among themselves as well as performing an autonomous measurement with high degree of reliability^(7,9,10).

The benefit of AI is the ability to automate data analysis and assist the decision-making processes. AI models are trained to identify patterns and anomalies in data, classify events or signals, and regress data based on the input. This reduces the need for human intervention and improves the speed and accuracy. The hidden information can be extracted from the raw data, enabling new measurands without using extra sensing elements. For example, in industrial condition monitoring, structural damage can be detected and classified based on the sensor data^(8,11,12).

In many applications, the data is communicated to a central database to be processed and analyzed offline. This requires large amounts of data to be transferred from the measurement system to the database. This results in latency thus prohibiting timely and real-time operations. AI can create fully contained measurement systems processed locally, including data acquisitions and data analysis.

In instrumentation and measurements, supervised learning has gained particular attention. The regressions and classification capabilities provide natural analyses of the collected data. These AI solutions are implemented using different models based on different methods, such as artificial neural networks (ANNs), decision trees, and support vector machines (SVMs).

Instrumentation and measurement applications can be highly specific. In wearables, for example, models might be trained to provide acceptable performance for any potential user, while the performance may be improved in cases of a more user specific model. This creates a trade-off situation between how specific the model is, with the effort and cost this entails, and the performance of the final model. An alternative solution is on-device learning, which allows for the base model to continuously adjust to the specific application conditions. While on-device inference has come a long way, on-device learning still attracts significant research and development efforts^(8,11,13–15).

Uncertainty quantification and visualization are crucial in AI-assisted measurements because they enable better risk management, in turn increasing the trustworthiness and adoptability of such systems in the real world. This is especially important for neural networks and deep learning, which are widely applied in instrumentation and measurement and are known to be overconfident^(13–15).

3.16 SMART INSTRUMENT SYSTEMS

Interconnected and intelligent instrumentation enables the collection of timely, high-quality data through embedded sensors that communicate over wireless networks. Instrumented devices such as smart meters for gas, electricity, and water continually monitor the supply and demand for these utilities and actuate strategies devised by intelligent components.

Smart connected objects, whether embedded, mobile, or wearable generate massive data to enable development, deployment, and use of smart products and services. However, this data combined with analytical models proves to be much useful. Smart instruments improve transportation, manufacturing, energy and water management, environmental sustainability, medical diagnostics and treatments, education, and commerce enable better outcomes and lower costs⁽¹⁶⁾.

3.17 IIOT AND MEASUREMENTS

IIoT (Industrial Internet of Things) is an approach for distributed devices connected over a network to collect data, collaborate, and perform tasks without human intervention. IIoT is one of the cores enabling technologies behind the fourth industrial revolution. A typical IIoT structure and dataflow is illustrated in [Figure 3.6](#). In this application, the basic level consists of large number of connected devices. Intercommunications of these distributed devices are realized in the next level. Once the intercommunication is completed the information gathered is sent to Edge and Cloud computing platform via Edge computing to fulfill application needs at the top level^(17,18).

The IEEE defines an IIoT system as “a system of entities (including cyber-physical devices, information resources, and people) that exchange information and interact with the physical world by sensing, processing information, and actuating.” The term “Thing” in IIoT can be defined as: “an IIoT component has functions, properties, and ways to exchange information.” The components of an IIoT system interact with each other to fulfill the goals which the system has been designed for.

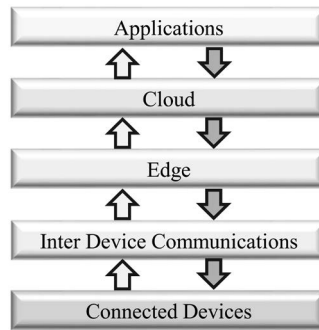


FIGURE 3.6 Typical IoT structure and dataflow.

Some examples of IoT are smart homes, transport systems, and industrial systems. Smart homes consisting of interconnected alarms, cameras, sensors and actuators, and switches. Actuators control lighting, temperature, ambience, and security issues. Transportation systems consist connected vehicles, drivers, pedestrians, and traffic infrastructure leading to efficient traffic routing, road assistance, emergency response, parking support, and toll collection. Industrial IoT integrate machines, robots, sensing of process variable, communication, and collaboration.

IoT presents many benefits in 1) simultaneous wide area measurements, 2) continuous measurements, and 3) real-time measurement analysis. *Simultaneous wide area measurements* operate on many embedded devices with ubiquitous connectivity, which can simultaneously gather measurements over large areas. *Continuous measurements* enable measurements by fusing the instruments which leads to collaborative operations. *Real-time measurement analysis* streamlines collection and analysis of measurement data. The analysis is carried out in the smart instruments, or at a nearby edge node, or a combination of the two^(17–20).

3.18 SENSORS AND TRANSDUCERS

Sensors and transducers produce electrical outputs in response to physical variations. Outputs are processed suitably by analog and digital methods to have desired information about the physical process. An actuator converts an electrical signal into an external action.

The sensors and transducers can be categorized in several ways depending on energy inputs and outputs, input variables, sensing elements, electrical operations, materials used, and physical constructions. For example, from the energy input and output point of view, there are three fundamental types of sensors: modifiers, self-generators, and modulators.

In *modifiers*, a particular form of energy is modified rather than being converted another form of energy. Therefore, the same form of energy exists in both the input and the output stages. In *self-generators*, electrical signals are produced from non-electric inputs without the application of external energy. Self-generators produce

small signals hence they often need extensive signal conditioning processes. Typical examples of self-generators are piezoelectric transducers, and photovoltaic cells. *Modulators*, produce electric outputs from non-electric inputs, but they require an external source of energy. Strain gages are typical examples of such devices.

3.19 SENSOR TECHNOLOGY

Sensors are the primary and foremost necessary elements of instruments and instrumentation systems. They produce output signals in response to physical variations. Signals generated by sensors are processed suitably by analog and/or digital techniques to extract the useful information thus representing the behavior of the physical variations.

3.19.1 SENSING MATERIALS

Sensors are fabricated using numerous materials such as metals, polymers and plastics, ceramics and glass, biological materials, semiconductors, and combinations of two or more of these materials.

3.19.1.1 Metals

Metals are commonly used for manufacturing sensors. As sensing elements, metals exhibit many different properties and characteristics. They can be electrically conductive, nonconductive, or semi-conductive; magnetically ferrous or nonferrous; thermally conductive or nonconductive; mechanically elastic, brittle, strong, lustrous, or opaque; chemically catalyst, chemo-absorbent, reactive, or non-reactive, and so on. Most of these properties are utilized in sensor technology depending on the requirements of the physical variations. Some of the most common metals used extensively are aluminum, beryllium, silicon, magnesium, beryllium, iron, copper, lead, platinum, nickel, gold, and zinc. Aluminum exhibits good electrical, electromagnetic, and optical properties. Beryllium finds applications in optical, and x-ray measurements. Silicon is the prime element used in semiconductors and IC circuits. Iron is used in magnetic applications. Magnesium, platinum, and nickel are often preferred as the basic elements in temperature sensing. Copper and gold have good electrical and thermal properties thus finding applications primarily in semiconductor and microsensors. Lead has good piezoelectric, sound, and vibration properties. Zinc is used as one of the main additives to improve the sensing properties of some alloys.

Metal alloys in various combinations are used in sensor technology, particularly in temperature sensing. A good example of such alloys is the chromel (90% Ni, 10% Cr), which forms one of the most basic components of temperature sensors. Metals and their alloys find applications as they exhibit strong dielectric and polarization properties. Examples are electromagnetic and nuclear radioactivity measurements for their sensitivity to radiation, and chemical and biological measurements due to ionization properties. Metal alloys are also used as piezoelectric, pyroelectric, electrolyte, and ferroelectric materials responding to mechanical stress and strain, light sensing, chemical stimuli, and magnetism.

3.19.1.2 Polymers and Plastics

Polymers and plastics find many applications in sensor technology. They are made from *monomers*, which are basically chemical raw-synthetic materials. Monomers react with each other to form long chains of polymers and polystyrenes. The primary element of polymers and plastics is the carbon. Carbon atoms combined with other elements to produce hundreds of different types of materials. Apart from carbon, several other elements are used to produce many forms of polymers and plastics, such as hydrogen, nitrogen, oxygen, fluorine, silicon sulfur, and chlorine. By suitably combining these elements and application of suitable technology, large number of molecules can be produced exhibiting many different properties that suitable for using in sensors. Plastics find extensive applications in sensors. For example, *alkyd* has good electrical as well as moisture absorption properties and *epoxy* has electrical properties. Many forms of plastics exhibit other properties such as piezoelectricity that find applications in stress, strain, vibration, and sound measurements.

3.19.1.3 Ceramics and Glasses

Ceramics and glasses are useful materials for producing a range of sensors. Ceramics are made from several forms of metal carbides and nitrates, which exhibit thermal stability, light in weight, resistant to chemicals, structural strength, and good electrical properties. For instance, silicon carbide has a high dielectric constant therefore finds applications in capacitive sensors. Ceramics are also used as basic elements in microsensor technology.

Glasses are based on silicate systems and made from three major components, silica (SiO_2), lime (CaCO_3), and sodium carbonate (Na_2CO_3). Glasses are produced in many different forms and shapes to give different optical properties particularly in mid- and far-infrared, visible, and ultraviolet ranges.

3.19.1.4 Biological Materials

Biological materials perform vital functions in sensor technology. They are used for sensing chemical and biochemical species. Most biosensors consist of two parts in their structures, the biologically active component, and the transducer component. The biologically active component provides a specific interaction with the target molecules. This interaction releases energy that can be sensed by the transducer component that produces representative electrical signals. Biosensors are manufactured and used for their different physical and chemical properties, such as antibody-based biosensors, enzyme-based biosensors, and biological micro-resistors.

3.19.1.5 Semiconductor Materials

Semiconductor materials are manufactured primarily from silicon, germanium, and other doping elements (e.g., arsenic, phosphorus, lithium, antimony). Interconnect elements like copper and aluminum are extensively used in modern sensor technology. However, silicon is the most common material used in sensors technology, particularly in microsensors, intelligent sensors, and system-on-chip devices. Silicon exhibits favorable mechanical, chemical, and electrical properties. It is a relatively

inert element that can only be attacked by a handful of chemical substances such as halogens, dilute alkali, and hydrofluoric acid.

Properties of silicon and its applications in sensor technology is well researched area. Silicon has good properties in radiation, temperature, and optical measurements, as well as many useful properties in mechanical, electrical, chemical, and magnetic measurements. For example, in radiation and optical measurements, silicon exhibits photovoltaic, photoelectric, photoconductive, and photo-magneto-electric properties. In temperature measurements, it exhibits Seebeck effect, Nernst effects, and temperature dependent *pn* junctions. In chemical measurements, silicon exhibits ion sensitivity to certain chemicals by changing resistance, currents, voltages, and other properties. It also has various magnetic properties such as Hall-effect, magnetoresistance, and Suhi effect.

Silicon has good mechanical properties too. These properties are widely utilized to fabricate pressure transducers, temperature sensors, force, and tactile sensors. Thin film and photolithographic technology allow silicon to be fabricated in small, high precision mechanical structures suitable for microsensor, nanosensor, and quantum technologies. Micromachining allows the use of semiconductors to form structures suitable for sensing acceleration, force, pressure, and mechanical impulses.

One major problem with silicon is temperature dependence. Many of the properties in response to physical variables such as strain, light, and magnetic fields tend to be non-linear responses and temperature dependent. The physical properties of silicon can be enhanced considerably by depositing different elements and materials in the crystal structure. For example, addition of ZnO thin films into a silicon structure enhances piezoelectric properties and can be used for producing accelerometers, and surface acoustic wave (SAW) filters.

Apart from the oxygen, silicon is the second most abundant material on Earth's crust, existing as oxides and silicates. Some oxides are sand, quartz, rock crystals, mica, and so on. Pure silicon is obtained by heating silica and carbon in an electric furnace in the presence of carbon electrodes. Although there are many other methods, Czochralski process (a form of crystal growth method) is a common method to produce crystals of silicon used in solid-state semiconductor devices and micro-machined sensors.

3.19.2 DEVELOPMENT OF SENSORS

Sensors are developed using materials, as explained above, and by employing a diverse range of processing techniques. Methods for the fabrication of sensors vary depending on the specifications, size, and complexities.

Today, modern microsensors are fabricated by making full use of properties of semiconductors and their associated technologies. In microsensors, the use of other materials and deposition of thick and thin films are often required to give the sensing materials useful properties otherwise they would not normally have. For example, piezoelectric material films applied on silicon wafer provide piezoelectric properties. There are several methods of depositing thin and thick films on substrates or semiconductor wafers. Some of the methods are spin-casting, vacuum deposition, sputtering, electroplating, and screen printing.

3.19.2.1 Spin-Casting

Spin-casting is often used for deposition of organic materials for humidity measurements and chemical applications. This process involves the use of a thin film material dissolved in a volatile liquid solvent. The solution is poured onto the sample and the sample is rotated at high speeds thus evenly spreading the material over the sample due to centrifugal force. As the solvent evaporates, a thin layer of film of the material remains on the sample. Although this method is adequate for most applications, in some cases the thin film may not be uniformly spread on the surface, which may cause non-linearities, defects, and other problems in the sensor operations.

3.19.2.2 Vacuum Deposition

Vacuum deposition is realized by converting the metal to form the thin film into gaseous form and then depositing it onto the surface of the sample. A vacuum deposition system consists of a chamber and a heating mechanism to put the metal into gaseous form. For this heating of the material above the melting point is necessary. The gaseous metal atoms are allowed into the chamber in a controlled manner to be deposited on the sample. The thickness of the deposition on the unmask regions of the sample is controlled by evaporation time and the vapor pressure of the metal. However, as in the case of spin-casting, distribution of the film on the sample may be non-uniform thus causing problems associated mainly with non-linearities.

3.19.2.3 Sputtering

Sputtering is like vacuum-deposition method. In this method, an inert gas such as argon or helium is introduced into a chamber that contains anode and cathode electrodes supplied by an external high voltage source. The anode contains the sample to be deposited on and the cathode contains the deposited material. The principle is that the high voltage ignites a plasma effect on the inert gas and the gas ions bombard target containing the material to be deposited. When the kinetic energy of the bombarding ions is sufficiently high, some of the atoms from the target surface are freed and carried by the gas to the surface of the sample. Sputtering techniques yield better uniformity particularly in the presence of a magnetic field with desired intensity. Sputtering does not require high temperatures therefore virtually any type of material, including organic materials, or mixture of different materials can be sputtered.

3.19.2.4 Chemical Vapor Deposition

Chemical vapor deposition (CVD) is one of the most common methods used for the fabrication of semiconductor-based sensors. It is a widely applied technique particularly in production of optical and optoelectronic devices, and many others. The CVD process takes place in a reaction chamber where substrates or wafers are positioned on stationary or rotating tables. The dopants are allowed to enter the chamber mixed with a carrier gas such as hydrogen. The substrate is kept at an elevated temperature that helps the additives to be deposited on the surface of the sample. The thickness of the deposition is controlled by the amount of dopant in the gas, pressure of the inlet, and temperature of the substrate.

3.20 IC SENSORS

The sensor technology has significantly shifted toward IC sensors in the form of Microsystems, intelligent sensors, and nanosensors. Microsystems refer to dimensions of devices in micrometer (10^{-6} m) ranges whereas nanotechnology refers to dimensions of devices in nanometer (10^{-9} m) ranges. The Micro-Systems Technology (MST) is well established and simply known as the MST. A subset of MST is the *micro-electromechanical systems* (MEMS), *micro-electro-optical systems* (MEOMs), and *systems-on-chip* (SOC) devices.

Data obtained from sensors and transducers are interpreted into forms that humans can understand by associated interface circuits. The very large scale integrated (VLSI) circuits have been extensively used to realize complex sensor modules. IC technology allows integration of many components in the same chip. This makes IC sensors to be interfaced directly to digital devices.

Semiconductor-based sensors are produced by using *microfabrication* techniques, which refers to the collection of processes used by the electronic industry for manufacturing ICs. IC sensors provide easy interface, lower cost, and reliable outputs. A typical example of the IC sensors is the photodiode in which an incident light falls on a reverse biased *pn*-junction and the photonic energy carried by the light creates electron-hole pair on both sides of the junction causing a current to flow. The output voltage of photodiodes is highly non-linear thus requiring suitable linearization and amplification circuits, which could be done in the same chip.

IC sensors may be grouped according to the signal domains:

Chemical domain sensors constitute a large section of commercially available semiconductor sensors. They are based on techniques such as ion-sensitivity field-effect transistors (ISFETs), chemically sensitive thin films, and polymers.

Magnetic domain sensors are made from magnetically sensitive semiconductors that are obtained by using doping techniques and thin films such as Ni-Fe. Majority of sensors use principles of Hall effects, magnetoresistance, and Suhi effects.

Mechanical domain sensors constitute a large range of devices from MEMs to tactile sensors. Some examples are accelerometers, piezoresistive, photoelectric- and photovoltaic-based sensors, and micromachined devices.

Radiant domain sensors contain wide spectrum of electromagnetic radiation, visible spectrum, and nuclear radiation. Some examples of are photovoltaic, photoelectric, photoconductive, and photomagneto-effect sensors.

Thermal domain sensors are largely semiconductor-based devices that exhibit sensitivity to temperature. Although sensitivity to temperature is undesirable in many IC applications, the temperature dependence of semiconductors can be put into useful forms for temperature measurements and control. Some of these devices are based on Seeback effects, and Nernst effect.

The usefulness of semiconductor-based IC sensors is enhanced considerably by the integration of microprocessors, microcontroller, converters, logic circuits,

internal memories, and other digital components in the same chip. Further, micromachining techniques combined with semiconductor processing technology provides a multiple range of sensors all integrated in the chip for mechanical, optical, magnetic, chemical, biological, and other types of measurements. Undoubtedly, advances in digital technology and cost-effective manufacturing techniques of IC sensors, and productions in large scale reflect positively on the future development of new types and wide range of instruments.

An examples of IC sensors are the *power ICs* (PICs). PICs are electronic devices that are already equipped with embedded internal sensors. They are produced by combining bipolar and metal oxide semiconductor (MOS) circuitry with the power metal oxide semiconductor field effect transistors (MOSFET) technology. The Power ICs is based on the consolidation of several circuit elements into one single device. In single chip devices, some circuit elements (e.g., operational amplifiers, comparators, regulators) are best implemented by the bipolar IC technology. MOS circuitry handles logic, active filters, and time delays. Some circuits, such as A/D converters and power amplifiers can be implemented by either bipolar or MOS technologies.

Many other IC sensor systems consist of discrete or multiple sensors combined with ASICs or some components of the printed circuits boards (PCBs). This leads to the manufacturing of a diverse range of sensors requiring various forms of interface electronics. The interface requirements are dependent on the quantity to be measured, types of physical effects, overall system architecture, and application specifications.

Sensing arrays include several sensors for different measurands, such as pressure, flow, temperature, and vibration. The multiple sensors are be used for many reasons, such as increasing the range, providing redundancy, or capturing information at different spatial points. A good example of sensor arrays is in the chemical applications where the same chip can measure different chemicals. For example, multi-element gas analyzers are developed by using thin film detectors in which four-element gas analyzer produced by chip hybrid configurations.

Photodiode arrays are another example of sensing arrays, and they prove to be highly cost-effective in many applications. The popular silicon charge-coupled device (CCD) with many sensing elements is typical example of such devices. A complete spectral measurement can be made by a single unit. Some array sensors use pyroelectric and piezoelectric effects in zinc oxide thin films for gas flow, chemical reactions, and cantilever beam acceleration, tactile, and SAW vapor sensing.

Microinstruments and microsensors are typical example of modern digital instruments. Many single-chip implementation of microinstrumentation system are based on CMOS technology. Such devices can incorporate voltage, current, and capacitive sensor interfaces, temperature sensors, multiple channel analog-to-digital (A/D) converters, microcontrollers with hardware multipliers and accumulators. The microcontroller can have multiple programming capabilities such as debugging, interrupts, direct memory access (DMA), clock manager, and so on.

3.21 SENSORS

Humans can sense physical variables such as temperature, light, odor, and acoustics quite well within extremely limited ranges. We need sensors and instruments that can convert physical variables into observable forms having much larger ranges, hence a diverse range of sensors and instruments have been developed to extend sensing capabilities.

Sensors are classified in many ways depending on:

1. Output signals: analog or digital,
2. Physical parameters: optic and ionization,
3. Power requirements: active or passive,
4. Applications: chemical, biological, IoTs, and wearables,
5. Conversion phenomenon: photoelectric, and electromagnetic,
6. Conversion principles: resistive, inductive, and magnetic,
7. Proprioceptive or exteroceptive: internal sensing, and external sensing,
8. Manufacturing methods: IC sensors, nanosensors, quantum sensors, and smart sensors, and
9. Sensor technology and construction: Sensor arrays, soft-sensors, and virtual sensors.

3.21.1 OPERATION OF SENSORS

In this book, we will not follow any of the above listed classifications. We will discuss sensors in an alphabetical order except for voltage measurements. Almost all sensor outputs are converted to voltage for the ease of signal processing and digital applications, hence it is worth providing information on the fundamental understanding on the “voltage.”

There are virtually thousands of different sensors, it is not possible to discuss all in one book.

3.21.1.1 Voltage Measurements

Voltage measurements involve determination of the electric potential difference between two points. The measurement unit for the voltage, in the International System of Units (SI), is the “volt,” which takes its name in the honor of the Italian physicist Alessandro Volta (1745–1827). The term voltage is also used to indicate the *electric potential*. The potential is the amount of work needed to move a unit charge from a reference point to another point against an electric field. The work done in moving the charge is always relative to another charge, hence the potential difference is always relative to some reference point. Often, the reference point is the earth, although other references can be used.

Since the potential difference is taken as the work per unit charge, the unit of potential (the volt) is related to the unit of work (joule – J) and the unit of charge (coulomb – C) by:

$$1 \text{ volt} = 1 \text{ joule} / \text{coulomb} \quad (3.1)$$

Although the concept of electric potential is useful in understanding electrical phenomena, it is worth noting that only the differences in potential energy are measurable. In general, it is commonly understood that the term voltage refers to the potential difference.

3.21.1.2 Acceleration Sensors

Acceleration sensors are also known as accelerometers. Accelerometer technology is well advanced, and they are manufactured as intelligent sensors using advanced IC technologies. Accelerometers are manufactured to be small, light in weight, and rugged and robust in operating in harsh environments. Acceleration sensors can be classed as *active* or *passive* accelerators. An active accelerometer (e.g., piezoelectric) gives an output without the need for an external power supply, while a passive accelerometer only changes its electric properties (e.g., capacitance) and requires an external electrical power supply.

Accelerometers can be classified in several ways, to be dynamic or kinematic types, and mechanical or electrical types. Dynamic accelerometers are based on the principles of measuring the force by a seismic mass tracking the motion of an accelerated base. Both mechanical or electrical types depend on the restoration forces that are based on mechanical properties, for example law of motion, or distortion of a spring, or fluid dynamics, or upon electrical or magnetic properties.

Seismic accelerometers make use of a seismic mass that is suspended by a spring or a lever inside a rigid frame. The frame carrying the seismic mass is firmly connected to the vibrating source whose characteristics are to be measured. As the system vibrates, the mass tends to remain fixed in its position so that the motion can be registered as a relative displacement between the mass and the frame. An appropriate transducer senses this displacement, and the output signal are processed further.

Piezoresistive accelerometers are essentially semiconductor strain gauges with large gauge factors. High gauge factors are utilized since the material resistivity is dependent primarily on the stress not only on the dimensions of the device. This effect can greatly be enhanced by appropriate doping of semiconductors such as silicon. The increased sensitivity is critical for vibration measurements since it allows miniaturization of accelerometers. Most piezoresistive accelerometers use two or four active gauges arranged as a Wheatstone bridge. Extra precision resistors are used, as part of the circuit, in series with the input to control the sensitivity, balancing, offsetting, and compensation of temperature effects.

Piezoelectric accelerometers are used widely for general-purpose acceleration, shock, and vibration measurements. They are basically motion transducers with large output signals for comparatively small sizes. They are available with very high natural frequencies and suitable for high-frequency applications and shock measurements.

Microaccelerometers are produced by planar processing and IC technology. Accelerometers and pressure sensors were among the first IC sensors produced. The technology has been progressing steadily and now an extremely diverse ranges IC-based accelerometers are readily available. Most sensors use bulk micromachining rather than using surface micromachining techniques. In bulk micromachining

the flexures, resonant beams, and all other critical components of the accelerometer are made from bulk silicon to exploit the full mechanical properties of the silicon crystals. With proper design and film processing, the bulk micromachining yields in stable and robust accelerometer.

An example of microaccelerometers is the vibrating-beam accelerometers, also termed resonant-beam force transducers which are made in such a way that acceleration along a positive input axis places a vibrating beam in tension. Thus, the resonant frequency of the vibrating beam increases or decreases with the applied acceleration. An example is mechanically coupled beam structure microaccelerometer also known as a double-ended tuning fork (DETf). An electronic oscillator capacitively couples the energy into two vibrating beams to keep them oscillating at their resonant frequency. The beams vibrate 180° out of phase to cancel reaction forces at the ends. The dynamic cancellation effect of the DETf design prevents energy from being lost through the ends of the beam. The acceleration signal appears as an output from the oscillator in the form of a frequency-modulated square wave, which can be used for direct digital interface.

There are many other IC accelerometers with operational principles very similar to capacitive force-balance or vibrating-beam accelerometers. The manufacturing techniques may change from one manufacturer to other. Nevertheless, in general, vibrating-beam accelerometers are preferred because of their superior air-gap properties and good bias performance characteristics.

The selective etching of multiple layers of deposited thin films, or surface micromachining, allows all kinds of movable microstructures to be fabricated on silicon wafers. A surface micromachined structure is shown in [Figure 3.7](#). The structure is formed by polysilicon and sacrificial materials such as silicon dioxide. The sacrificial material acts as an intermediate spacer layer and is etched away to produce a freestanding structure. Surface machining technology also allows smaller and more complex structures to be built in multiple layers on a single substrate.

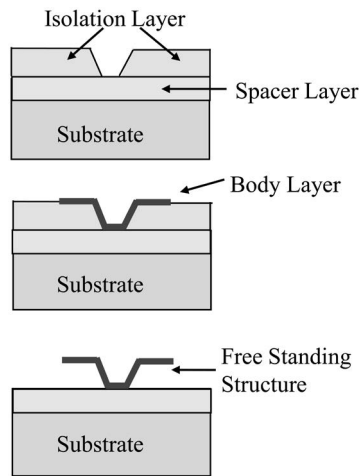


FIGURE 3.7 Steps of surface machining.

Vibrating beam accelerometers are similar in philosophy to the vibrating string accelerometers with an advantage that the frequency output provides easy interface with digital systems. Also, measurement of output voltage phase leads to velocity determinations by a simple step of integration. Static stiffness eliminates the tension and makes the device much smaller. The recent trend is that most vibrating beam accelerometers are manufactured as micromachined devices. With differential frequency arrangements, many common mode errors can be eliminated including the clock errors within the chip.

Surface micromachining has also been used to manufacture specific application accelerometers, such as those suitable for air-bag applications in the automotive industry. In one type of such accelerometers, a three-layer differential capacitor is created by alternate layers of polysilicon and phosphosilicate glass. A silicon wafer serves as the substrate for the mechanical structure.

Differential-capacitance accelerometers are based on the principle of the change of capacitance in proportion to applied acceleration. They come in different shapes and sizes. In one type, the seismic mass of the accelerometer is made as the movable element of an electrical oscillator as shown in Figure 3.8. The seismic mass is supported by a resilient parallel-motion beam arrangement from the base. The system is set to have a certain defined nominal frequency when undisturbed. If the arrangement is subjected to acceleration, the output voltage frequency varies above and below the nominal value depending on the direction of acceleration.

3.21.1.3 Acoustic Sensors

Acoustic sensors make use of vibrations of acoustic (e.g., sound, ultrasound) pressure in air, liquid, or solids. Sound vibrations are commonly sensed by piezoelectric sensors. But there are many other sensors based on capacitive or inductive properties of devices. For example, moving coil inductive sensors give good responses at low frequencies compared to piezoelectric types. Apart from the vibration characteristics, the sound has unique properties that needs to be explained.

Sound is defined as the vibrations of solid, liquid, or gaseous medium in the frequency range of 20 Hz to 20 kHz, which can be detected by the human ear. Sound travels in a media by obeying the laws of shear and longitudinal forces. In contrast to the solids, the liquid or gaseous media cannot transmit shear forces, therefore in

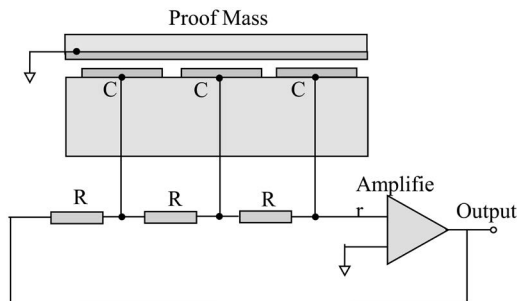


FIGURE 3.8 A typical differential capacitive accelerometer.

these media the sound waves are longitudinal, that is the particles move in the direction of the propagation of the wave.

In sensing of sounds, there are three important parameters: velocity, frequency, and wavelength. The frequency is determined by the source, but the velocity and the wavelength are determined by the medium of propagation. The velocity of sound depends on the density and elasticity properties of the material. The velocity is high in the dense solids and relatively lower in the gaseous environment.

In practice, the sound pressure is almost always the only parameter measured directly. All other parameters such as sound power, particle velocity, reverberation time, and directivity are derived from the pressure measurements. The sound pressure measurements are performed by microphones in the gaseous media and hydrophones in the liquid media. Microphones can be produced as capacitive, piezoelectric and electret, fiber-optics, carbon, moving iron (variable reluctance), and moving-coil microphones.

The characteristics of all microphones are analyzed for both acoustical and electrical properties. The overall sensitivity is expressed in millivolts or microvolts per unit intensity of the sound wave. Microphones can be directional or omni-directional. If a microphone responds to the sounds coming from any direction it is said to be omni-directional.

The most used microphones for sound measurements are capacitor types, piezoelectric, electret and fiber-optic microphones. They tend to have good stability and sensitive, precise, and good temperature properties. Other types of acoustic sensors are carbon, moving-iron and moving-coil types, geophone, hydrophone, seismometer, and sound locator, but they will not be discussed further in this book.

3.21.1.4 Amorphous Materials

Amorphous materials are used as sensors since they have good magnetic properties, that can be used in acceleration, vibration, pressure, rotation, torque, and temperature sensing. Amorphous sensors are manufactured in the form of powders, ribbons, or flakes by using methods such as evaporation and vapor deposition, ion mixing or ion implantation, and liquid quenching. The vapor deposition technique is used in semiconductor-type sensors. Amorphous materials can be classified as amorphous alloys (Fe, Co, Ni), amorphous rare earths, and amorphous superconductors. Amorphous alloys have good soft material properties and are extensively used in magnetic heads as sensors. Thermal dependence of amorphous alloys is a major drawback thus preventing their wider applications.

On the other hand, amorphous superconductors are a class of superconducting materials such as bismuth and gallium. These materials are used in high field magnets, memory devices, and some computer applications. Many of the amorphous metallic films are based on rare earth 3D transition metal alloys, such as Gd_xCo_x (gadolinium, cobalt).

3.21.1.5 Biomedical and Biological Sensors

Biomedical and biological sensors are well developed and extensively used in biomedical and biological applications. Traditionally, majority of biological detectors required human intervention in a laboratory environment, but in recent years, fully

automatic devices and robots are involved in biological applications. As an example of biomedical and biological sensors, detection of micro-organisms in air can be determined in three different methods:

1. *Biochemical*, which detect a DNA sequence and protein that are unique to a bioagent through its interaction with test modules.
2. *Chemical*, such as the mass spectrometry, which work by breaking down a sample into its components such as amino acids and then comparing their weights with those of known bioagents and other molecules.
3. *Biological tissue-based systems*, in which a bioagent or biotoxin affects live mammalian cells, causing measurable responses.

Some classes of detectors rely on comparing the DNA taken from an organism with the DNA of a known agent. These detectors are built on the prior knowledge of the bioagent. In automatic systems, a detector collects samples, then the multiple copies of the DNA are made by using a polymerize chain reaction principles (PCR). In this method, the samples are heated and cooled periodically to produce copies of the DNA. As the DNA is copied, the resulting strands are mixed with fluorescent DNA probes. The probe bound with the bioagent glows under ultraviolet light. By using different probe markers several PCR reactions occur at the same time, thus enabling detection of more than one agent simultaneously.

In IC-based devices, there are microfluidic devices, which contain tiny channels, valves, and chambers in a single chip. Once the microorganism is in the chip, their cells can be open ultrasonically. The PCR is applied by means of small thin film heaters.

Apart from sonic, optical, and fluorescent methods, magnetic methods, illustrated in Figure 3.9, are used to detect DNA. These devices comprise an array of wire-like magnetic field microsensors. These sensors are coated with single-stranded DNA probes specific for a gene from a bioagent. Once a strand of bioagent DNA in a sample bind with a probe, the resulting double strand binds a single magnetic microbead. The resistance of sensor decreases in proportion to the number of microbeads.

Some sensors use fluorescent antibodies to bind to bacterial cells. In these devices the sample passes through a portable flow cytometer, which counts cells by measuring their fluorescent or other properties as they move in a liquid.

Another commonly used method in biodetectors is the live tissues. Many toxins trigger measurable or differentiable reactions in living cells. Mammalian cells, such

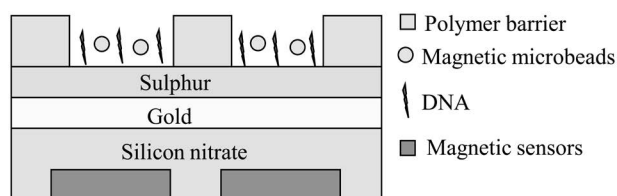


FIGURE 3.9 A magnetic biosensor.

as heart cells, those are cultured in a lab and then seeded into a cartridge containing a microelectrode array. When biotoxin is introduced, the cells create voltages in millivolts detectable at the electrodes.

3.21.1.6 Capacitive Sensors

Capacitive sensors are based on the characteristics of electrical charges and associated electric field properties. Capacitive sensors rely on the changes in capacitances in response to physical variations. The changes in the capacitance can occur as a response to variations such as the area or the distance between the plates or the variations in the dielectric properties of the material between the plates. In instruments, capacitive sensors find many diverse applications from humidity and moisture sensing to displacement sensing.

Integrated circuit capacitors are made primarily as MOS ICs and monolayer versions contain tantalum or other suitable deposits. Two heavily doped polysilicon layers formed on a thick layer of oxide generally form the plates of capacitors. The dielectric is usually made from a thin layer of silicon oxide. Important parameters for IC capacitors are the tolerances, voltage coefficients, temperature coefficients, and the capacitance values. IC capacitive sensors are achieved by incorporating a dielectric material sensitive to physical variables.

Capacitive position, displacement and proximity sensors consist of two parallel plates with capacitance C . The capacitance is a function of the distance d (cm) between the electrodes of a structure, the surface area A (cm²) of the electrodes and the permittivity ϵ (=8.85 pF/m for air) of the dielectric between the electrodes, therefore:

$$C = f(d, A, \epsilon) \quad (3.2)$$

Variations due to an external stimulus in distance between the plates or area of the plates or dielectric constant can be detected.

Capacitive proximity sensors are based on stray capacitance that exists between a metal plate and earth, a quantity that can be altered by the presence of materials, which can be solid, liquid or powder. Often, bridge or resonant circuits are needed for signal processing.

3.21.1.7 Chemical Sensors

Chemical sensors are mostly based on IC technology. A commonly used sensor makes use of changes in resistivity of metal oxides in response to reducing or oxidizing properties of the chemical environment around the sensor.

Chemical sensors constitute large and important classes of instruments and instrumentation. They find wide applications in air and water quality measurements, pollution level determination, detection of chemical and gas leaks, determination of toxicity levels, prospecting of minerals in mining and metallurgy industries, detecting explosives, sensing drugs, fire warning, and many different types of domestic, medical, industrial, and health and safety related applications. They are used for identifying chemical compounds and elements in environmental applications, and food processing.

Chemical sensors are classified as contact types and non-contact types. The contact types involve the physical interaction of the sensor with the chemicals under observation. The chemicals can be in gas, or liquid, or solid, or in mixture forms. The non-contact types use remote sensing techniques, which are based on the analysis of gaseous part of the chemical substances under investigation.

Chemical sensors have properties of *selectivity*, which is defined as the ability to primarily respond to only one chemical element or species (compounds) in the presence of other species. Therefore, the performance of chemical sensors is evaluated by its selectivity or its ability to eliminate or minimize the effect of interfering species.

There are many different methods of sensing chemicals, some of which are: catalytic sensors, enzyme sensors, chemosensors, chromatographic sensors, electrochemical sensors such as potentiometric and amperometric sensors, radioactive sensors, thermal sensors, optical sensors, mass sensors, concentration sensors, titration sensors, and semiconductor sensors. Out of this vast repertoire of chemical sensors only few examples are discussed below.

Enzymes sensors are used to sense enzymes that are special type of catalyst, proteins found in living organisms. Enzymes exist in aqueous environment in the form of immobilization matrices as gels or hydrogels. As sensors, enzymes tend to be very effective in increasing the rate of some of chemical reactions and are strongly selective to a given substrate. Figure 3.10 illustrates a typical enzyme sensor. The sensing element can be a heated probe, or an electrochemical, or an optical sensor. The enzyme, acting as a catalyst, immobilized inside the layer into which substrate diffuses, hence it reacts with the substrate and the product is diffused out of the layer into the sample solution.

Catalytic chemical sensors are based on liberation of heat because of catalytic reaction taking place at the surface of the sensors. For example, when the combustible gas reacts at the catalytic surface, heat increases the temperature to be detected by suitable temperature sensors such as platinum types or other resistive thermo detectors (RTDs). As the generated gas contacts the surface of the platinum filament it becomes oxidized without combustion. The heat of the oxidation raises the filament temperature, and the heat change is detected as an increase in its electrical

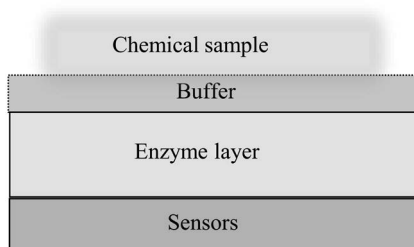


FIGURE 3.10 Arrangement of an enzyme sensor.

resistance. Some of the catalytic sensors contain heating elements and temperature sensors employed side by side.

A different type of catalytic sensor is the Taguchi sensor that is used often to detect carbon monoxide or natural gases. It consists of a heating element in a hot tin-oxide pellet. When gases encounter the hot pellets, their combustion causes electrical resistance to change. The current flow through the pellets becomes proportional to the combustion rate.

There are catalytic sensors that can detect *flowing gases* and *fluids*. A typical example is the thermal conductivity sensor, which is a warm filament in a stream of flowing gas. The loss of heat from the filament depends on the flowrate of the gas or fluid and thermal conductivity properties of the arrangement.

Electrochemical sensors are the most used chemical sensors. They can be classified as voltage sensors (potentiometric), current sensors (amperometric), resistivity or conductivity sensors (conductometric), and ion sensitive sensors. In all cases special electrodes are used in a closed electrical circuit form. Modern electrochemical sensors are produced from semiconductors.

Chromatographic sensors sense the chemical samples are introduced in the form of a gaseous or vaporized mixtures into a carrier gas stream, which passes in a suitable column over a solid adsorbing surfaces or over non-volatile liquids applied to solid surfaces. The individual components of the sample move through the column at rates corresponding to their different distribution coefficient factors between fixed and mobile phases. Detectors that are based on different operational principles such as: thermal conductivity, electron capture, and so on.

Radioactive chemical sensors rely radioactive methods of analyte analysis. They are mostly used in smoke alarms. In a typical arrangement, two metal plates are located inside a chamber, a small radioactive source emits alpha radiation that ionizes the air in between the metal plates and allows a current to flow. If chemicals/particles enter the space between the plates and bind with the ions, then the amount of normal current is altered thus triggering an alarm. This method may be applied only to certain analytes.

Chemosensor is a molecule that can bind selectively and reversibly to the analyte of interest. A chemosensor binds or/and unbinds easily with the dissolved gas molecules of interest. The constitutes and physical properties of chemosensor may be unique to each analyte depending on the chemical composition. That is, a different chemosensor will be required for each analyte. The analyte can be made from various compounds, for example methane.

Light-based chemical sensors use the properties of interaction of electromagnetic radiation with matter. As a result of this interaction some properties of light is altered, such as intensity, polarization, and velocity. Optical chemical sensors are designed and built-in variety of ways. A typical semiconductor-based optical sensor is illustrated in [Figure 3.11](#). This sensor consists of two chambers, one acting as the reference containing a reference concentration chemical and the other as the sensor. These chambers are illuminated by a common light emitting diode (LED). The surfaces of chambers are metallized to improve internal reflectivity and the bottom of chambers are covered with glass. One of the chambers has slots covered with gas

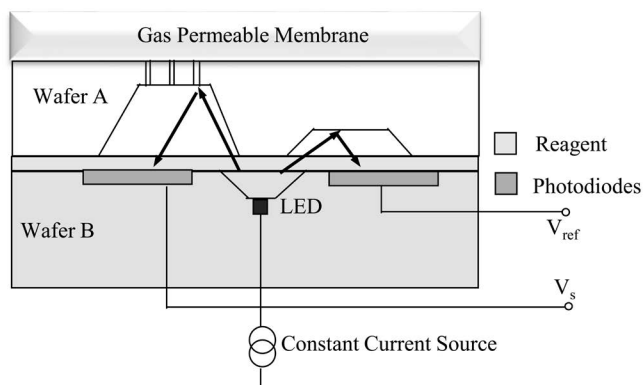


FIGURE 3.11 An optical semiconductor chemical sensor.

permeable membrane. The slots allow the penetration of gases for being measured (e.g., CO_2) into the chamber. Wafers A and B form optical waveguides. The chamber filled with reagent is used to monitor the optical absorbency for comparison with that obtained from the reference chamber.

Luminescence principle is one method of chemical detection. In this case, a solution is used that allows “analyte” to be dissolved into the solution. In the bound state certain molecules become luminescent. This means that the bound molecules will absorb and re-emit certain spectrum of light. When the solution is exposed to light the amount re-emission is directly proportional to the number of bound molecules, which is proportional to the presence analyte in the medium. By measuring this luminescence with a photometer, the presence of analyte can be determined. In the predictions, the probability theory is used and the known concentration of the chemosensor are related to the estimated amount of analyte.

Photoionization is another type of optical chemical sensors respond to many different gases. They operate on the principles of ultraviolet light generated by the gases under investigation. Although they require high voltages and specially constructed tubes, they still find wide applications in gas analyzes.

Absorption and emission chemical sensors act as a binding agent of analytes to some chemosensors. The absorption and emission spectrum gives distinctive bands of colors, which may be analyzed and compared against known spectra for the determination of the chemical composition. This method is also used in analysis of gases, such as oxygen.

Spectrometric chemical sensors consist of a beam of light aimed through the gas or fluid. The intensity of light emerging from the other side is measured. The decrease in the intensity is related to the color or cloudiness of the sample under investigation. Some typical applications are the measurement of the sooth and smoke in air or sediments in flowing waters.

Colorimetric chemical sensors convert the substances into colored compounds, and the intensity of color is related to the amount of chemical under investigation. A large number specific and very sensitive ranges color reactions are available for

comparative and accurate measurements. The measurements may be conducted with spectrophotometry of pure monochromatic light, which is obtained by high resolution of white light with the aid of suitable prism mechanisms.

Other chemical sensors worth mentioning are breathalyzers, chemical Hall effect sensors, fluorescent sensors, holographic sensors, microwave chemistry sensors, olfactometers, zone sensors, redox electrodes, zinc oxide nanorod sensors, and others.

3.21.1.8 Density Measurements and Density Sensors

Density measurements and density sensors measure the mass per unit volume ($\rho = m/V$) of materials under fixed conditions. The term applicable to solids, liquids, and gases. The density depends on temperature and pressure. This dependence is much greater in gases. Although, there are many different units in use, usually the values of densities are given in terms of grams per cubic centimeter.

Specific gravity (SG): is an abstract number expressing the ratio of the density of one substance to the density of another reference substance, both obtained at the same temperature and the same pressure. For solids and liquids water is taken as the reference substance. Air under specified conditions is the reference for gases. The specific gravities of solids, liquids, and gases may be expressed by:

$$\text{Liquid (or solid) SG} = \frac{\text{density of liquid (or solid)}}{\text{density of water}}$$
$$\text{Gas SG} = \frac{\text{density of gas}}{\text{density of air}}$$

Care must be taken to define standard conditions under which the densities or specific gravities are measured, so as not to introduce unwanted errors. Commonly accepted sets of conditions are *normal temperature and pressure* (NTP) and *standard temperature and pressure* (STP). NTP is usually used for solids at the temperature of 0°C with a pressure of 760 mm mercury. The STP is used for solids and fluids at a temperature of 15.6°C with a pressure of 1.0 atm (or 101.325 kPa).

In many industrial processes the density is taken as the controlling parameter for the rest of the process, therefore accurate measurements are necessary. Density measurements are made for at least two important reasons: 1) to determine the mass and volume of products and 2) to assess the quality of products. In some industrial applications density measurements ascertain the value of the product.

The densities are commonly determined by sampling techniques. There are two basic sampling techniques the *static density measurements* and *dynamic (on-line) density measurements*. Within each concept there are many different methods available depending on physical principles and process characteristics. In many cases, application itself and conditions determine the best suitable method to be employed. Generally, static methods are well developed, lower in cost and more accurate. Dynamic samplers can be expensive requiring a high degree of automation. Nevertheless, many dynamic methods are computerized, offering easy to use, flexible, and self-calibrating features.

There is no single universally applicable density measurement method. Measurements of densities of fluids are much more complex than solids, therefore there are many different techniques developed. Hydrometers, pycnometers, hydrostatic weighing, flotation methods, drop methods, radioactive methods, and optical methods are typical examples of measuring liquid densities. Flask, gas balance, optical, and x-ray methods are typical techniques employed for gas density measurements.

Magnetic density sensors are preferred methods for liquids and gases. These methods allow the determination of the effects of pressures and temperatures down to cryoscopy range. Basically, these devices contain a small ferromagnetic cylinder, encased in a glass jacket. The jacket and ferromagnetic material combination constitutes a buoy or float. The cylinder is held at a precise height within the medium by means of a solenoid, which is controlled by a servo system integrated with a height sensor. The total magnetic force on the buoy is the product of induced magnetic moment by the solenoid and the field gradient in vertical direction. The magnetic force at a particular distance in vertical direction in the solution compensates the difference in the opposing forces of gravity (downwards) and buoyancy (upwards) exerted by the medium, through the Archimedes' principle. The magnetic force is directly proportional to the square of the current in the solenoid. If the buoyant force is sufficient to make the ferromagnetic assembly float on the liquids of interest, the force generated by the solenoid must be downward to add to the force of gravity for equilibrium.

Vibration density sensors make use of the changes in natural frequency of vibration of a body containing fluid in it or surrounded by it. The natural frequency of the vibrating body is directly proportional to the stiffness and inversely proportional to the combined masses of the body and the fluid. It also depends on the shape size and elasticity of materials, induced stresses, the total mass, and mass distribution of the body. There are different types of vibration density sensors, such as the vibrating tube, vibrating cylinder, and tuning fork densitometers. These densitometers make use of the natural frequency of a low mass tuning fork. In some cases, the liquid or gas is taken into a small chamber in which the electromechanically driven forks are situated. In the other cases, the fork is inserted directly into the liquid.

Pycnometers are static devices used for measuring densities of liquids and gases. They are manufactured as fixed volume vessels, which can be filled with sample fluids. The density of the fluid is measured by weighing the sample with the vessel. The simplest version consists of a glass vessel in the shape of a bottle with a long stopper containing a capillary hole. The capillary is used to determine the exact volume of the liquid thus giving high resolution. Many pycnometers are made from glasses, but some are made using metals for enough strength against extreme and high pressures. In some rugged processes, metal pycnometers are used to take samples during a dynamic operation.

Hydrostatic weighing densitometers are suitable for solid and liquid density measurements. The density of solid is often measured by weighing it first in air, and then weighing again after having submersed it in a suitable liquid of known density. The latter weighing is done by suspending the solid under the pan of a precision balance by means of a very thin wire.

The hydrostatic weighing methods of liquids give continuous reading for two phase liquids such as slurries, sugar solutions, and powders. They are rugged, yield accurate results, and are used for the calibration of the other liquid density sensors. However, they must be installed horizontally on a solid bases hence they are not flexible enough to adapt for all processes, thus the processes must be designed around it.

Hydrometers are direct reading instruments, most used devices for measurement of density of liquids. Buoyancy principle is used as the main technique of operation. Almost all hydrometers are made from high-grade glass tubing. The volume of fixed mass is converted to a linear distance by a sealed bulb shaped glass tube containing a long stem measurement scale. The bulb is ballasted with a lead shot and pitch, the mass of which is dependent on the density range of the liquid to be measured. The bulb is simply placed into the liquid and the density is read from the scale. The scale may be graduated in density units such as kg/m^3 .

Column type densitometers are used for liquid density measurements. There are several different versions of column type densitometers. In a version, a known head of sample liquid and water from their respective bubbler pipes are used. A differential pressure-measuring device compares the pressure differences, proportional to relative densities of the liquid and the water. By varying the depth of immersion of the pipes a wide range of density measurement can be obtained. Both columns must be maintained at the same temperature to avoid the temperature effects.

Refractometric method is suitable for density measurement of gases and clear liquids. They are essentially optical instruments operating on the principles of refraction of light traveling in liquid or gas media. Depending on the characteristics of the samples, measurement of refractive index can be made in a variety of ways; critical angle, collimation, and displacement techniques are few methods to mention. Usually, an in-line sensing head is employed whereby a sensing window, commonly known as a prism, is wetted by the product to be measured. In some versions, the sensing probes must be installed inside the pipelines or in tanks or vessels. They are most effective in reaction type process applications where blending and mixing of liquids takes place. For example, the refractometers can measure dissolved soluble solids accurately.

Infrared diodes, lasers and other lights may be used as sources. However, this measurement technique is not recommended in applications in processes containing suspended solids, high turbidity, entrained air, heavy colors, poor transparency, and opacity or extremely high flow rates. The readings are automatically corrected for variations in process temperature. The processing circuitry may include signal outputs adjustable in both frequency and duration.

Absorption-type densitometers are also used in specific situations, particularly in dynamic applications. x-rays, visible light, gamma rays, ultraviolet (UV) and sonic absorptions are typical examples of this method. Essentially, attenuation and phase shift of a generated beam going through the sample is sensed and related to the density of the sample. Most absorption type densitometers are custom designed suitable to characteristics of the process. Two typical examples are: 1) UV absorption or x-ray absorptions which are used for determining the local densities of mercury deposits in arc discharge lamps and 2) ultrasonic density sensors are used in connection with difficult density measurements such as: density measurement of slurries. The lime

slurry, for example, is a very difficult material to handle. It has a strong tendency to settle out, conglomerate, and to coat all equipment it meets. An ultrasonic density control sensor can fully be emerged into agitated slurry, thus avoiding the problems of coating and clogging. Since, the attenuation of the ultrasonic beam is proportional to the suspended solids, the resultant electronic signal is proportional to the specific gravity of the slurry. Such devices can give accuracy up to 0.01%. The ultrasonic device measures the percentage of the suspended solids in the slurry by providing a close approximation of the specific gravity.

3.21.1.9 Distance Sensors

Distance sensing methods are applied from picometer distance measurements to measurements of distances of galaxies, and indeed distances of the limits of observable universe. Distance measurement requires the determination of the position of an object relative to another object. Position of an object implies the location of the object in reference to a point. Determining the position of the object makes use of distance as well as directional information such that, the position can be specified in rectangular or polar coordinates in two or three dimensions and time.

There are many different methods of sensing distances in relation to stationary or moving objects or to stationary or moving reference points. If the distances are stationary relative to a fixed point, this requires different methods compared to objects in motion. The tools and techniques of distance measurement are one of humans longest running inventive pursuits. Figure 3.12 illustrates the range of distances that science and engineering have an interest in measuring. Generalization can be made based on these broad classes. Consequently, there is an extremely rich and diverse methods of distance determination depending on objects’ distances, physical properties, and opted measurement techniques.

In position determinations the *direction*, which is vector quantity, is an important property. In the large-scale position sensing, the sensing of direction on the Earth’s surface can be achieved by using the Earth’s magnetic field, by gyroscopes, radio signals, and satellite systems. One of the most ancient methods is the traditional *compass* that uses the effect of the Earth’s magnetic field on a small-magnetized needle, which is freely suspended. In this arrangement, the needle points along the line of the field, in the direction of magnetic north and south. The use of compasses is enhanced by electronic means. The needle position within the device can be sensed

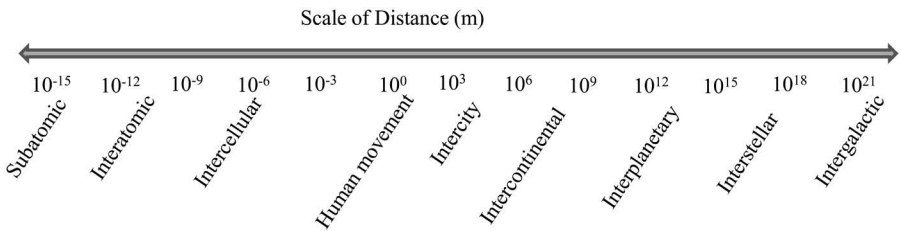


FIGURE 3.12 Range of measurable distances (Source: J. Ballantyne⁽²¹⁾).

by various methods such as the Hall-effect sensors to convert the position of the needle to electrical signals. By using A/D converters and digital techniques, directions can be read out numerically in degrees, minutes, and seconds.

Gyroscopes are alternative devices that are used for direction determination. A basic gyroscope does not have an inherent electric output. But the electric output can be obtained by placing slight mechanical loading on the gyro wheel or totally contactless by optical means.

Radio waves have been used for position determination and navigational purposes in the form of radio beacons that are used in much the same way as light beams.

Large-Scale Distance Sensors are designed to measure distances of objects relative to a fixed point. They are primarily based on the reflections of optical, radio, or sonar waves from the object. The basic principle is that a pulse of a few waves is sent out from the transmitter, reflected at some distant object and the returning signals are detected. Since the speed of the wave in the propagation media is known, the distance of the reflector can be calculated from the time that elapses between the sender and receiver. This time can be very short (few nanoseconds to few microseconds) so that the duration of the wave pulse must be measured accurately. Large-scale distance measurement leads to the sensing of distance traveled by the object. These types of devices are predominantly used in aviation for target detection such as the radar, land surveying, satellite-based detection systems, military applications, and road traffic control. The speeds of moving objects can easily be determined from continuous measurements of distances, as in the case of detection of speed of vehicles.

Small-Scale Sensors are mostly based on resistive, capacitive, or inductive principles in industrial types, while others make use of optical interferometers and millimeter-wave radar techniques.

A simple way of sensing small scale distance is using potentiometers. The moving object is connected to the slider of the potentiometer, so that each position along the axis corresponds to movement of the object. Alternative methods are capacitive, induction or other magnetic sensors. For very small distances, the strain gauges can be used. Laser interferometers are applicable when small precise changes must be sensed. One advantage of the laser interferometer is that the output can readily be converted to digital forms since it is based on counting the wave peaks.

Linear digital encoder gives an output, which is binary number that is proportional to the distance of the encoder relative to a fixed point. They provide a set of binary digital signals directly, but since they require good light sources, appropriate light detectors, and precise printed patterns on slides.

3.21.1.10 Eddy Current Sensors

Eddy current sensors are essentially inductive sensors. They have two coils in the shape of probes. One of the coils, known as the active coil, is influenced by the presence of the conducting target. The second coil, known as the balance coil, serves as an arm of a bridge that balances the circuit. The second coil also provides temperature compensation. The magnetic flux from the active coil passes into the conductive target from the probe. When the probe is brought close to the target, the flux from the probe links with the target, producing eddy currents within the target. The eddy

currents in the target alter the magnetic properties of the sensors, thus yielding in unbalanced conditions of the connected bridge on the balance coil.

3.21.1.11 Electric Current Sensors

Electric current sensors are based on charged bodies in an electric field. If a charged body is not restrained, it will start moving in an electric field. The result of the movement of charged bodies from one point to another point in the space is the electric current. If metallic conductors, such as copper, are used in electric circuits so that electrons can flow through the conductors. A basic property of an electron is its charge (1.603×10^{-19} C). The electric current is the rate of charge flowing in a cross section of the conducting element in one second measured in the SI units in “amperes” (A), which takes its name in the honor of the French physicist André-Marie Ampère (1775–1836).

One of the simplest methods of current sensing is the current-to-voltage conversion by means of resistors. In current measurements, the resistor is called the shunt resistor, despite it is connected in series with the load. The shunt resistor produces a voltage output that can be measured by a variety of secondary meters, such as analog meters, digital meters, and oscilloscopes. In addition, there is a range of non-contact electric current-based sensors that are available such as Hall effect probes and current transformers.

3.21.1.12 Electromagnetic Field Sensors

Electromagnetic field sensors are frequently used in aviation, communication systems, satellites, geomagnetic studies, navigation, manufacturing processes, and medicine. Radar dishes, satellite receivers, and antennas are some examples of electromagnetic field sensors.

3.21.1.13 Environmental Sensors

Environmental sensors are essential to monitor environmental condition for climate changes which are highly political and social sensitive issues. There are many methods, such as electromagnetic field sensors, humidity and moisture sensors, acidity, and alkalinity sensors, actinometers, ceilometers, hygrometer, pyrgeometer, pyranometer, weather radar, so forth, some of which are explained in chemical sensors.

3.21.1.14 Fire and Smoke Sensors

Fire and smoke sensors find many industrial and domestic applications. These sensors operate on two main principles: the ionization, and the optical detection. The ionization detector uses a radioactive source, usually americium-241, with a low activity level, typically 0.8 μ Ci. In the construction of sensors, there is a gap between source plate and an electrode normally conducting state due to emission of alpha particles. The ion current is very small, about 10 pA. In the presence of smoke, molecules entering the ionization chamber are struck by the alpha particles. The molecules capturing alpha particles cause a reduction in the current that activates the alarm.

The optical type of detector operates on Tyndall effect of scatter. When a beam of light passes through clean air, the beam is invisible, therefore looking along an axis at right angles to the beam results in no detectable light. In the presence of smoke, light is scattered, hence there is an appreciable amount of light visible on the transverse axis. The scattered light in the transverse direction is detected and resulting electrical signals are processed appropriately.

3.21.1.15 Force Sensors

Force sensors make use of physical behavior of the body under external force. Force can be measured by many different methods. Some of these methods are balancing the force against a standard mass through a system of levers, measuring the acceleration of a known mass, equalizing the force to magnetic force by suitable coil and magnet arrangements, measuring the pressure caused by the force on a surface, and converting the force into the deformation of some elastic elements.

Consequently, there are a vast range of sensor suitable for force measurements. Examples are strain gauge load cells, piezoelectric sensors, resistive methods, force sensing resistors, conductive polymers, magnetoresistive force sensors, inductive force sensors, magnetoelastic force sensors, piezotransistors, and capacitive force sensors.

3.21.1.16 Gas Sensors

Gas sensors could be included in the section of chemical sensors since gas, in chemical terms, is any matter that is in the gaseous state, characterized by high molecular kinetic energy and a tendency to expand out of an open container.

In practice, the word gas takes distinct and separate meanings. For example, natural gas is a fossil fuel that is collected in the gas phase, and the gasoline, which is also called gas, is a liquid mixture of hydrocarbon fuels that become gaseous in the combustion chamber of an engine. There are two types of gases *inorganic gasses*, and *organic gasses*.

1. *Inorganic gases* are gases, such as sulfur dioxide, hydrogen sulfide, nitrogen oxides, hydrochloric acid, silicon tetrafluoride, carbon monoxide and carbon dioxide, ammonia, ozone, and so on.
2. *Organic gases* are gases, such as hydrocarbons, terpenes, mercaptans, formaldehyde, dioxin, fluorocarbons, and others.

Modern gas detectors are based on semiconductor technology and microprocessors. In recent years, the functionality of gas detectors has increased considerably, and most gas detectors can detect simultaneously multiple types of gases. A MOS catalytic (CAT) hydrogen gas sensor is shown in [Figure 3.13](#). A hydrogen sensitive catalytic metal, such as platinum or palladium, is deposited on the oxide layer. In presence of hydrogen, the current-voltage (I-V) characteristic of the MOS transistor change. This change is detected indicating the presence of hydrogen.

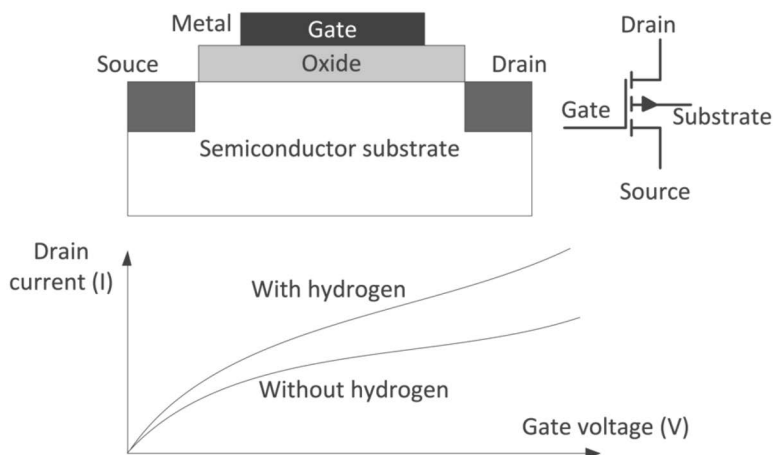


FIGURE 3.13 MOS transistor structure and the I-V for hydrogen detection (Source: H. Matbouly⁽²²⁾).

Popular methods of detection of gases are the infrared and fluorescent spectrometry, light absorption methods, colorimetry or photometry, chromatographic methods, flame ionization, indicator tubes, test papers, titration, semiconductor methods with electric conductivity and amperometric methods, and ring oven methods. Some of these methods have already been explained in the chemical sensors section. The most used other gas sensors will briefly be visited next.

Infrared spectrometry is based on the absorption of infrared energy. This method is suitable for both automatic and non-automatic measurements. This technique involves the determination of the differences in the absorption of infrared energy between gas samples. The sensor contains a gas sample path and a reference path. The sample and reference paths are separated in time or space. In one version, the device contains two cells one for sample gas and the other for the reference gas. A nichrome radiation source, an infrared optical system, a detector mechanism, and an appropriate electronic circuitry for signal processing is included in the same unit.

Fluorescence spectrometry on the other hand measures the fluorescent light emitted by certain molecules of gas when excited by a radiation source of appropriate energy or wavelength. Organic molecules containing conjugated double bonds are the most encountered fluorescing materials, hence are suitable for application of this method.

Indicator tube gas sensors are inexpensive devices that are used to detect gases in the atmosphere. A pollutant gas causes a change in the color of a reagent packed in a tube. Comparison of change in color with the standard colors indicates the amount of concentration of the pollutant gas. For quantitative results, a known volume of air must be drawn through the tube at a fixed rate. Influence of other gases must be eliminated by the correct choice of the reagent. Potential sources of errors are the variations in tube diameters, temperature changes and inaccurate estimation of color intensity. Possible inconsistencies in the manufacturing in the reagents must

also be considered. Indicators may contain tubes with several layers of impregnated compounds to remove other interfering substances. Many indicator tubes come in kit form containing pumps and other necessary ancillaries.

Semiconductor gas sensors are based on semiconductors to sense different types of chemicals and gases. They are classified as *conductometric*, *potentiometric*, and *amperometric* sensors.

Conductometric sensors are based on the changes of electrical resistance or conductivity of a material due to presence of some chemical elements or compounds. The conductivity changes due to traces of ions generated or carried by the chemical substance under investigation. Pure water for example is a poor conductor but a small trace of ions in the water increases its conductivity considerably. Many conductivity sensors can detect the concentration of ions as well as the type of ions since different ions affect conductivity differently. In many cases, only very small liquid volumes of samples are required for measurable changes in conductivity.

Amperometric gas sensors are based on the generation of electric current or potential by free elements or compounds. For example, sulfur dioxide and other reducing gases in air sample react with electrically generated free iodine or bromine in a detection cell. The cell has an anode and a cathode electrode and contains a constant titrant concentration. Any sulfur compound introduced into the cell reacts with titrant and changes the concentration to produce potential change in the solution to be picked up by electrodes. Range of a typical instrument can be 0.0001 ppm to 0.2 ppm.

Potentiometric chemical or gas sensors act as an electric battery. A typical example of potentiometric sensors based on semiconductors is given in Figure 3.14. These sensors are based on the use of ion selective field-effect transistors (ISFETs). They are derived from metal-oxide semiconductor FETs or MOSFETs by hydrogen ion sensitive elements. These silicon chips contain a pH responsive membrane much like glass electrodes. The signals are amplified and processed on the same chip.

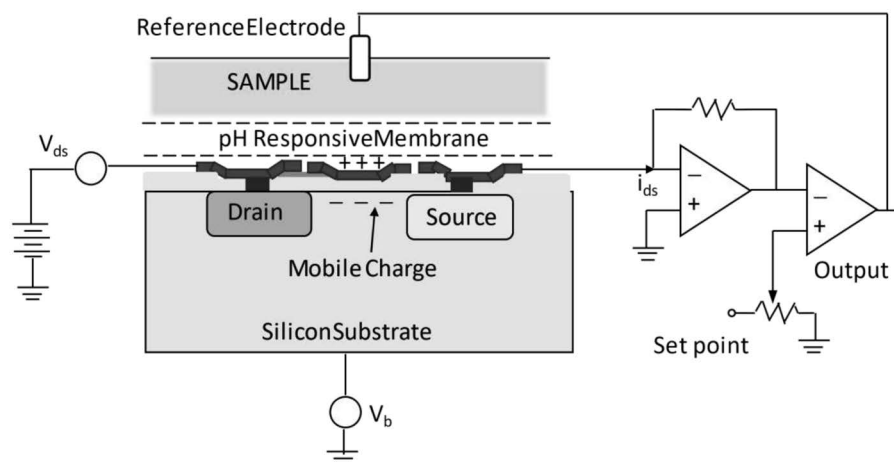


FIGURE 3.14 A pH sensor based on ion-selective FET.

In these sensors, the potential or current signals generated by the electrochemical process between the solution and the sensors are processed. Potentiometric sensors have two important characteristics: 1) very small current needs to be drawn during measurements thus implying alterations in the characteristics and 2) the response is not linear but usually logarithmic.

3.21.1.17 Global Positioning Systems

Global positioning systems (GPS) are the second-generation satellites evolved from the Naval Global Positioning System. Twenty-four operational satellites constitute the system. Each satellite orbit is circular, about 22,200 km high above the earth's surface. They are inclined at angles of 55° with respect to earth's axis.

The position determination using the GPS system is based on the ability of the receivers to determine accurately the distance to the GPS satellites above the user's horizon at the time of fix. When accurate distances of two such satellites and the heights are known, then the position on Earth can be determined. To do this, the receiver needs to know the exact time at which the signal was broadcasted and the exact time that it has received. If the propagation speeds of the electromagnetic waves generated by the satellites through the atmosphere are known, the resulting distance can be calculated. The measured ranges are called *pseudoranges*. Normally, information is received from at least four satellites leading to accurate calculations of the fix. The time errors plus propagation speed errors result in range errors, common to all GPS receivers. The time is the fourth parameter that is evaluated by the receiver if at least four satellites can be received at a given time. If a fifth satellite is received, an extra error matrix can be evaluated for better accuracy.

Each GPS satellite, broadcasts simultaneously on two frequencies for the determination and elimination of ionosphere and other atmospheric effects. The frequencies of transmission are at 1,575.42 MHz and 1,227.6 MHz designated as L1 and L2 in L-band of UHF frequency range. Both these signals are modulated by 30 sec navigation messages transmitted at 50 bits per second. The first 18 sec of each 30 sec frame contain *ephemeris* data for that satellite, which defines the position of the satellite as a function of time. The remaining 12 sec is the *almanac* data, which defines the orbits and operational status of all satellites. The GPS receivers store and use the ephemeris data to determine the pseudo range, and the almanac data to help determine the four best satellites to use for positional data at any given time. However, the "best four" philosophy has been overtaken slowly by an all-in-view philosophy.

Differential navigation is also applied where one Earth-bound user set is navigating relative to another user set via a data link. In some cases, one user has been at a destination at some prior time and is navigating relative to coordinates measured at that point. The true values of this receiver's navigation fix are compared against the measured values and the differences become the differential corrections. These corrections are transmitted to area user sets in real-time, or they may be recorded for post-mission use so that position fixes are free of GPS-related biases.

Differential navigation and GPS systems find applications in enroute navigations for commercial and civil aviation, military applications, navigation of ships especially in shallow waters, seismic geophysical explorations, land surveying,

autonomous vehicles, traffic control systems, electronic maps, orienteering, and route determinations.

GPS Receivers receive information from satellites. They operate on *slow sequencing*, *fast sequencing*, and *continuous tracking* principles. Receivers are integrated in cellphones and other communication devices finds many civilian uses. In majority of GPS receivers, three functional elements are implemented: 1) data-base management of satellite almanac, ephemeris, and deterministic correction data, 2) computation of precise satellite position and velocity for use by navigation software, and 3) using satellite and receiver position data to periodically calculate the constellation of satellites with optimum geometry for navigation.

3.21.1.18 Humidity and Moisture Sensors

Humidity and moisture sensors are used particularly in agricultural applications, domestic and in international food trading. Moisture is the amount of water or the amount of liquids in materials. The presence of moisture in a gas is termed humidity. The absolute humidity of a gas is the mass of water per unit mass of gas. The maximum humidity that can be attained is called saturation humidity and it depends on the temperature. Relative humidity is defined as the ratio of absolute humidity to saturation humidity at a particular temperature.

Another measure of relative humidity is the dew point. When a surface is cooled and in contact with gas, a temperature reaches to such level to deposit water onto the surface by condensation. This surface temperature is called the dew-point temperature.

There are many different methods for moisture and humidity sensing. These methods range from the changes in electrical properties to change in optical properties or changes in mechanical characteristics. Depending on the characteristics exhibited, for a specific application different technique are available.

The oldest method of measuring humidity is the *hygrometer*. Human hair, free from oil or grease, is strongly affected by humidity shrinking in dry conditions and elongating in moist conditions. Relative humidity sensors are constructed by slightly tensioned hair or a suitable material for detecting changes in length. Some of these methods may use of linear potentiometer, capacitive sensors, or oscillatory circuits.

Use of *lithium chloride* for moisture measurement is very common. Lithium chloride has a very high resistance in its dry state, but the resistance drops considerably in the presence of water with a reasonable linearity in proportion to the amount of water. The lithium chloride cell is made part of a measuring bridge, and the output of the bridge is processed suitably.

Another method for measuring humidity and moisture content is by using *microwave energy*. Microwave energy at various frequencies, e.g., 2.45 GHz, is strongly absorbed by water. The attenuations and phase shifts of the microwave across the moist media becomes a good way measuring the water content of the media. Microwave techniques are extensively used in agricultural products such as grain moisture determinations.

Conductive IC humidity and moisture sensors rely on the change of resistance of some semiconductor structures under varying humidity conditions. The sensing elements are made from materials such as polystyrene films (Pope cells) treated with

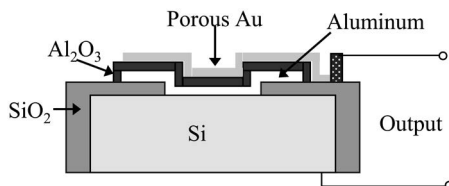


FIGURE 3.15 An aluminum oxide thin film conductive moisture sensor.

sulfuric acid. In some cases, solid polyelectrolytes are used. Solid-state humidity sensors are fabricated from silicon substrate as illustrated in [Figure 3.15](#).

In sensors in [Figure 3.15](#), the silicon substrate has a high conductance. The gold porous material on the top of the substrate allows the penetration of water molecules to provide electrical contact with the sensor. An oxide layer deposited on top of aluminum layer becomes anodized and gets affected from the porous oxide surface. In this way an electrical conduction mechanism is obtained from the porous gold to the silicon substrate. This yields the resistance of the system to be directly proportional to the number of water molecules in the porous material.

Capacitive humidity and moisture sensors are extensively used because of their ease of construct and the cost effectiveness. Capacitive methods utilize the changes in the permittivity of atmospheric air as well as the changes in the permittivity of the capacitors. The permittivity of dielectric materials between plates of capacitors are functions of the moisture contents and the temperature.

Thin film capacitive sensors can be fabricated on silicon substrates as shown in [Figure 3.16](#). In this construction, a SiO_2 layer is deposited on the n -type silicon substrate. The two metal electrodes, shaped in interdigitized patterns, are located on the SiO_2 . These electrodes may be made from aluminum, tantalum, chromium, or phosphorous-doped polysilicon. Since there are many types of capacitive humidity sensors operating on similar principles, here only the aluminum and tantalum types will be introduced as typical examples.

Aluminum type capacitive IC humidity sensors constitute most popular capacitive humidity sensors. In these types of sensors, high purity aluminum is chemically oxidized to produce a pre-filled insulating layer of partially hydrated aluminum oxide, which acts as the dielectric. A water-permeable but conductive gold film is deposited onto the oxide layer, usually by vacuum-deposition, which forms the second electrode of the capacitor.

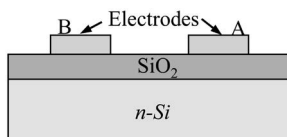


FIGURE 3.16 A capacitive thin-film humidity sensor.

In another type of aluminum-aluminum oxide, the sensor has a porous structure. The oxide with its pore structure forms the active sensing element. Moisture in the air, reaching the pores, reduces the resistance and increases the capacitance. The decreased resistance is due to an increase in the conduction through the oxide. An increase in capacitance can be viewed to be due to the changes in the dielectric constant. The quantity measured can be resistance, capacitance, or impedance. High humidity end is best measured by variations in capacitance since resistance changes small in this region.

Tantalum capacitive IC humidity sensors are made from plates one of which consists of a layer of tantalum deposited on a glass substrate. A layer of polymer dielectric is then added, followed by a second plate, which is made from a thin layer of chromium. The chromium layer is under a high tensile stress so that it cracks into a fine mosaic structure, which allows water molecules to pass into the dielectric.

Many capacitive Moisture Sensors, based on the changes in the permittivity of granular or powder type dielectric materials such as wheat and other grains containing water, are readily available. Usually, the sensor consists of a cylindrical chamber. The chamber is filled with samples under test. The variations in capacitance values with respect to water content are processed. The capacitor is incorporated as a part of an oscillatory circuit operating at a suitable frequency.

3.21.1.19 Infrared Sensors

Infrared sensors (IR) sense the heat energy radiated by objects. There are many different IR sensors, and they can be classified as active sensors or passive sensors. Thermopile sensors are good examples of the IR sensors.

Thermopile sensors are passive infrared, PIR, sensors. They operate on the same principles as thermocouples. That is, they have several serially connected hot junctions, shown in Figure 3.17, with improved absorptivity and cold junction that gives the temperature difference between the two. The sensor is hermetically sealed in a metal can with a hard infrared transparent window such as silicon, germanium, or zinc selenide. The operating frequency is determined by the thermal capacity and thermal conductivity of the membrane sandwiched between the hot junctions and cold junction. These sensors have good noise properties.

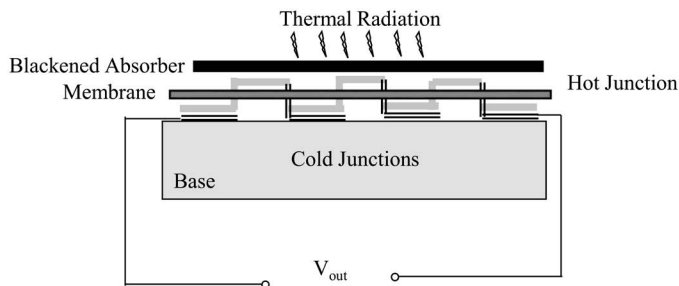


FIGURE 3.17 A thermopile.

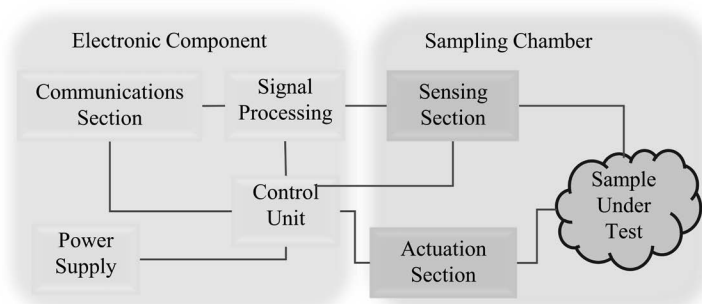


FIGURE 3.18 Structure of a typical Lab-on-chip chemical sensors.

3.21.1.20 Lab-on-Chip Sensors

Lab-on-chip sensors are composed of three main parts: sensing, processing, and communications (Figure 3.18). Depending on applications, many types of sensors are used such as optical, magnetic, thermal sensors, and so on. Mostly hybrid technologies are used to fulfill the functions of different components. CMOS technology allows the functional integration of all parts including sensors, signal conditioning, processing circuits, and power elements^(22–24).

3.21.1.21 LIDARs

Laser-based light detection and ranging sensors (LIDARs) are noncontact range finders that uses time-of-flight or frequency modulated continuous waves like radars, but they operate at light frequencies (200 THz–600 THz). They can provide high resolution detailed maps of the surroundings of the object. Lidars are gaining wider acceptance of usage in autonomous vehicles and unmanned aerial vehicle applications. Many LIDAR-based systems incorporate AI such as principal component analysis (PCA), ANN, and CNN in their operations⁽²⁵⁾.

3.21.1.22 Magnetic Sensors

Magnetic sensors find many applications in everyday life and in industry. They provide convenient, non-contact, simple, rugged, and reliable operational devices compared to many other sensors. The technology to produce magnetic sensors involves many different disciplines such as physics, metallurgy, chemistry, and electronics.

Generally, magnetic sensors are based on sensing the properties of magnetic materials. For example, magnetization which is the magnetic moment per volume of materials, is used in some measurement systems for sensing force, induction, field methods, and superconductivity. However, most sensors make use of the relationship between magnetic and electric phenomenon. A typical application is the position of ferromagnetic objects that are sensitive to magnetic fields (e.g., metals buried under surface) can be sensed magnetically.

The magnetic elements in sensors are produced in a wide range of forms: toroid, rods, films, substrates, and coatings. In sensors, some elements are essentially free

standing, whereas others are an integral part of more complex devices. To obtain maximum material response, the relative orientation and coupling between sensors and measurand determines the success of the applications.

Many different types of magnetic sensors are available, but they can broadly be classified as the primary sensors or the secondary sensors. In *primary sensors*, also known as the magnetometers, the physical parameters to be measured are the external magnetic fields. They are largely used in biological, geophysical, and extraterrestrial measurements. In *secondary sensors*, the external parameters are made from other physical variables such as force and displacement.

Magnetic thin film sensors are active devices made from deposited films. The thin films are usually made from amorphous alloys, amorphous gallium, and the like. The deposition of thin films can be done by several methods, such as thermal evaporation, electroplating, sputter deposition, or by some chemical means. The choice of technology depends on the characteristics of the sensors. For example, thin-film superconductors require low-temperature operations, whereas common semiconductors normally operate reasonably well at room temperatures. Thin-film technology finds applications in magneto-optics where erasable optical media for high-density magnetic storage is possible. Miniature magnetoresistive sensors for magnetic recording and pick-up heads, Hall effect sensors, and other magnetic semiconductors make use of thin-film technology extensively.

Different techniques can be employed for measuring magnetic fields. Each technique has unique properties that make it suitable for a particular application. These applications can range from sensing the presence of the magnetic field in a closed area to the determination of scalar and vector properties in open spaces. The vector type magnetic field sensors can be divided into two parts. Those that measure low fields, less than 1 mT range, called magnetometers and those that measure high fields that are greater than 1 mT are called gaussmeters.

In vector (direction) measuring instruments, the fluxgate and the induction coil sensors are used. Induction coil, fluxgate and fiber optic magnetometers are used in low field vector measuring. Hall effect sensors are used in gaussmeters, and they are the most common in high-field vector applications. Magnetoresistive sensors can be applied in both high and low field sensing. The GMRs are replacing the traditional fluxgate magnetometers for their improved sensitivities.

Another type of magnetometer is the proton (nuclear) precision magnetometer, which measures the scalar magnetic field strength particularly in geological exploration and aerial mapping of the geomagnetic fields.

Fluxgate magnetometer is one of the most used types in magnetic measurements. It is made from two coils wound on a ferromagnetic material. One of the coils is excited with a sinusoidal current, which drives the core into a saturation state. At this stage the reluctance of the core to external magnetic field increases thus repelling the external flux and reducing the effect of the external field on the second coil. As the core becomes unsaturated, the effect of external field increases. The increase and reduction of the effect of the external field on the second coil is sensed as harmonics of the induced voltage. These harmonics are then directly related to the strength and variations in the external field. The sensitivity of this type of device is dependent on the magnetic properties and the shape of the saturation curve of the core material.

The measurement range of a typical sensor can vary from 10^{-10} T to 10 mT operating up to 0 to 10 kHz frequency range.

Search coil magnetometers operate on the principle of Faraday's Law of induction. The flux through the coil changes if the magnetic field varies in time or coil is moved through the field. The sensitivity depends on the properties of the core material, dimensions of the coil, number of turns, and the rate of change of flux through the coil. These magnetometers can sense weak fields as low as 10^{-10} T within the frequency range of 1 Hz to 1 MHz. The upper limit is dependent on the relative magnitudes of resistance and inductance of the coil. During signal processing, they can be used as part of a bridge or part of a resonant circuit.

Other magnetic sensors are fiber optic magnetometers, Hall effect devices, magnetoresistive sensors, anisotropic magnetoresistors, giant magnetoresistive sensors, and proton precision magnetometers.

3.21.1.23 Mass and Weight Sensors

Mass and weight sensors are used extensively in everyday applications and trading goods. Mass is a quantitative measure of inertia of a body at rest. It is a product of density and volume. Volume is the amount of space that an object occupies. Weight on the other hand is the force with which a body is attracted toward Earth, and it is determined by the product of mass and acceleration of gravity. The mass of an object is constant, but its weight on the surface of the Earth is not constant, because gravity varies slightly from one place to another depending on the height of the object from Earth's surface. In space, outside the Earth, the planetary or Sun's influences on the weight of an object may be near zero or significant or insignificant depending on the location relative to the planetary and interstellar objects.

There are many different types of devices to measure the mass of bodies such as the equal arm balances, spring scales, beam balances, and electronic balances. The oldest method is the equal arm balance, which basically determines the force of gravity to find the difference in weights of two objects. The modified version of the equal arm balance is the beam balance. Other type of mass measurements is the spring scale, which again uses force of gravity as a reference scale. An adaptation of the spring scale is the torsion balance that is suitable to measure very small objects down to ranges of micrograms.

Electronic mass and weight sensors are largely based on load cells and strain gauges. The strain gauge is an electrically resistive wire element that changes resistance when a force is applied onto it. In the weight measurements, the strain gauge is bonded to a steel cylinder (or a cubic structure) that shortens when a body is placed on it. Since bonded to the base, the gauge is shortened or elongated in proportion to the direction of compression thus varying the electrical resistance. The change in the resistance of the gauge can electronically be made into digital data format.

3.21.1.24 Mechanical Variable Measurements

Mechanical variable measurements can be divided into two broad categories: solid variable measurements and fluid variable measurements. There are many instruments in both categories. Primarily, solid variable measurements consist of stress, strain,

mass, weight, density, acceleration, velocity, force, torque, and power. Measurements for fluids can be categorized as: pressure, velocity, flow, level, viscosity, and surface tension measurements.

3.21.1.25 Micro-Electromechanical Sensors

MEMS rely on miniaturized electromechanical elements in chips. They are suitable to manufacture single or 3D accelerometers and gyroscopes. An example is given in [Figure 3.7](#). They are based on silicon MEMS/CMOS technology on a monolithic structure. MEMS require dedicated analog/mixed circuitry to process and digitize the raw sensor outputs. Some MEMS are manufactured on electromechanical mass/spring principals while others rely on capacitive or piezoelectric technologies. Cost effective and mass manufacturing of MEMS makes them widely applied in consumer products⁽²⁶⁾

3.21.1.26 Navigation Sensors

Navigational sensors are ubiquitously used in aircrafts, ships, transport systems, location determinations, maps, hiking, and so on. The original meaning of the word *Navigation* comes from the word “ship driving.” In ancient times when sailing boats were used, navigation was a process of steering the ship in accordance with some navigational means such as stars and other celestial objects. The objective was to bring the vessel from location A to location B safely. In present days, navigation is a combination of science and technology. The term is no longer limited to the control of a ship in the sea, but applied to land, air, underwater, and space navigational systems.

Inertial navigation is a technique of using a self-contained system to measure a vehicle’s movement and to determine how far it has moved from a starting point. Accelerometers are commonly used to measure magnitudes of accelerations to determine the movements. Acceleration is a vector quantity involving magnitude and direction. Typically, accelerometers measure the component of acceleration along a predetermined line or direction. Mathematical integration of accelerometer signals leads to velocity and distance information. Other devices such as gyroscopes provide a reference frame for the accelerometer by supplying directional information.

The inertial navigation can be described as a process of directing the movement of vehicles, rockets, ships, aircrafts, humans, and robots from one point to another with respect to a reference axis. The object’s current position can be determined from “*dead reckoning*” with respect to a known initial starting (reference) position. On Earth’s surface, the conventional reference will be South, North, East, and West. This is referred as the *Earth’s fixed axes*. A vehicle such as an aircraft or a marine vessel will have its own *local axes* as Roll, Pitch, and Yaw.

Gyroscopes are often used as navigation devices. They can be classified in two broad categories: 1) mechanical gyroscopes and 2) the optical gyroscopes. Within these categories, there are many different types.

Optical gyroscopes are based on the properties of light instead of Newton’s Law of motion. In optical gyroscopes, two light waves circulate in opposite directions

around a path of radius R , beginning at source S . This arrangement produces interferometer fringe shift against rotation rate.

There are two different types of optical gyroscopes, which can be categorized as: active gyroscopes or passive gyroscopes, sometimes referred as resonant or non-resonant gyros. In passive gyrosensors the Sagnac phase is measured externally. In active gyros the Sagnac phase causes a frequency change internal to the gyro, which is directly proportional to the rotation rate of the object.

Satellite Navigation and Radiolocation are used in everyday life. They find extensive applications in position determination and movement detection of objects, wildlife, humans, military, commercial, and recreational activities. Modern electronic navigation systems can be classified by range, scope, error, and cost. The range classifications can be short, medium, and long ranges within which exact limits being rather indefinite. The scope classifications can be either self-contained or externally supported, or active (transmitting) or passive (not transmitting) mode of operation. Advanced navigation systems are based on satellites such as the GPS and Glonass as explained before.

Irrespective of the method selected, in electronic navigation systems three types of accuracy is important, that is: 1) *predictable* or *absolute accuracy*: the accuracy of a position with respect to the geographic coordinates of the earth, 2) *repeatable accuracy*: the accuracy with which the user can return to a position whose coordinates have been determined at a previous time with the same navigation system, and 3) *relative accuracy*: the accuracy with which a user can measure position relative to that another user of the same system at the same time.

Other navigation sensors are airspeed indicators, attitude indicators, depth gauges, machmeter, sextants, variometers, yaw rate sensors, and more.

3.21.1.27 Optical Sensors

Optical sensors are light-based devices and can be grouped as *quantum* sensors and *thermal* sensors.

Light, basically, is an electromagnetic radiation, which consists of an electric field and a magnetic field component. Compared to radio waves, light has a short wavelength and high frequency. Many different properties of light are used starting from cosmos explorations to photographic imaging and high-speed data transmission via fiber cables. Once the light is propagated from a source, it can be expanded, condensed, collimated, reflected, polarized, filtered, diffused, absorbed, refracted, and scattered thus giving a rich repertoire of configuration of possible sensor structures. Some of these properties of light are manipulated on purpose to serve application needs, and sometimes manipulations are not necessary since properties happen naturally due to optical properties of the media and physical characteristics of light.

One of the properties of light is the polarization as in the case of any other radiated wave. Normal unpolarized light consists of waves whose direction of oscillation can be in any plane at the right angles to the direction of the motion. When such light is passed through a polarizing material, it can become plane polarized. Materials with polarized light in this way, are generally crystals, which contain lines of atoms aligned at critical spacing to allow or block the passage of light in a particular plane.

Like the other waves in the electromagnetic range, light has a velocity, c , in space 3×10^8 m/s so that relationship between frequency, f , and wavelength, λ , can be written as:

$$c = \lambda \times f \quad (3.3)$$

Also, the energy is carried by photons. The energy of a single photon is given by:

$$E = h \times f \quad (3.4)$$

where h is the Plank's constant ($h = 6.63 \times 10^{-34}$ J or 4.13×10^{-15} eV s)

When a photon strikes a surface of a conductor, part of the photon energy generates free electrons in the conductor. This is known as the photoelectric effect, and it can be described by:

$$h \times f = \phi + K_m \quad (3.5)$$

where ϕ is called the work function of the emitting surface and K_m is the maximum kinetic energy attained from the electron.

When light travels through transparent materials, the velocity is reduced by a factor termed as the *refractive index*. For visible light, values of refractive index range just above unity to about a value of 2.0 in gemstones and diamonds. The refractive index for infrared radiation can have greater values. Some materials which may be opaque to visible light may be quite transparent to infrared.

There are many different types of light sensors which operate on *photoconductive* and *photovoltaic* principles. The phototransistors, charge coupled sensors, and photoemissive sensors are few to mention operating on thermal principles. Photovoltaic sensors operate on quantum principles. These sensors are available in many different configurations, some built with powerful lenses and high-speed responses to monochromatic light, and others having different color sensitivities to chromatic light.

Image sensors are devices that transform optical images into electrical signals. A standard image sensor converts the light photons falling onto the image plane into the corresponding spatial distribution of electric charges. The accumulated charge at the point of generation is transferred suitably and converted to useable voltage signals. Each of these functions can be accomplished by a variety of approaches, such as the vacuum tube technology and solid-state technology. Solid-state sensors are based mainly on photodiodes and CCDs.

The simplest image sensors are the parallel types, in which a matrix of photodiodes is used in small-scale applications such as the optical character recognition. A photodiode consists of a thin surface region of p -type silicon formed on an n -type silicon substrate. A negative voltage applied to a surface electrode reverse the biases of the p - n junction. This creates a depletion region in the n silicon, which contains only immobile positive charge. Light penetrating into the depletion region creates electron-hole pairs, which discharges the capacitor linearly in time.

Solid-state image sensors arrays are considerably complex and manufactured mostly in IC forms, and there are two basic types, the serially switched photodiode arrays and the charge coupled photodiode arrays. In both arrays the basic principle is to use light intensity to charge capacitors and then to read the capacitor voltages. Reading is done by shifting through the integrated registers.

The most developed example of such devices is the CCD. The CCD is a form of MOS device in which a clock pulse can pass a charge from one plate to another like analog shift register. In these devices, small areas of silicon are exposed to the light thus causing a local charge, called a "packet" to be generated. The synchronized switching of electrodes on the surface of CCD, the packets are shifted in a serial fashion to an output device. CCDs are fundamentally linear devices; therefore, they are opted in array arrangements. CCDs are used in one-dimensional or two-dimensional arrays. The one-dimensional arrays are useful for scanning of light spectra, and two-dimensional arrays are useful for low-resolution cameras.

Spectrometers use spectroscopy principles. The term spectroscopy has few meanings. It applies to radiation of energy of any wavelength and sometimes to the analysis of material particles (mass spectroscopy). The electromagnetic radiation emitted by matter is measured by spectrometers. The correct selection and use of appropriate spectrometers for a particular region of spectrum is essential. Spectrometers find diverse range of applications, such as: emission and absorption of light by matter, determination of atoms and molecules present in samples of matter, electronic structures of atoms and molecules, and characteristics of the terrestrial bodies such as mass, temperature, size, and velocity.

Spectroscope is an optical device that separates light into its constituents to produce a spectrum. It uses prisms, diffraction gratings, or interferometers to observe the spectrum visually, or record photographically, or to detect by photoelectric or by radiometric means in spectral regions. The wavelength of radiation can be read directly or indirectly from a dial. If the radiation is recorded photographically is called a spectrograph, and the photograph produced is called the spectrogram.

There are three methods of wavelength separation by spectroscopes. These are 1) Refractive dispersion by prisms made from glass, 2) diffractive dispersion by gratings (made from engraved diamonds and mirrors), and 3) Interferometric dispersion, such as the Fabry-Perot, the Lummer-Gehrke plate, and the Michelson echelon interferometers. They generally consist of optical parts such as: slits, dispersing devices such as prisms, grating or interferometers, focusing lenses, and mirrors that concentrate monochromatic beams.

Generally, photoelectric cells (e.g., barrier layer and phototube types) are used for measuring spectra in the visible and ultraviolet regions. In the phototube types of transparent envelopes of quartz or special ultraviolet transmitting glasses are used. Cesium hydride or other suitable metallic materials can be selected to serve as photosensitive surfaces. Photoconductive cells such as the thalofide and zinc sulfide are also used for the near infrared region. The photocell is usually held fixed, while the spectrum emerging from an exit slit is scanned by rotation of the prism or grating.

Fiber optic sensors are used as sensors in varying forms, such as temperature sensors, displacement sensors, and optical gyroscopes. Fiber optic sensors are based on

the properties of light as well as properties of optical materials suitably constructed for specific applications. The operation principles of the sensors vary considerably. Some of the sensors make use of scattering properties of light while others make use of attenuations and losses.

An optical fiber is a circular dielectric glass waveguide that can efficiently transport optical energy by means of total internal reflections. It consists of a central glass core surrounded by a concentric cladding material with a slightly lower ($\approx 1\%$) refractive index. Since the core has higher index of refraction than the cladding, light is confined to the core if the angular condition of total internal reflection is met. The attenuation of the light begins on the moment when light enters the fiber. The acceptance and transmission of light depends greatly upon the angle at which the light enter the fiber.

Other optical sensors are angle sensitive pixels, Nichol's radiometers, scintillometers, Shack-Hartmann wavefront sensors, avalanche diodes, Nanowire detectors, transition edge sensors, and visible light photon detectors.

3.21.1.28 Photosensors

Photosensors are photon detectors sensitive to photons with energies greater than a given intrinsic gap energy of the sensor. Some of the most common photon detectors make use of properties of cadmium and lead compounds, e.g., cadmium sulfides (CdS) and cadmium selenide (CdSe), lead sulfide (PbS), and lead selenide (PbSe) devices. The main classes of photosensors are photodiodes/phototransistors, photoresistors/photoconductors, and photovoltaic cells.

Photodiodes and phototransistors are semiconductor devices that are sensitive to light. The incident light falls on a reverse biased *pn*-junction, the photonic energy carried by the light creates electron-hole pair on both sides of the junction causing a current to flow in a closed circuit. The output voltage of photodiodes may be highly non-linear thus requiring suitable linearization and amplification circuits. The photodiodes are constructed like any other diode, but without the opaque coating, so that light can fall on to the junction. The current generated at the junction can be linearly related to the amount of illumination falling on the junction. The response time of photodiode sensors is very short, typically 250 ns. Wide ranges of photodiodes are available such as *pn*-photodiodes, *PIN* photodiodes, Schottky photodiodes, and avalanche photodiodes.

Phototransistors are the most used light sensors. They have current gain properties in addition to converting photons into charge carriers. Particularly, Darlington phototransistors possess high sensitivity and high current gain. A phototransistor is a form of transistor in which the base-emitter junction is sealed with a transparent package. This junction is affected by incident light to acts as a diode. The light at the junction promotes electrons from valence to conduction band. The current in this junction is then amplified by the normal transistor action to provide a much larger collector current. The response of the phototransistor can be adjusted by applying an appropriate bias voltage to the base. The response times of phototransistors are much larger than photodiodes, in microseconds. Phototransistors are manufactured in the IC forms, containing light sensors integrated together with amplifiers and other processing circuits.

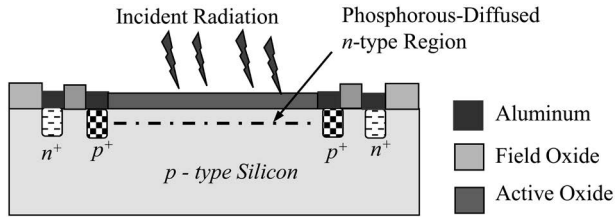


FIGURE 3.19 A Si photodiode.

Photovoltaic sensors are based on silicon cells that generate voltages in response to a beam of light. The response time of these sensors is slow, and it takes longer time to stabilize the output. The wavelength response of a photovoltaic silicon cell covers the whole visible spectrum, which makes them very useful for environmental light measurements.

The first form of photovoltaic device was the selenium cell, which produced voltage across the cell in proportion to the illumination. Modern photovoltaic devices are constructed from silicon, thus acting as a silicon photodiode with a large area junction as illustrated in [Figure 3.19](#). These devices find applications in camera exposures and controls of light levels in manufacturing processes. With some suitable filtering, the response photodiode curves can be made very similar to the response human eye to the light.

Photoemissive sensors are vacuum tubes with treated metal surfaces. Light falling on the surface ejects electrons to be picked up by an anode. Photoemissive sensors are available having different spectral sensitivities. They have low power demands but require high voltages.

Photomultipliers gather emitted electrons which are attracted to a nearby plate with high voltage bias. The electrons are accelerated to a high energy level so that when they strike the plate, they splash many electrons from it. These electrons are attracted to another plate biased at an even higher voltage than the previous one. Each stage causes amplification therefore the more stages the device has the higher the resulting signal becomes. This is the reason; photomultipliers are the most useful devices for detecting extremely weak lights, but they require high voltages up to 1,000 V.

3.21.1.29 Piezoelectric Sensors

Piezoelectric sensors produce electric charge when a force is applied on to it. This effect exists in natural crystals such as quartz (SiO_2), or manmade polarized ceramics and polymers. A piezoelectric sensor can be viewed to have a crystal structure with a dielectric material. This dielectric material is arranged like a capacitor with electrodes thus behaving like an electric charge generator when stress is applied on it. The piezoelectric effect is a reversible physical phenomenon. That is, applying a voltage across the terminals produces a mechanical strain.

In most common piezoelectric sensors is the lead zirconate titanate (PZT). Ceramic-based sensors are used to construct biomorph sensors for force, sound,

motion, vibration, and acceleration sensing. The biomorph consists of two layers of different PZT formulations bonded together in rectangular strips. Rectangular shape structure can conveniently be mounted as a cantilever on structures. The motions perpendicular to the surface of the sensor generates signals to be used in measurements.

3.21.1.30 Piezofilm Sensors

Piezofilm sensors are made from polymer films that generate charge when they are deformed. Piezofilm sensors also exhibit mechanical motions when a charge is applied. Surface machining techniques are combined with piezoelectric thin-film materials (e.g., zinc oxide) to produce semiconductor pressure sensors particularly useful in low-level acoustic applications.

3.21.1.31 Pollution Sensors

Pollution sensors is a part of environmental sensors, but pollutions sensors have specific characteristics. Historically, pollution sensors constituted an important part of environmental monitoring systems for decades, and they still do with an increased importance to assess climate change and environmental degradations.

Air pollution is defined as an unwanted change in the quality of the earth's atmosphere caused by human-related emission of gases, solids, liquid, and particulates. Air pollution is one of the major reasons of changes in climate, greenhouse effects, and ozone depletions. These changes may soon have series consequences on all living organisms. Polluted air in a region is carried everywhere by winds and air currents, hence not confined within national boundaries. Therefore, air pollution is a concern for every living irrespective of what and where the sources are, hence accuracy and consistency of sensors, instruments, and instrumentation for air pollution measurements is carries a vital importance.

Air pollutants can be classified according to their physical and chemical composition as: *inorganic gases* (sulfur dioxide, hydrogen sulfide, nitrogen oxides, hydrochloric acid, carbon monoxide, carbon dioxide, ammonia, ozone, and others), *Organic gases* (hydrocarbons, terpenes, mercaptans, formaldehyde, dioxin, fluorocarbons, and others), *inorganic particulate* (asbestos, lime, metal oxides, silica, antimony, zinc radioactive isotopes, and others), and *organic particulate* (pollen, smuts, fly ash, and others).

Accurate measurement of air pollution is necessary to establish acceptable levels of pollutions and to configure control mechanisms against offending sources. Accurate prediction of pollution helps in setting policies and regulations, as well as in observing its effect on humans, plants, vegetation, animals, environment, and properties. Nevertheless, precise estimation of all offending substances responsible for air pollution is difficult due to geographical, physical, and seasonal variations. Currently, many studies are taking place for understanding the processes in the formation, accumulation, diffusion, dispersion, and decay of air pollution and the individual pollutants causing it. Effective national and international control programs depend on the full understanding of pollution and its effects.

There are two basic types of methods used in determining air pollution: the spot sampling method and the continuous sampling. These techniques can be

implemented by a variety of instruments. For example, the *automatic* instruments are based on one or more of methods such as electrolytic conductivity, electrolytic titrimetry, electrolytic current or potential, colorimetry, turbidimetry, photometry, fluorimetry, infrared or ultraviolet absorption, and gas chromatography. Some of these methods have been explained above. *Nonautomatic* instruments are based on absorptions, adsorptions, and condensations.

Water pollution is important to monitor since water is a resource fundamental to all life, and it is important in both quantity and quality, particularly for humans. Fresh water is essential for life, and clean, unpolluted water is necessary to health and the preservation of nature. Water is used for many purposes besides human consumption, and in arid and semiarid countries large quantities of water are used for irrigation due to lack of rain.

Soil pollution refers to the addition of soil constituents, due to domestic, industrial, and agricultural activities that were originally absent in the system. Soil contamination takes place in two forms 1) the slow but steady degradation of soil quality (e.g., organic matter, nutrients, water-holding capacity, porosity, purity) due to contaminants such as domestic and industrial wastes or chemical inputs from agricultural operations, and 2) the concentrated pollution of smaller areas, mainly through dumping or leakage of wastes. The sources of contamination include the weathering of geological parent materials, where element concentrations exceed the natural abundances in wet or dry deposition forms. Soil pollution is determined by chemical methods as explained previously. Well-defined sampling methods and mathematical techniques are used in the assessments.

3.21.1.32 Position, Angle, Proximity, and Displacement Sensors

Position, angle, proximity, and displacement sensors are used for measuring position and displacement. Many of the related sensors have been explained. In addition to those, other popularly used magnetic position and displacement sensors can be listed as inductive sensors, Hall-effect sensors, magneto resistive sensors, and eddy current proximity sensors.

Hall-effect sensors are generally fabricated as monolithic silicon chip and encapsulated into epoxy or ceramic packages. There are two popular types: linear Hall-effect sensors and threshold Hall-effect sensors. A *linear Hall-effect sensor* incorporates an amplifier and other electronic circuits for interfacing with peripherals. Compared to basic Hall-effect sensors, they tend to be more stable and operate over wider ranges of voltages. The *threshold sensors* have Schmitt trigger detector with a built-in hysteresis to eliminate spurious oscillations.

Position sensitive sensor (PSD) operate near infrared region. They can be a very effective way sensing short- and long-range displacements and positions. As in the case of cameras, PSDs are used in conjunction with LEDs. A PSD operates on the principle of photoelectric effect. The surface resistance of a silicon photodiode changes depending on the light fallen on to it. They can be single-dimensional or two-dimensional sensor. A sensor high resistance silicon with p and n^+ -type layers is illustrated in Figure 3.20. The sensor has two electrodes to form contact with the p -type. The photoelectric effect occurs in the upper pn -junction.

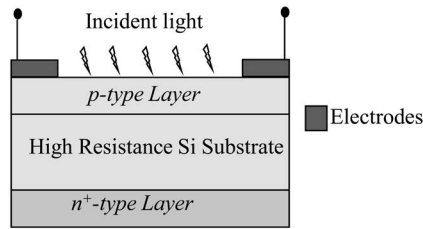


FIGURE 3.20 A silicon-based position sensitive detector.

Microwave and Doppler radar sensors Gunn diode oscillators (about 10 mW output) are in launching horns, and the receiving horns contain mixer diodes. The microwave beam, typically 10.7 GHz is generated by the Gunn diode and a low frequency signal is generated when any object in the field of view moves. This signal is extracted suitably to work as a proximity detector of a moving object.

Some of the other position, angle, proximity, and displacement sensors are aux-anometers, flex sensors, gravimeters, impact sensors, inclinometers, range finders, odometers, angular rate sensors, selsyns, tilt sensors, tachometers, variable reluctance sensors, and velocity receivers.

3.21.1.33 Pressure Sensors

Pressure sensors are used for the measurements of one of the most fundamental variables in mechanical, civil, and environmental engineering applications. Pressure is defined as the normal force exerted on unit surface area of an object. The SI unit of pressure is the Pascal (Pa), in honor of Blaise Pascal (1623–1662), which is defined as the newton per square meter (Nm^{-2}). There are three basic types of pressure measurements:

1. *Absolute pressure* represents the pressure difference between the point of measurement and perfect vacuum where pressure is zero,
2. *Gauge pressure* is the pressure difference between the point of measurements and the ambient pressure, and
3. *Differential pressure* is the pressure difference between two points.

The type of sensors suitable for a particular application depends on characteristics of the object under investigation, such as solids, gases, air, or liquids. Also, there are different types of sensors for low-pressure, high-pressure, and/or differential pressure.

Pressure can be sensed indirectly or directly. *Indirect* methods rely on the action of pressure to cause displacement of a diaphragm, a piston, or some other device, so that the electronic sensing of displacement can be related to the pressure. Indirect pressure sensors are known as the aneroid barometers. There are many different versions aneroid barometers depending on the sensing elements of the electrical components. [Figure 3.21](#) illustrates a semiconductor piezoresistive indirect pressure sensor.

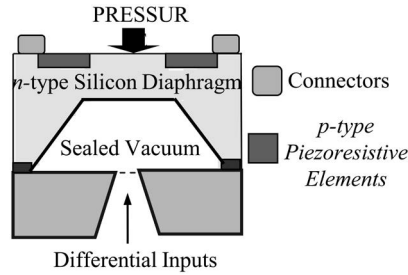


FIGURE 3.21 A semiconductors piezoresistive pressure sensor.

In this sensor, *p*-type doping material is introduced into *n*-type silicon diaphragm using ion implementation technology. Another method of sensing pressure is using piezoelectric crystals such as the PZT. These sensors are useful to measure pressures varying in time.

A common method of measuring pressure based on IC technology is the *aneroid barometer*, which is a sealed bellows that expands and contracts as the external pressure changes. Some suitable displacement sensors detect the movement of the bellows.

The pressure sensors can also be divided into two main groups; the low-pressure sensors and high-pressure sensors that will be discussed next.

Low-pressure sensing involves much more specialization and tailoring for the characteristics of a particular application. The most common sensors are the Pirani gauges and the ionization gauges.

Pirani gauges use the principles of thermal conductivity of gases that decreases in proportion to pressure for a wide range of low pressures. It uses a hot wire element. The principle of the device is that less heat is conducted through the gas as the gas pressure around the wire is reduced. The resistance of the wire, which is dependent on the heat transferred by gas, is measured by suitable electrical bridge networks.

Ionization gauges use a stream of electrons to ionize the gas of which the pressure is measured. The positive gas ions are then attracted to a negative charge electrode and the amount of current carried by these ions is measured. Since the number of ions per unit volume depends on the number of atoms per unit volume, the readings of currents can be related to the pressure. Ionization gauges are suitable for very low-pressure measurements down to 10^{-7} Pa. There are many different types of ionization gauges, but the principles of the operation are the same.

High-pressure sensors are based on capacitance manometer principles. A capacitance manometer contains a large diaphragm serving as one plate of the capacitor. The deformation of the diaphragm thus causes the capacitance to change. The changes in capacitance are detected and processed by electronic circuits on the same chip. Capacitive manometers are available for measuring pressure differences as well as absolute pressures.

Many semiconductors capacitive pressure sensors are made from two parallel metals plates. One of the metal plates is fixed and the other plate forms a flexible diaphragm. The flat diaphragm is clamped around its circumference and is bent into a curve by applied pressure P .

There are many semiconductor pressure sensors that operate entirely on silicon chips. These ICs contain sensors and associated signal processing electronics. They can measure pressure in the range of 1 psi to 100 psi.

Other less frequently used pressure sensors are barometers, boost gauges, Bourdon gauges, Oscillating U-tube, and Pirani gauges.

3.21.1.34 Radars

Radars detect the range, velocity, and angle of objects in its field of view. There are two types: pulsed radars and frequency modulated continuous wave radars. In pulsed radars, the range is calculated by using the time delay between the transmitted and received signal. Frequency modulated radar transmits a continuous wave, and the reflected signal is mixed with the transmitted signal to generate an intermediate frequency (IF) signal to calculate the range.

3.21.1.35 Radiation Sensors

Radiation sensors are frequently used in handheld instruments to monitor radiation in different locations, because of the health and safety implications of radiation on humans and other living organisms.

Many radiation sensors use ionization as the physical principle behind the operations. There are two basic types of ionizing radiation: 1) particle radiation such as alpha and beta particles, neutrons, and cosmic rays and 2) electromagnetic radiation such as x-rays and gamma rays.

Particle radiation constitutes alpha particles, beta particles, neutrons, and cosmic rays. Alpha and beta particles are emitted during the normal radioactive decay phenomenon, but the neutrons are emitted in a nuclear reaction. Also, they are produced when a deuteron (a proton and a neutron) and a triton (a proton and two neutrons) collide during atomic reactions. An intense source of neutrons results from the fission reaction in nuclear reactors. But the neutrons are highly unstable, and they change spontaneously to protons, electrons, and neutrinos.

Alpha particles also are spontaneously emitted from heavy atomic nuclei, called *alpha decay*. They are positively charged and consist of two protons and two neutrons, which is identical to the nucleus of helium 4. *Beta particle radiation* refers to the emission of electrons and positrons known as the beta decay. A positron is a positively charged electron. There are two types of beta particles, known as beta-minus and beta-positive particles. The energy of beta particle has the same magnitude as gamma rays. Beta particles ionize atoms in a similar way as heavy positive particles. But they can interact with matter in excitation of electrons by displacing electrons to more energetic orbits without entirely removing them.

Electromagnetic radiation constitutes x-rays and gamma rays. The x-ray radiation is emitted when an electron drops from an outer orbit into a lower orbit.

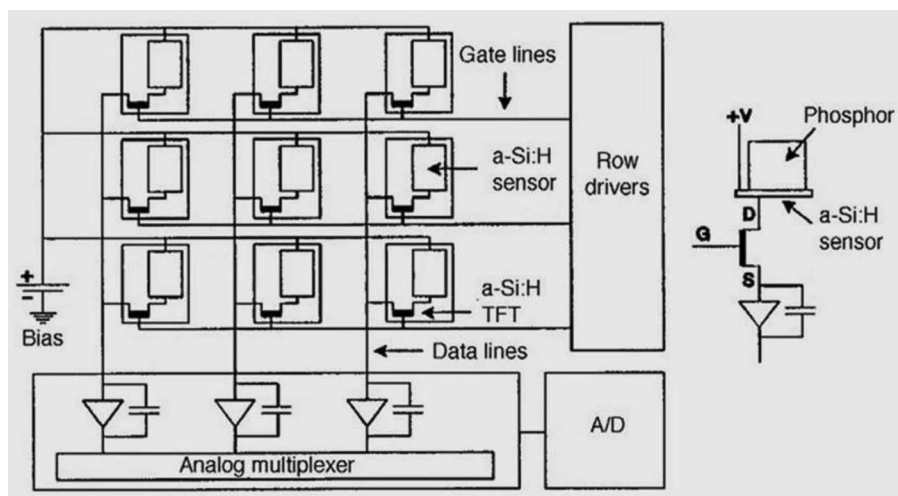


FIGURE 3.22 Schematic representation of a 2D x-ray imager consisting of a (Si:H) $p-i-n$ diode connected to a Si:H TFTs for readout (Source: M.R. Squillante⁽²⁷⁾).

Similarly, a gamma ray is emitted when the nucleus of an atom existing in an excited state decays to a lower energy state. They both can be visualized as photons, which are a form of traveling packets of electromagnetic energy without mass or electrical charge. x-rays and gamma rays have shorter wavelengths; therefore, they carry greater energies than other forms of electromagnetic radiation. A 2D x-ray imager consisting of hydrogenated amorphous silicon (Si:H), $p-i-n$ diodes connected to Si:H TFT (thin film transistor) is illustrated in Figure 3.22. Silicon $p-i-n$ diodes are extension of standard $p-n$ junction diodes with reduced capacitance by incorporating an intrinsic region between p and n region. In these arrays, pixels consist of $p-i-n$ sensors connected to a TFT. The entire array reads in matrix form. They are used medical x-ray imaging applications with phosphorous.

Gamma and x-rays interact with matter by either photoelectric effect, or Compton scattering process, or pair production process. In the photoelectric effect, all the photon energy is transferred to orbital electrons. In Compton scattering, the photon passes a portion of its energy to the electrons. In pair production the gamma ray disappears, and an electron and positron are emitted in its place.

Accuracy in radiation measurements is important necessary since radiation is not perceivable by human senses but can have significant adverse health effects on human health. Humans are exposed continuously to naturally occurring radiation or radiation generated by human activities. The main sources of the natural radiation come from cosmic radiation or/and soils and rocks on the earth's crust, such as uranium-238, thorium-232, and potassium-40. Man-made sources of radiation are very common. Radiation is used often in medicine for diagnosis, curing, and physical examinations. Nuclear reactors generate beta, gamma,

and neutron radiation originating in the fission process. Other sources of man-made radiation include detonations of nuclear weapons. In this case, radioactive materials enter the atmosphere and transported by the winds and carried other parts of Earth. Then, they are brought down to the surface by rain known as the “fallout.”

It is important to understand the units of radiation to make sense of the results displayed by the instruments. At first sight, a confusing number of units are used to measure radiation, but the logical arrangements of units are as follows: The radiation activity refers to the disintegration rate of a sample of radioactive material. This is measured by curie (Ci), Marie Curie (1867–1934) that is equal to 37 billion disintegration per second. The Système International unit for the radiation activity is the Becquerel (Bq), Antoine Becquerel (1788–1878). One Bq is the quantity of the radioactive element in which there is one atomic disintegration per second, therefore ($1 \text{ Bq} = 2.7 \times 10^{-11} \text{ Ci}$). The term radiation intensity is the strength of a radiation field at a given point in space. It describes the number of radiation particles passing through a unit area in per unit time. Exposure describes the strength of a gamma ray or x-ray field falling on a body at a point in space. It is measured in terms of the number of ions produced in the atmosphere adjacent to the body. Exposure is sometimes measured in units of coulombs per kilogram of air. However, a more common unit is the roentgen, defined as the charge of one electrostatic unit produced by photons in one cubic centimeter of air at standard temperature, humidity, and pressure conditions. One roentgen, after Wilhem Conrad Roentgen (1845–1923), equals 0.0088 joules per kilogram.

The units of absorbed dose are the gray (Gy), after Stephen Gray, and the rad (rad). One gray = 1 joule of radiation absorbed per kilogram of tissue and 1 rad = 100 erg per gram of tissue, therefore 1 rad = 0.01 Gy. The unit “rad” is applied to any type of radiation and to any material. The biological effects of radiation are not directly proportional to energy and the way that this energy is deposited in an organism. The damage depends on the energy of the radiation, the nature of the tissue, the absorbed dose and dose rate. For instance, alpha particles cause more biological damage than gamma rays, for the same level of absorbed dose. That is because the alpha energy is deposited over more localized volumes.

The biological effect of different types of radiation depends on the quality factors. These factors are as follows: x-rays and gamma rays = 1, beta particles = 1–1.7, neutrons 2–11, alpha particles = 10, and heavy nuclei = 20. The unit of radiation used to describe biological effects is the dose equivalent, which is the product of the quality factor and the absorbed dose, sometimes called the biological dose. The units of dose equivalent of radiation are the sievert (Sv) and rem (rem). That is, dose of radiation in gray multiplied by the quality factor. Therefore, 1 Sv is the amount of radiation that is equivalent to biological effectiveness of 1 Gy of gamma rays. Similarly, one rem is the amount of radiation that is equivalent to biological effectiveness of one rad of gamma rays, hence, 1 Sv = 100 rem. In one year, a radiation worker should not be allowed to accumulate a dose more than 5 rems to the whole body, 30 rems to the skin, and 75 rems to hands, forearms, feet, and ankles. It is possible for a human to recover from a dose of several hundred rems.

There are four basic types of radiation detectors: gas-filled types, scintillation types, semiconductor types, and thin-film types. In this book, the first three types of detectors will be explained. The film-detectors primarily consist of a chemical thin film, which is sensitive to the incoming radiation. The intensity of the darkening of the film indicates the amount of radiation that a person wearing the detector has been exposed to.

Gas-filled detectors consist of cylindrical containers filled with inert gases (e.g., argon). A voltage is applied to a wire placed in the cylinder and the cylinder wall. Radiation entering the container ionizes some of the gas atoms to give negative charged and positive charged ions. Free electrons travel to the anode (the wire), and the cathode (the cylinder wall). The resulting electronic pulse is amplified and recorded as a count. In some gas-filled detectors, electrons traveling to the anode can cause further ionization, magnifying the incoming radiation. Instruments based on these principles can be very sensitive to small amounts of incoming radiation.

A typical and widely used example of the gas-filled detectors is the Geiger-Müller detector also known as the Geiger counter, shown in [Figure 3.23](#). In this arrangement, the anode is tungsten or platinum wire, while the cylindrical tube acts as the cathode for the circuit. The tube is filled with argon with a mixture of hydrocarbon gas. Internal pressure is kept slightly less than atmospheric pressure. The radiation particle is transmitted through the cathode material. Interaction of the particle produces ionization of the gas molecules thus causing a voltage pulse for each particle.

Semiconductor detectors are widely used and made from semiconductor materials. When an electric field is placed across the semiconductor, a region is created such that no free electrons exist. Incoming radiation ionizes atoms in this region producing free electrons, which can be collected and counted. The semiconductor sensors are highly sensitive, accurate, and have high resolution compared to gas-filled counterparts. They can distinguish between radiation particles that are close together in energy contents.

The semiconductor sensors can be manufactured from germanium or silicon. Silicon sensors can operate at room temperatures, and they are fabricated from slices of a single silicon crystal. The basic principle is that the energetic heavy charged particles lose their kinetic energy along a linear path in an appropriate absorbing material.

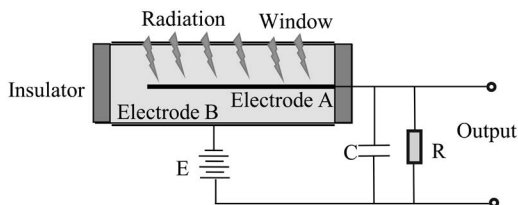


FIGURE 3.23 Geiger-Müller radiation detector.

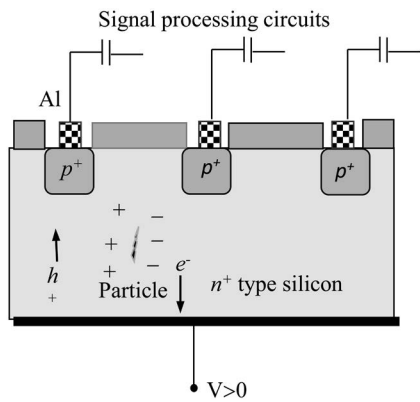


FIGURE 3.24 A semiconductor radiation detector.

There are many different types of semiconductor radiation sensors: diffused-junction, ion implanted, strip types, drift types, and CCD sensors. As a typical semiconductor sensor, a position-sensitive silicon strip type sensor is illustrated in [Figure 3.24](#). This sensor is suitable to detect radioactive particles with better than 5 μm localized hit rates. They are often used in particle physics experiments.

Scintillation detectors work on the principle of fluorescence. In this case, the energy is absorbed by a substance and re-emitted as visible or near-visible light. A photomultiplier tube senses the amount of light. An example is illustrated in [Figure 3.25](#). In this figure, the scintillator is attached to the front end of a photomultiplier. The front end contains plates called *dynodes* positioned inside a photo multiplier tube in an alternating pattern. Each dynode is attached to an increasing voltage source having highest positive potential on the last one.

The most widely used scintillators are inorganic alkali halide crystals (e.g., sodium iodine), some plastics, and suitable organic-based liquids. Organic scintilla-tors are sensitive but slow. A general limitation of scintillation detectors is that they have relatively inferior energy resolution and thus find limited but highly specialized applications.

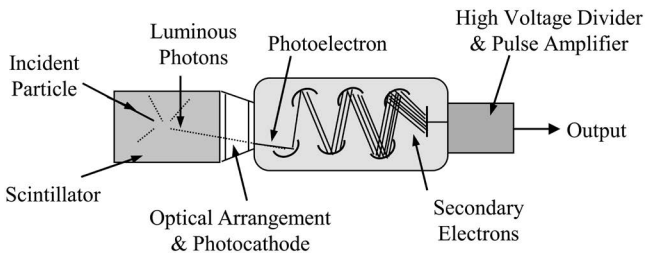


FIGURE 3.25 A scintillation detector.

Other radiation and ionization sensors to mention are bubble chamber, cloud chamber, ionization chamber, neutron detector, particle detector, wire chamber, and thermoluminescent dosimeter.

3.21.1.36 Resonant Sensors

Resonant sensors are based on mechanical resonant structures. For example, a micro-machined silicon can act as a mechanical resonant element if it is designed with a membrane or tuning fork. Electric activation for resonant operations can be achieved by various methods such as piezoelectric films like ZnO. Most resonant structures are implemented in pairs, one of which acts as a reference and not interfaced to inputs of digital devices. Comparison of the output frequency of the sensing element to the frequency of a reference element reduces the influence of the unwanted parameters. This technique is often used in chemical sensors using surface acoustical wave (SAW) delay-line oscillators.

3.21.1.37 Rotation Sensors

Rotation Sensors are used in many industrial applications. Simplest form of sensing angular velocity is by using AC or DC generators, commonly known as the tachogenerators.

In some applications, signals are generated for each revolution of the rotating object by optical or by piezoelectric or magnetic pulsing arrangements. These pulses enable the determination of the angular velocities as well as the positions of the rotating object.

Angular methods for displacement measurements can be grouped to be small angular measurements or large angular measurements. Small displacements are measured using capacitive sensors, strain gauges, or piezo sensors. For larger displacements potentiometric, or inductive techniques are more appropriate. The rotary digital encoders which can directly be interfaced also find a wide range of applications.

3.21.1.38 Satellite Relay Systems

Satellite relay systems are important for instrumentation systems since they allow the communication of instruments with a base station. The uses of satellites are highly developed due to advance technologies utilized extensively throughout the world. In the past five decades, satellite relay systems have progressed from quasi-experimental nature to the ones with routine provisions of all types of new services. They take advantage of unique characteristics of *geostationary satellite orbits* (GSO). The design of satellite systems is well understood, highly researched, and the supporting technology is dynamic. The satellites are useful for long distance communication services, for services across oceans or difficult terrains, and point to multipoint services such as television broadcasting.

There are many satellite systems operated by private organizations and by different countries. Every year some new ones are added while others become

obsolete and taken out of service. Some of the operational satellite systems are Iridium of Motorola, Globalstar of Loral Corporation, Intelsat, CS-series of Japan, Speedcast, Aussat, Galaxy and Satcom, Anik, Telespazio, TDF, Intelsat, Eutelsat, and others. Many of these satellite systems are developed mainly for communication purposes and data transmissions, which has important implications in remote and rapid deployment of instrumentation systems. Some of these communication satellites are suitable for navigation purposes. However, systems that specifically designed for navigation are only handful in numbers. The most established and readily accessible by civilian and commercial users are the GPS system of the United States, Galileo of Europe, Beidou of China, IRNSS of India, QZSS of Japan, and Glonass of Russia.

3.21.1.39 Semiconductor Sensors

Semiconductor sensors make use of properties of silicon and other semiconductor elements. Silicon-based sensors find many applications such as: Radiation measurements (photovoltaic effect, photoelectric effect, photoconductivity, photomagneto effects), mechanical measurements (piezoresistivity, photoelectric, and photovoltaic effects), thermal measurements (Seebeck effect, temperature, Nernst effect), magnetic measurements (Hall effect, magnetoresistance, Suhi effect), Chemical measurements (ion sensitivity).

Developments of silicon-based sensors are enhanced considerably due to advances in embedded controllers, microelectronics, and micromachining. Micromachining technology combined with semiconductor processing technology provides sensors for mechanical, optical, magnetic, chemical, biological, medical, and other phenomenon as listed above. Digital logic provided by either logic circuits, Dips, and microcontrollers plays a vital role in the performance, design, and construction of semiconductor sensors.

Batch processing techniques are ideally suited for making high-volume, low-cost sensors. Because of mass production, the characteristics of individual sensors may have some deviations from ideal values; therefore, most specifications are given in terms of full-scale values. In some cases, typical specifications are given, which indicate a certain degree of spread in the performance from one sensor to another.

Semiconductor sensors provide easier-to-interface, lower cost, and more reliable inputs to digital systems.

3.21.1.40 Soil Acidity Sensors

Soil acidity and alkalinity are important for water supplies, industrial and domestic water users, agriculture and horticulture, power generation, and in numerous material processing industries. The acidity and alkalinity of water is measured on the pH scale, which is based on free hydrogen ions in the water. Natural water has a pH value of 7, strong acid solutions have a pH of 2 and strong alkaline solutions have a pH of 12.

The standard sensing of acidity uses glass electrode system as shown in [Figure 3.26](#). In this arrangement, a thin glass bulb contains a mildly acidic solution

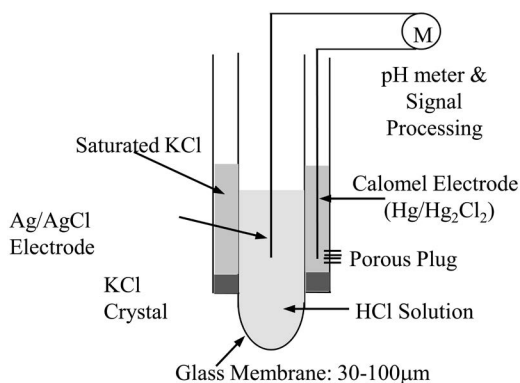


FIGURE 3.26 A glass electrode acidity measurement system.

with good conducting properties. A suitable reference electrode, calomel, and a sensing element inside the bulb provide electrical signals for pH levels.

3.21.1.41 Smart Sensors

Smart sensors have created new opportunities for wider applications advanced technologies such as wireless instrument networks (WINS). These opportunities depend on the development of scalable and low-cost sensor network architectures as well as cost effective RF communication technologies. Distributed signal processing such as the EDGE computing and AI decision-making enables events to be identified accurately by sensors. The information obtained by smart sensors is conveyed in message packets to a host station to make decisions.

3.21.1.42 Soft Sensors

Soft sensors (software-based sensors) are virtual sensors based on the data collected from multiple or single hardware sensors controlled by digital systems. A soft sensor creates hardware-sensor-like signals and function in making real-time analysis, monitoring processes, and making calculations. Algorithms for software sensor development require high quality historical real-life data. For this, historical data often manually treated to remove impurities generated by the hardware sensors. Good quality historical information extracted from the hardware devices is achieved by AI, machine learning, and data mining methods. During the operations, data driven methods are applied for soft sensor modeling. Some of these methods are principal component regression (PCR), PCA, SVM, neuro fuzzy systems, multilayer perceptron (MLP), and partial least square (PLS) methods. A soft sensor development process is illustrated in [Figure 3.27 \(a\)](#) and [\(b\)](#).

3.21.1.43 Stress and Strain Sensors

Stress and Strain sensors are used to sense the fractional changes in the dimensions of an object. An example of the strain measurement is the change in the length of an object divided by its original length; therefore, the strain does not have any

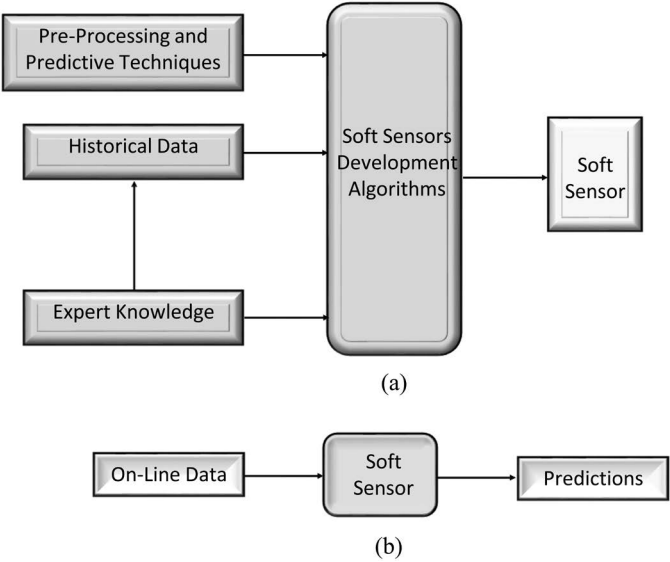


FIGURE 3.27 Soft sensor development process a) Data obtained from hardware sensors, and b) Data collected from the internet (Source: P. Kadlec⁽²⁸⁾).

dimensions. The stress on the other hand is the force applied on an object divided by the area. Consequently, strain takes place because of stress. There are different types of stresses, such as the tensile stress and the bulk stress. The *tensile stress* is the force divided by the area over which it is applied. The *bulk stress* on the other hand is the force per unit area. The most common strain sensors are produced for tensile strain measurements. The measurement of strain also allows calculation of the amount of stress by using the knowledge on the elasticity of the object. The information of elastic modules such as the Young’s modulus, shear (twisting) modulus, and the bulk (pressure) modulus of the objects are useful in performance calculations.

Sensing tensile strain involves the measurement of small changes in the length of a sample. The most common forms of strain measurement use resistive sensing, semiconductors, piezoelectric strain gauges, and optical interferometry.

The principles behind the resistive, semiconductor and piezoelectric types are explained elsewhere in this book. An advanced method measuring the strain is achieved by interferometry and fiber optic sensors. In these methods, high sensitivities are achieved particularly when laser technologies are used. Laser constitutes a coherent light beam resulting in good interference effect. Other methods involve various types of fiber optic sensors.

3.21.1.44 Temperature and Thermal Heat Sensors

Temperature and thermal heat sensors measure the intensity of heat. Temperature measurement and associated sensors are the most established, most diverse, and most used technologies. Much information in literature can be found on temperature

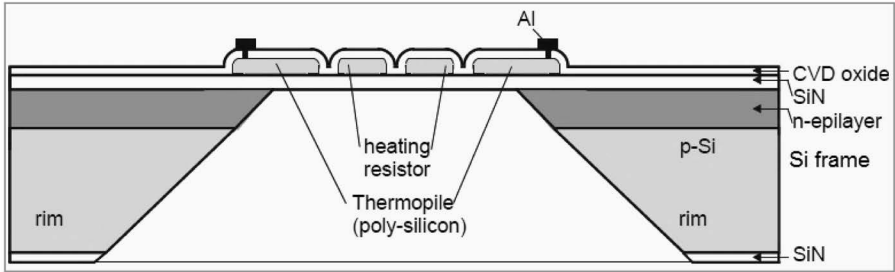


FIGURE 3.28 Schematic representation of a calorimeter performing three different functions. (Source: S. van Herwaarden ⁽²⁹⁾).

sensors (Webster and Eren). In this book only a few examples will be given. Simplest method of measuring temperature is the thermometers that make use of thermal expansion of materials such as liquid-in-glass. In electronic temperature measurements, there are many different methods and sensors, such as resistive sensors, thermoelectric sensors, thermocouples, thermostats, IC temperature sensors, piezoelectric sensors, and fusible links. A silicon-based calorimeter is illustrated in [Figure 3.28](#). This calorimeter consists of a heater and a temperature sensor on a thin-film frame connected to the silicon frame. The silicon frame gives mechanical stability to the calorimeter and acts as a thermal heat sink. This calorimeter performs three functions, the measurement of heat, the thermal effects and the temperature.

Other sensors are bolometers, Golay cells, Gardon gauges, phosphor thermometry, pyrometers, and others.

3.21.1.45 Time Measurements

Time measurements traditionally relied on mechanical methods. Mechanical methods have severe limitations in precision and accuracy particularly in measurements of sub-second time such as microseconds and below. Therefore, recently many electronic methods are developed. Majority of modern time measurements make use of crystal-controlled oscillators, exemplified in [Figure 3.29](#).

[Figure 3.29](#) illustrates a crystal oscillator with quartz and a coupling mechanism. Generally, the piezoelectric properties of the quartz crystals make it a good vibration

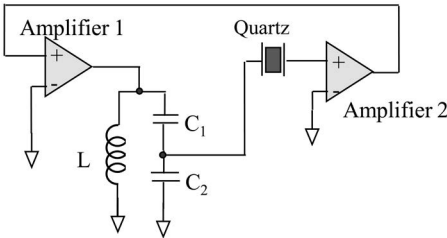


FIGURE 3.29 Block diagram of a crystal oscillator.

transducer, and the crystal is cut such that it can vibrate mechanically. This arrangement acts as an electrical resonator with practically no losses. In these resonators, the frequency is stable, and the oscillations can be maintained with little energy input. Crystal oscillators are used in domestic electronic clocks and watches where precision may not be all that important. For more precise timing, the crystals should be kept at a constant temperature. For utmost stability and precision maser oscillators are available.

3.21.1.46 Ultrasonic Sensors

Ultrasonic sensors make use of properties of sound. They are based on different principles such as Doppler shift, scattering, and time-of flight. Acoustic sensors are used in detection of defects, fluid and gas flow measurements, ultrasonic cleaners, security devices, elastic properties of solids, porosity and size estimations, distance measurements, acoustic microscopy, medical applications, bone and tissue scanning, motion and displacement measurements, and thickness measurements. Application of ultrasonics highly developed particularly in medical applications.

Ultrasonic sensors are developed to measure acoustic waves of frequencies higher than 20,000 Hz. They find wide applications including medicine, and industrial processes. Ultrasonic transducers are available over a wide range of frequencies, many sizes, and shapes. An ultrasonic transmitter and receiver in a single package are illustrated in [Figure 3.30](#). A common method of transduction is the piezoelectric material, but in some applications magnetostrictive materials can be used with some added advantages. Both piezoelectric and magnetostrictive transducers can function as transmitters and receivers at the same time. They can transmit and receive signals through air, solids, or liquids.

Ultrasonic detectors made from magnetoresistive materials exhibit Magnetostriction properties. Magnetostriction is the change of dimensions of magnetic materials as they are magnetized and/or demagnetized. They consist of a magnetostrictive metal core inside a coil. The electric waveform is applied to the coil that causes the core to vibrate due to magnetic induction. Several types of nickel alloys or alfenol have strongly magnetostrictive properties. Ultrasonic methods in the range of 20 kHz to 100 kHz are used in underwater communications and other related applications.

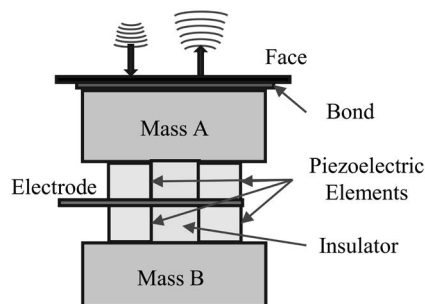


FIGURE 3.30 An ultrasonic transmitter and receiver in a single package.

3.21.1.47 Viscosity Sensors

Viscosity sensors find applications in measuring viscosities of liquids or gases. Viscosity of a fluid is a measure of its resistance to deformation at a given rate. It is a property of liquid describing how fast or slow liquid flows. Viscosity is temperature dependent. Under standard conditions, viscosity of water is 1.0016 mPa sec (milli Pascal per second), whole milk is 2.12 mPa sec, and viscosity of mercury is 1.526 mPa sec. Viscosity sensors make use of damping of mechanical oscillations by a viscous liquid or gas. A typical example involves the vibration of a plunger by a piezo-electric transducer, and amplitude of vibration is sensed by another transducer. This arrangement constitutes part of an oscillating circuit; the damping effect of liquid on the amplitude of oscillation is dependent on the viscosity of the liquid.

3.21.1.48 Web Sensors

Web sensors enable web connections without the need for any other intermediate devices. They are particularly useful for remote data acquisitions, environmental monitoring, wildlife monitoring, and so on. Internet-accessible sensors have IP addresses and communication capabilities so that users can access the sensor directly by wireless means or via a website. A typical arrangement of wireless web sensor node is illustrated in [Figure 3.31](#).

3.21.1.49 Other Important Sensors

There are two significant and fast developing and highly researched classes of sensors: nanosensors and quantum sensors. **Nanosensors** are produced from materials at nanoscale levels, and they are used in medical applications and environmental monitoring. Most of the nanosensors are made of nanoparticles, nanoclusters, quantum dots, carbon nanotubes, graphene sensors, metallic organic frameworks (MOFs), and plasmonic nanosensors. **Quantum sensors** make use of quantum properties in quantum mechanics. There are various types of quantum sensing such as quantum entanglement, quantum interference, and quantum state squeezing. Currently, quantum sensors are intensive research subjects. However, quantum sensing finds its roots in well-established techniques such as magnetic resonance imaging (MRI) by making use of nuclear or electronic spins as qubits. These newly developing sensing techniques have proven potentials to find applications in microbiology, microscopy, magnetic sensing, medical imaging, atomic clocks, gravitational wave detections, neuroimaging such as brain scanning, and in many other fields.

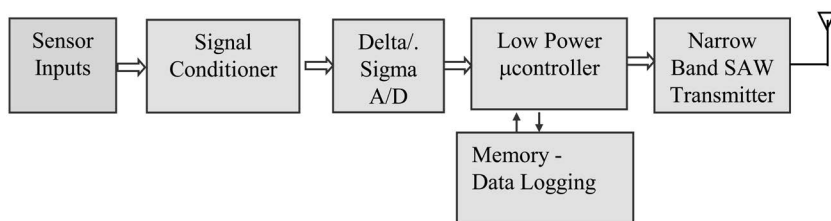


FIGURE 3.31 A wireless web sensor.

Other important sensors but are used less frequently are actigraphy, atomic force microscopy, catadioptric sensors, compressive sensors, hyperspectral sensors, Moire deflectometers, molecular sensors, smart dust, and so on

3.22 INTELLIGENT SENSORS

The term *smart sensor* was adopted in the mid-1980s to differentiate a new class of electronic-based sensors from the conventional ones. Smart sensors were neither supported by intensive software nor were intelligent enough as we understand intelligence today. They were assembly of electronic devices based on CPUs and functioned as smart instruments rather than stand-alone sensor devices. Smart sensors today computational sensors that can convert a raw signal into useful information and AI provides value added features increasing the quality of that information. Smart sensors can perform functions such as self-identification, self-testing, use of look-up tables, adjusting to predetermined calibration curves as well as communicating with other devices. Smart sensors also include intelligent internal operations such as signal amplifications, signal conditioning, signal processing, and A/D and D/A conversions.

Over the past several decades the dramatic increase in the availability of computational resources, coupled with the maturation of machine learning, has profoundly impacted sensor technology. By leveraging inverse design and AI techniques the data acquisition hardware can fundamentally be redesigned to “lock-in” to the optimal sensing data with respect to a user-defined cost function or design constraint. Computational sensors enabled by AI can therefore foster new and wide distributed applications that benefits from “big data” analytics to create powerful sensing networks, impacting various fields, including for example biomedical diagnostics, environmental sensing, and global health monitoring, and many others⁽³⁰⁻³³⁾.

The integration of sensors with complex analog and digital signal processing circuits and microprocessors in the same chip led to improved smart sensors and enabled extensive development of the supporting software. The use of digital signal processing circuits and integration of intelligent techniques such as the ANNs makes the sensors to serve as non-linear signal processing tools. On-board operating system and additional decision-making software and complex logic circuits result in much faster and much efficient, fault tolerant and reliable systems. A general scheme of a smart sensor is illustrated in [Figure 3.32](#).

A variety of smart sensors are manufactured with the neural networks and other intelligence techniques programmed and held on-board the chips. These sensors are capable of assimilating large quantity of data hence they can take autonomous and appropriate actions to achieve goals in dynamically changing environment. They are adaptable in anticipating events and complexities of the process therefore sensing, learning, and self-configurations are the key elements. Intelligent sensors appear in the marketplace as pressure sensors and accelerometers; biosensors; chemical sensors; optical sensors; magnetic sensors, and so on.

ANNs are applied in single sensor systems, redundant sensor systems, multiple sensor systems, and fully integrated decision and control systems. In *single sensor system*, application of AI results in improved systems linearity beyond the

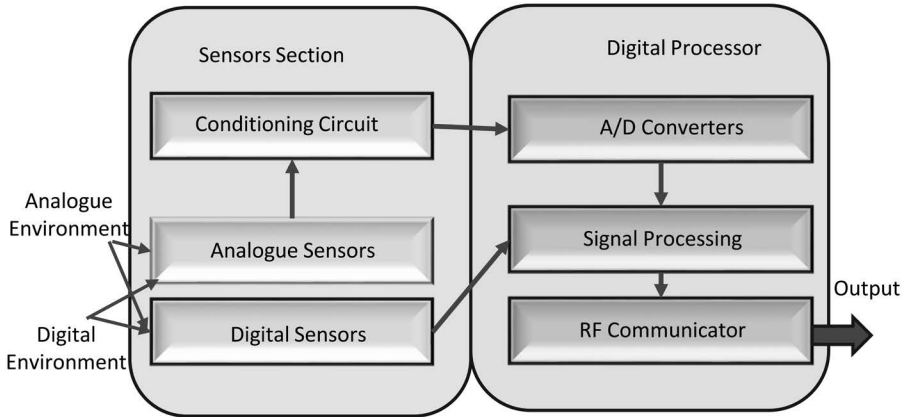


FIGURE 3.32 A smart sensor architecture.

conventional IC compensation. In *redundant sensor systems* where, multiple identical sensors are involved, the measurement accuracy and robustness are improved significantly. In *multiple sensor systems* where different types of sensors are used for measuring different physical parameters, AI improves linearity of individual sensors and assists in complex decision-making by sensor fusion. In *fully integrated decision and control systems*, AI performs both sensor enhancement as well as intelligent control. *Fully integrated* systems find extensive applications in aerospace and defense industries, consumer products, and other industrial needs.

In many cases, the network topologies, training algorithms, optimize parameters are readily available in the AI packages. AI can also be implemented by hardware and software combinations using CPUs array processors, and by ASICs. Manufacturers like Intel, Motorola, and other organizations have released specialized ICs for general purpose neural network implementations.

The NC3002 is an example of such sensors, which is based on the digital VLSI parallel processing technique. These types of sensors find applications in machine learning and image recognition supported by ANNs in quality determination and inspection applications. The architecture of NC3002 is structured in a way to implement the Reactive Tabu Search learning algorithm, which a competitive alternative to back-propagation method. This algorithm does not require derivatives of the transfer functions. The chip is suitable to act as a fast parallel number-crunching engine intended for operation with a standard CPU housed in single-chip or multiple-chip configurations^(30,34–39).

3.23 SENSOR PERCEPTION

An autonomous sensor system can collect data in a complex and dynamic environment, interpret, fuse data, and make decisions. The system may be regarded as an autonomous agent. If the time is involved the process is called active perception. An

active perception is time dependent and requires sensation and perception to become an active perception.

Sensation is the process where physical stimulus, or sensed data, enters the system. After the sensation, *perception* process interprets the stimulus and gives it a meaning. Perception takes place when an object and a subject encounter each other in such a way that the subject becomes conscious of the object. When time is added to the process perception becomes an *active perception*. In an artificial sensor system, active perception includes reasoning, decision-making, and acting upon the received information⁽⁴⁰⁾.

3.24 SENSOR ARRAYS AND MULTISENSORS

In many measurements, more than one sensor of the same type or different sensors are used, forming sensing arrays to provide sufficient information about the different aspects of a physical process.

3.24.1 SENSOR ARRAYS

Sensing arrays include several sensors for different measurands, such as pressure, flow, temperature, and vibration. These arrays are used for many reasons, for example, to increase the measurement range, to provide redundancy, or to capture information at different times and/or under different conditions. A good example of sensor arrays is in the chemical applications where a single chip with multiple sensors can be used to measure different types of chemicals. Multi-element gas analyzers are developed by using thin film detectors containing numerous sensors. Currently, there is considerable research and development effort spent on multiple sensors or sensing arrays integrated in the same chip together with signal conditioning components and computational capabilities.

CMOS technology allows integration of many sensors in a single chip, and it is a common technology applied in sensing arrays. Some examples of CMOS sensing arrays are photodiode arrays, ion detectors, moisture sensors, electrostatic discharge sensors, strain gauges, edge damage detectors, corrosion detectors, and so on. A basic structure of combinational CCD/CMOS technology is illustrated in [Figure 3.33](#).

Photodiode arrays are typical examples of AI related sensor arrays in image sensing. A photodiode consists of a thin surface region of *p*-type silicon formed on an *n*-type silicon substrate. A negative voltage applied to a surface electrode reverses the bias of the *p-n* junction. This creates a depletion region in the *n* silicon, which contains only immobile positive charge. Light penetrating into the depletion region creates electron-hole pairs, which discharges the capacitor linearly in time. There are two basic types of photodiodes: the serially switched photodiode arrays and the charge coupled photodiode arrays. In these arrays the basic principle is to use light intensity to charge a capacitor and then to read the capacitor voltage by shifting it through registers.

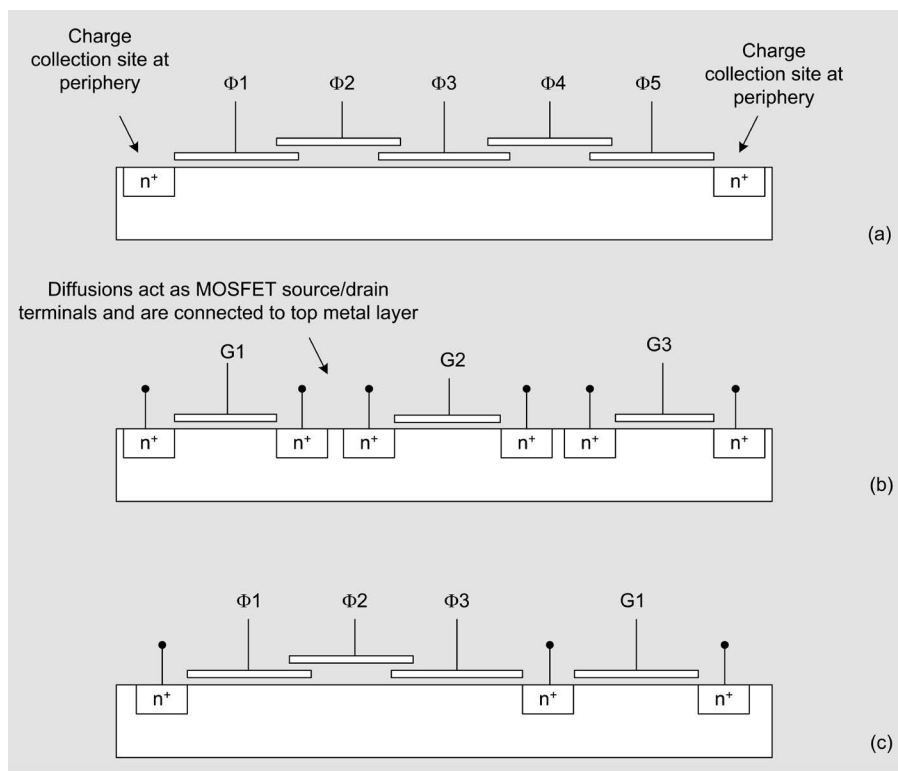


FIGURE 3.33 A basic structure of (a) CCD, (b) CMOS, and (c) combinational CCD/CMOS technology (Source: M. Habibi ⁽⁴¹⁾).

3.24.2 MULTISENSOR

Integrated multisensor chips attract attention of research and development. Multipurpose integrated sensor chips have been manufactured for simultaneous measurements of physical and chemical variables. IC technology allows design of complex systems that incorporate high performance analog subsystems such as op-amps and data converters on the same die with the digital circuits. These devices, generally manufactured by MOS technology, include signal conditioning, array accessing, and output buffering along with many sensors such as infrared sensing array, chemical sensors, accelerometers, vapor sensors, and tactile sensing arrays. Some of these multi sensing functions of these chips utilize both the pyroelectric and piezoelectric effects in ZnO thin films.

AI integrated microsensor chips constitute complex microsystems requiring high performance microelectronic components together with nonelectronic miniaturized subsystems. These chips can be categorized as MOEMS, MEMS, Lab-on-chip, RF MEMS, System-on-chip (SoC), Data Storage MEMS, and so on.

3.25 SENSOR FUSION

Multiple sensors are fused to obtain complete information about a process. For example, sensors in environmental applications can be pressure sensors, acoustic sensors, chemical, electromagnetic, thermal, proximity, and optical sensors. Each of these sensors has independent modalities and has different characteristics. The system configuration of a multisensor chip is shown in [Figure 3.34](#).

Various AI methods such as Deep Learning, CNN, RNN, Graph Neural Networks, Deep Belief Networks, Autoencoders, Hidden Markow Models, and Neural Networks are applied in sensor fusion. AI is used for data collection, uploading synthetic images, and post-mission analysis. Often, the goal is to identify individual sensor operations, to reduce uncertainty from multiple data sources, and to coordinate sensors for overall reliable and robust operations. Sensor fusion is a critical task in a wide range of applications, such as border security, healthcare, weather forecasting, IoTs, defense and military applications, navigation, autonomous vehicles, and communication systems^(42,43).

An important aspect of classifying fusion between sensors is the nature of the sensor creating different data. The source of data can be categorized as heterogeneous or homogeneous or both in large systems. Homogeneous data are typically simpler to fuse together since the data is already in a compatible form. If the sensor data are heterogeneous, most fusion methods require some level of preprocessing and a compatible means of feature extraction.

There is not a common model of sensor fusion in singular comprehensive system. Most existing models propose partitioning of the information fusion architecture based on how the source input information undergoes in preprocessing, classification, feature extraction, situation assessment, and decision-making. Classification is used in many multiple sensor systems. A level-based classification can be adopted as low-level, mid-level, and high-level sensor fusion corresponding to the data-level,

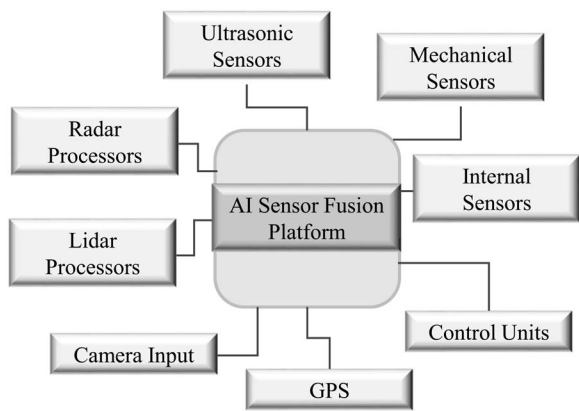


FIGURE 3.34 A multisensor chip.

feature-level, and decision-level fusion. Sensor fusion can also be split between preclassification and postclassification. The *preclassification* sensor fusion takes place before any classification occurs, while *postclassification* fusion takes place afterwards. Postclassification is a decision-level fusion having several variations such as abstract level fusion, rank-level fusion, measurement level fusion, and dynamic level fusion^(42,44–46).

3.25.1 ABSTRACT-LEVEL SENSOR FUSION

Abstract-level fusion includes methods like majority voting and weighted majority voting. Other fusions are rank level fusion, measurement-level fusion, and dynamic level fusion. *Rank-level fusion* generates, and assigns ranks to classes. It uses class set reduction or class set reordering methods to select subsets of decisions. *measurement-level fusion* uses classification and combination approaches. This method uses multiple classification approach to consolidate scores from each method. *Dynamic level fusion* uses winner takes all approach.

Sensor fusions are also categorized to be competitive, complementary, and cooperative fusion.

3.25.2 COMPETITIVE AND COMPLEMENTARY SENSOR FUSION

Competitive sensor fusion uses each individual modality to deliver independent measurements of the same property by using voting. This method improves reliability and accuracy. *Complementary sensor fusion* combines information from a phenomenon. In this methods, various algorithms are used including deep learning, hidden Markov models, and SVM. These algorithms can handle end-to-end learning from heterogeneous sensor inputs. *Cooperative sensor fusion* uses the information provided by two or more independent sensors to derive information that would not otherwise be available from individual sensors if they operated independently. Cooperative fusion is most suitable for the heterogeneous data.

Another classification for sensor fusion is decentralized (distributed) centralized, and hierarchical. Most sensor fusion architectures are *centralized* to provide measurements to a common processing unit. Pooling the information for classification and decision-making has advantages for efficient resource utilization. In a *decentralized* architecture there is no communication between the sensor nodes and each node uses its own processing abilities to fuse local information with the information coming from its peers. Decentralized sensor fusion operates autonomously. *Hierarchical methods* are a hybrid of the centralized and decentralized techniques. The system still centralizes the data, but some data is already processed in individual sensors thus reducing the communication needs.

Sensor fusion is the process of combining measurements from multiple nodes, each of which has a certain level of uncertainty. When these sources of information are fused together, it is not always clear how these uncertainties will interact and influence the overall performance of the sensor fusion algorithm. Uncertainty is important in sensitive applications. Using deep learning can be fooled by adversarial data and prone to generalizing poorly or overfitting the data. To improve such issues with uncertainty, further algorithms are implemented^(42,45–47).

3.26 SENSOR STANDARDS

There are numerous standards that are emerging for hardware architecture, software, and communications of modern intelligent sensors, instruments, and instrumentation. For example, the IEEE-1451 is a set of standards that define interfaces network-based data acquisition and control of smart sensors and transducers. These standards aim to make easy to create solutions using existing networking technologies, standardized connections, and common software architectures. The standard allows application software, field network, and transducer decisions to be made independently. It offers flexibility to choose the products and vendors that are most appropriate for a particular application.

At the process connection level, the IEEE-1451 provides standard ways of creating self-describing measurement and control devices. Sensors complying with these standards have on-board information on the serial numbers, calibration factors, and accuracy specifications. During the installation, location information can also be loaded so that the system can have self-describing system properties. The IEEE 1451 standards come in seven sections.

The first section of the 1451.1 is termed network capable application processor (NCAP) Information model. This section is concerned with the software architecture that moves the intelligence to device level. The 1451.1 approaches enable the benefits of objected-oriented technology. It creates flexible, natural software modules that allow engineers to think at the level of real-world systems, not at the level of programming languages. Object-oriented systems are built by assembly and produce systems that are much easier to adapt to new demands. The object-oriented technology makes open systems possible. This could have the same liberating effect in software configuration as open systems had on hardware. In this way, flexibility is achieved so that those systems can be assembled, reassembled, or modified quickly.

The second section, 1451.2 concerns with the transducers and Microprocessor Communication Protocols and Transducer Electronic Data Sheet (TEDS). The IEEE-1451.2 standard defines the transducer data and electronic interface of digital information direct from sensors, thus creating a modular architecture. The modular architecture allows embedding of the module to any field networks automatically and transparently.

The third section, 1451.3 is the Digital Communication and Transducer Electronic Data Sheet (TEDS) Formats for Distributed Multidrop Systems.

The fourth section, 1451.4 is called the Mixed Mode Communication Protocols and Transducer Electronic Data Sheet (TEDS) Formats. The 1451.4 standard defines a standard interface that uses the connecting wires for analog signals or for digital communications. IEEE 1451.4 offers a standard interface and protocol by which a sensor can describe itself over a network. With the advent and adoption of intelligent networked and wireless sensors, the notion of self-identifying devices may seem elementary, but this has taken more than a decade to happen with analog sensors⁽⁴⁸⁾.

The fifth section, 1451.5 IEEE standards for smart transducer interface for sensors and actuators – wireless communication protocols and TEDS format.

The sixth section, IEEE 1451.6, IEEE standards for smart transducers interface with sensors and actuators – Transducers to radio frequency identification systems communications protocols and TEDS formats. IEEE 1451.7 addresses RFID systems communication protocols.

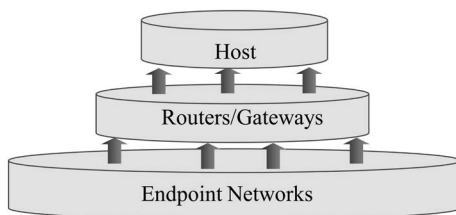


FIGURE 3.35 Wireless instrument network layers.

3.27 WIRELESS SENSOR NETWORKS

Wireless sensors and instrument enjoy a wide variety of network options. Apart from well-established networks such as the fieldbuses, there are many other alternatives either researched heavily and/or offered by many vendors.

The networks for wireless sensors and instruments have three layers (shown in [Figure 3.35](#)). These layers are networks at the endpoints at the physical levels, networks by using routers and gateways, and networks at the global level. All these optional networks can be realized by *sensor-sensor*, *sensor-computer*, and *human-system* communications.

Wireless sensor and instruments networks are arranged as structured networks or as self-organized networks. *Structured networks* represent the wireless distributed sensor networks (WDSN), and *Self-organized networks* represent smart wireless autonomous microsystems (SWAMs), wireless sensor networks (WSNs), and the wireless integrated sensor networks (WINS).

3.27.1 WIRELESS DISTRIBUTED SENSOR NETWORKS

WDSN require routers and gateways. A typical example is illustrated in [Figure 3.36](#). Routers enable to extend network coverage area and the gateways interface the network to host computer, LAN, or Internet.

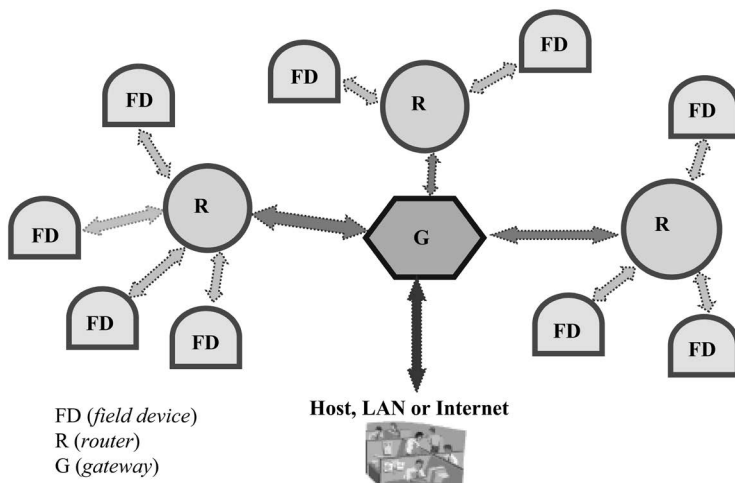


FIGURE 3.36 A typical smart wireless distributed sensors network (WDSN).

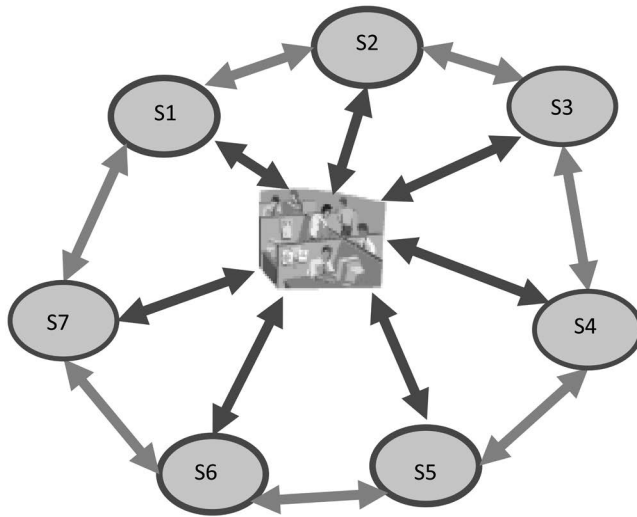


FIGURE 3.37 A typical smart wireless autonomous microsystem (SWAMS).

3.27.2 SMART WIRELESS AUTONOMOUS MICROSYSTEMS

In SWAMS, each sensor can be a master – transmits and receives by own initiative. This leads to *ad hoc* networking as illustrated in [Figure 3.37](#). This method is suitable for short distance configurations operating ISM frequency bands.

3.27.3 WIRELESS SENSOR NETWORKS

WSNs contain multiple sensors arranged structurally or self-organized form. WSN are made up from multiple fixed sensors or multiple mobile sensors or combination of the two.

If an instrument is wireless, it does not necessarily mean it is mobile. Paradoxically, if an instrument is fixed, it does not mean it is not mobile in its data handling capability. Nowadays, we can regard all sensors and instruments to be mobile if they have communication capabilities.

For wireless instrument systems, there are many techniques available including the public and private telephone networks, satellite networks, and fieldbus systems in industrial applications. The protocols need to support handover capability from one device to the next to realize mobility. Bluetooth for example uses Gaussian frequency shift keying (GFSK). Each device has 6-bytes device address using the same hop frequency from a Piconet. When a device is present in more than one Piconet, scatternet is established. A typical example of such networks is illustrated in [Figure 3.38](#).

3.27.3.1 Sensor Clouds

Sensor clouds are a new computation paradigm for WSNs. It allows coupling of multiple WSN for a single or multiple applications transparent to users. A sensor cloud is a heterogeneous environment that brings together multiple networks of different sensors. Smart sensors are used in sensor clouds are devices that sense the variables and

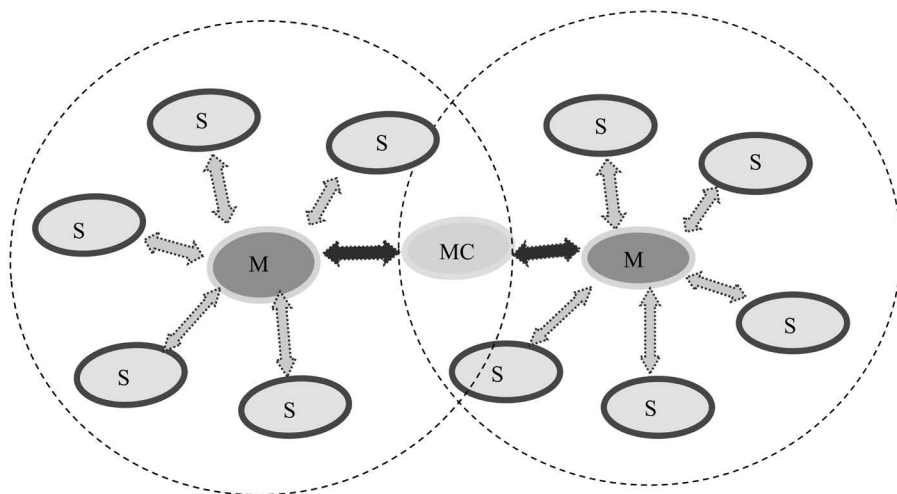


FIGURE 3.38 A typical scatternet-based network.

interprets the results making ready for the computerized systems such as Fog, Edge, or cloud computing. When it uses AI, it is known as *sense in science*. Applications include sound recurrence, temperature, pressure, camera pictures, shading, synthetic creation, and so on^(49–51).

An essential requirement in WSN is the localization in a distributes sensor network. Sensors are generally distributed randomly, therefore finding positions accurately is necessary. Locations can be determined by classical approaches or AI assisted methods. *Classical methods* two methods 1) range-based localization such as angle of arrival (AoA) or time of arrival (ToA), or time difference of arrival (TDoA), and 2) range-free localization such as pattern recognition and HOP count. *AI assisted methods* include Fuzzy Logic, AAN, SVM, and probabilistic reasoning in self-determined localization⁽⁶²⁾.

WSN is considered as the core IoT, and other wide range of applications where devices need to be connected dynamically to the internet. It has many novel applications in healthcare, military, environmental monitoring, ocean monitoring, smart homes, traffic control systems, disaster protection, engineering and science research, and so forth. WSN is tipped to have big potential in improving the living standards and better understanding of the environment. WSN has attracted considerable attentions in the applications of various AI techniques. Figure 3.39 shows some of the techniques applied in WSN that include Fuzzy Logic, evolutionary computation, multiagent systems, deep learning, and swarm intelligence^(53,54).

3.27.4 WIRELESS INTEGRATED NETWORK SENSORS

In wireless integrated network sensors (WINS), the micro-sensors are organized in nodes to fulfill the functions multi-hop routing, and self-organization, as illustrated in Figure 3.40. This method is used often in military applications, and it is gaining

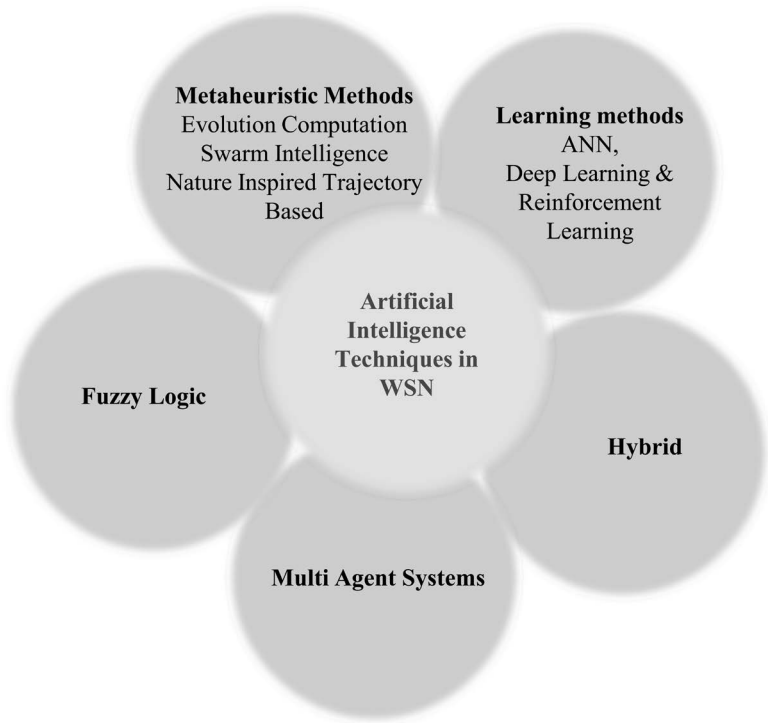


FIGURE 3.39 AI techniques applied in WSN.

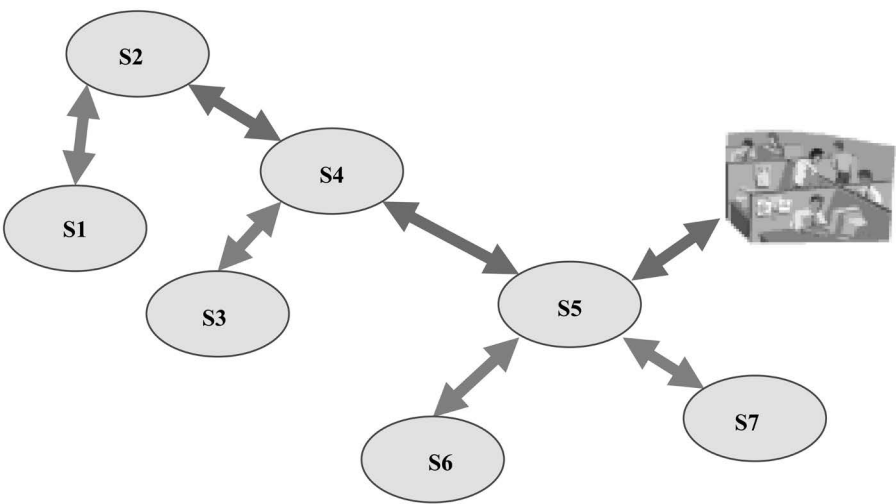


FIGURE 3.40 A typical wireless integrated network sensor (WINS).

wide range of industrial and civilian applications, therefore further information will be provided here. In this system, self-organization takes place by each device communicate with its nearest neighbor, thus forming a chain to pass information from one device to the next.

Wireless Integrated Network Sensors provide distributed network and Internet access to sensors. WINS development was initiated in 1993 at the University of California, Los Angeles; the first generation of field-ready WINS devices and software was demonstrated three years later. Later, the Defense Advanced Research Projects Agency (DARPA) sponsored low-power wireless integrated microsenors (LWIM) project demonstrated the feasibility of multi-hopped, self-assembled, wireless networks. This network also demonstrated the feasibility of algorithms for operating wireless sensor nodes and networks at the micropower levels.

Today, the basic network structure and supporting electronic components for WINS are not much different than WSN or WDSN, as explained in the previously. Therefore, the details of the components to configure WINS systems and network topologies will not be repeated here, instead, underlying characteristics and surrounding issues for WINS will be highlighted. However, most of these underlying principles and surrounding issues are equally valid to WSN and WDSN.

One of the main characteristics of WINS is that it originally was intended to facilitate monitoring and control in large scale applications such as transportation industry, manufacturing, healthcare, environmental monitoring, security systems, border controlling, traffic control in metropolitan areas, and so on. WINS aim to combine sensor technology, signal processing, decision capability, computation, wireless networking capability in a compact low-power system. The main feature of WINS is the RF communication capability, which generally but not always provides bidirectional network access for low-bit rate short range communications of sensors and instruments among themselves or with other host devices.

Most WINS systems aim to combine sensor technology with low-power sensor interface, signal processing, data conversion, computing, control functions, and RF communication circuits. The need for low-cost and low power consumption can present significant engineering challenges in the design and implementation of these systems particularly if the conventional digital CMOS technology is selected as the building block of the system. In some applications, the sensor component are required to operate continuously for event recognition, while the network interface is operating at low duty cycles.

In some systems based on WINS architectures, the sensors are required to detect the events continuously. All the components, sensor, data converters, buffer and other elements operate at micropower level. Upon detection of an event, the microcontroller issues a command for signal processing. Protocols for node operation then determine whether a remote user or neighboring WINS node should be alerted. The WINS node then supplies an attribute of the identified event, for example, the address of the event from a look-up table stored in all network nodes.

Multihop communications (shown in [Figure 3.41](#)) yield large power and scalability advantages for WINS networks. In this case, WINS node data is transferred over the asymmetric wireless links to an end user or to a conventional wireline or wireless IP network service through a network bridge. However, RF communication path loss has been a primary limitation for wireless networking in some operational ranges.

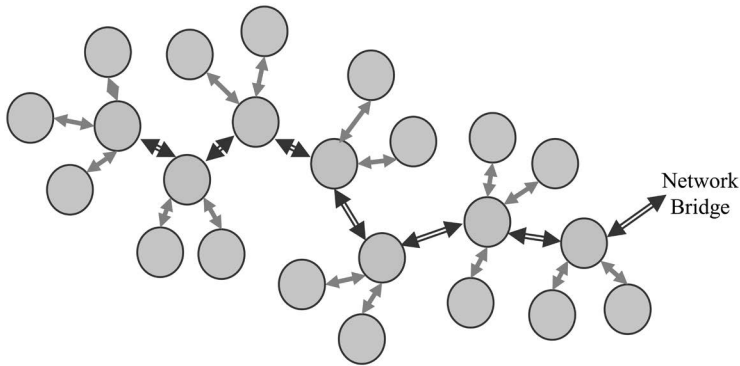


FIGURE 3.41 Multihop communication of WINS.

3.27.4.1 Communications and Network Protocols in WINS

Communication and networking protocols in WINS for the embedded radio network includes spread spectrum signaling, channel coding, and multiple access techniques such as the FDMA, TDMA and CDMA network protocols. The operating bands for the embedded radio are generally selected in the unlicensed bands at 902–928 MHz and 2.4 GHz. These bands provide compromise between the power cost associated with the high frequency operation and the penalty in the antenna gain reduction with decaying frequency for compact antennas^(53,54).

3.28 CONCLUSIONS

This chapter provides background knowledge on measurements, instruments, sensors, and communication of instruments and sensors. Essential components of instruments and sensors are explained. Wireless communication methods in instruments have been discussed and applications of various forms of AI has been highlighted. It has been shown that AI finds rich applications in wireless sensors, instruments, and instrumentation. The use of combined use of these modern technologies improves our living standards, protects the environment, and offers many new and novel application areas.

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4 AI and Wireless Systems Applications

4.1 APPLICATIONS OF AI

Artificial intelligence (AI) is applied extensively in many areas. Detailed applications relevant to sensors, instruments, and instrumentation are given throughout this book. Some of the common applications areas are agriculture, archeology, astronomy, automation, autonomous vehicles, biomonitoring, chatbots, climate monitoring, climate change, communications systems, computer vision, control systems, cyber security, education, entertainment, environmental monitoring, earth monitoring and earthquakes, facial recognition and image processing, financial services, governments, healthcare, image processing, industrial processes, instrumentation and measurements, lifestyle, marketing and sales, health and medicine, navigation, research and design, robotics, smart cities, smart homes, social media, space explorations, trade and commerce, text mining, travel and transportation, video and gaming, wireless communication networks, and many more. A brief overview of some of the applications is as follows:

Automation or Robotics uses AI techniques in automation and robotics for control and decision-making purposes.

Biomonitoring involves AI programs patient profiles to identify aberrant symptoms and their trends. Intensive research is ongoing to incorporate some additional data into the system, such as the history of medical records, to provide more training data.

Computer Vision has been very successful in biometric identification and classifying pictures using AI. Video games use AI extensively.

Financial Services use AI as companies are able to identify key insights into business data and verify financial fraud occurrences using AI. For example, through AI-based cyber-surveillance companies can predict financial risks and take measures against frauds.

Governments use AI for utilities, public protection, and the like. Various sources of data are used to identify insights for governmental decision-making.

Healthcare applications involve AI, particularly with the proliferation of wearable sensors and devices. Wearable sensors offer patient details, such as overall health status, blood pressure, pulse, and vital parameters in real-time. Such data are used scientifically to assess health condition by drawing trends from the patient's past and projecting the potential presence of any diseases. AI technology also allows medical professionals to evaluate data and learn patterns, helping to make better diagnoses and offer relevant therapies.

Marketing and Sales utilizes AI technology to analyze their customers.

Buying history of customers provides personalized product support for their next purchase and future.

Text Mining refers to the use of machine learning techniques for text analysis.

The text analysis process involves information retrieval and lexicon analysis from the text. The text mining applications like security applications, sentiment analysis, online media applications, business, and marketing applications. The purpose of text mining techniques is to transform the text into analytical data using the natural language processing approach.

Transportation based on the history and travel patterns across different routes.

AI helps the organization of transport to avoid route problems and traffic congestions. Transport typically uses AI techniques to interpret knowledge and model data for decision-making and to help clients make decisions during their journey.

4.2 AI AND WIRELESS COMMUNICATIONS

Wireless technologies and wireless networks are applied ubiquitously in data flow, mobile devices, machine-to-machine communications, network access in isolated areas, environmental monitoring, health and medicine, home and office networks, infrastructure health monitoring, plants and factories, smart cities, transport systems, military, border control, space explorations, wildlife and habitat monitoring, and many other areas.

Autonomous vehicles (AV) use AI and wireless systems in vehicles to run autonomously without human interaction. AI provides decision-making capabilities to control the vehicle. AI makes decisions by using the vehicle information, such as steering wheel control, required engine power, applications of breaks for automatic control, as well as communicating with other vehicles.

Autonomous driving combined with vehicle-to-infrastructure (V2I) communications enables automated vehicles to receive up-to-date information about neighboring vehicles and traffic conditions in the vicinity of the vehicles. This direct, dynamic connectivity is likely to enhance both safety and traffic efficiency. For example, deep reinforcement-learning (DRL) models are trained by interacting with their environment and have been developed to optimize the behaviors of vehicles.

Cyber-physical systems (CPSs) are engineered systems that integrate physical and digital components with real-time feedback and control loops. CPSs are applied in many areas, such as transportation, intelligent buildings, industrial processes, and healthcare. CPSs rely on advanced sensors, actuators, and software to track and control physical processes in dynamic environments.

Industrial control system (ICS) largely use deep neural networks (DNNs). DNN is investigated to increase the security and efficient flow of industrial processes. In some cases, DNN is trained on probability-based feature vectors that are gathered from process network packets.

Internet of things (IoT) is a network of devices connected through the Internet such that the connected devices can send and receive data from each other and from other devices. IoT plays an important role in adding value to human living standards.

IoT contribution to social well-being can be classified as tangible and abstract capabilities. The *tangible capabilities* provide quantitative metrics on the system it has been applied for, such as temperature, humidity, radiation, position, speed, information on intelligent building, and so on. The *abstract capabilities* are services not available by instruments, such as quality of living in smart cities, smart transportation systems, and traffic condition predictions.

Short-range communication protocols for IoT allow communication between user devices and neighboring devices, as well as long-range communication with clouds and cloudlets. For example, recent developments in narrowband IoT (NB-IoT) and cloud radio access network (C-RAN)-based IoT developments are expanding the communications with the cellular backbone.

IoT software platforms can be classified as M2M platforms, infrastructure platforms, hardware-specific platforms, and enterprise-related platforms. *M2M platforms* concentrate on the connectivity of devices to telecommunication networks. *Infrastructure platforms* are intended to provide processing power and hosting space for IoT applications. *Hardware-specific software platforms* are used to build specific IoT hardware items. *Enterprise platforms* cater to the consumer-specific software packages and integration of IoT devices in their systems. Some of the off-the-shelf programming languages and software that can be used are C, C++ Python, Java, NodeJS, NET, HTML5, CSS3, Android software development kits (SDKs), JavaScript, representational state transfer (REST), constrained application protocol (CoAP), and JavaScript object notation for linked data (JSON-LD).

Personalized medical care uses AI and wireless systems for personalized medical applications. For example, smartphone-based healthcare monitors physiological parameters and employs wearable devices for data acquisition. Sensors may be placed in the proximity of a person or on the person to monitor environmental parameters, body temperature, movement, ECG, and status of implantable devices. Wearable devices are organized into a wireless body sensor network (WBSN). Unlike traditional wireless sensor networks (WSNs), WBSNs comprise a limited number of sensors of a heterogeneous variety. Each sensor output is a signal with peculiar features in terms of bandwidth, dynamics, shape, and relevance. A smartphone is configured as a part of network.

Radar applications involve target surveillance in civilian as well as in military uses. Self-organizing mapping (SOM) networks are applied to find clusters in the target samples, which represent the minority class. The distribution characteristics of the original dataset are maintained for a more representative sample generation. Subsequently, the synthetic minority oversampling technique (SMOTE) is applied based on a given sample, and the cluster centers adjust the number of targets to the number of clutters by generating synthetic samples through interpolation. The transformed training set is fed into a random forest classifier comprised of in excess of 1,000 or more decision trees using the Gini index as a split criterion. The SOM-SMOTE random forest is compared to a regular random forest, a support vector machine (SVM), and a SOM-SMOTE SVM on the test set.

Remote control in the utility industry uses instrumentation, intelligent electronic devices, and Ethernet-based communication media, leading to widespread automation of power systems in transmission and distribution.

Remote sensing uses AI and wireless systems to extract information and knowledge from the stream of remote sensing data. Numerous AI algorithms have been applied to process and analyze remote sensing images for detection, classification, and regression. For this, DL architectures, e.g., convolutional neural networks (CNNs), achieve good results in many applications.

Smart Healthcare makes use of intelligence and knowledge where data is reliable and the rules of biology are well understood. Deep learning proves to be very effective for solving a range of biological variables, such as genes, RNA, proteins, biome, clinical, wearable, imaging, and symptoms. It can identify multiple forms of health preservation and disease. The data requirement is very high and may rely on the sharing of sensitive health data on a regular basis by many. In fields such as nutrition, exercise, and skin status, progress looks to be advancing quickly.

5G and 6G Networks offer many promising applications, including healthcare, industry, autonomous cars, e-voting, smart houses, and supply chain management.

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